Online Appendix to *Political Entrenchment and the Persistence of Economic Sanctions*

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This appendix accompanies the manuscript *Political Entrenchment and the Persistence of Economic Sanctions*. Part I holds the general results, which hold for abstract payoffs and transitions; Part II holds the parametric families and the numerical example economies; Part III holds the literature. In Parts I and II, what the main text needs comes first and derivative material follows. Part I opens with maps of how the results are proved (Appendix A). The Lean 4 development and the paper sources are public at `https://github.com/qidaolake/political-entrenchment-sanctions-lean`. Results and equations are numbered locally (Lemma A.1, equation (B.2), ...). Cross-references of the form "Theorem 4", "Section 3", or "(12)" point into the manuscript, not into this document.

How to read this appendix

Where the main text's proofs are. Each subsection of the main text opens with a footnote that points here; the map collects those pointers.

- Section 2 (the model): the bounds of the class and the precise conditions in panel A0 of Appendix C.4 and in Appendix B.1; the closed-form constants in Appendix N; the parametric examples in Appendix J.1; the example economies in Appendices J and L, and the numerical ones, among them the dual-basin economy, in Appendices K.1 and K.2; existence in Appendix F.
- Section 2.3 (the myopic benchmark) and Section 3.1 (foresight against it): panel F of Appendix C.4; the component statements in Appendix B.2.
- Section 3.2 (the target's and the sender's parts): Appendix C.5 and Appendix C.2; the strict comparison in Appendix B.3.
- Section 3.3 (lock-in): Appendix D; the low basin in Theorem D.5; an equilibrium with both basins in Appendix K.2.

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- Section 3.4 (patience): Appendices C.6 and C.7, and the general conditions in Appendix B.5.
- The end of Section 3.4 (minmax and uniqueness): the statements in Appendix B.4, proved in Appendix C.7.
- Section 4 (welfare): the proofs in Appendix C.9; Appendix L and Appendix B.6 for the interest-group polity and the family's microfoundations.
- Section 5.1 (one government forward-looking): the component statements in Appendix B.2, proved in Appendix C.4; the welfare reversal across the four cases in Appendix C.9.
- Section 5 (discussion): Appendices D and E.

How the appendix is organized. The appendix has three parts, and each of the two larger parts puts what the main text needs before derivative material. Part I holds the general results, which hold for abstract payoffs and transitions in the class, with no functional form. What the main text needs comes first: maps of how the results are proved (Appendix A); the conditions in full and the component statements of the main results (Appendix B); the proofs (Appendix C), among them the construction on the class under the lifting condition with its class-membership panel (Appendix C.4); persistence, temporary shocks and lock-in, with the low basin, relief and the exit duration (Appendix D); and the two-channel decomposition and the path of the state (Appendix E). The derivative and supplementary results follow: the existence machinery (Appendix F), the assumption-discharge routes (Appendix G), the robustness and boundary results (Appendix H) and further results on persistence and the long run (Appendix I). Part II holds the examples: the ones the main text uses, the parametric families with the benchmark economy (Appendix J), the numerical example economies (Appendix K) and the microfounded interest-group example (Appendix L); then further example economies (Appendix M) and the closed-form constants of the parametric family (Appendix N). Part III places the paper in the literature (Appendix O). The formalization map, which lists for every statement the machine-checked declaration that proves it, is in the formal verification supplement. Every cross-reference in the manuscript resolves into this document.

Notation and symbols

This list collects the symbols used repeatedly in the manuscript and appendix. Symbols confined to a single proof, proposition, or certificate table are defined locally where they first appear.

Symbol	Meaning
$t;\ x_t,\ a_t,\ s_t$	Date $t \in \mathbb{N}$; political state, target compliance, and sender sanction at date t, each of the three taking values in $[0, 1]$.
$F(x, s)$	General state-transition law. Conditions (F1)–(F5) impose continuity, monotonicity, diagonal fixed points, and strict adjustment toward the current sanction.
δ	Adjustment rate in the partial-adjustment law $F(x, s) = (1-\delta)x+\delta s$.
β_T, β_S	target and sender discount factors.
U_T, U_S	target and sender one-period government objectives.
V_T, V_S	target and sender continuation-value functions.
T_T, T_S	target and sender Bellman operators.

Symbol	Meaning
γ_F	Optional inherited-state Lipschitz modulus of F at a fixed sanction.
$J_S(x, a, s; V_S)$	Dynamic sender objective $U_S(x, a, s) + \beta_S V_S(F(x, s))$.
σ^{M}	The sender's myopic best response.
$\sigma^{\mathrm{dyn}}_{V_S}$, $\sigma^{\mathrm{dyn}}_{V_S}$	The sender's dynamic argmax and its greatest selection, the dynamic best response, at continuation value V_S.
α, α^{M}	Equilibrium target policy and the independently optimized myopic target policy.
α_{V_T,V_S}	target argmax policy induced by a value pair in $\mathcal{C}_T \times \mathcal{C}_S$.
E, $\mathcal{E}$	A stationary feedback Stackelberg equilibrium $E = (V_T, V_S, \alpha)$ and the set of all such equilibria.
$M = (\alpha^{\mathrm{M}}, \sigma^{\mathrm{M}})$	Myopic equilibrium. This roman M is distinct from the bold row-Lipschitz matrix $\mathbf{M}$ below.
(x^E_t, s^E_t), (x^M_t, s^M_t)	Realized state and sanction paths of the equilibrium and of the myopic equilibrium from the common initial state x_0.
$\bar{x}_T, \bar{x}_S$; $\bar{x}$; $\mathcal{B}$	target and sender tipping thresholds; their joint pivot $\bar{x} = \max\{\bar{x}_T, \bar{x}_S\}$; and the high basin $\mathcal{B} = [\bar{x}, 1]$.
$\bar{s}$	Sanction-floor parameter used to construct the flip sets.
$\mathcal{C}_T, \mathcal{C}_S$	target and sender value sets.
$L^{\alpha}_{S,x}$, K_S	Composite sender stage-payoff state modulus and the resulting Lipschitz constant of the sender's value set. The latter is distinct from the sans-serif commercial-lobby label K_S.
m_S, c_σ	sender stage lower-slope level and strict-concavity modulus.
$\mathcal{F}_T(\bar{s})$, $\mathcal{F}_S(\bar{s}; m_S)$	target and sender upper flip sets used to derive the basin thresholds.
x^0_S, $x_S(a)$	Static sender flip and the sender's per-action flip threshold.
$\mathbf{M}$	Joint Bellman row-Lipschitz matrix with entries $\beta_T, L_{TS}, L_{ST}, \beta_{S,\mathrm{row}}$.
$\rho(\mathbf{M})$	Spectral radius of the row-Lipschitz matrix.
$R(x)$, $G(x)$	Sanction rule of the equilibrium and closed-loop state-update map, $R(x) = \sigma^{\mathrm{dyn}}_{V_S}(x, \alpha(x))$ and $G(x) = F(x, R(x))$.
00, 01, 10, 11	The four cases of one economy, target-first: the first index is 1 when the target is forward-looking ($\beta_T > 0$) and 0 when it is myopic, the second the same for the sender; R^{ij}, L^{ij} and a^{ij} are the sanction rule, the limit of the political state and the target policy in case ij (Section 5.1).
w_{lo}, w_{hi}, q, $\bar{Q}$, r_{lo}, r_{hi}, k_{lo}, k_{hi}, u_{lo}, u_{hi}, σ_x, τ_x	The class's sender rates: state slope, sanction curvature, state–sanction cross rate, substitution rate and marginal return to the first unit of sanction, each as a lower and an upper bound, and the state semiconcavity and semiconvexity moduli (panel A0 of Appendix C.4).
ℓ_{lo}, ℓ_{hi}, p_{lo}, c, v_{lo}, n_{lo}, v_{hi}, n_{hi}	The target's rates: compliance cost, sanction pain, and the state slope $v + ns$ at sanction s, each as a lower and an upper bound (panel A0).
$\underline{s}$	The lower bound $\sigma^{\mathrm{M}}(0, 0)$: the myopic sanction at full noncompliance at the lowest state (u_0/q on the quadratic–linear family when $u_0 \le q$, as under interiority).

Symbol	Meaning
$(q, u_0, k, r, w, \ell, c, n, v)$	The quadratic–linear family's nine payoff coefficients: sanction-cost curvature, baseline marginal return to sanctioning, substitution rate, the sender's state–sanction cross term, sender state valuation, compliance cost, target sanction pain, the target's state–sanction cross term and target state valuation (Appendix J.1); the transition's slope bounds and moduli $g_{\mathrm{lo}}, g_{\mathrm{hi}}, d_{\mathrm{lo}}, d_{\mathrm{hi}}, M_x, M_m, M_s, M_{\mathrm{cx}}, M_{\mathrm{mix}}, M_{\mathrm{cs}}$ are defined in Appendix C.4.
$x_F,\ x_L$	Least basin fixed point of G and the limit selected from the basin's lower endpoint.
σ_{real}	A sanction rule held fixed in a transition-law comparison.
$d_F, \eta_F, \widehat{N}_F$	Full-sanction drift, its minimum below the basin, and the resulting conservative shock-duration bound.
$P_3,\ h$	The labeled parameter point P_3 of the mechanism-active example economy (a point, distinct from the persistence condition P3 of Theorem D.2) and the half-width of the box transition shock used in the purification construction.
$\mathsf{J}_S, \mathsf{K}_S, \mathsf{J}_T, \mathsf{K}_T$	Pro-sanction coalition, sender commercial lobby, target regime bloc, and target opposition in the interest-group polity. Sans-serif lettering distinguishes these group labels from value-function constants.
$W_g, \theta_g,\ W$	Welfare of group g, its political weight, and aggregate welfare, obtained by setting political weights to one.
$g_J, q_K, b_J, b_K, k_S, d_s$	Interest-group welfare coefficients: sender sanction benefit and loss, target state valuations, sender sanctioning-cost curvature, and target sanction-cost rate.
β (planner)	Planner discount factor; it is local to that comparison and is distinct from the government discount factors β_T, β_S.

Part I
General results: abstract payoffs and transitions

What the main text needs: the conditions, the construction and the proofs

A How the results are proved: maps of the arguments

The figures in this section show how the paper's results are built. Each box says in words what a condition requires, what a step constructs, or what a result states. A solid arrow means that the box at its tail is used to obtain the box at its head. A dashed arrow means that two results are compared. A dotted arrow marks a step that the result does not need. Shaded boxes are conditions, plain boxes are steps, and boxes with a heavy border are results. Below each figure, three lines say what the result proves, what it uses, and the route of the proof; the proofs themselves are in the sections the figures cite.

A.1 The overall structure

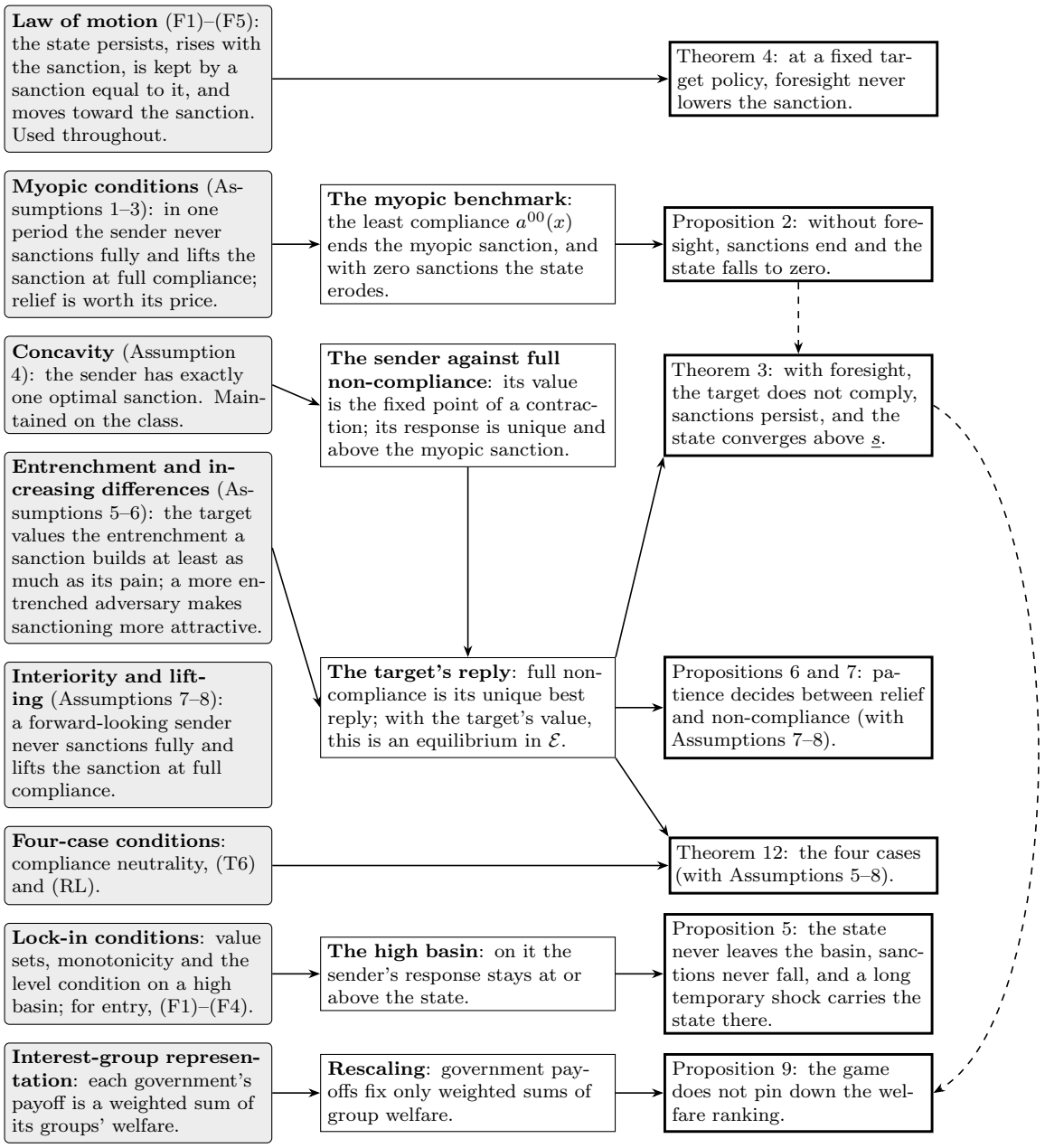


Figure 1: The structure of the argument. Left: the conditions, in words. Middle: what the proofs construct from them. Right: the paper's results. The law of motion is used by every construction; its one arrow marks the result that needs nothing else from the class (Theorem 4). Interiority and lifting are named in the result boxes that use them. The dashed arrows mark comparisons: Theorem 3 compares with the benchmark, and Proposition 9 ranks the two equilibria.

Reading. Three conditions matter most. Concavity makes the sender's response unique, the entrenchment condition makes full non-compliance the target's best reply, and increasing differences make the sender's response rise with the state. The myopic conditions give the benchmark. The other conditions each serve particular results.

A.2 The constructed equilibrium, Proposition 2 and Theorem 3

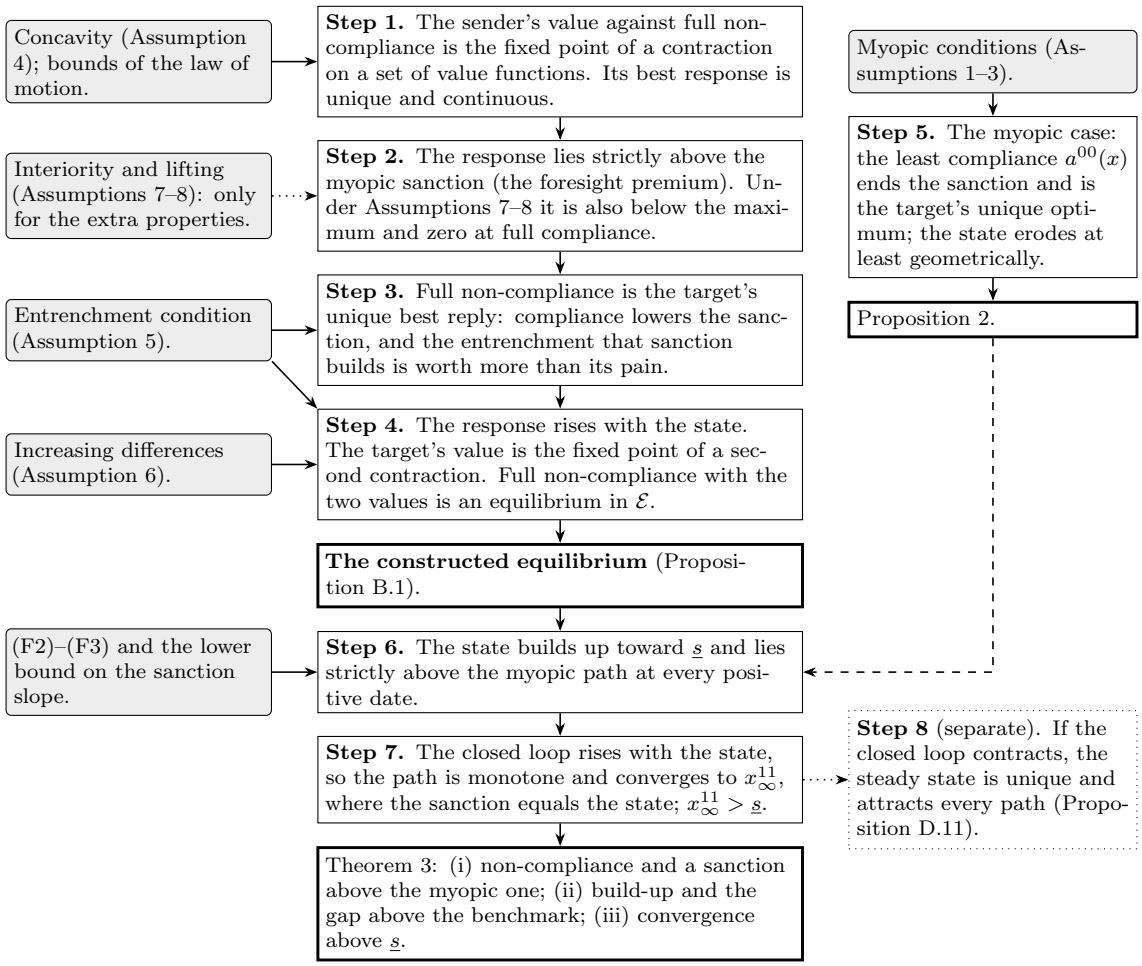


Figure 2: The proof of Proposition 2 and Theorem 3 (Appendix C.4, panels E and F). Steps 1–4 construct the equilibrium with full non-compliance; Steps 6–7 follow its path. Step 5 solves the benchmark, and the dashed arrow is the comparison in (ii). The dotted arrows mark what the theorem does not need: the extra properties of Step 2, and Step 8.

Proves. Both forward-looking: non-compliance, positive sanctions, and a state that converges above $\underline{s}$; both myopic: relief and a state that falls to zero.
Uses. Assumptions 1–6, the law of motion and the bounds of the class.
Route. Construct the sender's value against full non-compliance first, check that full non-compliance is then the target's best reply, and close the equilibrium with the target's value; then follow the path.

A.3 The comparison at a fixed target policy (Theorem 4)

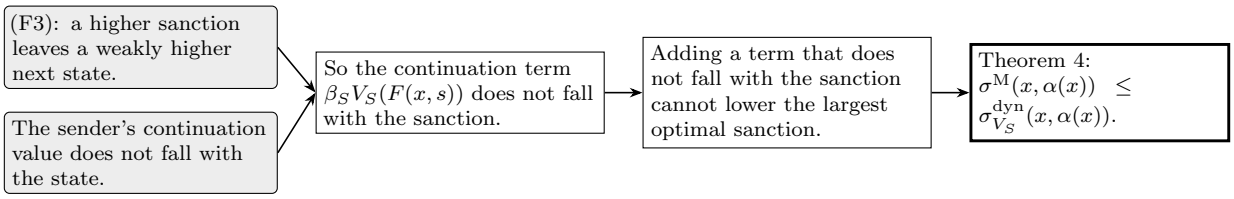


Figure 3: The proof of Theorem 4 (Appendix C.2). It needs no concavity, no equilibrium and none of the bounds of the class.

Proves. At any fixed target policy and any state, a sender that values a more entrenched adversary sanctions at least as much as a myopic sender.
Uses. (F3) and a sender value that does not fall with the state.
Route. A monotone comparison of the largest maximizers of two objectives that differ by a term rising in the sanction.

A.4 Lock-in (Proposition 5)

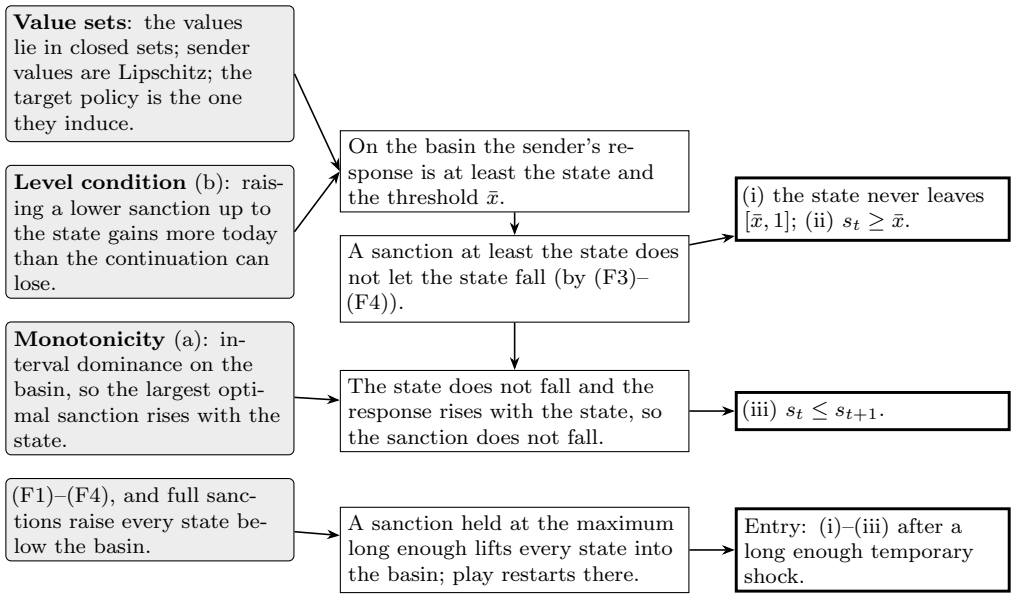


Figure 4: The proof of Proposition 5 (Theorem D.2 in Appendix C and the shock theorems of Appendix D). The level condition gives no exit; the monotonicity condition adds the rising sanction; entry needs only the law of motion.

Proves. Once the state is in a high basin, equilibrium play keeps it there with sanctions that never fall, and a long enough temporary shock carries it there.
Uses. The value sets, the level condition, the monotonicity condition, and for entry (F1)–(F4) with full sanctions raising every state below the basin. No condition on how the target ranks sanctions.
Route. A level argument for the sanction, then the law of motion for the state, then monotonicity for the sanction path.

A.5 The four cases (Theorem 12)

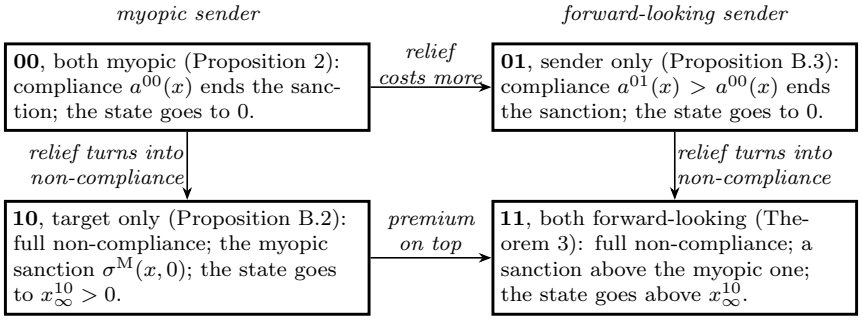


Figure 5: The four cases of one economy (Theorem 12). A downward arrow adds the target's foresight; a rightward arrow adds the sender's. Conditions: Assumptions 5–8, compliance neutrality, (T6) and (RL); the components are in Appendix B.2 and their proofs in Appendix C.4.

Proves. The target's foresight decides between relief and non-compliance; the sender's foresight changes the price of relief, or the level at which the state settles.
Uses. Assumptions 5–8, compliance neutrality, (T6) and (RL).
Route. Solve each case on its own, then compare neighboring cases, which differ in one discount factor.

A.6 Welfare non-identification (Proposition 9)

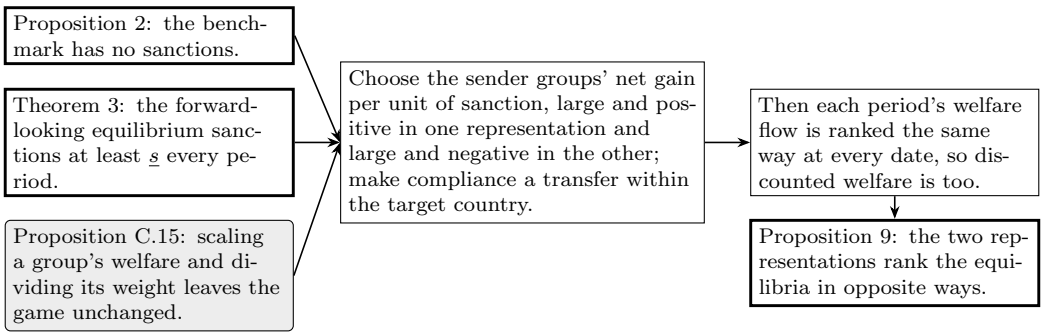


Figure 6: The proof of Proposition 9 (Appendix C.9).

Proves. Two interest-group representations of the same government payoffs rank the benchmark and the forward-looking equilibrium in opposite ways, at every start and every common planner discount factor.
Uses. Assumptions 1–5 and a sender response at full non-compliance that does not fall with the state.
Route. The game fixes only weighted sums of group welfare, and the two equilibria differ by a sanction of at least $\underline{s}$ in every period.

B The conditions in full and the component statements

The main text states its most important results in full and the others in words. This section holds the precise statements of the others, with their discussions, and the precise forms of the conditions that the main text names in words. Every cross-reference to them resolves here.

B.1 The bounds of the class and the conditions in full

Except where a statement names the quadratic–linear family of Appendix J.1, the results use no functional form for the transition or for either payoff. They hold on the *general class* of economies defined by (F1)–(F5), by bounds on the transition's slopes and curvature, and by bounds on the payoffs' rates. Subscripts lo and hi mark a lower and an upper bound. The transition is constrained only through its chords, with no derivative assumed. Its *slope bounds* bound the slope, or difference quotient, of every chord, with the state slope in $[g_{\mathrm{lo}}, g_{\mathrm{hi}}]$ at every sanction and the sanction slope in $[d_{\mathrm{lo}}, d_{\mathrm{hi}}]$, $d_{\mathrm{lo}} > 0$. The transition also satisfies a joint semiconcavity bound. Some results read two further curvature bounds (Appendix C.4). One is a semiconvexity modulus M_{cx}, which limits how far the transition can bend in the state. The other is a two-sided bound M_{mix} on mixed differences, which limits how much the state changes a sanction's effect on the next state. These two moduli are read only by results that bound the curvature of the sender's value from above and below, and by one route from bounds on the target's policy. So is the sender's semiconvexity modulus τ_x below. Among those results is the clause of Proposition B.1 that uses the increasing-differences condition (B.4). Membership of the class does not require these three moduli.

The rate bounds fall into four groups. *The sender's stake*: its payoff rises with the state at a rate of at least $w_{\mathrm{lo}} > 0$, and at full non-compliance at most w_{hi}, with a semiconvexity modulus $\tau_x \geq 0$ in the state there. *The sender's sanction choice*: the curvature of its sanction cost lies between q and $\bar{Q}$, and $q > 0$ follows from the concavity condition of Section 2.3. At full non-compliance its marginal return at the first unit of sanction lies in $[u_{\mathrm{lo}}, u_{\mathrm{hi}}]$, with $u_{\mathrm{lo}} > 0$. A higher state raises that return at a rate of at least $r_{\mathrm{lo}} \geq 0$, the sender's *state–sanction complementarity*. Compliance lowers it at a substitution rate of at least $k_{\mathrm{lo}} > 0$, and some results also assume an *upper substitution rate* k_{hi}. *The target's costs*: compliance costs between $\ell_{\mathrm{lo}} > 0$ and ℓ_{hi} per unit, and the sanction's marginal pain lies in $[p_{\mathrm{lo}}, c]$, with $p_{\mathrm{lo}} > 0$. *The target's stake*: its payoff rises with the state at a rate between $v_{\mathrm{lo}} + n_{\mathrm{lo}} s$ and $v_{\mathrm{hi}} + n_{\mathrm{hi}} s$ at sanction s, with $v_{\mathrm{lo}} > 0$ and $n_{\mathrm{lo}} \geq 0$.

The theorems use five constants built from these payoffs and bounds. Two of them define the entrenchment condition and its strong form. Write $K_S = w_{\mathrm{hi}}/(1 - \beta_S g_{\mathrm{hi}})$ for the Lipschitz constant of the sender's value in the state, and $A = u_{\mathrm{hi}} + \beta_S K_S d_{\mathrm{hi}}$ for an upper bound on the sender's dynamic marginal return to sanctioning at full non-compliance. Write $\underline{s} = \sigma^{\mathrm{M}}(0, 0)$ for the sanction that maximizes the sender's per-period payoff against full non-compliance at $x = 0$ (the myopic best response σ^{M} is defined in Section 2.3). Finally, write

$$g_T^0 = \frac{v_{\mathrm{lo}}}{1 - \beta_T g_{\mathrm{lo}}}, \qquad g_T = \frac{v_{\mathrm{lo}} + n_{\mathrm{lo}} \underline{s}}{1 - \beta_T g_{\mathrm{lo}}} \tag{B.1}$$

for a lower state slope of the target's value and its refined version, which uses $\underline{s}$ as a floor on the sender's response at non-compliance: the target's own stake per period, discounted over the persistence of the state. The two slopes give the *entrenchment condition* (Assumption 5)

$$c \leq \beta_T \, d_{\mathrm{lo}} \, g_T, \tag{B.2}$$

under which the discounted value of the entrenchment a unit of sanction builds covers the incumbent's current pain from it, and the *strong entrenchment condition* $c \leq \beta_T d_{\mathrm{lo}} g_T^0$ (Section 3.4). The strong form uses the unrefined slope, so it is more demanding. On the sender's side, write $g_S = w_{\mathrm{lo}}/(1 - \beta_S g_{\mathrm{lo}})$ for the least rate at which a sender value in the invariant set rises with the state. Under interiority the forward-looking sender's sanction at non-compliance exceeds the myopic sanction $\sigma^{\mathrm{M}}(x, 0)$ at every state by at least

$$\frac{\beta_S \, g_S \, d_{\mathrm{lo}}}{\bar{Q}} > 0, \tag{B.3}$$

the premium bound of Section 3.2: the numerator is the least that a unit of sanction adds to the sender's discounted value through the next state, and $\bar{Q}$ converts that marginal gain into a sanction increase.

Two remarks of the earlier main text belong here. The formalization maintains (F2) and (F3) on all of $\mathbb{R}$ at no cost in generality: a transition satisfying (F1)–(F4) on the square extends globally, unchanged on the square, with the same properties, and a machine-checked constructor builds the formal economy from square-defined data (Appendix H.7). For the closed-form constants of Appendix N the payoffs are also twice continuously differentiable on a neighborhood of $[0,1]^3$; every theorem in the formalization holds without this (Appendix H.7).

Of the conditions listed below, only the concavity condition is part of membership of the class. Every other condition is a hypothesis of the results that read it. The concavity condition says that the curvature q of the sender's sanction cost exceeds the curvature that the continuation term can add. This curvature is computed with a curvature bound σ_V on the sender's value that the Bellman operator preserves. Such a bound exists exactly when an explicit inequality on the primitives holds (Appendix C.4). Under this condition the sender's dynamic objective is strictly concave in the sanction. Interiority $A < q$ implies the myopic interiority $u_{\mathrm{hi}} < q$, under which a myopic sender never sanctions fully, and some results read only that (Theorem 3 and Proposition B.2(iv)). The lifting condition $A \le k_{\mathrm{lo}}$ implies $u_{\mathrm{hi}} < k_{\mathrm{lo}}$, under which full compliance ends a myopic sender's sanction. That is all the myopic equilibrium reads of it. The myopic relief condition $\ell_{\mathrm{hi}}\bar{Q} < k_{\mathrm{lo}}p_{\mathrm{lo}}$ is a one-period margin with the sender's response substituted. The reversal toward non-compliance is a continuation-value result. Five bounds are not required of every economy in the class. For the target, they are the upper bound ℓ_{hi} on its compliance cost, the lower bound p_{lo} on its pain and the upper bound $v_{\mathrm{hi}} + n_{\mathrm{hi}}s$ on its stake. For the sender, they are the bound away from non-compliance on its state–sanction complementarity and the upper substitution rate k_{hi}. Propositions C.7, C.8 and C.12 hold without them, and the other results that use k_{hi} take it as a hypothesis. Every example economy in the class has an upper substitution rate and the moduli M_{cx}, M_{mix} and τ_x. Each theorem-specific condition does one economic job and is read by the results named with it.

- *Concavity condition* (membership): a well-defined dynamic best response and controlled continuation values.
- *Interiority* $A < q$: an interior dynamic best response at non-compliance. The myopic and target-only cases, and the result that the sender's forward-looking incentive changes behavior, read only the myopic interiority $u_{\mathrm{hi}} < q$ it implies.
- *Myopic relief condition* $\ell_{\mathrm{hi}}\bar{Q} < k_{\mathrm{lo}}p_{\mathrm{lo}}$: at the margin, a unit of compliance costs at most ℓ_{hi} and buys at least $k_{\mathrm{lo}}/\bar{Q}$ units of sanction reduction worth at least p_{lo} each, so the myopic target accommodates. It is read by the myopic case and by the threshold discount factor of Proposition 6, and implied by the forward-looking relief condition (RL) of Proposition B.3.
- *Lifting condition* $A \le k_{\mathrm{lo}}$: full compliance drives the sanction to zero. It is read by cases 00 and 01 (myopic target) among other results, case 00 only through the inequality $u_{\mathrm{hi}} < k_{\mathrm{lo}}$ it implies (full compliance ends a myopic sender's sanction).
- *Compliance neutrality* (2): it determines the sender's value along relief before the target policy is known, and only the results that construct relief read it.
- *Entrenchment condition* $c \le \beta_T d_{\mathrm{lo}} g_T$: the discounted value of the entrenchment a unit of sanction builds covers the incumbent's current pain from it. It is read by cases 10 and 11 (forward-looking target) among other results; its strong form $c \le \beta_T d_{\mathrm{lo}} g_T^0$ is read by the minmax-payoff and uniqueness results.

- *Increasing-differences condition* (B.4): a nondecreasing sender response at non-compliance, read by the constructed full-non-compliance equilibrium of Proposition B.1.
- $\ell_{\mathrm{hi}} < \underline{s}\, p_{\mathrm{lo}}$: full compliance beats full non-compliance in the one-period problem. It is read by the clause of Proposition B.1 that the forward-looking incentives change behavior and by the two results of Appendix C.4 that derive the forward-looking relief condition (RL) from it.

B.2 The components of the four-case theorem

Proposition B.1 constructs the full-non-compliance equilibrium used in case 11; Theorem 3 and Propositions B.2 and B.3 are the components of Theorem 12 in the main text.

Proposition B.1 (The sender's response under interiority and the lifting condition)**.** *For every economy in the class satisfying interiority $A < q$, the sender's value against full non-compliance is the Banach fixed point of its Bellman operator on an invariant set of Lipschitz, lower-sloped, curvature-bounded value functions, and its dynamic best response never reaches the ceiling: every maximizer of J_S lies in $[0, A/q]$ with $A/q < 1$, and at full non-compliance it is strictly interior and strictly above the myopic sanction $\sigma^{\mathrm{M}}(x, 0)$ ($(u_0 + rx)/q$ on the family); under the lifting condition it is zero at full compliance. If in addition the transition has a state-section semiconvexity modulus M_{cx} and a bound M_{mix} on mixed differences, and the sender's payoff a semiconvexity modulus τ_x in the state (Appendix B.1), and the economy satisfies the entrenchment condition and the increasing-differences condition*

$$\beta_S\big(\sigma_C g_{\mathrm{hi}} d_{\mathrm{hi}} + (K_S + 2\sigma_C) M_{\mathrm{mix}}\big) \le r_{\mathrm{lo}}, \qquad \sigma_C = \frac{\tau_x + \beta_S K_S M_{\mathrm{cx}}}{1 - \beta_S g_{\mathrm{hi}}^2}, \tag{B.4}$$

then the response at non-compliance is non-decreasing in the state, the target's value is the Banach fixed point of its Bellman operator against that response, full non-compliance is the target's unique optimum at every state, and $(V_T, V_S, \alpha \equiv 0) \in \mathcal{E}$. At that equilibrium, at every state, no maximizer of J_S is a stage maximizer; and when $\ell_{\mathrm{hi}} < \underline{s}\, p_{\mathrm{lo}}$, so that full compliance beats full non-compliance in the one-period problem, no forward-looking target maximizer is a stage maximizer either.

The increasing-differences condition (B.4) says that the stage complementarity between state and sanction covers two forward-looking forces that can make the sender's response fall with the state. They are the curvature of the sender's continuation value (the σ_C term) and the dependence on the state of the sanction's effect on the next state (the M_{mix} term). Both transition terms, M_{cx} and M_{mix}, vanish for every additively separable transition with convex state sections, such as the partial-adjustment law (Appendix J.1). With a sender payoff convex in the state ($\tau_x = 0$, as on the family), the σ_C term vanishes too, and the condition reads $0 \le r_{\mathrm{lo}}$. The construction uses (B.4) only to make the response at non-compliance non-decreasing. The appendix states Theorems 3 and 12, Proposition B.2(iv), the long-run gap of Proposition C.5 and the welfare results of Section 4.2 under that monotonicity instead. It is strictly weaker than (B.4) with the supplied moduli: it holds at a member that meets the condition with a tighter modulus but not with the supplied ones (Appendix C.4). We state (B.4) because, unlike that monotonicity, it can be checked from the primitives and the supplied moduli.

Relief and non-compliance are constructed by different routes, which is why the results use different conditions. At non-compliance, the sender's value against full non-compliance is constructed first and the target's value against the resulting response next. Only then do the entrenchment and increasing-differences conditions verify that non-compliance is the target's optimum, so that the construction against a fixed full-non-compliance policy becomes an equilibrium. In cases 00

and 01, where the myopic target accommodates, compliance neutrality pins down the sender's value along relief, and the target's optimal compliance is the least that makes the selected sanction zero. The minmax bound on the target's value and the uniqueness of full non-compliance among equilibria with a non-decreasing sanction rule therefore hold without the conditions that construct relief. Conversely, a relief equilibrium requires solving its own continuation problem. The myopic and forward-looking cases are Proposition 2 and Theorem 3 of the main text; the target-only and sender-only cases follow (Propositions B.2 and B.3).

The separation of the myopic and the forward-looking cases is in the main text: the myopic case is Proposition 2, and the forward-looking case, with the comparison of the two paths, is Theorem 3, both under the one-period forms of interiority and the lifting condition.

The two main-text results are statements about the constructed equilibria, and their parts rest on different conditions. The forward-looking equilibrium of Theorem 3 is constructed under the increasing-differences condition. Given it, the lower bound on the build-up of the state and the forward-looking path lying above the myopic one use only the transition's monotonicity and sanction-slope bounds. The convergence also uses the response monotonicity that the condition delivers. The convergence to zero in Proposition 2 uses the sanction-slope bounds alone. The limit in Theorem 3 may depend on the initial state unless the equilibrium law of motion is a contraction (Proposition D.11). Its bounds $\underline{s} < x^{11}_{\infty} \leq A/q$ have closed forms on the family (Appendix J.1), and $A/q < 1$ under interiority. The comparison changes both discount factors together, holding the other primitives fixed.

Proposition B.2 (Target-only case)**.** *For every economy in the class satisfying the entrenchment condition, with $\beta_S = 0$ and β_T as given:*

(i) the sender's sanction rule is the myopic sanction $\sigma^{\mathrm{M}}(x, 0)$ ($(u_0 + rx)/q$ on the family under interiority) and its value is the stage value there;

(ii) full non-compliance is the target's unique optimum at every state, and its value is the fixed point of its Bellman operator against the myopic best response; the pair with $\alpha \equiv 0$ is an equilibrium in $\mathcal{E}$ of this case;

(iii) from every initial state $x_t \geq \underline{s}\big(1 - (1 - d_{\mathrm{lo}})^t\big)$ and x_t converges to a fixed point $x^{10}_{\infty} \geq \underline{s}$ of the myopic sanction rule; on the family under interiority the fixed point is unique, $u_0/(q-r)$, and $\big|x_t - u_0/(q-r)\big| \leq \big(1 - d_{\mathrm{lo}}(1 - r/q)\big)^t \big|x_0 - u_0/(q-r)\big|$ (7/12 per period at the partial-adjustment example);

(iv) under the increasing-differences condition (B.4) *and $u_{\mathrm{hi}} < q$ (a myopic sender never sanctions fully), the limit x^{10}_{∞} from a start is strictly below the limit x^{11}_{∞} of the forward-looking equilibrium of Theorem 3 from the same start.*

The target's forward-looking incentive alone therefore switches it from partial compliance to non-compliance, and the sender's forward-looking incentive then raises the steady state. The entrenchment condition does that work against the myopic best response as against the dynamic one. The steady state rises because the forward-looking sanction exceeds the myopic sanction at every state. On the family, the rate in (iii) bounds each path's distance to the steady state at each date. Across economies of the family that share u_0 and q, a larger sender state–sanction cross term r strictly raises the myopic sanction at every positive state where it is below full sanction. It strictly raises the target-only steady state too when the economy with the larger r satisfies interiority (Appendix H.7).

Proposition B.3 (Sender-only case)**.** *For every economy in the class satisfying interiority, the lifting condition and compliance neutrality whose transition also has semiconvexity in the sanction, condition* (T6) *of Appendix C.4, satisfying the forward-looking relief condition* (RL) *there, with* $\beta_T = 0$ *and* β_S *as given:*

(i) the sender's value is the value of the zero-sanction path, $\sum_{t\geq 0} \beta_S^t U_S(y_t, 0, 0)$ *with* $y_0 = x$ *and* $y_{t+1} = F(y_t, 0)$ *($U_S(x, 0, 0) = wx$ on the family); it is constructed before the target's policy is determined, because along the constructed equilibrium the sanction is zero and at sanction zero the sender's stage payoff (by compliance neutrality) and the transition do not depend on compliance;*

(ii) the target's unique optimum at every state is $a^{01}(x)$*, the least compliance at which the sender's dynamic best response vanishes; it is continuous in the state, lies strictly above* $a^{00}(x)$*, the compliance that ends a myopic sender's sanction, and with it the pair* $(U_T(x, a^{01}(x), 0), V_S, a^{01})$ *is an equilibrium in* $\mathcal{E}$ *of this case ($U_T(x, a^{01}(x), 0) = v\,x - \ell a^{01}(x)$ on the family);*

(iii) the equilibrium sanction is zero and from every initial state $x_t \leq (1 - d_{\mathrm{lo}})^t x_0 \to 0$*.*

The forward-looking sender is more demanding than the myopic one, so the myopic target pays more for relief, but it still buys it. Because of the foresight premium, the forward-looking sender still sanctions at the compliance $a^{00}(x)$. The forward-looking relief condition (RL), $\ell_{\mathrm{hi}} M_{\mathrm{tot}} < k_{\mathrm{lo}} p_{\mathrm{lo}}$, says that a unit of compliance avoids more sanction pain, valued at the lowest price p_{lo}, than it costs. Here M_{tot} bounds the curvature of the sender's dynamic objective in the sanction (Appendices C.4 and H.9). Relief can require full compliance (Appendix H.7).

B.3 The strict comparison at a fixed target policy

Proposition B.4 (Strict comparison at a fixed target policy)**.** *Fix* x *and a target action* a*, write* $\Gamma(s) = V_S(F(x, s))$ *for the sender's continuation value at the successor state* $F(x, s)$*, and let* m *be the greatest stage maximizer of* $U_S(x, a, \cdot)$ *on* $[0, 1]$*, which exists by continuity. If no feasible sanction below* m *carries a larger continuation value,* $\Gamma(s) \leq \Gamma(m)$ *for every* $s \in [0, m]$*, and there is one sanction* s' *at which the stage loss measured from* m *is outweighed by the discounted continuation gain,*

$$U_S(x, a, m) - U_S(x, a, s') < \beta_S\big(\Gamma(s') - \Gamma(m)\big),$$

then every stage maximizer lies weakly below m *and strictly below every maximizer of* J_S*.*

Proposition B.4 needs no concavity, differentiability, uniqueness or positive discount, and Γ need not be monotone: a non-decreasing Γ is one way to meet the comparison. The proposition orders two sets of maximizers, and the stage loss must be measured from the greatest stage maximizer. At an economy in Appendix H.9, a loss measured from a lower tied stage maximizer orders only that one. The proof closes Appendix C.2.

Compared with the myopic equilibrium at their own target policies, the dynamic equilibrium sanctions at least as much at every state when its sender value is non-decreasing in the state, its target complies no more and compliance placates the sender. It does so through two separate effects. Under the compliance ordering $\alpha(x) \leq \alpha^{\mathrm{M}}(x)$,

$$\sigma^{\mathrm{M}}(x, \alpha^{\mathrm{M}}(x)) \leq \sigma^{\mathrm{M}}(x, \alpha(x)) \leq \sigma^{\mathrm{dyn}}_{V_S}(x, \alpha(x)).$$

The first inequality is the indirect effect of the target's lower compliance on the myopic best response. The second is the direct effect of the sender's foresight at that compliance (Theorem 4).

The compliance ordering is an explicit condition, met by the example economies. Under the general transition, whole sanction paths are ordered under other conditions, without the compliance ordering or any condition on how the target ranks sanction levels. This is so for an equilibrium constructed from stated invariant value sets, when the myopic sanction stays below a ceiling on the high political basin and the myopic rule preserves that basin (Corollary I.6).

B.4 The target's minmax payoff and uniqueness among monotone rules

The benchmark for the target's security is permanent relief: zero sanctions at every date, with the target maintaining non-compliance throughout. Let $y_0 = x_0$ and $y_{t+1} = F(y_t, 0)$ be the state path under permanent zero sanctions, and define the permanent-relief payoff

$$\underline{V}_T(x_0) = \sum_{t=0}^{\infty} \beta_T^t \, U_T(y_t, 0, 0). \tag{B.5}$$

Proposition B.5 (The target's minmax payoff)**.** *Take any economy, whatever the sender's objective, in which the transition's zero-sanction section rises in the state at rate at least g_{lo}, a sanction s raises the successor state by at least $d_{\mathrm{lo}}s$ against zero sanctions, and the target gains at least $v_{\mathrm{lo}} \geq 0$ per unit of state at full non-compliance and zero sanctions, loses at most cs to a sanction s at full non-compliance, and gains nothing from compliance against a zero sanction (used only in (i) and (ii)). Under the strong entrenchment condition $c \leq \beta_T d_{\mathrm{lo}} g_T^0$, and from every initial state x_0:*

(i) $\underline{V}_T(x_0)$ is the lowest payoff to which a feasible pure history-dependent sanction strategy can hold the target against every feasible pure history-dependent compliance strategy; permanent relief attains it;

(ii) $\underline{V}_T(x_0)$ is the highest payoff that the target can guarantee itself with a feasible pure history-dependent compliance strategy against every feasible pure history-dependent sanction strategy; full non-compliance at every date attains it;

(iii) every equilibrium in $\mathcal{E}$ satisfies $V_T(x_0) \geq \underline{V}_T(x_0)$.

So $\underline{V}_T$ is the target's minmax payoff in pure strategies, which by (ii) is also the most it can guarantee itself, and the minmax over sanction sequences fixed in advance is its special case. Every economy in the class has the five properties the proposition reads, from the bounds g_{lo}, d_{lo}, v_{lo} and c and the positive compliance cost. Inequality (iii) holds at every equilibrium in $\mathcal{E}$, beyond the constructed equilibrium and the monotone-rule equilibria of Proposition B.6.

The bound holds because, under the strong entrenchment condition, the entrenchment that sanctions build is worth at least as much to the target as their pain. Under any sanction sequence, the lower bounds on the transition's slopes make the state's excess over the zero-sanction state a discounted accumulation of past sanctions. Valuing it at least at its lower state-valuation rate, against a pain of at most c per unit, the target gains from full non-compliance at every date at least $\big(\beta_T d_{\mathrm{lo}} v_{\mathrm{lo}}/(1-\beta_T g_{\mathrm{lo}}) - c\big) \sum_t \beta_T^t s_t$ over permanent relief. The strong entrenchment condition makes this nonnegative. Against permanent relief, compliance only incurs its cost, so full non-compliance attains the upper bound. With deterministic play and perfect monitoring, date-indexed actions reproduce the path of any pair of history-dependent plans, so both bounds extend to history-dependent strategies. Equilibrium optimality gives the target at least the payoff of deviating to full non-compliance at every date (Appendix C.7). Politically, under the strong entrenchment condition relief is the harshest treatment the sender can impose, so threats cannot coerce the target on the non-compliance side.

The same strong entrenchment condition makes full non-compliance, with the associated values, the only possible equilibrium with a non-decreasing sanction rule, and excludes a sanction rule equal to zero at every state.

Proposition B.6 (Uniqueness of non-compliance among monotone rules)**.** *On an economy in the class satisfying the strong entrenchment condition* $c \leq \beta_T d_{\mathrm{lo}} g_T^0$*:*

(i) every equilibrium in $\mathcal{E}$ *whose sanction rule is non-decreasing in the state has full non-compliance at every state, the constructed sender and target values, and the constructed response as its sanction rule;*

(ii) no equilibrium in $\mathcal{E}$ *has a sanction rule equal to zero at every state.*

Both clauses of Proposition B.6 concern non-decreasing sanction rules: (i) pins down every equilibrium with such a rule, and (ii) excludes the rule equal to zero at every state. Clause (i) requires no monotonicity of the target's policy. Its proof rules out positive compliance in the class through a lower state slope that the equilibrium value equation gives the target's value (Proposition C.12). Clause (ii) covers every equilibrium: a globally zero rule is non-decreasing, so it would fall within (i), which implies a positive response at full non-compliance. Under the increasing-differences condition the constructed equilibrium belongs to the class of (i), which is therefore not empty.

B.5 Relief, non-compliance and the threshold discount factor in general

For the forward-looking target, three inequalities on its discount factor each give one conclusion (the formulas are in Proposition C.9). (i) *Sufficient condition for non-compliance.* The most the target can save by buying off a unit of sanction, its pain c less the least compliance cost of that unit, is below the discounted entrenchment the unit builds. That entrenchment is valued at a lower slope that the target's value carries when the sender's response is monotone. The slope is g_T under the entrenchment condition and, in general, under two further conditions, a slope at most g_T. Then full non-compliance is the target's unique optimum at every state, and with the increasing-differences condition the constructed full-non-compliance equilibrium exists. (ii) *Sufficient condition for relief.* The most a unit of compliance costs, ℓ_{hi}, is below the least value of what it buys. This is $k_{\mathrm{lo}}/M_{\mathrm{tot}}$ units of sanction reduction, each worth the pain p_{lo} less the most discounted entrenchment it can cost in the relief equilibrium. Then, under interiority, the lifting condition and compliance neutrality, on economies in the class with semiconvexity in the sanction (T6), the relief equilibrium, in which the target complies at $a^{01}(x)$, exists. For existence the weak forms of (i) and (ii) suffice. The strict forms add uniqueness of the target's optimum. (iii) *An inequality that rules out non-compliance.* It compares as in (ii), measuring entrenchment by the Lipschitz constant of the target's value at the full-non-compliance profile. With interiority, the lifting condition and the target's discount factor times a bound on the Lipschitz constant of the state map at full non-compliance below one, it makes full non-compliance not a best reply at that profile's values. Since every slope in (i) is at most g_T, each sufficient condition for non-compliance implies the inequality for non-compliance at g_T, which excludes (ii) and, under the same bound on the state map, (iii). The sufficient condition for non-compliance and the inequality at g_T pass to a more patient target and (ii) to a less patient one. In strict form, the inequality at g_T and (ii) are open conditions on β_T.

In general these inequalities leave a gap between the two sufficient conditions. Above every discount factor at which the sufficient condition for relief holds and below every discount factor at which the sufficient condition for non-compliance holds, neither applies. They determine no equilibrium there. The gap can reflect conservative bounds as well as genuine differences in behavior. Each end has an exact counterpart, a threshold necessary and sufficient for existence, under the

conditions of Proposition C.10. At the relief end these are (T6), interiority, the lifting condition, compliance neutrality and a target state valuation at zero sanction that covers the rise of its compliance cost along the relief policy. At the non-compliance end they are a non-decreasing sender response at full non-compliance and the slope condition of Proposition C.9(i) at slope zero. The discount factors with an equilibrium without sanctions then form an interval from zero up to a largest one, and those with a full-non-compliance equilibrium an interval from a smallest discount factor up to one. At the partial-adjustment example with β_T varied (Appendix J.1), where an exact evaluation finds the gap open (Appendix C.6), both apply, and both exact thresholds lie strictly inside the gap. There, at one discount factor between them, a continuum of equilibria with compliance rising in the state exists (Proposition C.11).

Nine equalities, the linear–quadratic special case of the main text, close this gap: with them the sufficient condition for relief and the sufficient condition for non-compliance meet at a single discount factor. They make the transition affine on the square by hypothesis (a constant sanction slope, with $F(x, x) = x$); they are listed in the proof of Proposition 6 (Appendix C.7). They also set two curvature bounds on the sender's side to zero, rule out state–sanction cross terms and make five marginal rates constant. The five further equalities of Proposition C.9(iii), tight bounds on the transition's state slope and four zero transition moduli, concern the bounds supplied for the economy and leave the game unchanged (Appendix C.7). In symbols, the transition's constant sanction slope is δ (so that $F(x, s) = (1 - \delta)x + \delta s$), the constant curvature of the sender's cost of sanctioning is q, and the constant substitution rate, compliance cost, sanction pain and state valuation of the target are k, ℓ, c and v. Write $D = c - \ell q/k$ for the target's net cost of a sanction, its pain less the compliance cost of ending it. The threshold discount factor of Proposition 6 is

$$\beta_T^* = \frac{D}{\delta v + (1 - \delta)D}, \tag{B.6}$$

which rises with D and falls with v; under the nine equalities the myopic relief condition reads $\ell q < kc$, that is, $D > 0$.

The non-existence direction ranges over every equilibrium in $\mathcal{E}$: below β_T^* no equilibrium has full non-compliance. At $\beta_T = \beta_T^*$ the equilibrium exists, because existence passes to the limit in β_T (Proposition C.13). Two example economies lie on the two sides of the threshold (Appendix C.7). When $\beta_T^* \geq 1$, which holds exactly when $D \geq v$, no discount factor supports full non-compliance. The target's pain from a unit of sanction, net of the compliance cost of buying it off, is then at least its valuation of the political state. The nine equalities suffice for an exact threshold, and Proposition C.10 gives exact thresholds without them, as at the partial-adjustment example.

The sender's patience strictly raises the bounds on its own marginal return to sanctioning, but with these nine equalities it does not move the target's threshold. Sender patience changes the level of the sender's response, and so the compliance that relief requires, while the target's comparison turns on its cost, pain, state valuation and discount factor. On the class as a whole, sender patience also moves the curvature and Lipschitz bounds in both sufficient conditions. The same structure brackets the sender. If the upper substitution rate k_{hi} is below the lower bound $u_{\mathrm{lo}} + \beta_S g_S d_{\mathrm{lo}}$ on the sender's dynamic marginal return to sanctioning, and a larger concession is never worth more to the sender at zero sanction, a positive sanction remains in every equilibrium whose compliance does not rise with the state, full compliance included. If the lower substitution rate k_{lo} is at least the upper bound A on that return (the lifting condition), full compliance ends the sanction. Acemoglu and Jensen (2015, Lemma 2, Proposition 1) give robust discount-factor comparative statics for a single agent's dynamic program and for aggregates of price-taking agents. Ours concern one government's patience in a two-player game. A myopic against a forward-looking target is thus one cut of a continuous comparison. Along it the constructed equilibrium switches

from relief to non-compliance as the target's discount factor rises from the sufficient condition for relief to that for non-compliance.

Patience, like every other target parameter, can change whether an economy has a full-non-compliance equilibrium, and leaves the path of such an equilibrium unchanged. Take two economies that share the transition, the sender's discount factor and the sender's stage payoff at full non-compliance ($U_S(x, 0, s)$ on the square). In any full-non-compliance equilibria they have the same sender value and the same state and sanction paths from every start, whatever the target's payoff and discount factor and the sender's payoff away from non-compliance (Proposition C.7).

B.6 Welfare: the interest-group polity in full

The interest-group polity's accounting identity, the planner results and the rescaling argument.

In the interest-group polity of Appendix L, political weights account exactly for the gap between the sum of the two governments' objectives and aggregate welfare. In the tradition of Kaempfer and Lowenberg (1988), each country has two interest groups. The sender has a pro-sanction group and a commercial lobby harmed by sanctions. The target has a regime-aligned group and an opposition. Let $W_g(x, a, s)$ be group g's welfare and θ_g the political weight its government places on it (Grossman and Helpman, 1994). Becker (1983) supplies the pressure contest behind the weights. Aggregate welfare W sets every political weight to one and keeps the resource terms that enter the government objectives at unit weight. Four sign conditions on the coefficients deliver the game's stage assumptions. The sender's sanctioning cost is strictly convex, a higher political state raises the sender's marginal value of sanctioning, and compliance does not raise it. On the target side, the regime-aligned group's gain in political effectiveness under sanctions, weighted by its valuation of the political state, outweighs the opposition's gain, weighted by the opposition's loss from the state. For every coefficient vector, whether or not these conditions hold,

$$U_S(x, a, s) + U_T(x, a, s) = W(x, a, s) + \sum_{g \in G} (\theta_g - 1) W_g(x, a, s), \tag{B.7}$$

where G is the set of the four groups. The identity is accounting at common actions.

At one example economy of this polity (Appendix J.1), equilibrium play follows the corner path of full sanctions against full non-compliance from every start, while the zero-sanction path attains the planner's optimum over all feasible paths. The planner's value exceeds equilibrium welfare at every start and every planner discount factor (Proposition L.3).

The zero-sanction path attains the planner's optimum in every interest-group polity and under every transition satisfying (F2)–(F3), provided a sufficient condition holds: the first unit of sanction must not pay socially. Write g_J and q_K for the pro-sanction group's gain and the commercial lobby's loss per unit of sanction. The target's direct cost per unit is d_s, and B is the regime-aligned group's valuation of the political state less the opposition's. The zero-sanction path with the best constant compliance attains the optimum from every start if $B \leq 0$ and $g_J - q_K \leq d_s$. It also does so if $B \geq 0$ and

$$g_J - q_K + \frac{\beta B L}{1 - \beta\lambda} \leq d_s,$$

where L bounds the rise in next period's state per unit of sanction and λ, with $\beta\lambda < 1$, bounds the slope of the zero-sanction section $F(\cdot, 0)$ (Proposition L.3). The condition says that the sender-side groups' net gain plus the planner's discounted value of the higher state the sanction leaves behind does not cover the target's direct cost.

The comparison of the two paths extends beyond the example economy as well. For every vector of group coefficients and every transition satisfying (F2)–(F3), the welfare difference between the

zero-sanction path at full non-compliance and the corner path is the capitalized long-run gap plus B times a discounted residual. The residual is one minus the distance between the two paths, which starts at one. The sender's sanctioning cost is k_S, so the resource cost of a full sanction to both countries is $k_S + d_s$. Under (F1)–(F5) the difference is positive at every planner discount if and only if this cost exceeds the sender-side groups' net benefit $g_J - q_K$ and is at least that benefit plus B. The closed form under the partial-adjustment law of Appendix J.1 and the stationary comparison are in Appendix L (Propositions L.1 and L.2). For other transitions, the appendix bounds the gap below in the transition's own rates, more sharply given a lower bound on its state slope along the zero-sanction and full-sanction sections.

The example shows that politically sustained sanctions *can* be socially wasteful, and neither the four sign conditions nor the game determines the sign of the welfare gap. Multiply the pro-sanction group's per-unit benefit g_J by $\phi > 0$ and divide its political-weight schedule by ϕ. Both government payoffs are unchanged. So under every transition the rescaled polity still satisfies the four sign conditions and has the same game and equilibria. The planner, who counts that group's benefit at unit weight, sees the gap between the zero-sanction path and the corner path fall by $(\phi - 1)g_J/(1 - \beta)$. Hence, for every polity whose pro-sanction group gains from sanctions, some rescaling makes the corner path strictly better for the planner than the zero-sanction path, at every planner discount, every start and every transition satisfying (F2)–(F3). Where that group loses from sanctions, some rescaling makes the corner path strictly worse (Appendix L.6). Each statement goes one way. Where the group gains, the game never shows the zero-sanction path to be at least as good for the planner; where it loses, never the reverse.

Two polities next to the example economy reverse the welfare ranking, the first with the same game and equilibrium. The first rescales the example economy in this way and keeps its equilibrium, hence its play of the corner path. The second differs from it only in a lower direct sanction cost. The first fails the sufficient condition above at every discount, and the second above a threshold discount. For the first at every discount, and for the second above a higher threshold, the corner path is worth strictly more to the planner than the zero-sanction path. There the planner's optimum is not the zero-sanction path (Appendix L.6).

Maintained restrictions on the microfoundation narrow this non-identification: they partially identify the ranking of cases 10 and 11, in which the target maintains non-compliance. With the target side of the microfoundation and the commercial lobby's exposure held fixed, and the pro-sanction group's political weight confined to a band, the welfare gap between the two constructed equilibria ranges over an attained closed interval. A uniform strict ranking is identified exactly when that interval excludes zero. Both strict rankings occur when its endpoints have opposite signs. The exposure and the target-side inputs are maintained from outside the governments' game. A weight band alone need not identify the sign (Proposition L.7).

C Proofs of the results on the general class

This appendix proves the three main theorems.

C.1 Proof of Theorem D.2 (persistence on the basin)

Step 1: Threshold characterization of the basin. Define the one-sided sanction-preference thresholds $\bar{x}_T, \bar{x}_S$ as the smallest values such that the following two monotonicity conditions hold for all states at and above the respective threshold ($x \in [\bar{x}_S, 1]$ for the sender condition, $x \in [\bar{x}_T, 1]$ for the target condition):

- (Sender, ordinal) after substituting the target policy induced by the value sets, the family $\widetilde{J}_{V_T,V_S}(s,x)$ satisfies interval dominance in (s,x) on $[0,1]\times\mathcal{B}$; single crossing in (s,x) is a sufficient discharger.
- (Target combined slope condition) $U_T(x,\alpha(x),\cdot)+\beta_T V_T(F(x,\cdot))$ is non-decreasing in s on $[\max(x,\bar{x}_T),1]$.

The basin is $\mathcal{B}=[\max(\bar{x}_T,\bar{x}_S),1]$. By Assumptions 1 and 2, both monotonicity conditions hold for every $x\in\mathcal{B}$.

Step 2: Forward invariance. Fix $x_0\in\mathcal{B}$. We must show $\sigma_{V_S}^{\mathrm{dyn}}(x_0,\alpha(x_0))\geq\max(x_0,\bar{x})$: the sender's dynamic best response sits at or above the basin pivot. This is a statement about the *level* of the sender's upper argmax, and we state exactly which condition delivers it. It is *not* Assumption 1. An ordinal condition, Milgrom-Shannon single crossing or the weaker interval dominance the certificate actually carries (Remark G.2), constrains how the upper argmax *moves* as a parameter rises; it does not say where that argmax *lies* relative to $\max(x_0,\bar{x})$ when the continuation V_S is an arbitrary, possibly non-monotone, member of the K_S-Lipschitz value set $\mathcal{C}_S$: a member whose continuation payoff may fall by as much as $K_S|F(x,s_2)-F(x,s_1)|$ when the sanction rises from s_1 to s_2.

What Step 2 uses is the *cardinal transition-adjusted slope condition* (D.1), instantiated *below* the pivot in the pivot-star form of (P5): at each basin state x, the sender's stage-payoff gain from raising any sanction $s<\max(x,\bar{x})$ to the pivot must exceed $\beta_S K_S|F(x,\max(x,\bar{x}))-F(x,s)|$. (Remark G.5 records the same strict inequality on the complementary range $[\max(x,\bar{x}),1]$, where what it discharges is the *ordinal* condition. The level argument needs the inequality on the sub-pivot range instead, and the sub-pivot range is where the machine-checked certificate carries it. The two are distinct conditions, and neither is used in place of the other.) Under the sub-pivot slope condition the stage gain strictly exceeds the worst-case Lipschitz absorption of the continuation, so every sanction *below* the pivot gives a strictly lower value of $J_S(x_0,\alpha(x_0),\cdot;V_S)$ than the pivot itself, for every $V_S\in\mathcal{C}_S$, and hence no maximizer lies below it (Lemma C.1(b) below). That gives $\sigma_{V_S}^{\mathrm{dyn}}(x_0,\alpha(x_0))\geq\max(x_0,\bar{x})$.

Two disclosures follow, and we state them plainly. First, the level clause is proved under a cardinal condition that Assumption 1 does not supply; the machine-checked certificate carries that condition (a stage lower-slope field on the sub-pivot range, a Lipschitz field for the value set, and the transition-adjusted slope condition) as explicit hypotheses, alongside its ordinal field. A reader who wishes to keep the sender-side assumption set purely ordinal must take the level bound itself as a primitive: an ordinal condition on the parametrized family fixes the comparative static, not the location of the argmax, and against a K_S-Lipschitz continuation that may decline in s the location has to be won by a cardinal comparison of the stage gain with the worst-case continuation loss. Second, the cardinal condition does *not* displace the ordinal condition: the certificate carries both, as separate fields, and they discharge different steps: the cardinal condition the *level* (this step); the ordinal interval-dominance condition of Remark G.2 the *direction*, namely the monotone upper argmax in the state used in Step 4 below. The comparison at a fixed target policy (Theorem 4) instead uses monotonicity of the equilibrium sender value over successor states.

Hence $s_0^E\geq\max(x_0,\bar{x})\geq x_0$. By sanction monotonicity and the diagonal condition,

$$x_1^E=F(x_0,s_0^E)\geq F(x_0,x_0)=x_0\geq\bar{x},$$

so $x_1^E\in\mathcal{B}$. This is the general-F argument. Under (J.2) the displayed inequality becomes $(1-\delta)x_0+\delta s_0^E\geq x_0$. Induction yields $x_t^E\in\mathcal{B}$ for all $t\geq 0$.

Step 3: Sanction floor. The Step-2 floor argument is stated at an arbitrary basin state, not only at x_0: for every $x \in \mathcal{B}$ we have $\sigma_{V_S}^{\mathrm{dyn}}(x, \alpha(x)) \geq \max(x, \bar{x})$. By Step 2 the equilibrium path stays in $\mathcal{B}$, so $s_t^E \geq \max(x_t^E, \bar{x}) \geq x_t^E$ for every $t \geq 0$. Feeding $s_t^E \geq x_t^E$ into the transition (1) and using the monotonicity (F3) together with the diagonal normalization (F4) gives $x_{t+1}^E = F(x_t^E, s_t^E) \geq F(x_t^E, x_t^E) = x_t^E$; hence the state path is non-decreasing and $x_t^E \geq x_0$ for every t. Therefore $s_t^E \geq \max(x_t^E, \bar{x}) \geq \bar{x}$, which is conclusion (ii). The proof actually establishes the stronger inherited-state bound $s_t^E \geq \max(x_0, \bar{x})$. This argument uses only the Step-2 floor and the transition; in particular it does not presuppose Step 4.

Step 4: Sanction monotonicity. We show $s_t^E \leq s_{t+1}^E$. The state path is non-decreasing, $x_{t+1}^E \geq x_t^E$, by Step 3. The sender's dynamic best response $\sigma_{V_S}^{\mathrm{dyn}}(x, \alpha(x))$ is non-decreasing in x on the basin. This is the one sender-side step that is purely *ordinal*: it is the monotone-upper-argmax conclusion applied to the parametric variation in the state x, and in the formalization it is run off the interval-dominance field the certificate carries (Remark G.2), for which single crossing is sufficient but not necessary. No cardinal bound enters here. Hence $s_{t+1}^E \geq s_t^E$.

Single-crossing lemma.

Lemma C.1 (Sender dynamic objective: direction and level)**.** *Fix a value pair $(V_T, V_S) \in \mathcal{C}_T \times \mathcal{C}_S$, with $V_S \in \mathrm{Lip}_{K_S}$, and write $\widetilde{J}_{V_T,V_S}$ for* (D.2)*. Then:*

(a) (Direction; ordinal.) *Under Assumption 1, the greatest maximizer of $s \mapsto \widetilde{J}_{V_T,V_S}(s,x)$ is non-decreasing in $x \in \mathcal{B}$. Single crossing in (s,x) is a sufficient stronger condition.*

(b) (Level; cardinal.) *Write $\hat{x} = \max(x, \bar{x})$. Under the transition-adjusted slope condition* (D.1)*, in its pivot-star instance $s_1 = s < \hat{x} = s_2$, the upper argmax of $s \mapsto \widetilde{J}_{V_T,V_S}(s,x)$ on $[0,1]$ lies in $[\hat{x}, 1]$.*

Neither clause implies the other. Clause (a) is silent below the pivot, so it cannot deliver (b); and the stage slope in clause (b) holds only below the pivot, whereas deriving single crossing on the whole choice axis $[0,1]$ from the cardinal condition requires the stage lower slope on all of $[0,1]$, not merely below the pivot. The certificate accordingly carries a field for each.

Proof. (a) This is the monotone-greatest-argmax implication of the interval-dominance condition in Assumption 1. Milgrom–Shannon single crossing in (s,x) upcasts to interval dominance with no side conditions (Remark G.2).

(b) For the level conclusion, that the upper argmax lies in $[\max(x, \bar{x}), 1]$, we use the cardinal transition-adjusted slope condition on the sub-pivot range, *not* clause (a) and *not* any monotonicity of V_S. For $V_S \in \mathcal{C}_S = \mathrm{Lip}_{K_S}$ and any $s < \hat{x}$, the continuation increment from s to the pivot is bounded by its worst-case Lipschitz absorption, $\beta_S \big| V_S(F(x,\hat{x})) - V_S(F(x,s)) \big| \leq \beta_S K_S |F(x,\hat{x}) - F(x,s)|$, while (D.1) says that the stage gain from s to $\hat{x}$ strictly exceeds this allowance; hence $\widetilde{J}_{V_T,V_S}(s,x) < \widetilde{J}_{V_T,V_S}(\hat{x},x)$ for every $s < \hat{x}$ and *for every* $V_S \in \mathcal{C}_S$ (using only the K_S-Lipschitz bound, valid even for non-monotone V_S), so no maximizer lies below $\hat{x}$; no two sub-pivot sanctions are compared. (This is why the certificate carries *both* sender-side conditions rather than one, and why the two clauses above are stated separately. The ordinal interval-dominance field (Remark G.2) delivers the comparative-statics direction: the monotone upper argmax of Step 4. The cardinal stack (the stage lower slope on the sub-pivot range, the Lipschitz constant K_S of the value set, and the actual transition displacement) separately pins the *level*. Under the affine law it is sufficient to verify a lower slope m_S and $\beta_S K_S \delta < m_S$. Neither does the other's job: the ordinal form does not

pin the level against a general, possibly non-monotone, Lipschitz continuation; and the cardinal condition, in the sub-pivot form the level argument uses, says nothing above the pivot. Only when the stage lower slope is strengthened to the whole choice axis $[0,1]$ does the cardinal condition also discharge the ordinal field.) □

Steps 2 and 3 use neither A1 nor the target condition: conclusions (i) and (ii) rest on the level condition alone, at any equilibrium with a K_S-Lipschitz sender value at which the sub-pivot slope condition holds (`standardCompliance_persistence_generalF_level_star`), and A1 enters only in Step 4. This completes the proof of Theorem D.2.

C.2 Proof of Theorem 4 (dynamic versus myopic sanction at a fixed target policy)

Fix any target policy $\alpha^\circ\colon [0,1] \to [0,1]$ and any state x; neither an equilibrium nor the basin enters the argument. We compare the sender's myopic best response and its dynamic best response at the *same* target action $\alpha^\circ(x)$, holding all other arguments fixed. The proof is Milgrom–Shannon monotone comparative statics in the sender continuation value V_S.

Step 1: The dynamic objective single-crosses the myopic one in s. Fix x and $a = \alpha^\circ(x)$. The sender's forward-looking objective and its myopic ($V_S \equiv 0$) limit are

$$J_S(x,a,s;V_S) = U_S(x,a,s) + \beta_S V_S(F(x,s)), \qquad J_S(x,a,s;0) = U_S(x,a,s),$$

so their difference is the continuation term alone,

$$J_S(x,a,s;V_S) - J_S(x,a,s;0) = \beta_S\, V_S(F(x,s)).$$

The value V_S is non-decreasing in the state by hypothesis (Step 4), and $F(x,\cdot)$ is non-decreasing by (F3); hence $s \mapsto V_S(F(x,s))$ is non-decreasing, and the difference $\beta_S V_S(F(x,s))$ is non-decreasing in s. A difference non-decreasing in s is exactly the increasing-differences / single-crossing condition that Milgrom–Shannon require (Milgrom and Shannon, 1994), and here it is sourced by the *monotonicity* of V_S, not by the pointwise level of any value function. (Pointwise order alone is insufficient: a value pointwise above the myopic 0 need not have larger s-increments; monotonicity of V_S is the property that does.)

Step 2: The upper argmax of the dynamic objective lies weakly above the myopic one. By Milgrom–Shannon Theorem 4 in its upper-extremal form (equivalently, Topkis' monotonicity-of-the-maximum theorem (Topkis, 1998)), when an objective differs from a comparator by a function non-decreasing in s, the upper element of its argmax is at least the comparator's. Specializing to $a = \alpha^\circ(x)$ with comparator $J_S(\cdot\,;0)$:

$$\max \arg\max_s J_S(x,\alpha^\circ(x),s;0) \;\le\; \max \arg\max_s J_S(x,\alpha^\circ(x),s;V_S).$$

Step 3: The myopic limit is $V_S \equiv 0$. The myopic best response $\sigma^{\mathrm{M}}(x,a)$ is by definition the element of the stage-payoff argmax. Because the continuation term vanishes when $V_S \equiv 0$, this is also the upper element of the dynamic-objective argmax evaluated at the zero continuation value. The dynamic best response $\sigma^{\mathrm{dyn}}_{V_S}(x,a)$ is the corresponding upper element at the continuation value V_S. Therefore, at $a = \alpha^\circ(x)$,

$$\sigma^{\mathrm{M}}(x,\alpha^\circ(x)) = \max \arg\max_s J_S(x,\alpha^\circ(x),s;V_S \equiv 0),$$

$$\sigma^{\mathrm{dyn}}_{V_S}(x,\alpha^\circ(x)) = \max \arg\max_s J_S(x,\alpha^\circ(x),s;V_S).$$

Step 4: Monotonicity of V_S (explicit, and derived on gain-ball invariant sets). The single-crossing in Step 1 rests on V_S being non-decreasing in the state, *not* on any sign normalization of U_S. Accordingly Theorem 4 states it explicitly. At the certificate-complete example economy it is a theorem about the geometry of the invariant set: the sender's invariant set is the zero-based gain ball $\{V : \|V\|_\infty \le M,\ \mathrm{Lip}(V) \le K,\ V(y_2) - V(y_1) \ge g_S(y_2 - y_1) \text{ for } y_1 \le y_2\}$ with $g_S \ge 0$, so every member is non-decreasing. Lean records the two-step implication in `monotone_of_mem_valLipGainBall` and `theorem2_dynamic_dominates_myopic_at_same_alpha_of_mem_valLipGainBall`, with the theorem at the concrete example economy `eeCert_same_alpha_dynamic_dominance_from_gain_carrier`. For other economies, Banach cone preservation gives an alternative discharge *if* $U_S(\cdot, \alpha(\cdot), s)$ is state-monotone; this is `V_S_monotone_of_integrand_monotone`. A bare Lipschitz value set alone supplies neither route, so A1–A3 by themselves must not be read as a proof of V_S monotonicity.

Step 5: Combining. Steps 1–4 give

$$\sigma^{\mathrm{M}}(x, \alpha^\circ(x)) \le \sigma^{\mathrm{dyn}}_{V_S}(x, \alpha^\circ(x)),$$

which is (7), at every state and for every bounded continuous non-decreasing V_S (`theorem2_dynamic_dominates_myopic_at_same_alpha`); in particular it holds at every equilibrium whose sender value is non-decreasing.

Every stage maximizer, not only the selected one. The comparison does not depend on how ties in the stage problem are broken. Under the same non-decreasing V_S, every maximizer m of the stage payoff $s \mapsto U_S(x, a, s)$ on $[0, 1]$ satisfies $m \le \sigma^{\mathrm{dyn}}_{V_S}(x, a)$: Step 2 with m in place of the selected myopic best response gives a maximizer of J_S at or above m, and $\sigma^{\mathrm{dyn}}_{V_S}$ is the greatest one (`exists_dynamic_argmax_ge_of_stage_max`, `stage_max_le_σ_dyn_top`). As rules, the greatest myopic best response to any target policy lies weakly below the dynamic best response at every state (`stageTopRule_le_dynamicTopRealizedRule`). Every comparison is against the dynamic best response, the greatest maximizer of J_S. The claim that every stage maximizer lies below every maximizer of J_S is false under a non-decreasing value alone: with a constant V_S the two objectives differ by a constant, so they have the same maximizers, and a tied stage argmax then has a stage maximizer above a maximizer of J_S. The strict ordering of all maximizers holds under the conditions of Proposition B.4.

Specialization to the different-α form. Taking $\alpha^\circ = \alpha^{\mathrm{M}}$ on the myopic side and $\alpha^\circ = \alpha$ on the dynamic side, the fixed-policy inequality (7) yields the different-α corollary under (a) $\alpha(x) \le \alpha^{\mathrm{M}}(x)$ at the states compared, and (b) pointwise monotonicity of $\sigma^{\mathrm{M}}(x, \cdot)$ non-increasing in the compliance argument. Condition (b) is a comparative static and needs the cross sign, not the own sign: it follows from decreasing differences of U_S in (s, a) (the "compliance placates" condition $\partial^2 U_S / \partial s\, \partial a \le 0$ of Section 2), which orders the unique stage maximizer under Assumption 3 and, without uniqueness, orders the greatest stage maximizer (`isMaxOn_max_of_monotone_diff`, an exchange inequality). A nonpositive own derivative $\partial U_S / \partial a \le 0$ does not imply it: $U_S = -(s - a)^2 - 2a$ has that sign and the unique maximizer $s = a$, increasing in compliance. Both conditions hold for the standard Kaempfer–Lowenberg payoff family, whose cross term carries the required sign, but are extraneous to the substantive Milgrom–Shannon content of Theorem 4.

Proof of Proposition B.4 (strict comparison at a fixed target policy). Fix x, a and V_S. Write $\Gamma(s) = V_S(F(x, s))$ for the sender's continuation value at the successor state $F(x, s)$

and $J(s) = U_S(x, a, s) + \beta_S \Gamma(s)$ for the forward-looking objective. The map $s \mapsto U_S(x, a, s)$ is continuous, so its maximizers on the compact interval $[0, 1]$ form a nonempty closed set with a greatest element m (`exists_greatest_stage_maximizer`). Every stage maximizer therefore lies weakly below m, which is the first conclusion. For the second, let t be a stage maximizer and $\hat{s}$ a maximizer of J on $[0, 1]$, and suppose $\hat{s} \leq t$, so that $\hat{s} \leq m$. Stage optimality of m gives $U_S(x, a, \hat{s}) \leq U_S(x, a, m)$, and the comparison of continuation values gives $\Gamma(\hat{s}) \leq \Gamma(m)$. The model maintains $\beta_S \in [0, 1)$, and $\beta_S \geq 0$ preserves the second inequality after discounting. So $J(\hat{s}) \leq J(m)$. Rearranged, the hypothesis at s' says $J(m) < J(s')$, so $J(\hat{s}) < J(s')$ at a feasible s', which contradicts the optimality of $\hat{s}$. Hence $t < \hat{s}$ (`all_stage_lt_dynamic_of_greatest_stage_improvement_of_wedge_le`). The argument compares Γ only at pairs (d, m) with $d \leq m$, and it uses the discount only through $\beta_S \geq 0$. It uses no concavity, differentiability or uniqueness, and it does not claim that a maximizer of J exists. A non-decreasing Γ is one way to supply the comparison (`all_stage_lt_dynamic_of_greatest_stage_improvement_mono`). □

C.3 Proof of Theorem D.8 (temporary shocks and their asymptotic effect)

Let $z_0 = 0$ and $z_{t+1} = F(z_t, 1)$ be the full-sanction path from the worst feasible initial state. The drift $d_F(x) = F(x, 1) - x$ is continuous by (F1) and strictly positive on $[0, \bar{x}]$ by full-sanction ascent below the basin, which (F5) implies. Compactness therefore gives an attained positive minimum

$$\eta_F(\bar{x}) = \min_{x \in [0, \bar{x}]} d_F(x) > 0.$$

As long as $z_t < \bar{x}$,

$$z_{t+1} - z_t = d_F(z_t) \geq \eta_F(\bar{x}).$$

Induction gives $z_t \geq t\eta_F(\bar{x})$ while all preceding iterates remain below $\bar{x}$. Consequently

$$\widehat{N}_F(\bar{x}) = \left\lceil \frac{\bar{x}}{\eta_F(\bar{x})} \right\rceil$$

must satisfy $z_{\widehat{N}_F(\bar{x})} \geq \bar{x}$: otherwise every earlier iterate would be pre-basin, the induction would apply through $\widehat{N}_F(\bar{x})$, and the definition of the ceiling would contradict $z_{\widehat{N}_F(\bar{x})} < \bar{x}$.

Now start the same temporary shock from any $x_0 \in [0, 1]$. Monotonicity of F in the inherited state implies inductively that the resulting shocked path lies weakly above z_t at every date. Thus $x_{\widehat{N}_F(\bar{x})} \geq z_{\widehat{N}_F(\bar{x})} \geq \bar{x}$ uniformly over x_0. The shocked path is non-decreasing, so every longer shock also brings the state into the basin. Conclusions (i)–(iii) of Theorem D.2, which do not read the target condition P4, give the stated post-shock conclusions from that date (`standardCompliance_min_drift_shock_generalF_from_sender_package_of_preBasinAscent`).

Under the partial-adjustment law (J.2), $1 - x_t = (1 - \delta)^t (1 - x_0)$. Therefore the $N_*(\bar{x}, \delta)$ displayed immediately after Theorem D.8 is a valid uniform choice, and is generally sharper than the minimum-drift bound because it uses the exact affine trajectory.

Proof of Corollary I.4. Write $d_i(x) = F_i(x, 1) - x$ and let $x_2 \in [0, \bar{x}]$ attain the minimum $\eta_{F_2}(\bar{x}) = d_2(x_2)$. The hypothesis $F_1(x, 1) \leq F_2(x, 1)$ is equivalent to $d_1(x) \leq d_2(x)$ on the pre-basin interval. Hence

$$\eta_{F_1}(\bar{x}) \leq d_1(x_2) \leq d_2(x_2) = \eta_{F_2}(\bar{x}).$$

If $F_1(x, 1) < F_2(x, 1)$ throughout the interval, the middle inequality is strict. The first law's ascent makes $\eta_{F_1}(\bar{x}) > 0$, and that hypothesis passes the ascent to the second law, so both minima are

positive with no adjustment hypothesis on F_2; since $\bar{x} \geq 0$,

$$\frac{\bar{x}}{\eta_{F_2}(\bar{x})} \leq \frac{\bar{x}}{\eta_{F_1}(\bar{x})}.$$

Monotonicity of the natural-number ceiling gives the stated weak duration comparison. It cannot generally be strengthened because the ceiling is locally constant.

Under the affine law,

$$d_\delta(x) = \big((1-\delta)x + \delta\big) - x = \delta(1-x).$$

This function is decreasing in x, so its minimum on $[0, \bar{x}]$ is attained at $\bar{x}$ and equals $\delta(1-\bar{x})$. Substitution gives the displayed minimum-drift duration and its monotonicity in δ. □

C.4 The class under the lifting condition: class-membership panel and proofs of Proposition B.1, Theorem 3, Proposition B.3, Proposition B.2 and Theorem 12

Every formula in this subsection is transcribed from the formalization at the implementation head, with the source's quantifiers and domains; the declarations are listed in Appendix B. The main text names the conditions; this is the complete certificate and the argument. The formalization states every result of Section 5.1 on the class of panel A0, whose stage payoffs are abstract (`Examples/ChordEconomy.lean`, the structures `ChordEconomy` and `ChordEconomyCs`). Panels A–G then write the argument for the quadratic–linear family, whose rates are exact and whose constants have closed forms; each of the family's rates is an instance of a one-sided rate of panel A0, and the formalization proves every step with the one-sided rates, as listed at the end of A0. The family is an instance of the class (`toChord`, `toChordCs`), so every family formula below is a theorem at the family's members. Panel H is written on the class itself, as Proposition B.2 is stated, with the family's closed forms in its Step 4; panel I assembles Theorem 12.

A0. The class

Stage payoffs. U_S and U_T are jointly continuous. Each condition holds for arguments in $[0,1]$ and, where an increment is signed, for ordered arguments; it is a difference-quotient inequality, so no differentiability is assumed. For the sender (the formalization's S2–S12):

(S1) *state Lipschitz at full non-compliance*: $|U_S(x_2,0,s) - U_S(x_1,0,s)| \leq w_{\mathrm{hi}}|x_2 - x_1|$;

(S2) *state lower slope at every compliance*: $U_S(x_2,a,s) - U_S(x_1,a,s) \geq w_{\mathrm{lo}}(x_2 - x_1)$ for $x_1 \leq x_2$, with $w_{\mathrm{lo}} > 0$;

(S3) *strong concavity in the sanction*, at every real compliance: $s \mapsto U_S(x,a,s)$ is $(-q)$-semiconcave on $[0,1]$; $q > 0$ is not a separate condition but follows from the concavity condition below;

(S4) *curvature ceiling*: $s \mapsto U_S(x,a,s)$ is $\bar{Q}$-semiconvex on $[0,1]$;

(S5) *joint semiconcavity at full non-compliance* in (state, sanction), with moduli $(\sigma_x, r_{\mathrm{hi}}, q)$ in the form of (T3), $\sigma_x \geq 0$;

(S6) *state semiconvexity at full non-compliance*, with modulus $\tau_x \geq 0$;

(S7) *increasing differences at full non-compliance*: $(U_S(x',0,s') - U_S(x',0,s)) - (U_S(x,0,s') - U_S(x,0,s)) \geq r_{\mathrm{lo}}(x'-x)(s'-s)$ for $x \leq x'$, $s \leq s'$, with $r_{\mathrm{lo}} \geq 0$;

(S8) *the cross rate is at most* r_{hi}, in absolute value and at every compliance;

(S9) *substitution*: $(U_S(x,a',s')-U_S(x,a',s))-(U_S(x,a,s')-U_S(x,a,s))\le -k_{\mathrm{lo}}(a'-a)(s'-s)$ for $a\le a'$, $s\le s'$, with $k_{\mathrm{lo}}>0$, and, where a result reads it, the *upper substitution rate* k_{hi}: the same cross difference is at most $k_{\mathrm{hi}}(a'-a)|s'-s|$ in absolute value;

(S10) *upper bound on the marginal return*: $U_S(x,0,t)-U_S(x,0,s)\ge (s-t)\big(q(s+t)/2-u_{\mathrm{hi}}\big)$ for $t\le s$;

(S11) *lower bound on the marginal return*: $U_S(x,0,s)-U_S(x,0,t)\ge (s-t)\big(u_{\mathrm{lo}}-\bar{Q}(s+t)/2\big)$ for $t\le s$, with $u_{\mathrm{lo}}>0$.

For the target (the formalization's T2–T7):

(P1) *compliance cost*: $-\ell_{\mathrm{hi}}(a_2-a_1)\le U_T(x,a_2,s)-U_T(x,a_1,s)\le -\ell_{\mathrm{lo}}(a_2-a_1)$ for $a_1\le a_2$, with $\ell_{\mathrm{lo}}>0$;

(P2) *sanction pain*: $p_{\mathrm{lo}}(s_2-s_1)\le U_T(x,a,s_1)-U_T(x,a,s_2)\le c\,(s_2-s_1)$ for $s_1\le s_2$, with $p_{\mathrm{lo}}>0$;

(P3) *state slope, indexed by the sanction*: $(v_{\mathrm{lo}}+n_{\mathrm{lo}}s)(x_2-x_1)\le U_T(x_2,a,s)-U_T(x_1,a,s)$ for $x_1\le x_2$, with $v_{\mathrm{lo}}>0$ and $n_{\mathrm{lo}}\ge 0$, and $|U_T(x_2,a,s)-U_T(x_1,a,s)|\le (v_{\mathrm{hi}}+n_{\mathrm{hi}}s)|x_2-x_1|$. These are envelopes on the value of entrenchment at each sanction, not cross rates (on the family $n_{\mathrm{lo}}=n_{\mathrm{hi}}=n$ is the target's state–sanction cross term, which softens the pain); the quartic member of Appendix J has $n_{\mathrm{lo}}>n_{\mathrm{hi}}$.

The formalization splits the class. (S8), the upper bound in (P1), the lower bound in (P2) and the upper envelope in (P3) form a relief extension (`ChordEconomyRelief`), read by the relief-side results, the Lipschitz estimates and the sharp sufficient condition for non-compliance; Proposition C.8, Proposition C.7, Proposition C.4 and Proposition C.12 hold on the base class (`ChordEconomy`), and the enriched subclass extends the relief extension. The base class leaves out three further sets of data. The two-sided curvature data, namely (S6) and the transition's state-section semiconvexity M_{cx} and bound M_{mix} on mixed differences of panel A, form a two-sided extension (`ChordEconomyTwoSided`), which the relief extension extends; only the two-sided invariant set (the modulus σ_C below, and through it the route through the increasing-differences condition to a monotone response and L_{resp}) and a route to a monotone rule from bounds on the target's policy (`Examples/PlacableSenderPolicyClass.lean`) read them. The upper substitution rate of (S9) is a condition of the base results that read it (`PlacationCeiling`), the sanction floor of Proposition C.4 among them, and a field of the relief extension. And $q>0$ is a theorem: the concavity condition puts q above $\beta_S(\sigma_V d_{\mathrm{hi}}^2+K_S M_s)$, which is non-negative because the Riccati closure gives $\sigma_V\ge 0$. Every member the formalization exhibits lies in the two-sided extension and has an upper substitution rate, so the further generality these changes give the base class is not yet illustrated by an example. The myopic best response $\sigma(x,a)$ ($\sigma^{\mathrm{M}}(x,a)$ in the main text), a field of the game (Section 2), is the maximizer of $U_S(x,a,\cdot)$ on $[0,1]$, unique by (S3); on the family it is the clipped vertex, and in general it has no closed form. The class also requires it to vary continuously with (x,a); this is a condition of the class rather than of the game (`ChordEconomy.σ_continuous`), kept because the class's results on the four cases compare with the myopic sender (Appendix H.7). *Compliance neutrality*, $U_S(x,a,0)=U_S(x,0,0)$, is a hypothesis of the relief-side results only (the sender-only case, Theorem 12, the relief equilibrium); the all-equilibria floor of Proposition C.4 reads its one-sided pairwise form (a larger concession is never worth more to the sender at zero sanction) and Proposition C.8 its form against full non-compliance. All three hold on the family, where $U_S(x,a,0)=wx$.

Constants. With M_U and $M_{T,0}$ the suprema of $|U_S|$ and $|U_T|$ on the cube (not fields),

$$K_S = \frac{w_{\mathrm{hi}}}{1-\beta_S g_{\mathrm{hi}}}, \quad g_S = \frac{w_{\mathrm{lo}}}{1-\beta_S g_{\mathrm{lo}}}, \quad M_S = \frac{2M_U}{1-\beta_S}, \quad M_T = \frac{2M_{T,0}}{1-\beta_T}, \quad \mu = q - \beta_S\big(\sigma_V d_{\mathrm{hi}}^2 + K_S M_s\big),$$

$$\varrho = r_{\mathrm{hi}} + \beta_S\big(\sigma_V g_{\mathrm{hi}} d_{\mathrm{hi}} + K_S M_m\big), \qquad \sigma_{\mathrm{img}} = \sigma_x + \beta_S\big(\sigma_V g_{\mathrm{hi}}^2 + K_S M_x\big) + \frac{\varrho^2}{\mu},$$

$$A = u_{\mathrm{hi}} + \beta_S K_S d_{\mathrm{hi}}, \qquad \underline{s} = \sigma(0,0), \qquad g_T = \frac{v_{\mathrm{lo}} + n_{\mathrm{lo}}\underline{s}}{1-\beta_T g_{\mathrm{lo}}}, \qquad \sigma_C = \frac{\tau_x + \beta_S K_S M_{\mathrm{cx}}}{1-\beta_S g_{\mathrm{hi}}^2},$$

$$M_{\mathrm{tot}} = \bar{Q} + \beta_S\big(\sigma_C d_{\mathrm{hi}}^2 + K_S M_{\mathrm{cs}}\big), \qquad L_{\mathrm{resp}} = r_{\mathrm{hi}} + \beta_S\big(\bar{\sigma} g_{\mathrm{hi}} d_{\mathrm{hi}} + (K_S + 2\bar{\sigma}) M_{\mathrm{mix}}\big), \ \ \bar{\sigma} = \max(\sigma_V, \sigma_C),$$

and $L_{\mathrm{loop}} = g_{\mathrm{hi}} + d_{\mathrm{hi}} L_{\mathrm{resp}}/\mu$. The lower bound $\underline{s}$ is the myopic sanction at full non-compliance at the lowest state, below which the response at non-compliance never falls.

The certificate. A member of the class satisfies (curvature) $\mu > 0$, the concavity condition, and (Riccati closure) $\sigma_{\mathrm{img}} \le \sigma_V$, and nothing else is a condition of membership. Interiority, $A < q$, and the myopic relief condition, $\ell_{\mathrm{hi}}\bar{Q} < k_{\mathrm{lo}} p_{\mathrm{lo}}$, are hypotheses of the results that read them, not fields (`Interior`, `StageRelief`). Interiority implies myopic interiority $u_{\mathrm{hi}} < q$, that a myopic sender never sanctions fully (`uhi_lt_q`), and the result that the sender's forward-looking incentive changes behavior, the strict foresight premium above the stage choice, the myopic equilibrium, the target-only case and the convergence of the forward-looking path read only that; the remaining results, among them the interior bound on the dynamic best response (the cap $A/q < 1$), the sender-only case, the relief side and the four cases, keep interiority. The myopic relief condition is read by the myopic equilibrium of Theorem 3 and by the threshold discount factor β_T^*, and in the four cases it follows from the forward-looking relief condition (`stageRelief_of_reliefSharp`). The hypotheses of the propositions that read them are (the lifting condition) $A \le k_{\mathrm{lo}}$, the entrenchment condition $c \le \beta_T d_{\mathrm{lo}} g_T$, the one-period ranking $\ell_{\mathrm{hi}} < \underline{s}\, p_{\mathrm{lo}}$, under which full compliance beats full non-compliance in the one-period problem, the increasing-differences condition (RM) with r_{lo} on its right, and the forward-looking relief condition (RL) $\ell_{\mathrm{hi}} M_{\mathrm{tot}} < k_{\mathrm{lo}} p_{\mathrm{lo}}$. The lifting condition implies the myopic lifting condition $u_{\mathrm{hi}} < k_{\mathrm{lo}}$ (`uhi_lt_k`), all that the myopic equilibrium reads; the sender-only case, the relief side and the four cases keep the lifting condition. The concavity condition $\mu > 0$ and the closure admit some $\sigma_V \ge 0$ exactly under the discriminant condition of panel A with $N = B_0 Q + C_0^2$, $L = (1-\beta_S g_{\mathrm{hi}}^2)Q + B_0 D - 2C_0\beta_S g_{\mathrm{hi}} d_{\mathrm{hi}}$, $B_0 = \sigma_x + \beta_S K_S M_x$ and $C_0 = r_{\mathrm{hi}} + \beta_S K_S M_m$ (`chSigV_feasible_iff`).

Replacements. Panels A–G use the family's exact rates. On the class the stage vertex $(u_0 + rx)/q$ is replaced by $\sigma(x,0)$ and u_0/q by $\underline{s}$; the linear bound $u_0 - ka + rx \le u_0 + r$ on the marginal return by (S10); the stage state rate $rs + w \in [w, r+w]$ by (S1) and (S2); the decreasing differences $-ksa$ by (S9); the exact cross rate r by (S7) from below and (S8) from above; the stage curvature q by (S3) from below and (S4) from above; the costs ℓa, the pain $c - nx$ and the target's state rate $v + ns$ by (P1)–(P3); and $U_S(x,a,0) = wx$ by compliance neutrality. The sharp rate of sanction reduction uses the argmax shift under a nonlinear tilt (`argmax_shift_ge_tilt_of_semiconvex_interior`) in place of the linear shift, the foresight premium uses the shifted-argmax bound under $\bar{Q}$-semiconvexity in place of the exact quadratic, and the zero level is the least zero of an antitone family (`Foundations/ZeroLevel.lean`). The family's closed forms that have no one-sided counterpart stay on the family: the target-only steady state $u_0/(q-r)$ and its rate, the comparisons across economies in r, the interest-group microfoundations, and the coefficient box.

A. Primitives

Stage payoffs. With state x, compliance a and sanction s in $[0,1]$,

$$U_S(x,a,s) = -\tfrac{q}{2}s^2 + s(u_0 - ka + rx) + wx, \qquad U_T(x,a,s) = -\ell a + s(nx - c) + vx.$$

Coefficient signs supplied: $u_0 > 0$, $r \geq 0$, $w > 0$, $\ell > 0$, $n \geq 0$, $k > 0$, $v > 0$, and $c > n$ (hence $c > 0$): the sanction hurts the target at every state. The lifting condition implies $k > 0$, the entrenchment condition with interiority implies $v > 0$, and the myopic relief condition implies $c > n$; since these three are hypotheses of the results that read them, not conditions of membership, the signs are supplied (see C). The sign $q > 0$ is *derived*, from the concavity condition (see C). Discount factors $\beta_S, \beta_T \in (0,1)$.

Transition. F satisfies (F1)–(F4) on the square and, in addition, five slope and curvature bounds with nonnegative moduli, each stated for arguments in $[0,1]$; a sixth, (T6), is carried by the enriched subclass on which the sender-only case, Theorem 12, the theorem that both forward-looking incentives change behavior without the one-period ranking (`both_channels_active_of_relief`, after Theorem 12) and the relief and no-non-compliance sides of Proposition C.9 are stated:

(T1) *State slope bound* $0 \leq g_{\mathrm{lo}} \leq g_{\mathrm{hi}} \leq 1$: for every s and every $y_1 \leq y_2$, $g_{\mathrm{lo}}(y_2 - y_1) \leq F(y_2, s) - F(y_1, s) \leq g_{\mathrm{hi}}(y_2 - y_1)$.

(T2) *Sanction slope bound* $0 < d_{\mathrm{lo}} \leq d_{\mathrm{hi}}$: for every x and every $s_1 \leq s_2$, $d_{\mathrm{lo}}(s_2 - s_1) \leq F(x, s_2) - F(x, s_1) \leq d_{\mathrm{hi}}(s_2 - s_1)$.

(T3) *Joint semiconcavity* with moduli $M_x, M_m, M_s \geq 0$: for all x_1, x_2, s_1, s_2 and $t \in [0,1]$,

$$\begin{aligned} &tF(x_1,s_1) + (1-t)F(x_2,s_2) - F\big(tx_1 + (1-t)x_2,\ ts_1 + (1-t)s_2\big) \\ &\qquad \leq \frac{t(1-t)}{2}\Big(M_x(x_1 - x_2)^2 + 2M_m|x_1 - x_2|\,|s_1 - s_2| + M_s(s_1 - s_2)^2\Big). \end{aligned}$$

In the source this is the predicate `JointSemiconcaveOn` with third modulus $-M_s$; the literal content is the display, which bounds how far F lies *below* its chords. Nothing about concavity of F is assumed; the affine law has all three moduli zero.

(T4) *State-section semiconvexity* with modulus $M_{\mathrm{cx}} \geq 0$: for every s, every x_1, x_2 and $t \in [0,1]$, $F(tx_1 + (1-t)x_2, s) - \big(tF(x_1,s) + (1-t)F(x_2,s)\big) \leq \frac{M_{\mathrm{cx}}}{2}\, t(1-t)(x_1 - x_2)^2$, which bounds how far a state section lies *above* its chords; zero for convex sections and for the affine law.

(T5) *Bound on mixed differences* with modulus $M_{\mathrm{mix}} \geq 0$: for all x_1, x_2, s_1, s_2, $\big|(F(x_2,s_2) - F(x_1,s_2)) - (F(x_2,s_1) - F(x_1,s_1))\big| \leq M_{\mathrm{mix}}|x_2 - x_1|\,|s_2 - s_1|$; zero for the affine law and for every additively separable transition.

(T6) *Sanction-section semiconvexity* with modulus $M_{\mathrm{cs}} \geq 0$: for every x, every s_1, s_2 and $t \in [0,1]$, $F(x, ts_1 + (1-t)s_2) - \big(tF(x,s_1) + (1-t)F(x,s_2)\big) \leq \frac{M_{\mathrm{cs}}}{2}\, t(1-t)(s_1 - s_2)^2$; zero for the affine law and for convex sanction sections, 1/5 at the non-affine member. It is a field of the enriched subclass (`ChordEconomyCs`; `PlacableEconomyFcs` on the family), not of the class: it is consumed by the sender-only case, by Theorem 12, where it bounds the curvature of the sender's objective from below, by the theorem after Theorem 12 that both forward-looking incentives change behavior without the one-period ranking, and by the relief and no-non-compliance sides of Proposition C.9, through M_{tot}. Proposition B.1, Theorem 3 and Proposition B.2 are stated without it.

A supplied modulus $\sigma_V \geq 0$ completes the data; it is closed by the Riccati inequality in C and appears on no paper-facing statement as a free parameter. Its existence is part of membership of the class: a member is one for which some σ_V satisfies (curvature) and (Riccati closure), so the pair is a restriction on the primitives, not a preference coefficient.

Remark C.2 (Why the modulus is supplied). Semiconcavity in the sense of Cannarsa and Sinestrari (2004) is the property the sender's invariant set keeps, and two machine-checked refutations say why it must be kept by a supplied modulus rather than derived: the maintained transition class alone admits members whose closed loop preserves no curvature cone (`maxJoinTransition_closedLoop_not_semiconcave`, `minMeetTransition_closedLoop_not_semiconvex`), and the affine extreme point $F(x, s) = s$ with a single-peaked sender loss, whose stage argmax is unique, continuous, monotone and Lipschitz, has a closed loop with two kinks of opposite sign that kill every semiconcavity and every semiconvexity modulus at once (`clampedPeakSender_closedLoop_not_semiconcave`, `clampedPeakSender_closedLoop_not_semiconvex`). The same phenomenon appears for bilevel programs in Example 3 of Chen et al. (2025), where the lower level's argmax destroys the curvature the upper level needs; the constrained case is left open there, and the supplied modulus with its Riccati closure is the answer this family gives.

The concavity condition and the Riccati closure, two inequalities in the supplied modulus σ_V, admit some $\sigma_V \geq 0$ exactly when $Q > 0$, $0 < L < 2Q$ and $L^2 \geq 4DN$ in the primitives alone, where, in this paragraph only,

$$Q = q - \beta_S K_S M_s, \qquad D = \beta_S d_{\mathrm{hi}}^2, \qquad N = \beta_S K_S M_x\, Q + (r + \beta_S K_S M_m)^2,$$

$$L = (1 - \beta_S g_{\mathrm{hi}}^2)\, Q + \beta_S K_S M_x\, D - 2\,(r + \beta_S K_S M_m)\, \beta_S g_{\mathrm{hi}} d_{\mathrm{hi}}$$

(`sigV_feasible_iff`, whose source also names the products inlined here; the vertex $\sigma_V = L/(2D)$ then certifies). The modulus need not be supplied. With the denominator cleared, closure is the quadratic inequality $D\sigma^2 - L\sigma + N \leq 0$, whose least root $\sigma_- = (L - \sqrt{L^2 - 4DN})/(2D)$ is nonnegative, satisfies curvature and closure, and is at most every feasible modulus (`sigVLeast_curv`, `sigVLeast_closure`, `sigVLeast_least`); the member re-certified at σ_- has a larger μ, a smaller ϱ, a smaller response Lipschitz constant and a smaller sharp loop constant (`LloopSharp_withLeastModulus_le`), so every quantitative bound below is tightest there. The members supply the rational $\sigma_V = 1$ for display ($\sigma_- = 10 - 7\sqrt{2}$ at the affine member); a different feasible modulus gives different constants (at the vertex $\sigma_V = 10$ the sharp loop constant of the affine member is exactly 1, `placableAffineVertex_LloopSharp`), and every bound uses one consistently chosen modulus. At the members $(Q, L, N, D) = (3, 5/2, 1/4, 1/8)$ and $(3, 1623/700, 25/28, 9/50)$.

B. Derived constants

$$K_S = \frac{r + w}{1 - \beta_S g_{\mathrm{hi}}}, \quad g_S = \frac{w}{1 - \beta_S g_{\mathrm{lo}}}, \quad M_S = \frac{q/2 + u_0 + k + r + w}{1 - \beta_S}, \quad \mu = q - \beta_S\big(\sigma_V d_{\mathrm{hi}}^2 + K_S M_s\big),$$

$$\varrho = r + \beta_S\big(\sigma_V g_{\mathrm{hi}} d_{\mathrm{hi}} + K_S M_m\big), \qquad \sigma_{\mathrm{img}} = \beta_S\big(\sigma_V g_{\mathrm{hi}}^2 + K_S M_x\big) + \frac{\varrho^2}{\mu},$$

$$A = u_0 + r + \beta_S K_S d_{\mathrm{hi}}, \qquad g_T = \frac{v + n u_0/q}{1 - \beta_T g_{\mathrm{lo}}}, \qquad M_T = \frac{c + v}{1 - \beta_T}, \qquad \sigma_C = \frac{\beta_S K_S M_{\mathrm{cx}}}{1 - \beta_S g_{\mathrm{hi}}^2},$$

$$L_{\mathrm{resp}} = r + \beta_S\big(\bar\sigma g_{\mathrm{hi}} d_{\mathrm{hi}} + (K_S + 2\bar\sigma) M_{\mathrm{mix}}\big), \quad \bar\sigma = \max(\sigma_V, \sigma_C), \qquad L_{\mathrm{loop}} = g_{\mathrm{hi}} + d_{\mathrm{hi}} \cdot \frac{L_{\mathrm{resp}}}{\mu}.$$

K_S is the state-Lipschitz modulus of the sender's value (stage rate $rs + w \leq r + w$ compounded through the upper bound on the transition's state slope); g_S its lower state slope (the stage state

rate w at $s = 0$, through the lower bound on that slope); M_S the sup-norm radius; μ the strong-concavity modulus of the sender's dynamic objective in the sanction; A an upper bound on the sender's dynamic marginal return to the first unit of sanction, at full non-compliance (a bound, not the derivative of a value that need not be differentiable); g_T the target's lower value slope: the direct valuation v plus the cross term n at u_0/q, compounded through the lower bound on the transition's state slope, and at $n = 0$ equal to $v/(1-\beta_T g_{\mathrm{lo}})$; when $u_0 \le q$, as under interiority, u_0/q is the lower bound $\underline{s}$ (`toChord_sfloor`), below which the response at non-compliance never falls on either invariant set.

C. The certificate

A member of the family satisfies

$$\begin{aligned} &\text{(curvature)} && \beta_S\big(\sigma_V d_{\mathrm{hi}}^2 + K_S M_s\big) < q \quad \text{(i.e. } \mu > 0\text{)}, \\ &\text{(Riccati closure)} && \sigma_{\mathrm{img}} \le \sigma_V, \end{aligned}$$

together with the signs $k > 0$, $v > 0$ and $n < c$. Four further conditions are hypotheses of the propositions that read them, not conditions of membership, so that each case of the family is stated on the same structure:

$$\begin{aligned} &\text{(interiority)} && A < q, \\ &\text{(myopic relief condition)} && \ell q < k(c-n), \\ &\text{(the lifting condition)} && A \le k, \\ &\text{(entrenchment condition)} && c \le \beta_T d_{\mathrm{lo}} g_T. \end{aligned}$$

As on the class of panel A0, interiority and the myopic relief condition are hypotheses rather than fields (the predicates `PlacableEconomyF.Interior` and `PlacableEconomyF.StageRelief`, hypotheses `hInt` and `hSR`); the family keeps only the sign of the second, $n < c$, which that condition implies (`hcn_of_stageRelief`). A result that reads interiority only through $u_0 \le q$, under which $\underline{s} = u_0/q$, takes that alone in its bridge to the class (`toChord_sfloor`). The lifting condition is read by the results in which full compliance ends the sanction (the myopic and sender-only cases, the zero level of Appendix C.4); the entrenchment condition by the results in which the forward-looking target maintains non-compliance (the target-only and forward-looking cases, the target's invariant set). In the source they are the predicates `ChordEconomy.Placable` and `ChordEconomy.LossOffset` (on the family `PlacableEconomyF.Placable` and `PlacableEconomyF.LossOffset`, to which they reduce through `toChord`), and the corresponding hypotheses are named `hP` and `hTM` throughout. Derived from the membership conditions and the slope bounds, with nothing separately supplied: $\sigma_V \ge 0$ (Riccati closure with $\mu > 0$), and hence $q > 0$ (the concavity condition puts q above a sum of non-negative terms); under interiority $q > A > 0$, as $A \ge u_0 > 0$, and so $u_0 < q$; under the lifting condition $k \ge A > u_0 + r$; under the entrenchment condition with interiority $v > c - n$ (the condition cleared of its denominator, using $g_{\mathrm{lo}} + d_{\mathrm{lo}} \le 1$ from the two lower slope bounds and the diagonal, and $u_0 < q$); $g_{\mathrm{lo}} \le g_{\mathrm{hi}}$ and $d_{\mathrm{lo}} \le d_{\mathrm{hi}}$ (the slope bounds between 0 and 1); $0 \le M_{\mathrm{mix}}$ (the mixed bound at the corners); $d_{\mathrm{lo}} \le 1$ (the sanction slope bound at state 0 between sanctions 0 and 1); and feasibility of F on the square (monotonicity and the diagonal). In the source six of these (feasibility, the two orderings, $0 \le M_{\mathrm{mix}}$, $q > 0$, $\sigma_V \ge 0$) are theorems of the structure, not fields (`PlacableEconomyF.derived_fields`, from `mem_Icc_of_mono_of_diag`, `StateChordBandAll.rates_le`, `sanctionChord_rates_le`, `MixedChordBound.nonneg`, `plSigV_nonneg_of_closure`, and for $q > 0$ the concavity condition, `PlacableEconomyF.hq`, in place

of interiority, `plq_pos_of_interior`, since); the signs $k > 0$, $v > 0$ and $n < c$, which the lifting condition, the entrenchment condition and the myopic relief condition used to imply, are fields since those became hypotheses; a member supplies the remaining conditions only. The one-period ranking $\ell q < u_0(c - n)$, the endpoint comparison under which full compliance beats full non-compliance in the one-period problem, is not a field: it implies the myopic relief condition through the lifting condition ($u_0 \le k$), and it is a hypothesis of the clause of Proposition B.1 that the forward-looking incentives change behavior (`targetStage_one_gt_zero`, `both_channels_active_of_margin`) and of the dischargers `reliefSharp_of_scaled_Mtot_le` and `reliefSharp_of_Mtot_eq_q`; on the enriched subclass (RL) implies the myopic relief condition ($q \le M_{\mathrm{tot}}$) and replaces the one-period ranking in that clause (`both_channels_active_of_relief`). The entrenchment condition is stated at the refined slope g_T; the strong entrenchment condition $c \le \beta_T d_{\mathrm{lo}} v/(1-\beta_T g_{\mathrm{lo}})$ implies it (`hTMargin_of_old`) and coincides with it at $n = 0$, and the inclusion is strict off the affine law: at $F =$ `quadDiagF` with $(q, u_0, k, r, w, \ell, c, n, v) = (3, 2, 3, \frac{1}{100}, \frac{3}{10}, \frac{3}{5}, 1, \frac{1}{100}, \frac{1199}{300})$ every field holds, the entrenchment condition reads $1 \le 1201/1200$, and the strong entrenchment condition fails, $1199/1200 < 1$ (`placableFloor`, `exists_floor_member_not_old`). Equivalently $v \ge c(1 - \beta_T g_{\mathrm{lo}})/(\beta_T d_{\mathrm{lo}}) - n u_0/q$ (`lossOffset_iff_v_ge`): a sufficient valuation threshold, not a behavioral boundary (`loss_offset_converse_counterexample`, cited in Section 3.1). The increasing-differences condition

$$\beta_S\big(\sigma_C g_{\mathrm{hi}} d_{\mathrm{hi}} + (K_S + 2\sigma_C) M_{\mathrm{mix}}\big) \le r \tag{RM}$$

is a hypothesis of the composed-equilibrium and separation theorems, never a field; at $M_{\mathrm{cx}} = M_{\mathrm{mix}} = 0$ it reads $0 \le r$. It is used only to make the sender's response at full non-compliance, computed at the constructed value V_S^*, non-decreasing in the state (Step 4 of panel E, `responsePath_monotoneOn_of_margin`), and the formalization states the following results under that condition instead: the composed equilibrium (`equilibriumTop`, `both_channels_active_of_response_monotone`), the separation and the convergence of Theorem 3 (`myopic_dynamic_equilibrium_separation_generalF_stage_of_response_monotone`, `dynamicPath_tendsto_of_response_monotone`), Proposition B.2(iv) (`target_only_regime_generalF_of_response_monotone`), case 11 of Theorem 12 (`four_cell_generalF'_of_response_monotone`), Proposition C.5, Proposition 9, Proposition L.5 and Proposition L.7; the forms under the condition are their corollaries, and Proposition D.11 is still stated under the condition. The non-decreasing response is a condition on an endogenous object, the maximizer of the sender's objective at a value that is itself a fixed point, so it cannot be checked from the primitives alone; (RM) is a condition on the primitives and the supplied moduli that implies it, and the main text states the condition for that reason. The condition is not empty on the class: at a transition with concave state sections (`quadDiagF` with its arguments swapped, `quadDiagFSwap`, not affine at any rate and not 0-semiconvex in the state) and $(q, u_0, k, r, w, \ell, c, n, v) = (3, 1, 3, \frac{9}{278}, 1, 1, 4, \frac{1}{100}, 20)$, every field holds and (RM) holds with equality, its left side $\beta_S \sigma_C g_{\mathrm{hi}} d_{\mathrm{hi}} = 9/278 = r$ strictly positive (`placableRallyBinding`, `exists_binding_rally_member`); the mixed term is zero there. The condition restricts the supplied moduli, not the game: `placableQuadDiag` re-certified with the state-section modulus $M_{\mathrm{cx}} = 4$ violates (RM), its left side $270/287$ exceeding $r = 1/2$, while its response at non-compliance, the same function as at the modulus 0, is monotone (`exists_member_response_monotone_not_margin`), so the non-decreasing response is strictly weaker than (RM) for the supplied moduli. No economy that fails (RM) under every choice of moduli is exhibited, and no member of the enriched subclass that fails (RM) is exhibited for the forms of Theorem 12 or of the welfare results; whether the response at the binding member would stay monotone under

a smaller r is open. The forward-looking relief condition

$$\ell\, M_{\mathrm{tot}} < k(c-n), \qquad M_{\mathrm{tot}} = q + \beta_S\big(\sigma_C d_{\mathrm{hi}}^2 + K_S M_{\mathrm{cs}}\big), \tag{RL}$$

is likewise a theorem hypothesis, of the sender-only case and Theorem 12 on the enriched subclass. It is the sharp form of the earlier two-branch relief condition $2\ell \max\{M_{\mathrm{tot}}, k\} < k(c-n)$, which implies it (`reliefSharp_of_relief`) and under which the machine-checked headline rows are still stated; at the two example economies of the subclass (RL) reads $3 < 12$ and $45/14 < 12$, on the whole affine coefficient box of Appendix M.2 it holds with $M_{\mathrm{tot}} = q$, and wherever $u_0 M_{\mathrm{tot}} \le kq$ (in particular wherever $M_{\mathrm{tot}} = q$, by the lifting condition), it follows from the one-period ranking (a hypothesis) alone, through the identity $q\,(k(c-n) - \ell M_{\mathrm{tot}}) = (kq - u_0 M_{\mathrm{tot}})(c-n) + M_{\mathrm{tot}}(u_0(c-n) - \ell q)$ (`reliefSharp_of_scaled_Mtot_le`, `reliefSharp_of_Mtot_eq_q`). The condition $u_0 M_{\mathrm{tot}} \le kq$ is the largest on the other coefficients under which the one-period ranking discharges (RL) for every compliance cost (`reliefSharp_uniform_iff`): outside it, the midpoint of $(k(c-n)/M_{\mathrm{tot}},\, u_0(c-n)/q)$ is a cost that passes the one-period ranking and fails (RL). The gap between the two forms is inhabited off the affine law: at $F =$ `quadDiagF` with $(q, u_0, k, r, w, \ell, c, n, v) = (3, 2, 3, \frac{1}{100}, \frac{3}{10}, \frac{3}{5}, 1, \frac{1}{100}, 5)$ every family condition holds, (RL) reads $6393/3500 < 297/100$, and the two-branch form fails (`placableSharpRelief`, `exists_sharp_relief_member_not_old`); there $u_0 M_{\mathrm{tot}} = 2131/350 \le 9 = kq$ while $M_{\mathrm{tot}} > q$, so the larger discharger applies and the $M_{\mathrm{tot}} = q$ one does not (`placableSharpRelief_relief_disch`).

The four restrictions that mention the cross term n, namely its sign, the one-period ranking, the entrenchment condition and (RL), hold together exactly when $\max\{0, n_{\min}\} \le n < \min\{c - \ell q/u_0,\, c - \ell M_{\mathrm{tot}}/k\}$ with $n_{\min} = (c(1-\beta_T g_{\mathrm{lo}})/(\beta_T d_{\mathrm{lo}}) - v)\, q/u_0$ (`rally_softening_interval_iff`; with the family's own condition, the myopic relief condition, in place of the one-period ranking, exactly when $\max\{0, n_{\min}\} \le n < c - \ell M_{\mathrm{tot}}/k$, `rally_softening_interval_iff'`): a larger cross term n relaxes the entrenchment condition and tightens the two accommodation conditions. The interval is one of sufficient conditions, not the set of economies in which the target maintains non-compliance.

Economic interpretation and cost. Curvature and closure say the sender's own cost curvature out-curves what the continuation can add, so the forward-looking sanction problem is strictly concave and its solution unique; they are regularity, not behavior. Interiority says that even at full non-compliance and the top of the state space the first unit of sanction is worth less than its marginal cost at $s = 1$, so the sender never sanctions fully. The lifting condition says the target's compliance can buy the whole of that marginal return, so full compliance is met with no sanction. The myopic relief condition says that at the stage, compliance is worth buying at the margin: below the relief level a unit of it purchases k/q units of sanction reduction worth at least $c - n$ each, more than its cost ℓ; the one-period ranking $\ell q < u_0(c-n)$ is the endpoint form of the same comparison, full compliance against full non-compliance. The entrenchment condition says the target's per-unit sanction pain is covered by the discounted value of the entrenchment that sanction builds, at the target's lower value slope: the direct valuation v plus the cross term n at u_0/q, the lower bound below which the sanction never falls when $u_0 \le q$, as under interiority; this is a substantive statement about a government that values political entrenchment more than it dislikes the sanction, and it is what makes the forward-looking target maintain non-compliance. The sender's positive valuation w of the state is a political objective of the sanctioning government, not a national-welfare gain. (RM) says the stage complementarity r between the state and the sanction covers two forward-looking forces against a monotone response: saturation of the continuation (bounded by σ_C) and cross-effectiveness of the transition (bounded by M_{mix}). It is the price of transition generality, paid only by transitions with concave state sections or cross-curvature.

D. Invariant sets of value functions

$$\mathcal{C}_S = \{V : \|V\| \le M_S,\ V \text{ is } K_S\text{-Lipschitz},\ V(y_2) - V(y_1) \ge g_S(y_2 - y_1) \text{ for } y_1 \le y_2,$$
$$V \circ \mathrm{clip} \text{ is } \sigma_V\text{-semiconcave on } [0,1]\},$$

$\mathcal{C}_S^2 = \mathcal{C}_S \cap \{V \circ \mathrm{clip} \text{ is } \sigma_C\text{-semiconvex on } [0,1]\}$, and $\mathcal{C}_T = \{V : \|V\| \le M_T,\ V(y_2) - V(y_1) \ge g_T(y_2 - y_1) \text{ for } y_1 \le y_2\}$. All three are closed in the sup norm and nonempty (the affine value $y \mapsto g\,y$ with the right slope lies in each).

Proposition C.3 (Two-sided target value)**.** *On the relief class, let $\gamma_T = g_{\mathrm{hi}} + d_{\mathrm{hi}} L_{\mathrm{resp}}/\mu$ be the sharp loop constant and suppose $\beta_T \gamma_T < 1$. Then the target value of every equilibrium whose sender value lies in the two-sided invariant set is K_T-Lipschitz in the state, with $K_T = L_U/(1 - \beta_T \gamma_T)$ and $L_U = \max\{v_{\mathrm{hi}}, v_{\mathrm{hi}} + n_{\mathrm{hi}}\} + c L_{\mathrm{resp}}/\mu$ ($v + n + c L_{\mathrm{resp}}/\mu$ on the family) (`equilibrium_V_T_lipschitz_of_carrier`; Hinderer 4.2(c) through the invariant closed set of K_T-Lipschitz values, `targetFix_lipschitz`). This covers the constructed equilibrium (`VTstar_lipschitz`) and, with no further condition, the non-compliance-profile value (`VTdef_lipschitz`). For $s_1 \le s_2$ the target's discounted continuation gain from the higher sanction is at most $\beta_T K_T d_{\mathrm{hi}}(s_2 - s_1)$, which needs only $\beta_T \gamma_T < 1$ (`VTdef_continuation_upper`); with a nondecreasing response it is at least $\beta_T g\, d_{\mathrm{lo}}(s_2 - s_1)$ at every slope $g \ge 0$ satisfying the slope condition, in particular at g_T under the entrenchment condition (`VTdef_continuation_lower`, `VTstar_continuation_bracket`). $\beta_T \gamma_T < 1$ is a hypothesis of the theorems that read it, not a field; K_T and L_U are closed forms in existing constants.*

Proposition C.4 (Certified equilibria)**.** *Let E be any equilibrium in $\mathcal{E}$ of an economy in the class whose target policy is nonincreasing in the state, on a member at which a larger concession is never worth more to the sender at zero sanction (automatic on the family). Then the sender's value has state slope at least $g_S = w_{\mathrm{lo}}/(1 - \beta_S g_{\mathrm{lo}})$ (`V_S_lowerSlope_of_antitone`), and, if the member also has an upper substitution rate k_{hi} (`PlacationCeiling`; the family has one, k), at every state the realized sanction is at least $\min\{(u_{\mathrm{lo}} + r_{\mathrm{lo}} x - k_{\mathrm{hi}} \alpha(x) + \beta_S g_S d_{\mathrm{lo}})/\bar{Q}, 1\}$; on the family this is the stage vertex at the policy's compliance plus the foresight premium $\beta_S g_S d_{\mathrm{lo}}/q$ (`equilibrium_sanction_ge_wedge_of_antitone`). The general theorem (`senderValue_lowerSlope_of_policy_chord`, `Theory/Equilibrium/CertifiedEquilibria.lean`) is stated over the abstract game: the sender operator at any continuous policy is a contraction on the whole value space, the lower-slope cone is closed, nonempty and invariant under a joint state-slope condition, so the unique fixed point lies in it; on the class, under the pairwise neutrality, that condition reduces to the policy being nonincreasing in the state (`chord_of_antitone`). Nothing is said about equilibria whose compliance rises with entrenchment, and no equilibrium beyond the constructed ones is asserted to exist.*

E. Proof of Proposition B.1

Step 1: the sender's value. The sender Bellman operator at full non-compliance,

$$(\mathsf{T}_S V)(x) = \max_{s \in [0,1]} U_S(x, 0, s) + \beta_S V(F(x, s)),$$

maps $\mathcal{C}_S$ into itself: the sup bound is the stage bound geometrically summed; the K_S-Lipschitz bound is the stage rate $r + w$ plus $\beta_S K_S g_{\mathrm{hi}}$ through (T1); the lower slope g_S is the stage state rate w plus $\beta_S g_S g_{\mathrm{lo}}$ through (T1), the maximum being taken pointwise in x; and the semiconcavity of the image is the Schur modulus $\sigma_{\mathrm{img}} = \sigma + \varrho^2/\mu$ of a supremum over s of a function that is jointly semiconcave with moduli (σ, ϱ, μ) in (x, s), which requires the concavity condition $\mu > 0$ in

the same clause (the modulus is undefined without it). In this modulus the stage term contributes $(0, r, q)$, the continuation contributes β_S times the composition moduli obtained from (T3) and the invariant set's σ_V, and the closure inequality makes it at most σ_V. The operator is a β_S-contraction in the sup norm, so it has a unique fixed point $V_S^* \in \mathcal{C}_S$ (Banach). Because $\mu > 0$, the objective $s \mapsto U_S(x, a, s) + \beta_S V_S^*(F(x, s))$ is μ-strongly concave, so the maximizer $\rho(x, a)$ is unique and, by the maximum theorem, continuous in x; it coincides with the dynamic best response of Section 2.

Step 2: the response is interior, zero at full compliance, and above the myopic sanction. Cap: for $t \le s$ the stage satisfies $U_S(x, a, t) - U_S(x, a, s) \ge (s - t)\big(q(s + t)/2 - (u_0 + r)\big)$ because the linear coefficient $u_0 - ka + rx$ is at most $u_0 + r$, while the continuation is $\beta_S K_S d_{\mathrm{hi}}$-Lipschitz in s by (T2); a maximizer of a q-strongly concave function with initial difference quotient at most $L_0 = u_0 + r$ plus an L-Lipschitz perturbation lies at or below $(L_0 + L)/q = A/q$, and $A/q < 1$ by interiority. The lifting condition: at $a = 1$ and any $s > 0$ the objective's change from $s = 0$ is at most $s(u_0 + r - k + \beta_S K_S d_{\mathrm{hi}}) - \frac{q}{2}s^2 = s(A - k) - \frac{q}{2}s^2 < 0$, so $\rho(x, 1) = 0$. Foresight premium: the continuation is nondecreasing in s ((T2) and $g_S \ge 0$), so $\rho(x, a)$ lies weakly above the stage maximizer; at $a = 0$ the stage maximizer is the vertex $(u_0 + rx)/q$, and moving the sanction up from it by $\varepsilon \le \min\{1 - \text{vertex}, \ \beta_S g_S d_{\mathrm{lo}}/q\}$ costs the stage $\frac{q}{2}\varepsilon^2$ and gains the continuation at least $\beta_S g_S d_{\mathrm{lo}}\varepsilon$, which is larger. So the vertex is not a maximizer of J_S, no maximizer of J_S is a stage maximizer (the sender's forward-looking incentive changes behavior), and $\rho(x, 0) > (u_0 + rx)/q \ge u_0/q$.

Step 3: the target maintains non-compliance, and the stage disagrees. Fix $V_T \in \mathcal{C}_T$ and $V_S \in \mathcal{C}_S$. The response is antitone in a (the stage has decreasing differences $-ksa$ in (s, a), and the argmax is unique), so raising a from a_1 to a_2 lowers the sanction by $\Delta\rho \ge 0$. The target's objective $U_T(x, a, \rho(x, a)) + \beta_T V_T(F(x, \rho(x, a)))$ changes by at most $-\ell(a_2 - a_1) + \Delta\rho\,(c - nx) - \beta_T g_T d_{\mathrm{lo}}\Delta\rho \le -\ell(a_2 - a_1) + \Delta\rho\,(c - \beta_T d_{\mathrm{lo}} g_T) < 0$, using (T2) for the state loss, the lower slope g_T for the value loss, and the entrenchment condition. So $a = 0$ is the unique maximizer at every value pair in $\mathcal{C}_T \times \mathcal{C}_S$. At the stage, full compliance gives $-\ell + vx$ (the sanction is zero), while full non-compliance gives $\rho(x, 0)(nx - c) + vx \le -(u_0/q)(c - n) + vx < -\ell + vx$ by the one-period ranking, the hypothesis of that clause of Proposition B.1; so no forward-looking target maximizer is a stage maximizer (the target's forward-looking incentive changes behavior). Under (RL) on the enriched subclass the same follows without the one-period ranking: a compliance $a \le \min\{1, \mu\rho(x, 0)/(2k)\}$ keeps the response positive, the sharp rate of sanction reduction gives $\rho(x, 0) - \rho(x, a) \ge ka/M_{\mathrm{tot}}$, and the stage gain is at least $a\,(k(c - nx)/M_{\mathrm{tot}} - \ell) > 0$ (`target_channel_active_of_relief`).

Step 4: the monotone response and the composed equilibrium. Under (RM), a condition on members of the two-sided extension of panel A0, to which the family belongs, on $\mathcal{C}_S^2$ the sender's objective has increasing differences in (sanction; state): the stage cross-difference is exactly $r\,\Delta x\,\Delta s$, and the continuation's is bounded below by $-\beta_S\big(\sigma_C g_{\mathrm{hi}} d_{\mathrm{hi}} + (K_S + 2\sigma_C) M_{\mathrm{mix}}\big)\Delta x\,\Delta s$ through the semiconvexity σ_C of $V_S^* \circ \mathrm{clip}$ and (T1), (T2), (T5). With a unique argmax, increasing differences make $\rho(\cdot, 0)$ nondecreasing on $[0, 1]$. This is the only use of (RM): from here on only that monotonicity is read, and the formalization builds the composed equilibrium from it (`equilibriumTop`; the form under the increasing-differences condition, `equilibriumTop_of_margin`, is its instance). Given that, the target operator at the response, $(\mathsf{T}_T V)(x) = U_T(x, 0, \rho(x, 0)) + \beta_T V(F(x, \rho(x, 0)))$, maps $\mathcal{C}_T$ into itself: the sup bound is direct, and the lower slope g_T is preserved because the stage payoff at non-compliance rises by at least $v + n\rho \ge v + nu_0/q$ per unit of state (the response never falls below u_0/q, `responsePath_ge_sfloor`), and a higher state gives a weakly higher sanction and hence, through (T1) and the entrenchment condition, a continuation gain that at least offsets the stage loss $-(c - nx)\Delta\rho$. It is a β_T-contraction; its fixed point V_T^* and $\alpha \equiv 0$ complete $(V_T^*, V_S^*, 0) \in \mathcal{E}$, with both forward-looking incentives changing behavior by Steps 2 and 3. $V_S^* \in \mathcal{C}_S^2$ because σ_C is the fixed point of the linear closure $\sigma \mapsto \beta_S(\sigma g_{\mathrm{hi}}^2 + K_S M_{\mathrm{cx}})$.

F. Proof of Proposition 2 and Theorem 3

Step 5: the myopic case. With $\beta_S = \beta_T = 0$ the sender's problem is the stage problem, whose unique maximizer is the clipped vertex $\max\{0, \min\{1, (u_0 - ka + rx)/q\}\}$; it is zero exactly when $a \geq a^{00}(x) := (u_0 + rx)/k$, which lies in $(0,1)$ because $u_0 + r < k$. For $a \geq a^{00}(x)$ the target's payoff is $-\ell a + vx$, strictly decreasing; for $a < a^{00}(x)$ it is $-\ell a + \frac{u_0 - ka + rx}{q}(nx - c) + vx$, whose slope in a is $-\ell + \frac{k}{q}(c - nx) > 0$ because $\ell q < k(c - n) \leq k(c - nx)$, the myopic relief condition (`myopic_target_lt`). So $a^{00}(x)$ is the unique optimum, the realized sanction is zero, and the values wx and $vx - \ell a^{00}(x)$ satisfy the two Bellman equations of Section 2 at zero discount, where the dynamic best responses reduce to the myopic best response because the stage objective is strictly concave in the sanction. Convergence to zero: (T2) at state x between sanctions 0 and x, with $F(x,x) = x$, gives $F(x,0) \leq (1 - d_{\text{lo}})x$, hence $x_t \leq (1 - d_{\text{lo}})^t x_0$.

Step 6: the forward-looking paths. Write $b = u_0/q$ and $\rho(y) = \rho(y,0) \in (b, A/q]$ by Step 2. The lower bound on the build-up of the state: when $y \leq b$, (T2) between sanctions y and b gives $F(y, \rho(y)) \geq F(y,b) \geq y + d_{\text{lo}}(b - y)$, and the comparison recursion $z_{t+1} = z_t + d_{\text{lo}}(b - z_t)$ from $z_0 = 0$ gives $x_t \geq b\big(1 - (1 - d_{\text{lo}})^t\big)$; the affine minorant is used only at $y \leq b$. From a common start the forward-looking state strictly exceeds the myopic one: at $t = 1$, $F(x_0, \rho(x_0)) - F(x_0, 0) \geq d_{\text{lo}}\rho(x_0) > 0$, and monotonicity of F in the state propagates the strict gap. None of this uses monotonicity of ρ.

Step 7: convergence. The response ρ is continuous without further conditions; when it is nondecreasing, as Step 4 gives under (RM), the closed loop $G(y) = F(y, \rho(y))$ is nondecreasing and continuous on $[0,1]$; its orbit from any x_0 is monotone, hence convergent to a fixed point x_∞^{11}. Since $F(x_\infty^{11}, \rho(x_\infty^{11})) = x_\infty^{11} = F(x_\infty^{11}, x_\infty^{11})$ and $d_{\text{lo}} > 0$, (T2) forces $\rho(x_\infty^{11}) = x_\infty^{11}$, and ρ's range gives $b < x_\infty^{11} \leq A/q$ (`dynamicPath_tendsto_of_response_monotone`). The limit may depend on x_0; what is unique is the *state* at which sanction equals state along that orbit, not the equilibrium.

Step 8: the contraction bound (separate). By strong concavity μ and the fact that the objective's dependence on the state is $L_{\text{resp}}|x - y|$-Lipschitz in the sanction (through r, (T1), (T2), (T5) and $\bar{\sigma}$), the maximizers satisfy $|\rho(x) - \rho(y)| \leq L_{\text{resp}}|x - y|/\mu$ (the sharp argmax stability bound, from the strong-concavity inequality at $t\rho(x) + (1-t)\rho(y)$ and $t \downarrow 0$; the midpoint inequality of the first formalization gave the factor four); then $|G(x) - G(y)| \leq g_{\text{hi}}|x - y| + d_{\text{hi}}|\rho(x) - \rho(y)| \leq L_{\text{loop}}|x - y|$. When $L_{\text{loop}} < 1$ the steady state is unique and globally stable (Proposition D.11); this is not an ingredient of Steps 1–7.

Two views of one concept. Write $E^{00} \in \mathcal{E}(p^{00})$ for the myopic case and $E^{11} \in \mathcal{E}(p)$ for the forward-looking one; the game is the same, only the discount factors differ. The comparison at a fixed target policy of Section 3 and this equilibrium comparison, in which both governments re-optimize, are distinct. The two mixed cases are Proposition B.2 and Proposition B.3, and Theorem 12 orders all four; no interaction theorem is claimed beyond those directions.

G. Proof of Proposition B.3

Step 1 (the zero-sanction value). Let $\mathcal{R}V(x) = w\,x + \beta_S\,V(F(x,0))$ on the two-sided sender invariant set of panel D (sup-norm at most M_S, state-Lipschitz K_S, lower slope g_S, σ_V-semiconcave and σ_C-semiconvex through the clip). $\mathcal{R}$ is a β_S-contraction (`relaxOp_contracting`) and maps that set to itself (`relaxOp_mem2`): the norm and Lipschitz legs use the upper bound on the transition's state slope at $s = 0$; the lower-slope leg uses the lower bound on that slope; semiconcavity of the composition $V \circ F(\cdot, 0)$ follows from the joint semiconcavity (T3) restricted to $s_1 = s_2 = 0$, with composed modulus $\sigma_x + \beta_S(\sigma_V g_{\text{hi}}^2 + K_S M_x) \leq \sigma_V$ by the Riccati closure (`chSig_le_SigV`); semiconvexity follows from the state-section semiconvexity (T4) through the composition lemma

of `Foundations/SemiconvexComposition`. Its fixed point V_S^{01} (`VS01`, `VS01_fix`) is the value of the path along which the sanction is zero forever. Nothing about the target has been used: this is why the sender's value can be written down before the target's policy is known.

Step 2 (the response and its two curvatures). At V_S^{01} the sender's dynamic objective in the sanction is $\psi_{x,a}(s) = U_S(x,a,s) + \beta_S V_S^{01}(F(x,s))$. It is μ-strongly concave, as in panel E, so its maximizer $\rho(x,a)$ is unique and, by the maximum theorem, jointly continuous on $[0,1]\times\mathbb{R}$ (`ρ01_continuousOn`). It is also M_{tot}-semiconvex in s with $M_{\text{tot}} = q + \beta_S(\sigma_C d_{\text{hi}}^2 + K_S M_{\text{cs}})$ (`senderObj_semiconvex`): the stage term contributes q, and the continuation term is the composition of a σ_C-semiconvex, K_S-Lipschitz value with a sanction section whose slopes lie in $[d_{\text{lo}}, d_{\text{hi}}]$ and which is M_{cs}-semiconvex (T6). This is the only step of this proof that uses (T6); the same ceiling M_{tot} enters the relief and no-non-compliance sides of Proposition C.9.

Step 3 (the rate of sanction reduction). Since U_S is linear in a with slope $-ks$, $\psi_{x,a} = \psi_{x,a'} + k(a'-a)\,s$ for $a < a'$. The shift lemma of `Foundations/SemiconvexArgmaxShift` says that if ψ is M-semiconvex on $[0,1]$ with an interior maximizer s', and s maximizes $\psi + \lambda u$ with $\lambda > 0$, then $s - s' \geq \min\{\lambda/M,\, 1 - s'\}$, and $s - s' \geq \lambda/M$ outright when $s < 1$: at an interior maximizer z of an M-semiconvex function the semiconvexity inequality through $z-\eta$ and y, read with $\eta \to 0$ on both sides of z, is the quadratic lower support $\psi(y) \geq \psi(z) - \frac{M}{2}(y-z)^2$; applying it to ψ at s' and to $\psi + \lambda u$ at s, each evaluated at the other maximizer, and adding gives $\lambda(s-s') \leq M(s-s')^2$, where $s - s' > 0$ by the coarse shift lemma, so that dividing gives $s - s' \geq \lambda/M$ (`argmax_shift_ge_of_semiconvex_linear_sharp`; the quadratic $-\frac{M}{2}(u-z)^2$ attains the bound, so nothing sharper holds on the class). Hence, for $a < a' < a^{01}(x)$, where both responses are interior ($\rho(x,a') > 0$ below the zero level and $\rho < 1$ by interiority),

$$\rho(x,a) - \rho(x,a') \;\geq\; \frac{k(a'-a)}{M_{\text{tot}}}$$

(`ρ01_shift_ge_linear`).

Step 4 (the zero level). Let $Z(x) = \{a \in [0,1] : \rho(x,a) = 0\}$. It is nonempty by the lifting condition ($\rho(x,1) = 0$) and closed by continuity, so $a^{01}(x) = \min Z(x)$ exists (`zeroLevel`, `zeroLevel_mem`). The response is antitone in a, so $\rho(x,a) = 0$ for $a \geq a^{01}(x)$ and $\rho(x,a) > 0$ for $a < a^{01}(x)$ (`ρ01_eq_zero_of_ge`, `ρ01_pos_of_lt`). At the myopic relief level $a^{00}(x) = (u_0 + rx)/k$ the stage marginal at $s = 0$ vanishes but the continuation marginal does not: V_S^{01} has lower slope $g_S > 0$ and the sanction section has lower slope bound $d_{\text{lo}} > 0$, so a small positive sanction beats zero (`ρ01_pos_at_myopicPolicy`). Hence $a^{00}(x) < a^{01}(x)$ (`myopicPolicy_lt_zeroLevel`); this is the foresight premium. Letting $a' \uparrow a^{01}(x)$ in Step 3, with $\rho(x,a') \geq 0$, gives for $a < a^{01}(x)$ (`ρ01_ge_of_lt_sharp`; no continuity of the response is needed)

$$\rho(x,a) \;\geq\; \frac{k\,(a^{01}(x) - a)}{M_{\text{tot}}}.$$

This is the untruncated per-unit rate of sanction reduction.

Step 5 (the target's optimum). With $\beta_T = 0$ the target maximizes $\tau_x(a) = -\ell a + \rho(x,a)(nx - c) + v\,x$ over $[0,1]$ (`tObj01`). For $a > a^{01}(x)$, $\rho = 0$ and $\tau_x(a) = -\ell a + v\,x < \tau_x(a^{01}(x))$ (`tObj01_lt_of_gt`). For $a < a^{01}(x)$, write $\Delta = a^{01}(x) - a \in (0,1]$; then, by Step 4 and $c - nx \geq c - n > 0$,

$$\tau_x(a^{01}(x)) - \tau_x(a) = (c - nx)\rho(x,a) - \ell\Delta \;\geq\; \Delta\Big(\frac{k(c-n)}{M_{\text{tot}}} - \ell\Big) > 0$$

by the forward-looking relief condition (RL) $\ell M_{\text{tot}} < k(c-n)$ (`tObj01_lt_of_lt'`). So $a^{01}(x)$ is the unique maximizer (`zeroLevel_isMaxOn'`, `zeroLevel_argmax_unique'`). (RL) is a

per-unit benefit–cost condition: the sanction pain avoided per unit of compliance, at the rate of sanction reduction k/M_{tot} and the lowest price $c-n$, exceeds the unit compliance cost ℓ. The two-branch form $2\ell \max\{M_{\text{tot}}, k\} < k(c-n)$ under which the headline rows are stated implies it (`reliefSharp_of_relief`), and those rows are recovered as corollaries (`tObj01_lt_of_lt`, `zeroLevel_isMaxOn`).

Step 6 (continuity of the zero level). Fix x_0 and $\varepsilon > 0$. If $a^{01}(x_0) - \varepsilon < 0$ the lower bound $a^{01}(x) > a^{01}(x_0) - \varepsilon$ holds by feasibility; otherwise $\rho(x_0, a^{01}(x_0) - \varepsilon) > 0$, and joint continuity gives $\rho(x, a^{01}(x_0) - \varepsilon) > 0$ for x near x_0, so $a^{01}(x) > a^{01}(x_0) - \varepsilon$. Conversely, if $a^{01}(x) > a^{01}(x_0) + \varepsilon$ then Step 4 gives $\rho(x, a^{01}(x_0)) \geq k\varepsilon/M_{\text{tot}} > 0$, while $\rho(x, a^{01}(x_0)) \to \rho(x_0, a^{01}(x_0)) = 0$; so $a^{01}(x) \leq a^{01}(x_0) + \varepsilon$ for x near x_0 (`zeroLevel_continuousOn`, from the general continuity of the least zero under a linear rate of sanction reduction, `ZeroLevel.level_continuousOn`). The policy is extended to $\mathbb{R}$ through the clip (α`01`, α`01_continuous`).

Step 7 (the two Bellman equations). Take $V_S = V_S^{01}$, $\alpha = a^{01}$ and $V_T(x) = v\,x - \ell a^{01}(x)$ (`VT01`). The sender's equation holds because the maximizer of $\psi_{x,a^{01}(x)}$ is $\rho(x, a^{01}(x)) = 0$, so the maximum equals $U_S(x, a^{01}(x), 0) + \beta_S V_S^{01}(F(x,0)) = w\,x + \beta_S V_S^{01}(F(x,0)) = V_S^{01}(x)$ by Step 1. The target's equation holds because $\beta_T = 0$ and $\tau_x(a^{01}(x)) = -\ell a^{01}(x) + v\,x$ is the maximum by Step 5, which is also the selector-consistency condition. The tuple is an equilibrium of the sender-only case (`senderOnlyEquilibriumTop`); the equilibrium is not assumed anywhere in Steps 1–6.

Step 8 (convergence to zero). The realized sanction is $\rho(x, a^{01}(x)) = 0$ (`senderOnlyRule_eq_zero`), so $x_{t+1} = F(x_t, 0)$. The lower bound on the transition's sanction slope, between 0 and x_t and with $F(x_t, x_t) = x_t$, gives $x_t - F(x_t, 0) \geq d_{\text{lo}} x_t$, hence $x_t \leq (1 - d_{\text{lo}})^t x_0 \to 0$ (`senderOnlyPath_le_geom`, `senderOnlyPath_tendsto_zero`). □

H. Proof of Proposition B.2

Clauses (i)–(iii) hold on the class of panel A0 under the entrenchment condition alone (`target_only_regime_generalF_core`); (iv) adds $u_{\text{hi}} < q$ and (RM), on the two-sided extension (`target_only_regime_generalF`, the corollary of `target_only_regime_generalF_of_response_monotone`); the closed forms of (iii) are on the family under interiority and the entrenchment condition (`PlacableEconomyF.targetOnlyPath_tendsto`, `targetOnlyPath_error_le_geometric`), and the family bundle `target_only_regime_closedForm` adds (RM) for (iv). (T6) is not used. This panel is written on the class: $\sigma(x, a)$ is the myopic best response and $\underline{s} = \sigma(0, 0)$.

Step 1 (the sender). At $\beta_S = 0$ the sender maximizes $U_S(x, a, \cdot)$, which is strictly concave by (S3) (`stage_strictConcaveOn`), so $\sigma_{V_S}^{\text{dyn}}(x, a) = \sigma(x, a)$ for every continuation value (`Theory/Mechanism/ZeroDiscountSelector.lean`). The sanction rule is therefore $\sigma(x, 0)$ (`targetOnlyRule_eq_stage`), and at $\alpha \equiv 0$ the sender's equation holds at the stage value $V_S^{10}(x) = U_S(x, 0, \sigma(x, 0))$ (`VS10`). None of the following needs a theorem-specific condition. $\sigma(x, \cdot)$ is antitone, by (S9) and uniqueness (`stage_`σ`_antitoneOn`). $\sigma(\cdot, 0)$ is continuous (panel A0) and nondecreasing, by (S7) and uniqueness (`stage_argmax_mono`), so it never falls below $\underline{s}$ (`sfloor_le`).

Step 2 (the target and the equilibrium). The argument of Step 3 of panel E applies with $\sigma(x, \cdot)$ in place of $\rho(x, \cdot)$. Take $a_1 < a_2$ in $[0, 1]$, $\sigma_i = \sigma(x, a_i)$ and $V_T \in \mathcal{C}_T$. By (P1) and (P2) the stage term changes by at most $-\ell_{\text{lo}}(a_2 - a_1) + c(\sigma_1 - \sigma_2)$, and by (T2) and the slope g_T the continuation changes by at most $-\beta_T d_{\text{lo}} g_T(\sigma_1 - \sigma_2)$. The objective $U_T(x, a, \sigma(x, a)) + \beta_T V_T(F(x, \sigma(x, a)))$ therefore changes by at most $-\ell_{\text{lo}}(a_2 - a_1) + (\sigma_1 - \sigma_2)(c - \beta_T d_{\text{lo}} g_T) < 0$, by the entrenchment condition (`targetObj10_strictAntiOn`), so $a = 0$ is the unique maximizer. The operator $(\mathsf{T}_T^{10} V)(x) = U_T(x, 0, \sigma(x, 0)) + \beta_T V(F(x, \sigma(x, 0)))$ is a β_T-contraction (`targetOp10_contracting`). It maps $\mathcal{C}_T$ into itself as in Step 4 of panel E with $\sigma(\cdot, 0)$ in place of $\rho(\cdot, 0)$ (`targetOp10_mem`). The sup bound is direct. For $y_1 \leq y_2$, now write $\sigma_i = \sigma(y_i, 0)$; the stage term rises by at least

$(v_{\mathrm{lo}} + n_{\mathrm{lo}}\underline{s})(y_2 - y_1) - c(\sigma_2 - \sigma_1)$ ((P3) at $\sigma_1 \geq \underline{s}$, then (P2)), and the continuation rises by at least $\beta_T g_T\big(g_{\mathrm{lo}}(y_2 - y_1) + d_{\mathrm{lo}}(\sigma_2 - \sigma_1)\big)$ ((T1), (T2)). With $g_T(1 - \beta_T g_{\mathrm{lo}}) = v_{\mathrm{lo}} + n_{\mathrm{lo}}\underline{s}$ and the entrenchment condition the sum is at least $g_T(y_2 - y_1)$. Step 4 of panel E gets the order $\sigma_1 \leq \sigma_2$ from (RM); here it comes from (S7). The fixed point $V_T^{10} \in \mathcal{C}_T$ (`VT10`), with V_S^{10} and $\alpha \equiv 0$, is an equilibrium in $\mathcal{E}$ of the target-only case (`targetOnlyEquilibriumTop`), and at its values every target maximizer is 0 (`targetOnlyEquilibriumTop_argmax_unique`). This proves (i) and (ii).

Step 3 (the path, on the class). The rule is at least $\underline{s}$, so Step 6 of panel F with $b = \underline{s}$ gives $x_t \geq \underline{s}\big(1 - (1 - d_{\mathrm{lo}})^t\big)$ (`targetOnlyPath_ge_approach`). The rule is nondecreasing and continuous, so, as in Step 7 of panel F, the orbit of $y \mapsto F(y, \sigma(y, 0))$ converges to some x_∞^{10}. (T2) with $d_{\mathrm{lo}} > 0$ then forces $\sigma(x_\infty^{10}, 0) = x_\infty^{10}$, and so $x_\infty^{10} \geq \underline{s}$ (`ChordEconomy.targetOnlyPath_tendsto`). No theorem-specific condition enters. The class result does not say the fixed point is unique, so x_∞^{10} may depend on x_0.

Step 4 (the family). Under interiority the myopic best response at non-compliance is the vertex $(u_0 + rx)/q \in [0, 1)$ (`targetOnlyRule_eq_vertex`). The family's entrenchment condition is the class's, because $u_0 \leq q$ gives $\underline{s} = u_0/q$ (`toChord_sfloor`, `toChord_lossOffset`). Since $r < q$, the vertex fixes only $u_0/(q - r)$ (`vertex_eq_self_iff`), so $x_\infty^{10} = u_0/(q - r)$ from every start (`PlacableEconomyF.targetOnlyPath_tendsto`). For the rate, the rule is affine about x_∞^{10} with slope $r/q \in [0, 1)$: $(u_0 + rx)/q - x_\infty^{10} = (r/q)(x - x_\infty^{10})$ (`vertex_sub_steady10`). Suppose $x \leq x_\infty^{10}$. Then $R = (u_0 + rx)/q$ lies in $[x, x_\infty^{10}]$, monotonicity of F and the diagonal put $F(x, R)$ in $[x, R]$, and (T2) gives $F(x, R) - x \geq d_{\mathrm{lo}}(R - x) = d_{\mathrm{lo}}(1 - r/q)(x_\infty^{10} - x)$. Hence $|F(x, R) - x_\infty^{10}| \leq \big(1 - d_{\mathrm{lo}}(1 - r/q)\big)|x - x_\infty^{10}|$; the argument for $x \geq x_\infty^{10}$ is symmetric. Iterating gives the centered bound of (iii) (`CenteredContraction.iterate_abs_le`, `targetOnlyPath_error_le_geometric`). The bound compares each state with x_∞^{10}; it does not compare two paths. This proves (iii).

Step 5 (the forward-looking limit is higher). Let $u_{\mathrm{hi}} < q$, and let the response $\rho = \rho(\cdot, 0)$ at V_S^* be nondecreasing, which Step 4 of panel E gives under (RM). From x_0 the forward-looking path converges to some x_∞^{11} with $\rho(x_\infty^{11}) = x_\infty^{11}$ (Step 7 of panel F, `dynamicPath_tendsto_of_response_monotone`). The strict foresight premium of Step 2 of panel E holds on the class under $u_{\mathrm{hi}} < q$ alone, with $\bar{Q}$ in place of q: $\sigma(x, 0) < \rho(x)$ at every state (`responsePath_gt_stage`). Write x_t^{10} and x_t^{11} for the two paths. If $x_t^{10} \leq x_t^{11}$, then $F(x_t^{10}, \sigma(x_t^{10}, 0)) \leq F(x_t^{10}, \rho(x_t^{10})) \leq F(x_t^{11}, \rho(x_t^{11}))$, because ρ is nondecreasing. By induction the target-only path stays below the forward-looking one (`rulePathF_le_of_rule_le_and_upper_mono`), so $x_\infty^{10} \leq x_\infty^{11}$. Equality would give $\sigma(x_\infty^{11}, 0) = x_\infty^{11} = \rho(x_\infty^{11})$, which the foresight premium rules out, so $x_\infty^{10} < x_\infty^{11}$ (`targetOnly_limit_lt_dynamic_limit_of_response_monotone`; under (RM), `targetOnly_limit_lt_dynamic_limit`). On the family, interiority gives $u_{\mathrm{hi}} < q$, and the comparison reads $u_0/(q - r) < x_\infty^{11}$ from every start (`targetOnly_steady_lt_dynamic_steady`). This proves (iv). □

I. Proof of Theorem 12

The theorem puts together results proved above and adds no new estimate. It is stated on the enriched subclass of panel A0 (`four_cell_generalF'`). Its form with a nondecreasing response in place of (RM) is `four_cell_generalF'_of_response_monotone`; `four_cell_generalF'` is that form at the response of Step 4 of panel E (`responsePath_monotoneOn_of_margin`), and `four_cell_generalF`, under the two-branch relief condition, follows from it through `reliefSharp_of_relief`. Each case uses the following hypotheses.

Case 00. The myopic case, Proposition 2, needs the myopic relief condition, and here that condition follows from (RL). The sender's stage curvature is at most its dynamic one, $\bar{Q} \leq M_{\mathrm{tot}}$ (`Qbar_le_Mtot`), so $\ell_{\mathrm{hi}}\bar{Q} \leq \ell_{\mathrm{hi}} M_{\mathrm{tot}} < k_{\mathrm{lo}} p_{\mathrm{lo}}$ (`stageRelief_of_reliefSharp`). Interiority and the lifting condition enter only through $u_{\mathrm{hi}} < q$ and $u_{\mathrm{hi}} < k_{\mathrm{lo}}$ (`uhi_lt_q`, `uhi_lt_k`). With these, Step 5

of panel F gives a zero realized sanction and $x_t \le (1-d_{\text{lo}})^t x_0 \to 0$ (`myopicPath_tendsto_zero`).

Case 01. This is Proposition B.3(iii), Step 8 of panel G (`senderOnlyPath_tendsto_zero'`). It uses interiority, the lifting condition, (RL), compliance neutrality, and (T6) through M_{tot} (Step 2 of panel G).

Cases 10 *and* 11. Apply Step 5 of panel H, with $u_{\text{hi}} < q$ from interiority and with the entrenchment condition (`targetOnly_limit_lt_dynamic_limit_of_response_monotone`). From each x_0 the target-only path converges to x_∞^{10} with $\sigma(x_\infty^{10}, 0) = x_\infty^{10}$, and the forward-looking path converges to x_∞^{11} with $\rho(x_\infty^{11}) = x_\infty^{11}$ and $x_\infty^{10} < x_\infty^{11}$. The fixed point is positive because a non-compliant target is sanctioned at every state: $x_\infty^{10} = \sigma(x_\infty^{10}, 0) > 0$ by (S11) with $u_{\text{lo}} > 0$ (`stage_argmax_pos`). Case 10 uses only the entrenchment condition (Steps 1–3 of panel H). Case 11 also uses $u_{\text{hi}} < q$ and, through the nondecreasing response, (RM). Neither case uses the lifting condition, (T6), (RL) or neutrality.

On the family (`PlacableEconomyFcs.four_cell_closedForm`), Step 4 of panel H gives $x_\infty^{10} = u_0/(q-r)$ from every start. This is positive because $u_0 > 0$ and interiority gives $r < q$ (`steady10_pos`), and Step 5 gives a forward-looking limit strictly above it. □

C.5 Consequences for the forward-looking equilibrium of the class under the lifting condition

Throughout this subsection $m = \beta_S g_S d_{\text{lo}}/\bar{Q}$ abbreviates the bound (B.3) on the sender's foresight premium, $\beta_S w d_{\text{lo}}/\big(q(1-\beta_S g_{\text{lo}})\big)$ on the family; it is a combination of the class's rates, not a parameter. V_S and V_T are the values of the constructed forward-looking equilibrium of Proposition B.1 and $R(x)$ its realized sanction at non-compliance.

The premium bound. The proofs below read the premium bound (B.3) of Section 3.2. Its proof follows the formalization.

Proof of (B.3). Let the member satisfy interiority. Fix a value $V \in \mathcal{C}_S$ in the sender's invariant set and a state $x \in [0,1]$, and write $\phi(s) = U_S(x, 0, s)$, $H(s) = \beta_S V(F(x,s))$ and $s^\circ = \sigma(x, 0)$. The response $\rho_V(x)$ at non-compliance maximizes $\phi + H$ on $[0,1]$ (`response_isMaxOn`); at $V = V_S$ it is $R(x)$. *The continuation's lower slope.* For $s_1 \le s_2$ in $[0,1]$ the lower bound (T2) on the transition's sanction slope gives $F(x,s_2) - F(x,s_1) \ge d_{\text{lo}}(s_2 - s_1) \ge 0$, and every value in the invariant set rises by at least g_S per unit of state (`clip_lowerSlope`). Hence $H(s_2) - H(s_1) \ge L(s_2 - s_1)$ with $L = \beta_S g_S d_{\text{lo}}$. Here $L > 0$, because $\beta_S > 0$, $d_{\text{lo}} > 0$, and $g_S > 0$ by (S2) and $\beta_S g_{\text{lo}} < 1$. *The base objective.* By (S4), ϕ is $\bar{Q}$-semiconvex on $[0,1]$. It is maximized at s°, and $0 < s^\circ < 1$: $s^\circ > 0$ by (S11) (`stage_argmax_pos`), and $s^\circ < 1$ by myopic interiority $u_{\text{hi}} < q$, which interiority implies (`uhi_lt_q`, `stage_argmax_lt_one`). *The shifted maximizer.* Every maximizer s of $\phi + H$ on $[0,1]$ satisfies $s \ge \tilde{s} := \min\{1, s^\circ + L/\bar{Q}\}$ (`min_one_shifted_argmax_le_of_semiconvex`). To see this, suppose $s < \tilde{s}$. If $s < s^\circ$, moving to s° does not lower ϕ and raises H by at least $L(s^\circ - s) > 0$. If $s^\circ \le s < \tilde{s}$, write $s = \lambda s^\circ + (1-\lambda)\tilde{s}$ with $\lambda = (\tilde{s} - s)/(\tilde{s} - s^\circ)$. The semiconvexity inequality at s and the quadratic lower support at the interior maximizer, $\phi(\tilde{s}) \ge \phi(s^\circ) - \frac{\bar{Q}}{2}(\tilde{s} - s^\circ)^2$ (this step uses $s^\circ > 0$; `quadratic_support_right_of_semiconvex_interior_max`), give $\phi(s) \le \phi(\tilde{s}) + \frac{\bar{Q}}{2}(\tilde{s} - s)(\tilde{s} + s - 2s^\circ)$. Because $\tilde{s} \le s^\circ + L/\bar{Q}$, $H(\tilde{s}) - H(s) \ge L(\tilde{s} - s) \ge \bar{Q}(\tilde{s} - s^\circ)(\tilde{s} - s)$. Together these give $(\phi + H)(\tilde{s}) - (\phi + H)(s) \ge \frac{\bar{Q}}{2}(\tilde{s} - s)^2 > 0$. Either way, s is not a maximizer. *The interior cap removes the clip.* By the cap in Step 2 of panel E of Appendix C.4, every maximizer at non-compliance is at most A/q (`dynamicArgmax_le_cap`, `responsePath_le_cap`), and $A/q < 1$ by interiority (`cap_lt_one`). So $\tilde{s} \le \rho_V(x) < 1$. Hence $\tilde{s} = s^\circ + L/\bar{Q}$ and $\rho_V(x) \ge \sigma(x,0) + \beta_S g_S d_{\text{lo}}/\bar{Q}$ (`responsePath_ge_stage_add_wedge`). Substituting $g_S = w_{\text{lo}}/(1 - \beta_S g_{\text{lo}})$ gives $\beta_S w_{\text{lo}} d_{\text{lo}}/\big(\bar{Q}(1 -$

$\beta_S g_{\mathrm{lo}})\big)$ (`wedge_eq`). On the family $\bar{Q} = q$ and $w_{\mathrm{lo}} = w$ (`toChord`), which gives the family form. *Along the paths.* Suppose in addition that the member satisfies the entrenchment condition and that its response at full non-compliance is non-decreasing in the state, as in Proposition C.5. These hypotheses name the two constructed equilibria. The target-only sanction rule is $\sigma(\cdot, 0)$ (Proposition B.2(i); `targetOnlyRule_eq_stage`) and the forward-looking one is R (`dynamicRule_eq_responsePath_of_response_monotone`), so $R(x) \geq \sigma(x, 0) + m$ at every state (`targetOnlyRule_add_wedge_le_dynamicRule_of_response_monotone`). From a common start, suppose $x_t^{10} \leq x_t^{11}$. Then $x_{t+1}^{10} = F(x_t^{10}, \sigma(x_t^{10}, 0)) \leq F(x_t^{10}, R(x_t^{11})) \leq F(x_t^{11}, R(x_t^{11})) = x_{t+1}^{11}$, using $\sigma(\cdot, 0) \leq R$, monotonicity of R, (F3) and (F2). So the forward-looking state path lies above the target-only one at every date (`rulePathF_le_of_rule_le_and_upper_mono`, `targetOnlyPath_le_dynamicPath_of_response_monotone`). The myopic sanction rule is non-decreasing in the state by (S7) and the uniqueness in (S3) (`stage_argmax_mono`), so $\sigma(x_t^{10}, 0)+m \leq \sigma(x_t^{11}, 0)+m \leq R(x_t^{11})$ at every date (`targetOnlySanction_add_wedge_le_dynamicSanction_of_response_monotone`; the form under the increasing-differences condition is `targetOnlySanction_add_wedge_le_dynamicSanction`). □

Proposition C.5 (The long-run gap)**.** *For every economy in the class satisfying interiority, every fixed point $x_\infty^{11} \in [0, 1]$ of the dynamic best response at non-compliance and every fixed point $x_\infty^{10} \leq x_\infty^{11}$ of the myopic sanction rule $x \mapsto \sigma(x, 0)$ satisfy*

$$x_\infty^{11} - x_\infty^{10} \;\geq\; \frac{\beta_S \, g_S \, d_{\mathrm{lo}}}{\bar{Q} - r_{\mathrm{lo}}},$$

which on the family, where $x_\infty^{10} = u_0/(q - r)$, reads $x_\infty^{11} - u_0/(q - r) \geq qm/(q - r) = \beta_S w d_{\mathrm{lo}}/\big((q - r)(1 - \beta_S g_{\mathrm{lo}})\big)$. Under the entrenchment condition and a response at full non-compliance non-decreasing in the state (as under the increasing-differences condition (B.4)*; Appendix C.4), from every initial state the path of the constructed forward-looking equilibrium converges to such a fixed point.*

Proof. The premium bound at x_∞^{11} reads $x_\infty^{11} = R(x_\infty^{11}) \geq \sigma(x_\infty^{11}, 0) + m$, and the myopic sanction rule's slope bound (S7) with (S4) gives $\sigma(x_\infty^{11}, 0) - x_\infty^{10} \geq r_{\mathrm{lo}}(x_\infty^{11} - x_\infty^{10})/\bar{Q}$; rearrange, using $r_{\mathrm{lo}} < \bar{Q}$: the cross difference over the unit square, at least r_{lo} by (S7), is at most the upper bound on the marginal return at the top state less the lower bound at the bottom one by (S10) and (S11), and myopic interiority $u_{\mathrm{hi}} < q \leq \bar{Q}$ with $u_{\mathrm{lo}} > 0$ then gives $r_{\mathrm{lo}} < \bar{Q}$ (`rlo_lt_Qbar`; `fixedPoint_ge_limit10_add_gap`). On the family $x_\infty^{11} = R(x_\infty^{11}) \geq (u_0 + r\, x_\infty^{11})/q + m$; multiply by $q > 0$ and divide by $q - r > 0$ (`fixedPoint_ge_steady10_add_gap`, `steadyGap_eq`). The limit exists and is a fixed point by Step 7 of the proof of Theorem 3, which reads only the monotone response (`dynamicPath_limit_ge_limit10_add_gap_of_response_monotone`, `dynamicPath_limit_ge_steady10_add_gap_of_response_monotone`; the form under the increasing-differences condition, `dynamicPath_limit_ge_steady10_add_gap`, is a corollary). The bound is a statement about the constructed equilibrium at fixed primitives; it is not a comparative static in β_S of equilibria in which both governments re-optimize, and the limit may depend on the start unless the closed loop contracts. □

Remark C.6 (The premium bound is attained)*.* At the affine member with rate and both discount factors $\frac{1}{2}$, $q = 2$, $u_0 = w = 1$ and $r = 0$, the constructed sender value is $\frac{4}{3}x + \frac{8}{9}$, the dynamic best response to non-compliance is $\frac{2}{3}$ at every state, the stage vertex is $\frac{1}{2}$ and $m = \frac{1}{6}$, so (B.3) holds with equality (`placableBoundary_wedge_attained`); the value is pinned down by uniqueness of the fixed point of the sender's Bellman contraction, not posited. Hence no constant larger than

one can multiply m in a bound valid at every member (`wedge_coefficient_cannot_be_raised`). This is an existence statement at one affine member: it says nothing about tightness elsewhere, and a transition supplied with looser slope bounds yields a smaller m for the same game.

Proposition C.7 (The path does not depend on the target's parameters)**.** *Let two games share the transition F on the square, the sender's discount factor β_S and the sender's stage payoff at full non-compliance, $U_S(x, 0, s)$ on the square (on the family, the coefficients q, u_0, r, w). Then any full-non-compliance equilibria in $\mathcal{E}$ of the two have the same sender value and the same sanction rule at every feasible state, and their state paths and realized sanctions coincide from every initial state and at every date, whatever the two games' target payoffs and discount factors and sender payoffs away from full non-compliance. On economies in the class satisfying their increasing-differences and entrenchment conditions the constructed equilibria are such equilibria.*

Proof. At non-compliance the two games' sender objectives $U_S(x, 0, s) + \beta_S V(F(x, s))$ are the same function of (V, x, s) on the square (on the family, because the compliance–sanction cross term $-kas$ vanishes), so their Bellman operators at the non-compliance policy coincide (`bellmanS_BCF_zero_eq_of_defiance_section_eq`). Each equilibrium's sender value is a fixed point of its operator at its own policy, which is full non-compliance, hence of this one (`V_S_eq_V_S_zero_of_policy_zero`); the operator is a contraction of the whole space of bounded continuous values, so the two values coincide (`senderValue_eq_of_defiance_section_eq`). The sanction rules are then the greatest maximizers of the same objective (`equilibriumRule_eq_of_defiance_section_eq`), and the paths coincide by induction (`equilibriumPath_eq_of_defiance_section_eq`, `defiance_branch_invariance_generalF`). No theorem-specific condition, invariant set or argmax uniqueness is read, and the statement is conditional on the two equilibria; for the constructed ones see `dynamicSanction_eq_of_sender_side_eq`. It does not say that a full-non-compliance equilibrium survives an arbitrary parameter change, it does not concern the target's value, which does differ, and off-path responses can differ when the payoffs away from non-compliance do. □

Proposition C.8 (Relief participation)**.** *Let an economy in the class satisfy the entrenchment condition, and let the sender's response at full non-compliance at its constructed value be non-decreasing in the state (as under the increasing-differences condition* (B.4)*). Let $E = (V_T, V_S, \alpha)$ be any equilibrium in $\mathcal{E}$ at which the target fully maintains non-compliance; under these hypotheses one exists, the constructed equilibrium. Write $s_0 = \min\{1, u_{\mathrm{lo}}/\bar{Q}\}$ and $G_S = s_0(u_{\mathrm{lo}} - \bar{Q}s_0/2) > 0$, the stage-gain floor ($u_0^2/(2q)$ on the family). Fix $x_0 \in [0, 1]$, a length $N \geq 1$ and a compliance sequence $a_t \in [0, 1]$, let $y_0 = x_0$, $y_{t+1} = F(y_t, 0)$ be the state under the zero sanctions of a temporary suspension of sanctions, and for $i \in \{S, T\}$ let*

$$J_i = \sum_{t<N} \beta_i^t \, U_i(y_t, a_t, 0) + \beta_i^N V_i(y_N)$$

be government i's value of the spell followed by E. Then full non-compliance, $a_t \equiv 0$, maximizes J_T over compliance sequences, and at it $(J_S - V_S(x_0)) + (J_T - V_T(x_0)) \leq -G_S$, so there is no number τ with $V_S(x_0) \leq J_S - \tau$ and $V_T(x_0) \leq J_T + \tau$. If moreover no compliance level raises the sender's payoff above full non-compliance's at zero sanction (automatic on the family), then for every compliance sequence

$$V_S(x_0) - J_S \;\geq\; G_S \sum_{t<N} \beta_S^t \;>\; 0, \qquad V_T(x_0) - J_T \;\geq\; \ell_{\mathrm{lo}} \sum_{t<N} \beta_T^t a_t \;\geq\; 0.$$

and the joint loss and the absence of τ hold at every sequence.

Proof. One period, sender. The myopic sanction $\nu = \sigma(y, 0)$ is feasible, so the Bellman inequality gives $V_S(y) \geq U_S(y, 0, \nu) + \beta_S V_S(F(y, \nu))$. The value is non-decreasing in the state and F in the sanction, so $V_S(F(y, \nu)) \geq V_S(F(y, 0))$; the stage gain of ν over zero is at least G_S by (S11) at s_0 (`stage_gain_ge`), and $U_S(y, a, 0) \leq U_S(y, 0, 0)$ for every a (`sender_relief_step`). On the family $\nu = (u_0 + ry)/q$ and $U_S(y, 0, \nu) - U_S(y, a, 0) = (u_0 + ry)^2/(2q) \geq u_0^2/(2q)$ (`toChord_stageGainFloor`). *One period, target.* The Bellman equality at non-compliance, with the sanction rule R, gives $V_T(y) = U_T(y, 0, R(y)) + \beta_T V_T(F(y, R(y)))$, hence

$$\begin{aligned} V_T(y) - U_T(y, a, 0) - \beta_T V_T(F(y, 0)) &\geq \ell_{\mathrm{lo}} a - cR(y) + \beta_T \big[V_T(F(y, R(y))) - V_T(F(y, 0))\big] \\ &\geq \ell_{\mathrm{lo}} a + \big(\beta_T d_{\mathrm{lo}} g_T - c\big) R(y) \geq \ell_{\mathrm{lo}} a, \end{aligned}$$

by (P1), (P2), the lower state slope g_T of V_T, the lower bound on the transition's sanction slope and the entrenchment condition $c \leq \beta_T d_{\mathrm{lo}} g_T$ (`target_relief_step`). *The spell.* Multiplying the one-period inequalities by β_i^t and summing over $t < N$ telescopes (`Foundations/FiniteDiscountedTelescope.lean`; `senderReliefValue_add_loss_le`, `targetReliefValue_add_cost_le`), and the remaining claims are arithmetic: at non-compliance the sender's step needs no condition at zero sanction (`targetReliefValue_le_defiance`, `relief_joint_loss_at_defiance`, `no_date_zero_transfer_supports_relief_at_defiance`), and at every sequence it reads the condition that no compliance level raises the sender's payoff at zero sanction (`relief_joint_loss`, `no_date_zero_transfer_supports_relief`, `relief_participation_of_margin`). Every full-non-compliance equilibrium has the constructed values, so the argument applies to each (`relief_participation_of_policy_zero_at_defiance`, `relief_participation_of_policy_zero`); the increasing-differences condition entered only through the non-decreasing response and to name the constructed equilibrium. □

The proposition excludes one arrangement: uncompensated zero sanctions for finitely many periods, then the same full-non-compliance equilibrium. It says nothing about a different continuation equilibrium, a third party's subsidy, payments at later dates under unequal discounting, or population welfare, and the last clause presupposes transfers valued one for one by both governments, which the game does not have. The target's half rests on the entrenchment condition and fails outside it: at the economy of Appendix B in which that condition fails and non-compliance is nonetheless the unique optimum, a one-period temporary suspension of sanctions is worth exactly $\frac{1}{18}$ more to the target than the equilibrium, at every state (`lossOffset_relief_period_benefits_target`): compliance is costly, but relief that is simply imposed requires none.

C.6 The bracket in the target's discount factor, and uniqueness in the monotone class

The lifting condition and the entrenchment condition are hypotheses of the propositions that read them, so on one structure both the relief and the non-compliance equilibria of the forward-looking target can be stated. The affine family without state–sanction cross terms (an exact threshold $D/(\delta v + (1 - \delta)D)$ in the target's discount factor) is the instance of the following proposition in which the general intermediate range closes to a point.

Write, on an economy in the class with semiconvexity in the sanction (T6), μ for the strong-concavity modulus and M_{tot} for the semiconvexity ceiling of the sender's dynamic objective in the sanction, L_{resp}/μ for the response's state Lipschitz rate, $L_\alpha = M_{\mathrm{tot}} L_{\mathrm{resp}}/(k_{\mathrm{lo}}\mu)$, $K_T^{\mathrm{rel}} = (v_{\mathrm{hi}} + \ell_{\mathrm{hi}} L_\alpha)/(1 - \beta_T g_{\mathrm{hi}})$ and $K_T = L_{U_T}/(1 - \beta_T L_{\mathrm{loop}})$ (Appendix N).

Proposition C.9 (The bracket). *(i)* Non-compliance side. *At any target value with lower state slope $g \geq 0$ and any value in the sender's invariant set, full non-compliance is a target optimum at every state whenever $c - \ell_{\text{lo}}\mu/k_{\text{hi}} \leq \beta_T d_{\text{lo}} g$, and the strict unique optimum when the inequality is strict; since $\ell_{\text{lo}} > 0$, the entrenchment condition implies the strict form at $g = g_T$. The target's value at the non-compliance profile carries the slope g whenever the sender's response at its constructed value is nondecreasing in the state (as under the increasing-differences condition* (B.4)*) and $g(1 - \beta_T g_{\text{lo}}) + \max\{0, c - \beta_T d_{\text{lo}} g\} L_{\text{resp}}/\mu \leq v_{\text{lo}} + n_{\text{lo}}\underline{s}$: g_T under the entrenchment condition, and $\min\{g_T, g_{\text{eff}}\}$ with $g_{\text{eff}} = (v_{\text{lo}} + n_{\text{lo}}\underline{s} - cL_{\text{resp}}/\mu)/(1 - \beta_T(g_{\text{lo}} + d_{\text{lo}} L_{\text{resp}}/\mu))$ in general, when $cL_{\text{resp}}/\mu < v_{\text{lo}} + n_{\text{lo}}\underline{s}$ and $\beta_T(g_{\text{lo}} + d_{\text{lo}} L_{\text{resp}}/\mu) < 1$, the condition that keeps the denominator of g_{eff} positive. With that non-decreasing response (under the increasing-differences condition* (B.4) *in particular) and the non-compliance side at that slope, in its weak form, the composed non-compliance equilibrium of Proposition B.1 exists with those values.*

(ii) Relief side and no-non-compliance side. *Under interiority, the lifting condition, compliance neutrality and the relief side $\ell_{\text{hi}} M_{\text{tot}} \leq k_{\text{lo}}(p_{\text{lo}} - \beta_T d_{\text{hi}} K_T^{\text{rel}})$, the zero level then being the unique target optimum when the inequality is strict, there is an equilibrium in $\mathcal{E}$ in which the forward-looking target complies at the zero level $a^{01}(x)$ at every state, the realized sanction is zero, the sender's value is the zero-sanction value and the target's is the relief value, which is K_T^{rel}-Lipschitz because the zero level is L_α-Lipschitz in the state. Under interiority, the lifting condition, $\beta_T L_{\text{loop}} < 1$ and the no-non-compliance side $\ell_{\text{hi}} M_{\text{tot}} < k_{\text{lo}}(p_{\text{lo}} - \beta_T d_{\text{hi}} K_T)$, full non-compliance is not a target best reply at the non-compliance-profile values, so no equilibrium with those values has the target maintaining non-compliance.*

(iii) An intermediate range, not a point. *Every slope that the target's value is shown in (i) to carry is at most g_T, so every sufficient condition for non-compliance implies the non-compliance side at g_T; the statements below are about that side, which is itself a sufficient condition where the value is shown in (i) to carry g_T: under the entrenchment condition, or where the response does not move with the state ($L_{\text{resp}} = 0$). The non-compliance side at g_T is disjoint from the relief side and, under $\beta_T L_{\text{loop}} < 1$, from the no-non-compliance side. Across members that differ only in β_T, the sufficient condition for non-compliance and the non-compliance side at g_T are inherited by a more patient target, the relief side by a less patient one, and the no-non-compliance side by a less patient one under $\beta_T L_{\text{loop}} < 1$ at the more patient member; in their strict forms, the non-compliance and relief sides are open conditions on β_T. Across members that differ only in the target's sanction-indexed state rates and pain floor, the non-compliance side is inherited by a larger lower envelope slope n_{lo}, the relief side by a larger pain floor p_{lo}, and, under the loop condition, the no-non-compliance side by a larger p_{lo} with a smaller n_{hi} (on the family, where $n_{\text{lo}} = n_{\text{hi}} = n$ and $p_{\text{lo}} = c - n$, the non-compliance side by the larger cross term n and the relief and no-non-compliance sides by the smaller). The non-compliance side at g_T depends on the sender only through μ, the upper substitution rate k_{hi} and $\underline{s}$, and the relief side only through M_{tot}, the lower substitution rate k_{lo} and K_T^{rel}; both are equalities of the conditions with those constants held fixed. Consider fourteen tightness conditions: tight slope bounds ($g_{\text{lo}} = g_{\text{hi}}$, $d_{\text{lo}} = d_{\text{hi}} = \delta$), zero moduli ($\sigma_V = M_s = \tau_x = M_{\text{cx}} = M_{\text{cs}} = M_{\text{mix}} = 0$), a zero upper envelope slope ($n_{\text{hi}} = 0$; then $r_{\text{hi}} = n_{\text{lo}} = 0$ follow) and the tight pairs $\bar{Q} = q$, $k_{\text{lo}} = k_{\text{hi}} = k$, $\ell_{\text{lo}} = \ell_{\text{hi}} = \ell$, $p_{\text{lo}} = c$, $v_{\text{lo}} = v_{\text{hi}} = v$. Under them, the tight slope bounds pin F to the affine law $F(x, s) = (1 - \delta)x + \delta s$ on the square (either tight pair alone does, with the fixed diagonal), $\mu = M_{\text{tot}} = q$ and $K_T^{\text{rel}} = K_T = g_T$, $L_{\text{resp}} = 0$ so that the value is shown in (i) to carry g_T, the tight slopes sum to one, and the non-compliance side holds exactly*

when $\beta_T > \beta_T^*$ and the other two exactly when $\beta_T < \beta_T^*$, with $\beta_T^* = D/(\delta v + (1-\delta)D)$ and $D = c - \ell q/k > 0$: the exact threshold of the affine setting without cross terms is where the intermediate range closes. The equalities are sufficient for this closing and are not shown to be necessary; the sender's w and u pairs need not be tight.

Proof. (i) The target's deviation from non-compliance to compliance a lowers the response by at most $k_{\mathrm{hi}}a/\mu$: a unit of compliance perturbs a μ-strongly concave objective with cross rate at most k_{hi} (`response_sub_le_of_le`, through `argmax_dist_le_of_strongConcave_lipschitz_perturbation_sharp`). The pain saved is at most c per unit of that fall, the continuation lost at least $\beta_T d_{\mathrm{lo}} g$ per unit through the lower bound on the transition's sanction slope, and the compliance costs at least $\ell_{\mathrm{lo}} a$ (`targetObj_strictAntiOn_of_sharp`, `targetArgmax_eq_zero_of_sharp`, `sharp_of_lossOffset`); with a weak inequality the objective is still non-increasing in compliance, so non-compliance is an optimum, which is all existence needs (`targetObj_antitoneOn_of_sharp_le`, `zero_isMaxOn_joint_of_sharp`). The target operator at a sender value in the two-sided invariant set, with a monotone response, preserves the lower-slope cone at slope g under the slope condition, the pain term along the response priced by L_{resp}/μ (`targetOp_mem_lowerSlopeCone`); the non-compliance-profile value is the fixed point of that contraction on the whole value space and lies in the cone (`VTdef`, `VTdef_lowerSlope`, Stokey–Lucas Corollary 1 to Theorem 3.2). Under the entrenchment condition g_T balances (`gT_balance`); in general $\min\{g_T, g_{\mathrm{eff}}\}$ does (`gSharp_balance`: when $L_{\mathrm{resp}} = 0$ any slope at most g_T; otherwise $g_T \le g_{\mathrm{eff}}$ is equivalent to the entrenchment condition at g_T, and $g_{\mathrm{eff}} < g_T$ makes the pain term at g_{eff} nonnegative). The equilibrium is `equilibriumTop_sharp`, `equilibriumTop_gSharp`. (ii) The zero level is defined at any value in the two-sided invariant set with the sharp rate of sanction reduction below it (`zeroLevelAt`, `response_ge_of_lt_zeroLevelAt`); across states the response at a fixed compliance moves by at most $L_{\mathrm{resp}}|\Delta x|/\mu$ (`response_dist_le_state`), so the zero level is L_α-Lipschitz (`zeroLevelAt_lipschitz`). The relief operator $V \mapsto U_T(y, a^{01}(y), 0) + \beta_T V(F(y,0))$ is a contraction preserving the K_T^{rel}-Lipschitz clause (`reliefOp_lipschitz`, `VTrel_lipschitz`); the zero level is the unique argmax at the relief values (`zeroLevel_argmax_unique_rel`) and the equilibrium is `reliefEquilibriumTop`, with `reliefEquilibriumTop_rule_eq_zero`; at the weak relief side the zero level is still an optimum and the equilibrium exists (`jointRel_le_of_lt`, `reliefBestReply_of_relief_side_le`, `exists_relief_equilibrium_iff`). The non-compliance-profile value is K_T-Lipschitz with no further condition (`VTdef_lipschitz`), and the deviation to the zero level at the constructed sender value is profitable under the no-non-compliance side (`defiance_not_maxOn_of_relief_side`, `no_defiance_equilibrium_of_relief_side`). (iii) A slope satisfying the slope condition of (i) is at most g_T, since the pain term is nonnegative (`le_gT_of_balance`, `sharp_gT_of_sharp_of_balance`); at $L_{\mathrm{resp}} = 0$ the pain term vanishes and g_T balances (`gT_balance_of_Lresp_eq_zero`), a situation that is affine by hypothesis, since a zero response slope pins the transition to $F(x,s) = F(1,0)\,x + (1 - F(1,0))\,s$ on the square (`F_eq_affineF_of_Lresp_eq_zero`); and a slope that balances and satisfies the non-compliance side at β_T does both at every larger β_T, both sides of the balance moving the right way and the sender's constants not depending on β_T (`certified_sharp_mono_βT`), while the sender's value, and so its response, is the same at both members (`VSstar_eq_of_sender_side_eq`). `bracket_existence_sides_disjoint` for the relief side: the lower slope bounds give $g_{\mathrm{lo}} + d_{\mathrm{lo}} \le 1$ (`glo_add_dlo_le_one`), (P2) and (P3) at compliance zero give $v_{\mathrm{lo}} + n_{\mathrm{lo}}\underline{s} \le v_{\mathrm{hi}} + (c - p_{\mathrm{lo}})\underline{s}$ (`vlo_nlo_floor_le`), and together they give $\beta_T d_{\mathrm{lo}} g_T \le \beta_T d_{\mathrm{hi}} K_T^{\mathrm{rel}} + c - p_{\mathrm{lo}}$, with no neutrality, the lifting condition or loop condition read; `bracket_sides_disjoint` for the no-non-compliance side (by $\mu \le M_{\mathrm{tot}}$, $d_{\mathrm{lo}} \le d_{\mathrm{hi}}$, $g_T \le K_T$); `sharp_side_mono_βT`, `relief_side_anti_βT`, `relief_side_notdef_anti_βT`; `sharpSideSet_isOpen`, `reliefSideSet_isOpen`; `sharp_side_mono_nlo`, `relief_side_anti_plo`, `relief_sid`

e_notdef_anti; sharp_side_eq_of_target_side_eq, relief_side_eq_of_target_side_eq; collapse_mu, collapse_Mtot, collapse_KTrel, collapse_KT, collapse_sides; the tight slopes sum to one (glo_add_dlo_eq_one_of_tight), and either tight pair pins the affine law on the square (eq_of_sanctionChords_tight, eq_of_stateChordBandAll_tight, add_eq_one_of_tight; on the class F_eq_affineF_of_tight_sanction, F_eq_affineF_of_tight_state); $L_{\text{resp}} = 0$ there (collapse_Lresp), and the threshold form is collapse_threshold. □

The three sides are inhabited, at the affine member and off the affine law: the members $\beta_T = \frac{1}{2}$ of Appendix B are on the non-compliance side, and the same coefficients at $\beta_T = \frac{1}{8}$ are on the relief and no-non-compliance sides (placableAffineLo_reliefEquilibrium, placableQuad DiagLo_reliefEquilibrium, placableQuadDiag_sharpEquilibrium, exists_bracket_b oth_sides_nonaffine). An exact-arithmetic evaluation of the sides on a grid of step 1/400 (docs/archive/workflow-scratch/bracket_probe3_2026-09-23.py) finds, at the affine member, the relief side up to 0.180, the non-compliance side at g_T from 0.185 and the sufficient condition for non-compliance, at $\min\{g_T, g_{\text{eff}}\}$ below the entrenchment condition, from 0.1875; and at the non-affine member the relief side up to 0.145, the non-compliance side at g_T from 0.2325 and the sufficient condition for non-compliance from 0.235. The intermediate range, between the relief side and the sufficient condition for non-compliance, is open at both. At the tight-modulus affine member without cross term ($\sigma_V = 0$, $r = 0$) the response does not move with the state, so the value is shown in (i) to carry g_T without the entrenchment condition (gT_balance_of_Lresp_eq_zero), and the sides are complementary at $\beta_T = 2/11$. The intermediate range is therefore not the price of the transition alone: it is open at an affine member, which has the sender cross rate $r = 1/2$ and the supplied modulus $\sigma_V = 1$, and it closes to a point at the tight-modulus one without cross term; in general its width depends, through the constants of (i)–(iii), on the spreads of the slope and rate bounds, the supplied moduli, the sender's cross rate and the target's envelope slopes. The sufficient condition for non-compliance holds on an upper set of discount factors (certified_shar p_mono_βT). So does the existence of a full-non-compliance equilibrium itself, under a monotone response and a monotone non-compliance value at one member, inside the intermediate range as well (Proposition 7). Where the intermediate range closes in (iii), the threshold is an equilibrium threshold: a full-non-compliance equilibrium exists exactly when $\beta_T \geq \beta_T^*$ (Proposition 6). On the intermediate range these sufficient conditions determine no equilibrium, and the equilibria of both sides are members of $\mathcal{E}$, so they inherit Proposition C.14. On the family, at a member with tight moduli and no cross terms the values are affine in the state, where the Bellman equation closes in one step (Puterman, 1994); off it, and on the class wherever a rate pair is split, the values are only slope- and Lipschitz-bounded, which is what the intermediate range measures.

The two ends of the intermediate range can be characterized exactly in general, as thresholds in the target's discount factor along members that differ only in it. The relief end needs one condition on the class: the state's lower rate at zero sanction covers the compliance cost along the zero level, $\ell_{\text{hi}} L_\alpha \leq v_{\text{lo}}$, where $L_\alpha = M_{\text{tot}} L_{\text{resp}}/(k_{\text{lo}}\mu)$ is the zero level's state Lipschitz rate.

Proposition C.10 (Exact thresholds in the target's discount factor)**.** *Consider economies in the class that differ only in the target's discount factor $\beta_T \in (0, 1)$; (ii) is stated on the relief class, and only (i) needs semiconvexity in the sanction* (T6)*.*

(i) With (T6)*, under interiority, the lifting condition and compliance neutrality, an equilibrium in $\mathcal{E}$ whose realized sanction is zero at every state exists if and only if the zero level is a target best reply at the relief values. If also $\ell_{\text{hi}} L_\alpha \leq v_{\text{lo}}$, the discount factors at which it exists form an interval $(0, \beta_R]$ whenever they are non-empty with supremum $\beta_R < 1$.*

(ii) If the sender's response at full non-compliance is nondecreasing in the state and the slope condition of Proposition C.9(i) holds at $g = 0$, $cL_{\text{resp}}/\mu \le v_{\text{lo}} + n_{\text{lo}}\underline{s}$, the discount factors at which an equilibrium in $\mathcal{E}$ with full non-compliance at every state exists form an interval $[\beta_D, 1)$ whenever they are non-empty with infimum $\beta_D > 0$.

(iii) Every discount factor at which an equilibrium in $\mathcal{E}$ has a zero realized sanction at every state is below $c/(d_{\text{lo}}v_{\text{lo}} + cg_{\text{lo}})$.

Proof. (i) An equilibrium with a zero realized sanction at every state is the relief equilibrium: its sender value is the zero-sanction value, its policy the zero level and its target value the relief value (`relief_equilibrium_of_zero_rule`), so its policy is a best reply there. Conversely the best-reply test builds the relief equilibrium (`reliefEquilibriumOfBestReply`, `exists_relief_equilibrium_iff`); the relief side of Proposition C.9(ii) is one sufficient condition for the test. Under $\ell_{\text{hi}}L_\alpha \le v_{\text{lo}}$ the stage payoff along the zero level rises with the state, so the relief value is nondecreasing (`VTrel_monotone`). Along the common zero-sanction path the relief values at $\beta_1 \le \beta_2$ solve $V_i = u + \beta_i V_i \circ z$, so $\beta_2 V_2 - \beta_1 V_1$ is nondecreasing. A compliance below the zero level draws a sanction and raises the successor state, so its gain is smaller at the less patient member; a compliance above it draws none, and its gain does not depend on the discount factor. The best reply at the more patient member therefore holds at the less patient one (`reliefBestReply_anti_βT`). The best-reply inequalities are weak and the relief values converge as the discount factor does, so the set is closed (`reliefBestReply_of_tendsto`), which gives $(0, \beta_R]$ (`reliefExistenceSet_eq_Ioc`). (ii) The balance at $g = 0$ makes the non-compliance value nondecreasing (`VTdef_monotone_of_weak_hvL`); Proposition 7 makes the set an upper set, and Proposition C.13 closes it (`defianceExistenceSet_eq_Ico`). (iii) At or above that bound the strong entrenchment condition holds (`lossOffset₀_iff_βT_ge`) and excludes such an equilibrium (`no_zero_rule_equilibrium`, `reliefExistence_lt_margin`). General in F. □

The proposition does not order the two thresholds: nothing in the class excludes a relief and a full-non-compliance equilibrium at the same discount factor. The sufficient conditions of Proposition C.9 lie inside the two sets, the sufficient condition for relief in the first and the sufficient condition for non-compliance, with the increasing-differences condition, in the second; so when $\beta_R < \beta_D$ the intermediate range they leave contains the exact intermediate range (β_R, β_D).

At the affine member of the family under the lifting condition every hypothesis of Proposition C.10 holds: $\ell_{\text{hi}}L_\alpha = 5/23$ against $v_{\text{lo}} = 30$, a response made monotone by the increasing-differences condition, and $cL_{\text{resp}}/\mu = 20/23$ (`aw_general_hypotheses`). There the two thresholds can be computed. Write $\beta_R = 36/197 \approx 0.18274$ and $\beta_D = (82 - 7\sqrt{2})/394 \approx 0.18300$.

Proposition C.11 (The exact intermediate range at the affine member)**.** *At the affine member of the family under the lifting condition, with coefficients $(q, u_0, k, r, w, \ell, c, n, v) = (3, 1, 3, \frac{1}{2}, 1, 1, 4, 0, 30)$, transition $F(x, s) = (x + s)/2$ and $\beta_S = \frac{1}{2}$ (Appendix M.2), and target discount factor $\beta_T \in (0, 1)$:*

(i) an equilibrium in $\mathcal{E}$ whose realized sanction is zero at every state exists if and only if $\beta_T \le \beta_R$;

(ii) an equilibrium in $\mathcal{E}$ with full non-compliance at every state exists if and only if $\beta_T \ge \beta_D$;

(iii) at $\beta_1 = 25248/138097$, which satisfies $\beta_R < \beta_1 < \beta_D$, neither exists, and for every $t \in [0, 4/9]$ there is an equilibrium in $\mathcal{E}$ with policy $\alpha(x) = t + 287x/4206$ and realized sanction $\frac{701}{2100} \cdot \frac{40-90t}{29} + x/10$; distinct t give distinct policies, and for $t < 4/9$ the sanction is positive at every state.

Proof. Every value is explicit. The dynamic best response is the clipped vertex of the sender's concave objective in the sanction (`σ_dyn_top_eq_clip`), and each Bellman equation is checked coefficient by coefficient, so each profile is an equilibrium of the game itself (`awReliefEquilibrium`, `awDefianceEquilibrium`, `awEquilibrium`). (i) If the realized sanction is zero at every state, the gap between the sender's value and $4x/3$ at y is half the gap at $y/2$, and it is bounded, so it vanishes (`eq_zero_of_eq_mul_comp`). The target's policy is then the compliance $4/9 + x/6$ at which the sanction just ends, and its value is affine with slope $179/(6 - 3\beta_T)$. Maintaining non-compliance at the zero state pays exactly when $179\beta_T/(6 - 3\beta_T) > 6$, that is, when $\beta_T > \beta_R$ (`aw_zero_rule_le`, `aw_zero_rule_iff`). (ii) At full non-compliance the sender's value is the unique fixed point of its Bellman contraction, $\frac{10-7\sqrt{2}}{2}x^2 + (4\sqrt{2} - 4)x + \frac{8(2-\sqrt{2})}{7}$, with response $\frac{8(\sqrt{2}-1)}{7} + (3 - 2\sqrt{2})x$. A unit of compliance costs the target one unit and, while the sanction is positive, lowers it by $3/D$, where $D = 7(2 + \sqrt{2})/8$ is the sender's curvature; a unit of sanction costs the target P, its pain less the discounted value of the state it builds. Full non-compliance is a best reply exactly when $P \le D/3$, and $(P - D/3)\Delta = \frac{197}{12}(\beta_D - \beta_T)$ with $\Delta > 0$ (`awd_threshold`); below β_D, complying at $1/10$ at the zero state pays (`aw_full_defiance_le`, `aw_full_defiance_iff`). So the two existence sets are the intervals of Proposition C.10, with endpoints β_R and β_D, and β_R is below the bound of that proposition's part (iii), here $4/17$ (`aw_exact_thresholds`). (iii) is `awβ1_mem_window`, `aw_continuum` and `aw_window_at_β1`. □

The proposition concerns one member, with the transition $F(x, s) = (x + s)/2$ fixed, and is affine-only. Its first two parts are instances of Proposition C.10. The continuum in (iii) is not an instance of a general result: it rests on the target's indifference among compliance levels, which needs the family's linear compliance cost and the particular discount factor β_1; with a strictly convex cost the indifference disappears (Proposition M.2). At that member the proposition settles the intermediate range in both directions. The sufficient conditions are conservative there: the exact bounds lie strictly inside the intermediate range, from about 0.180 to 0.1875, that they leave. And the exact intermediate range is not empty of equilibria: at one discount factor inside it, β_1, there is a continuum of equilibria with compliance rising in the state, while neither endpoint profile is an equilibrium. Other policies at β_1, and the equilibria at other discount factors inside the intermediate range, are not classified.

The same structure carries the sender-side counterpart. The lower bound $u_{\text{lo}} + \beta_S g_S d_{\text{lo}}$ on the sender's dynamic marginal return to sanctioning and the upper bound $A = u_{\text{hi}} + \beta_S K_S d_{\text{hi}}$ on that return bracket the lifting condition: when the lower bound exceeds an upper substitution rate k_{hi} that the member has, every equilibrium in $\mathcal{E}$ whose policy is nonincreasing in the state carries a sanction of at least $(u_{\text{lo}} + \beta_S g_S d_{\text{lo}} - k_{\text{hi}})/\bar{Q} > 0$ at every state, full compliance included, on a member at which a larger concession is never worth more to the sender at zero sanction (`nonplacable_floor_of_lowerAppetite`, from the premium floor of `equilibrium_sanction_ge_wedge_of_antitone`); when $A \le k_{\text{lo}}$ full compliance ends the sanction (`response_at_one`). The lower bound rises strictly with the sender's discount factor across members that differ only in it (`lowerAppetite_lt_of_βS_lt`): the floor moves with the sender's patience. The upper bound A rises strictly with it too (`A_lt_of_βS_lt`).

Proposition C.12 (Uniqueness in the monotone class)**.** *On an economy in the class satisfying the strong entrenchment condition* $c \le \beta_T d_{\text{lo}} v_{\text{lo}}/(1 - \beta_T g_{\text{lo}})$ *(the entrenchment condition at* $n_{\text{lo}} = 0$*, and stronger in general), every equilibrium in* $\mathcal{E}$ *whose sanction rule is nondecreasing in the state has full non-compliance at every feasible state, the constructed sender value, the constructed target value, and the constructed response as its rule. When the constructed response at full non-compliance is non-decreasing in the state as well (as under the increasing-differences condition* (B.4)*), the*

constructed non-compliance equilibrium exists and belongs to the class.

Proof. The target's value is its own joint integrand at its policy (`V_T_eq_at_policy`). At an ordered pair of states $y_1 \le y_2$ the deficit of the value below the slope $v_{\mathrm{lo}}/(1-\beta_T g_{\mathrm{lo}})$ is at least β_T times the deficit at a successor pair (`slopeDeficit_step`). The comparison sets the lower state at its own policy and sanction against the higher state's deviation to full non-compliance, which draws a weakly higher sanction, $\Delta s \ge 0$, because the rule is nondecreasing and the dynamic best response is antitone in compliance. The value at y_1 is its integrand at its policy and the value at y_2 is at least its integrand at full non-compliance. The stage difference is at least $v_{\mathrm{lo}}\Delta y - c\Delta s$: compliance at y_1 only costs, the pain of the extra sanction is at most c per unit, and the state is worth at least v_{lo} per unit at full non-compliance, the envelope term $n_{\mathrm{lo}} s\,\Delta y$ being nonnegative (P2, P3). The successor of y_2 exceeds that of y_1 by at least $g_{\mathrm{lo}}\Delta y + d_{\mathrm{lo}}\Delta s$, and the strong entrenchment condition covers the pain. So the infimum D of the deficit over ordered pairs, finite because the value is bounded, satisfies $D \ge \beta_T D$ and hence $D \ge 0$ (`target_lowerSlope_of_rule_monotone`; no operator on the value space, no continuity of the rule and no condition on the policy is used). The deviation from $\alpha(x)$ to full non-compliance raises the sanction, because the dynamic best response is antitone in compliance under decreasing differences with no uniqueness (`σ_dyn_top_antitoneOn_a_of_DD`), the pain of the rise is covered by the continuation at that slope through the lower bound on the transition's sanction slope, and the compliance saved is worth at least $\ell_{\mathrm{lo}}\alpha(x)$; so $\alpha(x) = 0$ (`policy_eq_zero_of_rule_monotone`). The two values are then the unique fixed points of the two contractions at non-compliance (`values_eq_of_policy_zero`, `unique_equilibrium_of_rule_monotone`). No compliance neutrality is read. For the last sentence, the constructed policy is constant and its rule is the constructed response, nondecreasing by hypothesis (under the increasing-differences condition by `responsePath_monotoneOn_of_margin`), which puts the constructed equilibrium in the class (`equilibriumTop_mem_monotone_class`). □

The class is a hypothesis on the quantifier and the proposition says nothing outside it; on the relief side, uniqueness among all equilibria is open. The non-compliance core holds over an arbitrary economy, not only the class: two slope bounds on the transition, decreasing differences of the sender's payoff in (sanction; compliance), four target rates at full non-compliance and the strong entrenchment condition give every equilibrium in $\mathcal{E}$ with a nondecreasing sanction rule a target value rising at least at rate $v_{\mathrm{lo}}/(1-\beta_T g_{\mathrm{lo}})$ and full non-compliance at every state, with no condition on the policy and none of the sender's curvature, invariant-set, Riccati or interiority conditions read (`lowerSlope_of_rule_monotone`, `policy_eq_zero_of_rule_monotone`); its extra generality over the class is illustrated by an economy whose compliance cost is not monotone (Proposition M.1). Under the same condition no equilibrium in $\mathcal{E}$ sanctions at zero at every state (`no_zero_rule_equilibrium`). The condition holds exactly when $\beta_T \ge c/(d_{\mathrm{lo}} v_{\mathrm{lo}} + c g_{\mathrm{lo}})$, a patience bound with no sender parameter that is sufficient and whose converse is refuted (`lossOffset₀_iff_βT_ge`).

C.7 Proofs of the patience, minmax-payoff, uniqueness and threshold propositions

Proof of Proposition 7. The argument runs on the relief class and reads no semiconvexity of the sanction sections (T6); the enriched-subclass statements of the headline rows are its corollaries. The two members share the transition, the stage payoffs and the sender's discount factor, so the sender's value and its response at full non-compliance coincide at both (`VSstar_eq_of_sender_side_eq`). A full-non-compliance equilibrium exists exactly when full non-compliance is a target best reply at the constructed values (`exists_defiance_equilibrium_iff`). Compliance lowers the response; with the response nondecreasing in the state and the non-compliance value nondecreasing at one

member, the discounted continuation difference that compliance gives up is nondecreasing in β_T (`monotone_sub_of_solves_of_monotone_right`), so the best reply at the less patient member is preserved at the more patient one (`exists_defiance_equilibrium_mono_βT`; headline row 53). The slope condition of Proposition C.9(i), which delivers the monotone non-compliance value, is itself inherited by the more patient member: its slope side is antitone in β_T (`balanceSide_anti`, `balance_of_le_βT`, `SameButβT.balance_of_balance`; headline row 57). A directional example on the quartic family shows the comparison is not vacuous.

Proof of Proposition B.5. The argument reads only the lower bound g_{lo} on the state slope of the transition's zero-sanction section, the lower bound d_{lo} on its sanction slope from zero, three target bounds (a state rate $v_{\mathrm{lo}} \geq 0$ at full non-compliance and zero sanction; pain at most c per unit at full non-compliance; and compliance that does not pay against a zero sanction, used only for the attainment in (i) and (ii)), and the strong entrenchment condition; it reads nothing of the sender's objective, so it holds over any economy with these properties (`Theory/Equilibrium/SecurityCore.lean`), and the class satisfies them. Let $r_0 = x_0$, $r_{t+1} = F(r_t, 0)$ and $\underline{V}_T(x_0) = \sum_t \beta_T^t U_T(r_t, 0, 0)$. Against any feasible pure sanction sequence (s_t), compare the state under full non-compliance at every date with the zero-sanction path: the lower slope bounds make the excess state a discounted accumulation of the sanctions, valued at least at rate v_{lo}, while the sanction pain is at most c per unit, so the payoff gap is at least $(\beta_T d_{\mathrm{lo}} v_{\mathrm{lo}}/(1-\beta_T g_{\mathrm{lo}}) - c)\sum_t \beta_T^t s_t$, which the strong entrenchment condition makes nonnegative (`reliefPayoff_le_defiancePayoff`; `Foundations/DiscountedChordSum.lean`). Against permanent relief, compliance only costs and moves neither the sanction nor the state, so no compliance sequence earns more than $\underline{V}_T$ (`reliefPayoff_isLeast`). In this deterministic game with perfect monitoring a pair of pure history-dependent plans generates an open-loop path with the same payoff (`jointState_eq_openLoopPath`, `jointPayoff_eq_openLoopPayoff`), so an open-loop pure saddle value is the minmax and the maxmin over pure history-dependent strategies on both sides (`isLeast_ceiling_history_of_openLoop_saddle`, `isGreatest_floor_history_of_openLoop_saddle`); under these properties $\underline{V}_T$ is both (`reliefPayoff_isLeast_history`, `reliefPayoff_isGreatest_history`; headline rows 55–56), the floor attained by full non-compliance at every date (`defiance_secures_reliefPayoff_history`). Equilibrium optimality against the deviation to full non-compliance at every date gives $V_T(x_0) \geq \underline{V}_T(x_0)$ in every equilibrium in $\mathcal{E}$ (`reliefPayoff_le_V_T`; headline row 52).

Proof of Proposition B.6. Part (i) is Proposition C.12 (`unique_equilibrium_of_rule_monotone`, `policy_eq_zero_of_rule_monotone`; headline rows 46 and 54). For (ii), a sanction rule that is zero at every state is nondecreasing, so an equilibrium with it would fall in the class of (i), which makes the constructed response its sanction rule; that response is strictly positive at full non-compliance (Proposition B.1), a contradiction (`no_zero_rule_equilibrium`; headline row 58).

Proof of Proposition 6. Five of the fourteen tightness conditions of Proposition C.9(iii) describe which bounds are supplied for the economy, not the game. A member of the relief class with $\sigma_V = 0$ and $d_{\mathrm{lo}} = d_{\mathrm{hi}}$ is the same game as an enriched member with the affine law's bounds, $g_{\mathrm{lo}} = g_{\mathrm{hi}} = 1 - d_{\mathrm{lo}}$ and $M_s = M_{\mathrm{cx}} = M_{\mathrm{mix}} = M_{\mathrm{cs}} = 0$, with the same β_T^*, and interiority and the lifting condition carry over (`collapseRecert`); so the nine tightness conditions $\sigma_V = \tau_x = n_{\mathrm{hi}} = 0$, $\bar{Q} = q$, $k_{\mathrm{lo}} = k_{\mathrm{hi}}$, $\ell_{\mathrm{lo}} = \ell_{\mathrm{hi}}$, $p_{\mathrm{lo}} = c$, $v_{\mathrm{lo}} = v_{\mathrm{hi}}$ and $d_{\mathrm{lo}} = d_{\mathrm{hi}}$, with interiority, the myopic relief condition and the lifting condition, give the threshold on the relief class, with no (T6) (`collapse_equilibrium_threshold_le_tight`). At the re-certified member the three

sides of Proposition C.9 collapse at β_T^* (`collapse_sides'`; headline row 45) and the transition is the affine law on the square (Proposition C.9(iii)). Above β_T^* the non-compliance side holds and the composed non-compliance equilibrium exists; below it, under interiority and the lifting condition, the no-non-compliance side holds, its loop condition $\beta_T L_{\text{loop}} < 1$ being automatic because $L_{\text{resp}} = 0$ there, so that $L_{\text{loop}} = g_{\text{hi}}$ (`collapse_Lresp'`, `βT_ghi_lt_one`), and no equilibrium in $\mathcal{E}$ has full non-compliance, so the non-existence direction ranges over every equilibrium in $\mathcal{E}$ (`collapse_equilibrium_threshold`, which excludes $\beta_T = \beta_T^*$). Members at $\beta_T = \frac{1}{2}$ and $\frac{1}{8}$ lie on the two sides, with $\beta_T^* = \frac{2}{11}$ (`Examples/PlacableSenderCollapseWitness.lean`). At $\beta_T = \beta_T^*$ the members just above the threshold have the equilibrium, and existence passes to the limit (Proposition C.13; `exists_defiance_equilibrium_at_betaStar`), so a full-non-compliance equilibrium exists exactly when $\beta_T^* \le \beta_T$, the threshold attained (`collapse_equilibrium_threshold_le`; headline row 59). So the discount factors with a full-non-compliance equilibrium are exactly $[\beta_T^*, 1)$ (`collapse_defianceExistenceSet`), and when $\beta_T^* < 1$ this is the interval $[\beta_D, 1)$ of Proposition C.10(ii) with $\beta_D = \beta_T^*$, the general theorem applying at the re-certified member, where $L_{\text{resp}} = 0$ discharges its two hypotheses (`collapse_exact_thresholds`): the threshold discount factor β_T^* is the closed form of the exact non-compliance threshold.

Proposition C.13 (Existence of non-compliance is closed in the target's discount factor)**.** *Consider economies in the class that differ only in the target's discount factor.*

(i) If members whose discount factors converge to β_T each have an equilibrium in $\mathcal{E}$ with full non-compliance at every state, so does the member at β_T.

(ii) The set of discount factors in $(0, 1)$ at which such an equilibrium exists contains its infimum whenever that infimum is positive.

Proof. A full-non-compliance equilibrium exists exactly when full non-compliance is a target best reply at the constructed values (`exists_defiance_equilibrium_iff`), and the sender's value and response do not depend on β_T (`VSstar_eq_of_sender_side_eq`). Along the common non-compliance path the target values solve $V_n = u + \beta_n V_n \circ z$, so $(1-\beta_n)\|V_n - V\| \le |\beta_n - \beta_T|\,\|V\|$; every deviation gain converges, and the weak best-reply inequalities pass to the limit (`defianceBestReply_of_tendsto`, `exists_defiance_equilibrium_of_tendsto`). Part (ii) applies (i) to members in the set whose discount factors decrease to the infimum (`defianceExistence_sInf_mem`). General in F. □

With Proposition 7 applied at the infimum, the existence set is then the interval from the infimum to one. Where the intermediate range closes, part (i) applied to the members just above β_T^* gives existence at $\beta_T = \beta_T^*$ in Proposition 6.

C.8 Pure history-dependent deviations

Proposition C.14 (Stationary equilibria survive pure history-dependent deviations)**.** (For the general theory of history-dependent plans against stationary ones in discounted dynamic programming see Strauch (1966) and Denardo (1967); the deterministic, perfect-monitoring setting used here reduces a history-dependent plan to a dated one.) *Let $E = (V_T, V_S, \alpha) \in \mathcal{E}$ and write $\sigma = \sigma_{V_S}^{\text{dyn}}$.*

(i) With the sender at σ, every pure target strategy that conditions compliance on the whole history of states and actions yields the target, from every initial state x_0, a discounted payoff of at most $V_T(x_0)$, which α attains.

(ii) With the target at α, every pure sender strategy that conditions the sanction on the whole history yields the sender at most $V_S(x_0)$, which $\sigma(\cdot, \alpha(\cdot))$ attains.

(iii) At every state x, after every compliance $a \in [0,1]$ on or off the equilibrium path, for every sanction s and every pure history-dependent continuation of the sender with discounted payoff Π from $F(x,s)$,

$$U_S(x,a,s) + \beta_S \Pi \;\le\; U_S\big(x,a,\sigma(x,a)\big) + \beta_S V_S\big(F(x,\sigma(x,a))\big).$$

Proof. The game is deterministic with perfect monitoring. Against a fixed feedback rule of the opponent, a pure history-dependent strategy generates from a given start one sequence of actions; the dated plan that replays that sequence, whatever the state, generates the same states, actions and payoff (`historyDiscountedPayoff_eq_replay`, in `Foundations/HistoryDependentDeviations.lean`; dated plans are themselves history-dependent strategies, `historyState_toHistoryPlan`, so the class is larger). The opponent's past actions are functions of the recorded states and own actions, so conditioning on them adds nothing. Parts (i) and (ii) are then the lifetime verification against dated plans of `Theory/Equilibrium/TopLifetimeVerification.lean` (`leader_optimal_against_history_deviation_top`, `follower_optimal_against_history_deviation_top`); the payoff and value bounds that verification needs are derived from continuity on the compact cube, not assumed. Part (iii) combines the sender's optimality at the node (x,a), which the equilibrium carries at every (x,a), with part (ii) from the successor state (`follower_node_optimal_against_history_deviation_top`). □

The proposition holds for every element of $\mathcal{E}$, hence for each of the four constructed equilibria. It says that a stationary equilibrium of the paper is unimprovable by such deviations. It does not characterize the other equilibria that history-dependent strategies may support, it does not cover randomized strategies, and the target's deviation is a unilateral change of its own actions against the sender's fixed contingent response.

C.9 Proofs of the welfare results of Section 4

The propositions of Section 4, the identity (8) and Proposition C.16 below are machine-checked in `Theory/Welfare/InterestGroup.lean`, `Theory/Welfare/PlannerZeroSanction.lean` and `Examples/ChordEconomyWelfareGeneral.lean` (Appendix B). None of them reads a functional form, a sign condition of an interest-group polity, or a parameter value. Proposition C.15 and the identity hold for any pair of payoff functions, Proposition 10 for any bounded welfare flow and any transition with the stated chords, and the others on the class.

The identity (8)**.** Substituting the two representation identities, $U_S + U_T = \sum_g \theta_g w_g = \sum_g w_g + \sum_g (\theta_g - 1) W_g = W + \sum_g (\theta_g - 1) W_g$ at every state and action (`payoffs_add_eq`). No sign condition and no transition is used.

The rescaling result, which Section 4.2 states in words, is the following.

Proposition C.15 (Rescaling one group)**.** *Take an interest-group representation and a sender group g, and multiply W_g by a positive factor while dividing θ_g by the same factor. The governments' payoffs, and so the game and its equilibria, are unchanged, and aggregate welfare changes by exactly the change in W_g. If along two paths of states and actions the group's welfare is higher on the first by a uniform positive margin at every date, and aggregate welfare and W_g are bounded along both, then for some factor the planner ranks the first path strictly above the second at every planner discount factor.*

Proof of Proposition C.15. Replacing (W_g, θ_g) by $(\varphi W_g, \theta_g/\varphi)$ for a factor $\varphi > 0$ leaves the product $\theta_g W_g$, and so the sender's representation identity, unchanged, and the new weight is positive; the result is a representation of the same payoffs (`rescaleS`). Aggregate welfare changes only in group g's term, from W_g to φW_g, and so becomes $W + (\varphi - 1)W_g$ (`rescaleS_welfare`).

For the ranking, write W_t^P, W_t^Q for aggregate welfare and $W_{g,t}^P, W_{g,t}^Q$ for group g's welfare at date t along the two paths, all at most M in absolute value, with $W_{g,t}^P \geq W_{g,t}^Q + \varepsilon$ for a margin $\varepsilon > 0$. Take $\varphi = 1 + (2M+1)/\varepsilon$. At every date

$$\big(W_t^P + (\varphi - 1)W_{g,t}^P\big) - \big(W_t^Q + (\varphi - 1)W_{g,t}^Q\big) \geq -2M + (\varphi - 1)\varepsilon = 1.$$

The rescaled flows are bounded, so their discounted sums converge absolutely at every $\beta \in [0, 1)$, and a strict inequality at every date between bounded flows survives discounting, since the date-zero term has weight one, also at $\beta = 0$ (`tsum_pow_mul_lt_of_forall_lt`, `exists_scale_ranks`, `exists_rescaleS_ranks`). □

Proof of Proposition 9. The representations. For a real η, give the sender two groups, with welfare $2\eta s$ at weight $\frac{1}{2}$ and $U_S - \eta s$ at weight one, and the target two groups, with welfare $-\big(U_T(x, 0, s) - U_T(x, a, s)\big)$ at weight two and $U_T(x, 0, s) + \big(U_T(x, 0, s) - U_T(x, a, s)\big)$ at weight one. The weighted sums are U_S and U_T at every state and action, and aggregate welfare is

$$W_\eta(x, a, s) = U_S(x, a, s) + U_T(x, 0, s) + \eta s$$

(`etaRep`, `etaRep_welfare`). Compliance is a transfer between the two target groups.

The flows. Along the myopic benchmark the sanction is zero at every date (Proposition 2), so its flow is $W_\eta(x_t, a^{00}(x_t), 0) = U_S(x_t, a^{00}(x_t), 0) + U_T(x_t, 0, 0)$ whatever η. Along the forward-looking equilibrium of Theorem 3 the target maintains full non-compliance, and the realized sanction s_t is at least $\underline{s}$ at every date: the forward-looking response is at least the stage response to full non-compliance, which is at least $\underline{s}$ because it is non-decreasing in the state (`responsePath_ge_sfloor`). Let M_b bound $|U_S(x, a, s) + U_T(x, 0, s)|$ on the cube, and let $\eta_+ = (2M_b + 1)/\underline{s}$. Writing W_t^{00} and W_t^{11} for the two flows under W_{η_+},

$$W_t^{11} - W_t^{00} \geq -2M_b + \eta_+ s_t \geq 1;$$

under $W_{-\eta_+}$ the same bound holds with the difference reversed. All flows are bounded by $M_b + \eta_+$, so the strict inequalities at every date survive discounting at every $\beta \in [0, 1)$ (`exists_reps_reversing_myopic_forward_ranking`, with the flows `repFlowMyopic` and `repFlowForward`). The hypotheses are those of Proposition 2 and of the construction in Theorem 3, with a non-decreasing response in place of the increasing-differences condition; compliance neutrality, semiconvexity in the sanction and the forward-looking relief condition are not used. The two representations, W_{η_+} and $W_{-\eta_+}$, depend on neither the initial state nor the planner's discount factor. □

The four cases. The same representations reverse the welfare ranking of all four constructed equilibria of Theorem 12.

Proposition C.16 (Welfare non-identification across the four cases)**.** *Take an economy in the class that satisfies the four-case conditions of Theorem 12, with a sender response at full non-compliance at its constructed value that is non-decreasing in the state in place of the increasing-differences condition. There are two interest-group representations of its government payoffs, each with two groups per country and positive constant weights, such that, at every initial state and every common*

planner discount factor $\beta \in [0,1)$, discounted aggregate welfare along the four constructed equilibria is ranked

$$11 \succ 10 \succ 00 \sim 01 \quad \textit{by one}, \qquad 00 \sim 01 \succ 10 \succ 11 \quad \textit{by the other}.$$

Proof. Take the representations W_η of the previous proof. The hypotheses are those under which the four equilibria are constructed in panel I of Appendix C.4, with a non-decreasing response in place of the increasing-differences condition. Along the equilibria of cases 00 and 01 the sanction is zero at every date and the two state paths coincide (`myopicPath_eq_senderOnlyPath_of_reliefSharp`); by compliance neutrality $W_\eta(x,a,0) = U_S(x,0,0) + U_T(x,0,0)$ does not depend on compliance, so the two flows coincide at every date and for every η, and the two cases tie. Along the equilibria of cases 10 and 11 the target maintains full non-compliance. The realized sanction s_t^{10} of case 10 is the stage response to full non-compliance, at least $\underline{s}$ because that response is non-decreasing in the state, and the realized sanction s_t^{11} of case 11 exceeds s_t^{10} by at least the foresight premium m of (B.3) at every date (`targetOnlySanction_add_wedge_le_dynamicSanction_of_response_monotone`). Let $\mu = \min\{\underline{s}, m\} > 0$, let M_b bound $|U_S(x,0,s) + U_T(x,0,s)|$ on the cube, and let $\eta_+ = (2M_b + 1)/\mu$. Writing W_t^{ij} for the flow of case ij under W_{η_+},

$$W_t^{11} - W_t^{10} \geq -2M_b + \eta_+\big(s_t^{11} - s_t^{10}\big) \geq 1, \qquad W_t^{10} - W_t^{00} \geq -2M_b + \eta_+ s_t^{10} \geq 1;$$

under $W_{-\eta_+}$ the same bounds hold with every difference reversed. All flows are bounded by $M_b + \eta_+$, so the strict inequalities at every date survive discounting at every $\beta \in [0,1)$ (`exists_reps_reversing_regime_ranking`). □

Proof of Proposition 10. Let $V(x)$ be the planner's value of the path with zero sanctions and compliance $\hat{\alpha}$ from x. It is bounded and satisfies $V(x) = W(x, \hat{\alpha}(x), 0) + \beta V(F(x,0))$ (`discountedPayoff_stationary_eq`).

Step 1: V rises with the state at a rate of at most $K = \max\{B,0\}/(1-\beta\lambda)$. For $x \leq x'$, the two zero-sanction paths x_t, x_t' stay ordered by (F2), and by (iii) their gap is at most $\lambda^t(x' - x)$ (`zeroPath_ordered`). By (ii), $V(x') - V(x) = \sum_t \beta^t\big(W(x_t', \hat{\alpha}(x_t'), 0) - W(x_t, \hat{\alpha}(x_t), 0)\big) \leq \sum_t \beta^t \max\{B,0\}\lambda^t(x' - x) = K(x' - x)$.

Step 2: V is a Bellman supersolution of the planner's problem. At every feasible (x,a,s), by (i), by Step 1 with $F(x,0) \leq F(x,s)$ from (F3), by (iii) and since $K \geq 0$,

$$W(x,a,s) + \beta V(F(x,s)) \leq W(x, \hat{\alpha}(x), 0) + \beta V(F(x,0)) + \big(\eta + \beta K L\big)s \leq V(x),$$

the last step by the condition on η, with equality at $a = \hat{\alpha}(x)$ and $s = 0$.

Step 3: verification. Summing the supersolution inequality along any time-dependent plan over the first N dates bounds that plan's discounted welfare up to N plus $\beta^N V(x_N)$ by $V(x_0)$; the terminal term vanishes because V is bounded, and the flows are bounded, so every plan's value is at most $V(x_0)$, which the zero-sanction path attains (`stationary_optimal_against_markov_deviation_of_global_bounds`, `zeroSanction_planner_optimal`). Its counterpart in the Kaempfer–Lowenberg polity, where welfare is linear in the state, is Proposition L.3(i); there the case $B \leq 0$ needs no slope bound on the transition. □

Proof of Proposition 11. The representation. Give the sender two groups, with welfare s and $U_S - s$, both at weight one, and the target two groups, with welfare U_T and $-U_T$, at weights two and one. The weighted sums are U_S and U_T, and aggregate welfare is $W = U_S$ (`senderWelfareRep`, `senderWelfareRep_welfare`).

Compliance never raises the sender's payoff. On the class compliance lowers the sender's marginal return to sanctioning, so $U_S(x,a,s) - U_S(x,0,s) \leq U_S(x,a,0) - U_S(x,0,0)$, and the right side is zero by compliance neutrality: $U_S(x,a,s) \leq U_S(x,0,s)$ (`U_S_le_defiance`).

Verification. Let V_S be the sender's constructed value against full non-compliance and ρ its dynamic best response. At every feasible (x, a, s),

$$W(x, a, s) + \beta_S V_S(F(x, s)) \leq U_S(x, 0, s) + \beta_S V_S(F(x, s)) \leq V_S(x),$$

with equality at $a = 0$ and $s = \rho(x)$. The flows and V_S are bounded, so the verification step of the previous proof makes the stationary plan $(0, \rho)$ socially optimal at β_S against every time-dependent plan over both actions (`exists_planner_efficient_rep`). Every equilibrium in $\mathcal{E}$ at which the target maintains full non-compliance has the constructed values and plays ρ at every feasible state (`values_eq_of_policy_zero`), so that plan is the equilibrium's play (`planner_efficient_rep_at_policy_zero`). On the quadratic–linear family at $r = n = 0$ the same argument, with a linear microfoundation, gives Corollary L.6. □

D Persistence, temporary shocks, and lock-in

This appendix collects the results that start from an equilibrium and explicit basin conditions: the general existence route, the conditional high-basin theorem and its long run, dual-basin path dependence, the effects of temporary sanction and suspensions of sanctions, the comparative statics of lock-in, and the contracting closed loop of the class under the lifting condition. Section 5 summarizes what they establish. A1–A3 and the basin, which only these results use, come first.

D.1 Assumptions and the basin

The high-basin and pathwise-ordering results of this appendix require only (F2)–(F4) of the transition; the shock-entry result (Theorem D.8) also uses continuity (F1) and full-sanction ascent below the basin, which (F5) implies. For the optional quantitative value-regularity route, we also use an inherited-state modulus γ_F:

$$\big|F(x_2, s) - F(x_1, s)\big| \leq \gamma_F |x_2 - x_1| \quad \text{for all } x_1, x_2, s \in [0, 1], \qquad \beta_S \gamma_F < 1. \tag{F6-Q}$$

Condition (F6-Q) is not part of the qualitative baseline (F1)–(F5), and it is stated on the feasible square. Under (J.2), $\gamma_F = 1 - \delta$: it is the coefficient on the inherited state, not the sanction pass-through δ (Appendix H.9).

We state three structural inputs and derive the high political basin. A1 and A2 restrict objectives uniformly over value sets; A3 is a curvature condition on the sender's stage payoff. The general existence theorem takes its invariant-set, selector and operator-bound conditions as hypotheses; the constructions of Section 3.1 and Appendix I.6 build their invariant sets and regularity constants from $(U_T, U_S, F, \beta_T, \beta_S)$ in closed form in the stated rates (Appendix N). The derived-basin route in Appendix G.1 derives the basin and the target condition from complementarity. The three assumptions also provide a fallback for specifications without global complementarity.

Basin pivots

Let $\bar{x}_T, \bar{x}_S \in [0, 1]$ denote the target and sender tipping thresholds. Under stage complementarity, each is the infimum of a flip set; Proposition G.1 establishes existence, interiority, and the upper-interval structure. Define the joint pivot $\bar{x} = \max\{\bar{x}_T, \bar{x}_S\}$ and the high basin $\mathcal{B} = [\bar{x}, 1]$.

Lipschitz constants

Write K_S for a Lipschitz constant, in the inherited state, of the sender's values in its value set. Under the optional regularity (F6-Q) the envelope recursion gives $K_S = L^{\alpha}_{S,x}/(1-\beta_S\gamma_F)$, where $L^{\alpha}_{S,x}$ bounds the state variation of the sender's stage payoff along a continuous target policy (Stokey et al., 1989; Bertsekas and Tsitsiklis, 1996); the target constant and the selector moduli are in Appendix N. The modulus γ_F does not bound the transition's response to the sanction, so the level condition used below is the transition-adjusted slope condition

$$U_S(x,a,s_2) - U_S(x,a,s_1) > \beta_S K_S \big|F(x,s_2) - F(x,s_1)\big|, \qquad s_1 < s_2, \tag{D.1}$$

on the range used below. It needs no separate modulus of F in the sanction and, under the partial-adjustment law, follows from $\beta_S K_S \delta < m_S$ for a stage lower slope m_S (Appendix H.9).

Three structural assumptions

We refer to Assumptions 1 to 3 as A1–A3. A1 controls the direction of the sender's response, A2 records the target's forward-looking preference on the basin, and A3 gives diminishing returns in the sender's stage problem. The high-basin theorem uses the direction and target conditions together with a separate sanction-level condition, the target condition only for its conclusion (vi); the shock theorem uses no target condition, and neither uses A3 directly. On the derived-basin route A2 can be derived rather than imposed, though the route's conclusions do not read it (Proposition G.4). Table 2, in Appendix H.7, records which result consumes which of these conditions: none is a maintained assumption on the whole model, and the constructed equilibria of Section 5.1 use none of A1–A3 as stated.

Write $\mathcal{C}_T$ and $\mathcal{C}_S$ for the target and sender value sets: closed, nonempty sets of bounded continuous functions on $[0,1]$ cut out by a sup-norm radius together with Lipschitz or slope bounds, with radii and moduli derived from the model primitives in Section 3 and Appendix N. The general existence theorem takes these sets and their invariance as inputs; the constructions discharge them, the target's invariant set on the class under the lifting condition by a sup-norm radius and a lower-slope condition with no Lipschitz clause.

Assumption 1 (Sender interval dominance after target-policy substitution)**.** For every value pair $(V_T, V_S) \in \mathcal{C}_T \times \mathcal{C}_S$, let α_{V_T,V_S} be its induced target policy and define the policy-substituted sender family by

$$\widetilde{J}_{V_T,V_S}(s,x) := J_S\big(x, \alpha_{V_T,V_S}(x), s; V_S\big). \tag{D.2}$$

On the sanction–state domain $[0,1]\times\mathcal{B}$, this family satisfies the interval dominance order in (s,x) of Quah and Strulovici (2009). Milgrom–Shannon single crossing in (s,x) is a sufficient, stronger condition.

A1 is ordinal: after the endogenous target policy is substituted, it makes the sender's dynamic best response non-decreasing in the state; single crossing is a stronger sufficient condition (Milgrom and Shannon, 1994). Direction does not determine level: keeping the response above $\max\{x,\bar{x}\}$ is the role of (D.1), and the proof of Theorem D.2 keeps the two arguments separate (Appendices C and G and Appendix H.9).

Assumption 2 (Target combined slope condition, value-set form)**.** For every $(V_T, V_S) \in \mathcal{C}_T \times \mathcal{C}_S$, let α_{V_T,V_S} be the induced target policy. For every $x \in \mathcal{B}$,

$$\boxed{s \longmapsto U_T\big(x, \alpha_{V_T,V_S}(x), s\big) + \beta_T V_T\big(F(x,s)\big) \text{ is non-decreasing on } [\max\{x,\bar{x}\}, 1].} \tag{D.3}$$

A2 is imposed uniformly over the value sets rather than at a presumed equilibrium. It combines the target's current payoff from sanctions with the political-coalition effect in its continuation value (Kaempfer and Lowenberg, 1988). Each component may be non-monotone; their sum must be non-decreasing on the displayed range. Strict concavity of the complete target objective in compliance, together with Berge's Maximum Theorem, supplies a continuous policy selector (Aliprantis and Border, 2006).

Assumption 3 (Uniform stage strong concavity in s)**.** There is $\mu > 0$ such that, for every $(x, a) \in [0,1]^2$, every $s_1, s_2 \in [0,1]$ and every $t \in [0,1]$,

$$t\,U_S(x,a,s_1) + (1-t)\,U_S(x,a,s_2) \le U_S\big(x,a,ts_1+(1-t)s_2\big) - \tfrac{\mu}{2}\,t(1-t)(s_1-s_2)^2.$$

This is the derivative-free form the formalization consumes. When U_S is twice differentiable in s it is the familiar $\sup_{(x,a,s)\in[0,1]^3} \partial^2 U_S/\partial s^2 \le -\mu < 0$.

A3 imposes diminishing returns to sanctions, as in standard public-choice models (Kaempfer and Lowenberg, 1988; Eaton and Engers, 1992), and makes the myopic best response unique. It does not make the maximizer of J_S unique, because the continuation term can offset stage curvature, and it is not what carries the substitution between compliance and the sanction (Appendix N and Appendix H.9).

A route that derives the basin. Under a stage-complementarity assumption the pivots $\bar{x}_T$ and $\bar{x}_S$ are not inputs: each is the infimum of a payoff flip set that is an upper interval, and the target condition A2 becomes a result (Assumption 4 and Propositions G.1 and G.4, in Appendix G.1). Sign conditions of this kind fix the signs of slopes and bound no magnitude, so they do not deliver persistence on their own; the magnitude condition is the sub-pivot slope condition (P5) of Theorem D.2. No result on the class under the lifting condition in Section 5.1 uses this route.

D.2 Equilibrium existence and persistence in the general game

The dynamic results of this subsection require a stationary feedback Stackelberg equilibrium, and they start from one.

D.2.1 Equilibrium existence

A joint contraction gives one route to existence in a smooth, gently coupled region. Let L_{TS} and L_{ST} measure how each Bellman row responds to the other player's continuation value, and let $\beta_{S,\mathrm{row}}$ be the sender row's own contraction modulus; $\rho(\mathbf{M})$ is the spectral radius of the coupling matrix these four numbers form (Appendix F).

Theorem F.1 (Appendix F) states the route and its Schur condition $L_{TS}L_{ST} < (1-\beta_T)(1-\beta_{S,\mathrm{row}})$, with no sign condition on the four numbers. Its uniqueness conclusion (every equilibrium with values in the invariant sets that plays the target selector those sets define coincides with the constructed one) is confined to the stated invariant-set and selector restrictions; it is not global equilibrium uniqueness. A summary of the statement is in Appendix H.7.

The economically transparent deterministic route is the high-basin corner. When full non-compliance is optimal for the target, the game reduces to sequential single-agent Bellman problems.

Theorem D.1 (Basin-corner feedback Stackelberg existence)**.** *Fix V_S at its Bellman fixed point against $\alpha \equiv 0$, and write $\sigma^* = \sigma^{\mathrm{dyn,top}}(V_S)$. Suppose, on the entire feasible state space, that:*

(i) the map $(x,a) \mapsto \sigma^*(x,a)$ is continuous at feasible states x and every target action a;

(ii) the target continuation value constructed against σ^* is non-decreasing in x;

(iii) for $a \in (0,1]$, $\sigma^*(x,a) \le \sigma^*(x,0)$: no positive compliance raises the sender's response above its response to full non-compliance; and

(iv) the substituted target stage payoff is highest at full non-compliance,

$$U_T\big(x,a,\sigma^*(x,a)\big) \le U_T\big(x,0,\sigma^*(x,0)\big) \qquad \text{for } a \in (0,1].$$

Then $\alpha \equiv 0$ is dynamically optimal and $E = (V_T, V_S, \alpha \equiv 0) \in \mathcal{E}$, with V_T constructed as the single-agent Banach fixed point.

Continuity gives a well-defined target Bellman operator, while the remaining conditions make full non-compliance optimal after the sender's response is substituted. Each condition is used once and only weakly: an antitone response and a strict stage decrease are a special case. This argument applies to the general monotone transition law and does not require purification.

Off the corner the deterministic joint iteration need not preserve order or contract; existence there is obtained for a transition-smoothed game (an atomless disturbance added to the state law; "purified" in earlier versions, a term this literature reserves for payoff noise) and at example economies, not for the primitive class (Appendices H.7 and H.8).

The dynamic results use the general transition law (1). Monotonicity and the diagonal condition deliver basin invariance; strict adjustment makes the long-run sanction equal to the limiting state. The partial-adjustment law (J.2) is needed only for the closed-form expressions stated separately below.

D.3 Persistence and the long run

Theorem D.2 (Persistence on the high political basin)**.** *Let $\bar{x}_T, \bar{x}_S \in [0,1]$ and $\bar{x} = \max\{\bar{x}_T, \bar{x}_S\}$. Fix the value sets $\mathcal{C}_T \times \mathcal{C}_S$ and let $E = (V_T, V_S, \alpha) \in \mathcal{E}$. Suppose:*

(P1) $V_T \in \mathcal{C}_T$, $V_S \in \mathcal{C}_S$, and $\alpha = \alpha_{V_T,V_S}$;

(P2) every sender value in $\mathcal{C}_S$ is K_S-Lipschitz;

(P3) A1 holds for every value pair, in the state-indexed form (D.2)*;*

(P4) the target combined slope condition A2 holds for every value pair; and

(P5) for every value pair (V_T, V_S), every basin state $x \in \mathcal{B}$, and every sanction $0 \le s < x$ below the pivot $\max\{x, \bar{x}\} = x$, the general-F sub-pivot slope condition holds in pivot-star form:

$$U_S(x, \alpha_{V_T,V_S}(x), x) - U_S(x, \alpha_{V_T,V_S}(x), s) > \beta_S K_S \big|F(x,x) - F(x,s)\big|.$$

Then for every $x_0 \in \mathcal{B}$, the equilibrium path satisfies:

(i) $x_t^E \in \mathcal{B}$ for every $t \ge 0$;

(ii) $s_t^E \ge \bar{x}$ for every $t \ge 0$;

(iii) $s_t^E \le s_{t+1}^E$ for every $t \ge 0$;

(iv) $\alpha(x_t^E)$ is feasible and maximizes the target's complete Bellman objective against the sender's dynamic best response;

(v) s_t^E *maximizes* $J_S(x_t^E, \alpha(x_t^E), \cdot\,; V_S)$*; and*

(vi) the map

$$s \longmapsto U_T(x_t^E, \alpha(x_t^E), s) + \beta_T V_T\big(F(x_t^E, s)\big)$$

is non-decreasing on $[\max\{x_t^E, \bar{x}\}, 1]$*.*

The proof separates the direction and the level of the sender's response: A1 makes the dynamic best response non-decreasing in the inherited state, and the transition-adjusted slope condition P5 keeps it above the basin floor, so that $s_t^E \geq x_t^E$ and the basin is preserved. P5 is read only at basin states and only against the pivot: it compares each sub-pivot sanction with the pivot, never two sub-pivot sanctions with each other, and it constrains no state below the joint pivot; the pairwise slope condition on $[\bar{x}_S, 1]$ implies it. The target condition P4 is logically separate and yields only conclusion (vi); conclusions (i)–(iii) hold under P1–P3 and P5 alone. A1 is used only for (iii): conclusions (i), (ii) and (iv), and a non-decreasing state path, follow from P1, P2 and P5, the level condition, without it. The theorem is conditional on the stated equilibrium and its value-set conditions; Theorem D.1 and Proposition I.5 provide unconditional applications under narrower primitive conditions. The proof is in Appendix C; a local form with no value sets, and an economy showing that these conditions select no limit, are in Appendix H.7, and the paragraphs this one replaces in Appendix H.9.

Corollary D.3 (Post-entry non-termination)**.** *Under the hypotheses of Theorem D.2, let* T *be a date with* $x_T^E \in \mathcal{B}$*. For every cutoff* $q < \bar{x}$ *and every* $n \geq 0$*,*

$$s_{T+n}^E > q.$$

Thus every sanction-reduction threshold strictly below the inherited floor is never reached after basin entry, and when $\bar{x} > 0$ *neither is relief.*

Relief remains feasible; the corollary says it is not chosen along the path of the equilibrium $E \in \mathcal{E}$ after the political state enters the basin. It is therefore a result about how long sanctions last after basin entry, not about which disputes enter sanctions in the first place. Only the level condition is used: at any equilibrium whose sanction rule is at or above the state on the basin, no cutoff below the inherited floor is reached after entry; neither A1 nor the target condition enters.

The rule and the closed loop. Two abbreviations make the long-run statements readable. Write

$$R_E(x) = \sigma_{V_S}^{\mathrm{dyn}}\big(x, \alpha(x)\big), \qquad G_E(x) = F\big(x, R_E(x)\big),$$

so that R_E is the sanction rule of E (the sender's dynamic best response evaluated at the target's equilibrium policy), and G_E is the closed loop it induces on the political state. The realized path of an equilibrium is then simply $s_t^E = R_E(x_t^E)$ and $x_{t+1}^E = G_E(x_t^E)$; these are the same objects as in Section 2, named rather than redefined. A *steady state* of the equilibrium is a fixed point of G_E: a political state the closed loop reproduces. Under strict directional adjustment (F5) that is equivalent to the sanction matching the state there, $R_E(x) = x$, so the two ways of saying it agree wherever (F5) holds.

Three long-run results follow, and they differ in what they assume about the sanction rule and in how precisely they name the limit. The first uses only the level condition of Theorem D.2 (the sanction rule at or above the state on the basin) and establishes convergence to an unspecified steady state. The second adds monotonicity of the closed loop and continuity of the rule and pins down the limit as an extremal steady state, which may differ across starting points. The third

retains those conditions but ends the interval at the first steady state above the pivot, yielding one limit for every start in that interval.

Proposition D.4 (Long-run lock-in level)**.** *Let $E \in \mathcal{E}$, let $\bar{x} \geq 0$, and suppose the sanction rule is at or above the state on $\mathcal{B}$, $R_E(x) \geq x$ for $x \in \mathcal{B}$, with no floor. With the state evolved by the general law* (1) *satisfying the maintained* (F1)–(F5)*, the state path from every basin start converges to some $x^* \in \mathcal{B}$. The realized sanction sequence converges to the same limit, and x^* is an equilibrium-consistent steady state: sanction x^* is optimal at state x^* and reproduces that state. The hypotheses of Theorem D.2 imply the level condition.*

This baseline result requires neither monotonicity nor continuity of the sanction rule, and neither A1 nor the target condition. Those conditions enter only when the limit itself is to be pinned down.

Monotonicity of the closed loop and continuity of the sanction rule pin down the limit (Proposition I.1, Corollary I.2 and Proposition I.3, stated in Appendix I). If the start lies below the rule's value there, the state converges to the least steady state of the closed loop at or above it, and if above, to the greatest steady state at or below it; under strict directional adjustment the realized sanction converges to the same value. Continuity of the rule is the requirement, and uniqueness of the maximizer of J_S is one sufficient condition for it (Appendix H.7). Order selection need not imply attraction of a whole basin: every start in an invariant interval converges to one common limit exactly when the interval contains a unique steady state, and the interval that runs from the basin pivot to the first steady state above it has that property by construction. In plain terms, the inherited political state selects, at date zero and within a single equilibrium, which self-sustaining steady state is reached (path dependence, not indeterminacy), and up to the first such steady state above the pivot every start reaches it. A steady state of G_E is a political state at which the sender's optimal sanction reproduces the entrenchment it inherits; none of the three results delivers a rate (Appendix H.9).

D.4 Dual-basin path dependence

The argument of Theorem D.2 has a sign-reversed counterpart on a low political basin, and the two can hold for one equilibrium. With $\varepsilon = +1$ the high orientation and $\varepsilon = -1$ the low one, the sender's sanction rule remains non-decreasing in the inherited state, and from every start in the corresponding basin the state stays there and the sanction moves monotonically on the basin's side; neither orientation reads a target condition (Theorem D.5; the high orientation reproduces conclusions (i)–(iii) of Theorem D.2). Because the two assumption blocks apply on disjoint state ranges, the same equilibrium can satisfy both: below a low threshold $\bar{x}_L$ the state stays below it with non-increasing sanctions, above a high threshold $\bar{x}_H$ it locks in with non-decreasing ones (Corollary D.6). This is selection by initial conditions within one equilibrium, not selection among equilibria; the appendix supplies an economy satisfying both sets of conditions. The region between the two thresholds is not left to assumption: if the closed loop strictly lowers the state on the low basin, strictly raises it on the high one and has a continuous sanction rule, it determines two of its own steady states $x_s^- \leq x_s^+$ strictly inside $(\bar{x}_L, \bar{x}_H)$, with convergence to zero from every start below x_s^- and lock-in from every start above x_s^+, and the two coincide exactly when there is a single steady state between the basins (Proposition D.7; Figure 7 draws the single-crossing case). The picture (two basins inside one equilibrium, with initial conditions selecting between them) is the known one of Section 1; what is new is that the basins are basins of a political state moved by a coercive instrument.

D.5 Persistence on a low basin and initial-condition selection

The persistence argument has a sign-reversed counterpart on a low political basin. Let $\varepsilon = +1$ denote the high orientation and $\varepsilon = -1$ the low orientation. In both the sender's sanction rule remains non-decreasing in the inherited state, and neither reads the target's sanction preference.

Theorem D.5 (Persistence with orientation ε)**.** *Fix $\varepsilon \in \{+1, -1\}$ and suppose $A1_\varepsilon$ holds, together with the orientation-specific level condition; neither orientation reads a target condition. Suppose $E = (V_T, V_S, \alpha) \in \mathcal{E}$ with $V_T \in \mathcal{C}_T$, $V_S \in \mathcal{C}_S$, and $\alpha = \alpha_{V_T, V_S}$. Then, from every starting state in the corresponding basin:*

(i) the state remains in that basin;

(ii) the realized sanction remains on the basin side of its corresponding extremum;

(iii) $\varepsilon(s^E_{t+1} - s^E_t) \geq 0$ for every t.

The high orientation reproduces conclusions (i)–(iii) of Theorem D.2; the low orientation keeps the state in the low basin with non-increasing sanctions. Because the two assumption blocks apply on disjoint state ranges, the same equilibrium can satisfy both.

Corollary D.6 (Initial-condition selection under a common equilibrium)**.** *Suppose the same equilibrium E satisfies the sender's high-basin conditions with state threshold $\bar{x}_H$ and sanction threshold $\bar{s}_H$ (at every state of $[\bar{x}_H, 1]$ some maximizer of J_S is at least $\max\{x, \bar{s}_H\}$, and the greatest one is non-decreasing in the state there), and the sender's low-basin conditions with thresholds $\bar{x}_L$ and $\bar{s}_L$ (on $[0, \bar{x}_L]$ the dynamic best response is at most $\min\{x, \bar{s}_L\}$ and non-decreasing in the state), where $\bar{x}_L < \bar{x}_H$. No target condition is used. Then the initial-state regions $[0, \bar{x}_L]$ and $[\bar{x}_H, 1]$ are disjoint. For every $x_0 \in [\bar{x}_H, 1]$ and every $t \geq 0$,*

$$x^E_t \in [\bar{x}_H, 1], \qquad s^E_t \geq \max\{\bar{x}_H, \bar{s}_H\}, \qquad s^E_{t+1} \geq s^E_t.$$

For every $x_0 \in [0, \bar{x}_L]$ and every $t \geq 0$,

$$x^E_t \in [0, \bar{x}_L], \qquad s^E_t \leq \min\{\bar{x}_L, \bar{s}_L\}, \qquad s^E_{t+1} \leq s^E_t.$$

This is selection by initial conditions within one equilibrium, not selection among equilibria. The appendix supplies an economy satisfying both sets of conditions.

Corollary D.6 leaves the region between $\bar{x}_L$ and $\bar{x}_H$ unclassified and takes both basin boundaries as given. Continuity of the equilibrium closed loop nevertheless derives bounds on the dividing region without imposing a sign in the middle.

Proposition D.7 (The dividing line is derived)**.** *Let $E \in \mathcal{E}$ and $0 < \bar{x}_L \leq \bar{x}_H < 1$, for instance the thresholds of Corollary D.6. Suppose that the closed loop strictly lowers the state at every state of $(0, \bar{x}_L]$ and strictly raises it at every state of $[\bar{x}_H, 1)$, and that the sanction rule is continuous on $[0, 1]$. The closed loop then maps $[0, 1]$ into itself, by feasibility of the transition and of the rule, and fixes both endpoints of the state space; the first formalization assumed both. Impose no restriction on the sign of $G_E(x) - x$ between $\bar{x}_L$ and $\bar{x}_H$; monotonicity of the sanction rule is also unnecessary. Then the equilibrium closed loop determines two of its own steady states,*

$$x^-_s = \inf\{\, x \in [\bar{x}_L, \bar{x}_H] : x \leq G_E(x) \,\}, \qquad x^+_s = \sup\{\, x \in [\bar{x}_L, \bar{x}_H] : G_E(x) \leq x \,\},$$

with $\bar{x}_L < x^-_s \leq x^+_s < \bar{x}_H$, such that every equilibrium path started anywhere in $[0, x^-_s)$ converges to 0 and every equilibrium path started anywhere in $(x^+_s, 1]$ converges to 1. Moreover $x^-_s = x^+_s$ exactly when G_E has a single steady state in $[\bar{x}_L, \bar{x}_H]$.

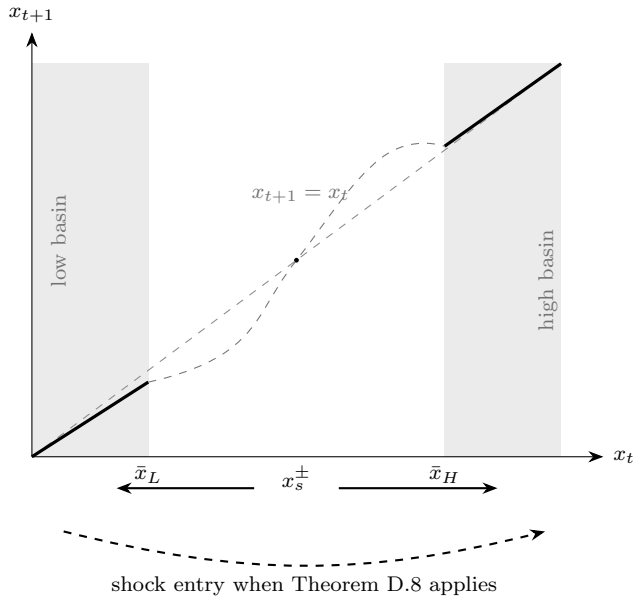


Figure 7: **Dual-basin selection.** The solid segments represent the equilibrium dynamics established on the low and high basins. The dashed middle segment is illustrative. By Proposition D.7 the middle region contains steady states $x_s^- \leq x_s^+$ derived from the closed loop, with convergence to zero below x_s^- and lock-in above x_s^+; the figure draws the case $x_s^- = x_s^+$ of a single crossing, which the proposition does not assert. Under the separate shock-entry conditions, a temporary sanction shock can move the state into the high basin, after which equilibrium dynamics resume.

The two convergence regions therefore extend strictly beyond the given basins, with endpoints derived from the equilibrium; the interval between them need not collapse to a point, and continuity of the sanction rule is what supplies a middle steady state (Appendix H.9).

D.6 Temporary-shock entry and persistence after entry

A temporary sanction shock can move the political state into the high basin, after which equilibrium policies keep the state there. The shock is not an equilibrium policy or an optimal sanction policy; each result describes resumed play under the same specified equilibrium, and every duration below is a sufficient ceiling that need not be minimal.

Theorem D.8 (Temporary-shock entry and persistence after entry)**.** *Maintain* (P1)–(P3) *and* (P5) *of Theorem D.2, with the same value sets, equilibrium, pivots, and sender Lipschitz constant; the target condition* (P4) *is not needed. In addition, suppose the transition follows the general law* (1)*, satisfies* (F1)–(F4)*, and rises under full sanctions below the basin, $x < F(x,1)$ for $x \in [0, \bar{x}]$ (so $\bar{x} < 1$). Define the full-sanction drift and its minimum below the basin by*

$$d_F(x) := F(x,1) - x, \qquad \eta_F(\bar{x}) := \min_{x \in [0,\bar{x}]} d_F(x).$$

Then $\eta_F(\bar{x}) > 0$. Define the conservative shock-duration bound by

$$\widehat{N}_F(\bar{x}) := \left\lceil \frac{\bar{x}}{\eta_F(\bar{x})} \right\rceil.$$

From any $x_0 \in [0,1]$, imposing $s_t = 1$ for $t = 0, \ldots, \widehat{N}_F(\bar{x}) - 1$ gives

$$x_{\widehat{N}_F(\bar{x})} \geq \bar{x}.$$

The same conclusion holds for every longer temporary shock at the full sanction. Consequently $x_{\widehat{N}_F(\bar{x})} \in \mathcal{B}$, and conclusions (i)–(iii) of Theorem D.2 hold along the equilibrium path resumed from that state.

Continuity and full-sanction ascent below the basin make the full-sanction drift strictly positive on the compact pre-basin interval, where its minimum is attained. Until the lowest shocked path enters the basin, it therefore rises by at least $\eta_F(\bar{x})$ each period. The bound is conservative because it uses only the smallest one-period drift.

The theorem establishes entry followed by persistence. It does not compare the shocked path with the path the same equilibrium would have followed unshocked, and so it does not assert that the shock selects a different long-run outcome; Proposition D.11 exhibits an economy in which it does not. The duration is a sufficient bound that depends on the economy and is not small in any uniform sense (Appendix H.9).

Under the partial-adjustment law (J.2), the sharper closed-form duration is

$$N_*(\bar{x}, \delta) := \left\lceil \frac{\log(1-\bar{x})}{\log(1-\delta)} \right\rceil, \qquad x_{N_*} = 1 - (1-\delta)^{N_*}(1 - x_0) \geq \bar{x}.$$

A transition that carries the state up faster under the full sanction has a weakly larger minimum entry drift and a weakly shorter sufficient duration; the drift comparison is strict when the transition comparison is, while the duration comparison stays weak because distinct drifts can share an integer ceiling (Corollary I.4, in Appendix I).

D.7 A temporary shock, then no endogenous exit

A confrontation may begin as an attempt to make the other government back down. The model does not solve that initial interaction, and it carries no types, beliefs, or signals. What it shows is what a temporary shock leaves behind. If the shock carries the political state into a region the resumed equilibrium preserves, the shock's original purpose ceases to matter for what follows: under unchanged primitives and that equilibrium, sanctions never fall below the region's floor. This is endogenous non-exit, not technological impossibility of relief (the transition still erodes entrenchment under low sanctions, and Theorem D.10 gives a sufficient duration for a temporary suspension of sanctions), and it presupposes that everything the shock leaves behind is represented by the scalar state.

Theorem D.9 (Temporary shock, then no endogenous exit)**.** *Maintain* (F1)–(F4)*. Let $(V_T, V_S, \alpha) \in \mathcal{E}$ with sanction rule $R(x) = \sigma^{\mathrm{dyn}}_{V_S}(x, \alpha(x))$, and let $\bar{x}$ and $\bar{s}$ satisfy $\bar{x} \leq F(\bar{x}, \bar{s})$ and $R(x) \geq \bar{s}$ for every $x \in [\bar{x}, 1]$; both hold whenever the sanction rule is at or above $\bar{x}$ on $[\bar{x}, 1]$ and $\bar{s} = \bar{x}$. Let $(u_t)_{t<T}$ be a temporary shock, a feasible sanction path set exogenously before date T, with the equilibrium resumed from date T:*

$$x_{t+1} = F(x_t, s_t), \qquad s_t = \begin{cases} u_t, & t < T, \\ R(x_t), & t \geq T. \end{cases}$$

(i) Splice. *For $t \geq T$ the path (x_t, s_t) is the equilibrium path of Section 2 restarted at x_T.*

(ii) Non-exit. *If $x_T \in [\bar{x}, 1]$ then $x_t \in [\bar{x}, 1]$ and $s_t \geq \bar{s}$ for every $t \geq T$, and for every cutoff $q < \bar{s}$ there is no date $t \geq T$ at which $s_t \leq q$.*

(iii) Entry under a non-maximal shock. *Let $0 \le \bar{x}$, let the sanction level h raise every state of $[0, \bar{x}]$, $x < F(x, h)$ there (held-sanction ascent, which forces $\bar{x} < h$ and which* (F5) *implies whenever $\bar{x} < h \le 1$), and let $u_t \ge h$ for every $t < T$. The drift $\eta_h(\bar{x}) := \min_{x \in [0,\bar{x}]} \big(F(x,h) - x\big)$ is attained and strictly positive, and for $T \ge N_h(\bar{x}) := \lceil \bar{x}/\eta_h(\bar{x}) \rceil$ every initial state $x_0 \in [0,1]$ gives $x_T \in [\bar{x}, 1]$. At $h = 1$, $N_1(\bar{x})$ is $\widehat{N}_F(\bar{x})$ of Theorem D.8. Where a lower bound $d > 0$ on the transition's sanction slope is available and $\bar{x} < h \le 1$, $\lceil \bar{x}/(d(h - \bar{x})) \rceil$ is an explicit, looser, sufficient duration.*

(iv) Ratchet tail. *If $x_T \in [\bar{x}, 1]$ and the sanction rule is at or above the state on $[\bar{x}, 1]$, the post-shock sanction satisfies $s_t \ge \max(x_T, \bar{x})$ for every $t \ge T$; if the rule is also non-decreasing on $[\bar{x}, 1]$, the sanction is nondecreasing in t. Both conditions hold under* (P1)–(P5) *of Theorem D.2 with pivots $\bar{x}, \bar{s}$.*

The basin conditions of Theorem D.9 are met on the class under the lifting condition at every equilibrium at which the target maintains full non-compliance, with pivot and floor both equal to $\underline{s} = \sigma^{\mathrm{M}}(0,0)$ (u_0/q on the quadratic–linear family when $u_0 \le q$, as under interiority); such an equilibrium exists under the entrenchment condition and a non-decreasing response at non-compliance, which the increasing-differences condition (B.4) implies, the constructed one. That is weak invariance and a floor, not the ratchet and not hysteresis (Appendix H.9).

The comparison behind (iii), two machine-checked boundaries of the statement (a floor equal to the pivot need not be reached in finite time, and no duration is uniformly small across transitions), and the instantiation at a non-affine transition are in Appendix H.7.

The theorem on a finite temporary shock, on the class under the lifting condition. *(From Appendix D.7, after Theorem D.9.)*

The basin conditions of Theorem D.9 are met on the class under the lifting condition at every equilibrium at which the target maintains full non-compliance, with pivot and floor both equal to $\underline{s} = \sigma^{\mathrm{M}}(0,0)$ (u_0/q on the quadratic–linear family when $u_0 \le q$, as under interiority): every such equilibrium realizes the constructed response, which is at least the floor at every state, and $F(\underline{s}, \underline{s}) = \underline{s}$, so a temporary shock ending in $[\underline{s}, 1]$, in particular one held at any level above $\underline{s}$ for the derived duration, is followed by permanent residence there with the sanction never below $\underline{s}$ (`placable_post_episode_nonexit_of_policy_zero`, `placable_highEpisode_nonexit_of_policy_zero`). Such an equilibrium exists under the entrenchment condition and a non-decreasing response at non-compliance, which the increasing-differences condition (B.4) implies, the constructed one; where none exists the statement is empty. This is weak invariance and a floor, not the ratchet and not hysteresis: at the affine member the closed loop contracts and the unshocked path settles above the floor as well.

D.8 Relief and the exit duration

The entry result has a counterpart on the exit side, which uses only the descent half of (F5), and only inside the basin: the zero sanction strictly lowers every state of $[\bar{x}, 1]$. Whether one equilibrium sustains both basins is a further question; Proposition K.1 is the one economy at which it is settled (Appendix H.9).

Theorem D.10 (Exit under a temporary suspension of sanctions)**.** *Maintain* (F1)–(F4)*, let $\bar{x} \le 1$, and suppose the zero sanction strictly lowers every state of the basin, $F(x, 0) < x$ for $x \in [\bar{x}, 1]$ (post-basin descent, which forces $\bar{x} > 0$ and which* (F5) *implies whenever $\bar{x} > 0$). Define the relief*

drift and its minimum inside the basin by

$$r_F(x) := x - F(x,0), \qquad \eta_F^R(\bar{x}) := \min_{x\in[\bar{x},1]} r_F(x).$$

The minimum is attained and $\eta_F^R(\bar{x}) > 0$. *Let*

$$\widehat{N}_F^R(\bar{x}) := \left\lceil \frac{1-\bar{x}}{\eta_F^R(\bar{x})} \right\rceil.$$

If the sanction is held at zero from an arbitrary feasible initial state $x_0 \in [0,1]$, *then the resulting state path satisfies* $x_n \le \bar{x}$ *at every date* $n \ge \widehat{N}_F^R(\bar{x})$.

As with Theorem D.8, the relief path is exogenous, not chosen: it is a temporary shock during which neither government optimizes, and the result is correspondingly not an optimal-relief claim. What it adds is the second threshold. Under the hypotheses of Theorem D.2 relief never arrives endogenously (inside the basin the equilibrium sanction stays above the state inherited at entry, so the set of dates at which the equilibrium grants relief is empty), and under the weaker floor of Theorem D.9 it never falls below the region's floor. Read together, the two statements say something neither says alone:

> The equilibrium never ends the sanctions regime on its own. A temporary suspension of sanctions of the stated sufficient duration crosses the threshold; whether sanctions then stay low depends on a separate low-basin condition for the resumed equilibrium.

The bound is a sufficient duration for a temporary suspension of sanctions, not a minimal one. What this implies for sunset clauses and review dates, and how the entry and exit durations compare, is in Appendices H.7 and H.9.

Two limits should be stated plainly. Theorem D.10 stops at the crossing: it establishes that the state re-enters $[0,\bar{x}]$ and by itself says nothing about whether it *remains* there once the shock ends. That gap is closed *conditionally*, with the same shape as the entry side:

Conditional on the low-basin level condition (the sanction rule at most the state and at most a level $\underline{s}$ on $[0,\bar{x}]$), a temporary suspension of sanctions of the stated length is followed by a state path that stays in $[0,\bar{x}]$ and is non-increasing (Proposition D.12). The level condition is the dual of the one Theorem D.8 assumes on the high basin: an assumption about the equilibrium in question, discharged at the dual-basin economy of Proposition K.1 and at no other economy in the development. Read with Theorem D.8, the two statements are two opposite-direction crossings, each followed under its own conditions by the state staying on its new side; they are not a same-equilibrium hysteresis unless both level conditions hold for one equilibrium. Two further limits, at the threshold itself and in the conservatism of the duration, are in Appendix H.9.

Relief in this section is set exogenously. At the constructed forward-looking equilibrium of the class under the lifting condition it does not arise endogenously, since the target maintains non-compliance there; against a myopic target, and below the intermediate range of the target's discount factor in Proposition C.9, compliance buys it (Table 3). Under the entrenchment condition and a non-decreasing response, at every equilibrium of the class at which the target maintains full non-compliance (the constructed one among them) a spell of zero sanctions of any length, followed by that equilibrium, is worth strictly less to the sender than the equilibrium itself, by at least a stage-gain floor per period ($u_0^2/(2q)$ on the quadratic–linear family) when no compliance level is worth more to the sender than non-compliance at zero sanction, and weakly less to the target, whose entrenchment condition makes the discounted entrenchment a sanction builds cover its pain;

their gains sum to a negative number, so no lump-sum side payment between them makes the spell acceptable to both (Proposition C.8). The obstacle to de-escalation there is participation before it is enforcement. Outside the entrenchment condition the target's half fails: a temporary suspension of sanctions can benefit a target that would still choose non-compliance.

At the dual-basin economy there is one equilibrium at which a temporary full-sanction shock of the stated length is followed by permanent residence in the high basin with the sanction at its maximum, and a temporary suspension of sanctions by permanent residence in the low basin with a zero sanction (Proposition K.1 in Appendix K.2). This is the same-equilibrium two-sided statement that Appendix D.8 declines to make in general. It is affine, it leaves the interval between the two basins unclassified, and it is an existence certificate (Appendix H.9).

Forward-looking incentives that change behavior do not imply hysteresis. On the class under the lifting condition (Section 3.1) both forward-looking incentives change behavior at a constructed equilibrium, and one might read that as history dependence. It is not, and the most transparent member of its quadratic–linear family shows why.

Proposition D.11 (A contracting closed loop)**.** *On every economy in the class under the lifting condition let ρ be the sender's dynamic best response at full non-compliance, evaluated at its constructed value, which exists on every such economy; under the entrenchment condition and the increasing-differences condition* (B.4) *the map $G(x) = F(x, \rho(x))$ is the state law of the equilibrium of Proposition B.1. Then G is Lipschitz on $[0,1]$ with constant $L_{\text{loop}} = g_{\text{hi}} + d_{\text{hi}} \cdot L_{\text{resp}}/\mu$, where $\mu = q - \beta_S(\sigma_V d_{\text{hi}}^2 + K_S M_s)$ is the strong-concavity modulus of the sender's dynamic objective, M_s the sanction curvature bound of the transition, and $L_{\text{resp}} = r_{\text{hi}} + \beta_S\big(\bar{\sigma} g_{\text{hi}} d_{\text{hi}} + (K_S + 2\bar{\sigma}) M_{\text{mix}}\big)$ with $\bar{\sigma} = \max(\sigma_V, \sigma_C)$ and $K_S = w_{\text{hi}}/(1 - \beta_S g_{\text{hi}})$. Two paths that resume the equilibrium at states x and y are within $L_{\text{loop}}^n |x - y|$ after n periods. On every member of the relief class with $L_{\text{loop}} < 1$ that satisfies myopic interiority $u_{\text{hi}} < q$, the entrenchment condition and the increasing-differences condition, the equilibrium has exactly one steady state in $[0,1]$, at which the sanction equals the state, and every initial state converges to it. At the affine member $K_S = 2$, $\mu = 23/8$, $L_{\text{resp}} = 5/8$, and $L_{\text{loop}} = 14/23$; at the non-affine member $K_S = 15/7$, $\mu = 141/50$, $L_{\text{resp}} = 17/25$, and $L_{\text{loop}} = 35/47$.*

So at both members both governments are forward-looking, both forward-looking incentives change behavior, and a temporary sanction shock has no asymptotic effect, though transient and cumulative effects are not excluded; the absence of an asymptotic effect holds on the whole relief class wherever $L_{\text{loop}} < 1$ and the stated conditions hold. The persistence in Theorem D.8 rests on its basin conditions, not on whether the forward-looking incentives change behavior. The two constants are evaluated upper bounds, not measured convergence rates (Appendix H.9).

D.9 Persistence after relief

Proposition D.12 (Post-relief persistence, conditional on the low-basin level condition)**.** *Maintain* (F1)–(F4)*, let $\bar{x} \leq 1$, suppose the zero sanction strictly lowers every state of $[\bar{x}, 1]$, $F(x, 0) < x$ there (which forces $\bar{x} > 0$), and let R be the sanction rule of an equilibrium's value and policy. Suppose the low-basin level condition holds: for some $\underline{s}$, $R(x) \leq \min(x, \underline{s})$ on $[0, \bar{x}]$. Then from every initial state, after a temporary suspension of sanctions of length $\widehat{N}_F^R(\bar{x})$, the state path evolved by F under R lies in $[0, \bar{x}]$ at every later date and is non-increasing.*

The level condition is the dual of the one Theorem D.8 assumes on the high basin: it is an assumption about the equilibrium in question, not a property of the model, and it is discharged

at the dual-basin economy of Proposition K.1 (where the rule is zero on the low basin) and at no other economy in the development. Read with Theorem D.8, the two statements are two opposite-direction crossings with conditional persistence on each side; they are not a same-equilibrium hysteresis unless both level conditions hold for one equilibrium, which is what Proposition K.1 exhibits. Two further limits, at the threshold itself and in the conservatism of the duration, are in Appendix H.9.

D.10 Reading lock-in

Whether persistent sanctions are permanent is a question about the dynamics, which lock-in answers equilibrium by equilibrium. Lock-in is a property of a given equilibrium: its direction and level conditions are imposed only at basin states, at every value pair in the stated sets, and the level condition only against the state itself (Proposition 5). These requirements can be met: on a class of corner economies, defined by inequalities among primitives, at which the constructed equilibrium sanctions fully at every state, and at an example economy whose continuation values change observable choices. Lock-in's comparative statics run in one direction: with the same transition and initial state, a non-decreasing sanction rule at least as high as another on the high basin yields a state path at least as high at every date.

Three notions of stability appear in these results, and they remain distinct. First, state paths converge. When the equilibrium law of motion is non-decreasing on a basin that it preserves and the sanction rule is continuous there, the start fixes the steady state reached: the least one at or above the start if the sanction there is at least the state, and otherwise the greatest one at or below it. Second, equilibrium uniqueness holds relative to stated restrictions: invariant sets of value functions, tie-breaking rules and non-decreasing sanction rules (Section 3.4). Uniqueness in dynamic games of this kind is open (Doraszelski and Satterthwaite, 2010). On every family the paper constructs, the sender's optimal sanction is unique at each state and feasible compliance level, so the greatest, the least and any other selection coincide there. Third, in a stochastic extension of the model studied elsewhere in this appendix, an invariant distribution of the state exists; its analytic existence argument and its formalized constructions have different verification scopes.

E The two-channel decomposition and the path of the state

This section holds three results that the main text states in words. The forward-looking gap splits exactly into the target's part and the sender's part (Section 3.2); when initial conditions matter, the limiting sanction is positive (Section 3.1); and a sanction rule that does not fall with the state produces no conflict cycle (Section 5.2).

E.1 The path of the state under a sanction rule

Dependence on initial conditions implies a positive limiting sanction.

Proposition E.1 (Limit dependence and the limiting sanction)**.** *Let the transition satisfy zero-sanction descent, $F(x,0) < x$ for every $x \in (0,1]$ (the erosion half of strict adjustment), and let R be a continuous feasible sanction rule. If the natural paths of R from two starts converge to two different limits, then from at least one of them the limiting sanction $R(x_\infty)$ is strictly positive: a limit state x_∞ of a path under R is a fixed point of $x \mapsto F(x, R(x))$, and a fixed point with $R(x_\infty) = 0$ can only be $x_\infty = 0$ (`rulePathF_limit_isFixed`, `exists_pos_limit_sanction_of_limit_dependence`, `Theory/LockIn/LimitDependence.lean`). The*

converse fails: at the affine member of the quadratic–linear family under the lifting condition the non-compliance equilibrium's closed loop has one globally stable steady state with a positive sanction (`placableAffine_unique_attracting_state`).

A non-decreasing rule produces no conflict cycle.

Proposition E.2 (No conflict cycles under a non-decreasing rule)**.** *Let R be a sanction rule that maps $[0,1]$ into itself and is non-decreasing in the state there, and let the state follow $x_{t+1} = F(x_t, R(x_t))$ from $x_0 \in [0,1]$, with sanctions $s_t = R(x_t)$. Then:*

(i) the state path and the sanction path are monotone in t, in the same direction, and both converge;

(ii) if $x_{n+k} = x_n$ for some $n \geq 0$ and some $k \geq 1$, then $x_{n+j} = x_n$ for every $j \geq 0$; in particular a periodic path is constant.

Both conclusions hold in every equilibrium in $\mathcal{E}$ whose sanction rule is non-decreasing in the state, and in the constructed equilibrium of each of the four cases of Theorem 12 at the hypotheses under which it is constructed, that of case 11 *under the increasing-differences condition* (B.4) *or with a non-decreasing response.*

Proof. By (F2), (F3) and the monotonicity of R, the closed loop $G(y) = F(y, R(y))$ is non-decreasing on $[0,1]$ (`ruleStepF_le_of_le`). If $x_1 \geq x_0$, induction gives $x_{t+1} = G(x_t) \geq G(x_{t-1}) = x_t$ for every t, and if $x_1 \leq x_0$ the reverse; the sanctions $R(x_t)$ move in the same direction (`rulePathF_monotone_or_antitone`, `rulePathF_ruleSanctionsF_monotone_or_antitone`). Both sequences are bounded, so both converge, with no continuity of R (`rulePathF_ruleSanctionsF_tendsto`); when R is continuous the limit state is a rest point of G, as in Step 7 of the proof of Theorem 3. For (ii), a monotone path with $x_{n+k} = x_n$ and $k \geq 1$ has $x_{n+1} = x_n$, so x_n is a rest point of G and the path stays there (`rulePathF_const_of_return`, `rulePathF_periodic_const`). The equilibrium form takes R to be the sanction rule of the equilibrium, whose values lie in $[0,1]$ (`equilibrium_no_cycles`). In the four cases the rule is zero in 00 and 01 (`myopicRule_eq_zero`, `senderOnlyRule_eq_zero'`), the myopic sanction $\sigma^{\mathrm{M}}(x,0)$ in 10, non-decreasing by (S7) (`targetOnlyRule_monotoneOn`), and the dynamic best response at full non-compliance in 11, non-decreasing under the increasing-differences condition (`responsePath_monotoneOn_of_margin`); the four statements are `myopic_no_cycles`, `senderOnly_no_cycles`, `targetOnly_no_cycles` and `dynamic_no_cycles` (`dynamicOf_no_cycles` with a non-decreasing response). General in F. □

Along deterministic equilibrium paths in this model, a conflict cycle therefore needs a sanction rule that falls with the state somewhere. No equilibrium in the paper is shown to have one: the dual-basin equilibrium's rule is shown to be non-decreasing only on its two basins, and Proposition B.6 does not cover a falling rule. The cycles of Acemoglu and Wolitzky (2014) rest on misperceived actions and limited memory, which this model does not have.

E.2 The forward-looking gap splits into two components

The dynamic–myopic comparison is described in the text as two forces: the target incumbent's own valuation of the political state its compliance leads to, and the switch from the sender's myopic best response to its dynamic best response. That description is exact, and the ordering condition the argument consumes is weaker than the one stated in Section 3. Both points are recorded here.

Three objectives and two components. Fix a feasible state x and a target continuation value V_T, and write, as functions of compliance $a \in [0,1]$,

$$J_x^{MM}(a) := U_T\big(x, a, \sigma^{\mathrm{M}}(x,a)\big), \tag{E.1}$$

$$J_x^{TM}(a) := U_T\big(x, a, \sigma^{\mathrm{M}}(x,a)\big) + \beta_T\, V_T\big(F(x, \sigma^{\mathrm{M}}(x,a))\big), \tag{E.2}$$

$$J_x^{FF}(a) := U_T\big(x, a, \sigma_{V_S}^{\mathrm{dyn}}(x,a)\big) + \beta_T\, V_T\big(F(x, \sigma_{V_S}^{\mathrm{dyn}}(x,a))\big). \tag{E.3}$$

J^{MM} is the objective the myopic target maximizes and J^{FF} is the integrand of the target Bellman equation (5), so both are objects the equilibrium already carries. The middle object J^{TM}, a target that values continuation but faces the *myopic* sender, is a device. It is not a third equilibrium concept, and no maximizer of it is constructed anywhere below.

The total forward-looking gap is $\Delta_x := J_x^{FF} - J_x^{MM}$. Its two components are the two steps between the three objectives:

$$C_x := J_x^{TM} - J_x^{MM}, \qquad S_x := J_x^{FF} - J_x^{TM}. \tag{E.4}$$

C_x is the *target-continuation* component (what the incumbent's own political-survival motive adds, holding the sender's behavior fixed), and S_x is the *sender-response-switch* component, what changes for the target purely because the sender it faces is forward-looking.

Proposition E.3 (Exact two-component decomposition)**.** *At every feasible x and every $a \in [0,1]$,*

$$\Delta_x(a) = C_x(a) + S_x(a), \tag{E.5}$$

with no residual and no cross term. Moreover $C_x(a) = \beta_T\, V_T\big(F(x, \sigma^{\mathrm{M}}(x,a))\big)$, so the target-continuation component is β_T times the target's continuation value at the successor state, which the separated-primitive route signs directly.

(E.5) is an identity, not an approximation; the content is that (E.1)–(E.3) are the right three objectives, and that the sender response is the only thing that moves between J^{TM} and J^{FF}.

The split earns its place because the two components have different logical status. C_x is signed under three conditions stated separately from the baseline transition primitives: compliance placates, sanctions pass through strictly to the successor state, and the target's continuation value is increasing on the reached successors. Only the first is close to baseline; the transition primitives give weak monotonicity in the sanction rather than strict, and an increasing target value is not a baseline property at all. S_x is not signed even under those. Both sender best responses falling in compliance, the dynamic one lying above the stage one, and the target's payoff rising in the sanction together do *not* determine whether S_x rises or falls. The component reads the target's payoff *difference* between the two responses, which turns on the *gap* between them and not on their level alone, and that gap may widen with compliance. Any route to the compliance ordering must therefore either sign S_x separately or dominate it. The switch from the stage to the dynamic best response of the sender thus has an *ambiguous* effect on the target's compliance incentive under the baseline; that is a scope delimiter, not a claim that the intended ordering typically fails.

Only the net movement is needed. Asking each component to behave separately is stronger than the ordering requires. Because (E.5) is exact, the comparison never sees the components apart: only their sum. The exact condition is that across any rise in compliance from a_L to a_H,

$$S_x(a_H) - S_x(a_L) \;<\; C_x(a_L) - C_x(a_H), \tag{E.6}$$

which permits S_x to *rise* (to work against the ordering), provided the target-continuation component gives up strictly more. Condition (E.6) is *equivalent* to a strictly decreasing Δ_x, so it is an accounting restatement of the gap condition rather than a new assumption: the route was always net, and reading it component by component conceals the trade-off. Quah and Strulovici (2012) is why a component-by-component sign argument is stronger than a condition on the aggregate (single crossing is not preserved by addition except under stringent conditions), but (E.6) is not theirs; it is a corollary of the exact decomposition above.

Two boundaries travel with it, and both are one-directional. The condition is *cardinal*, a strict inequality between differences, and it is therefore *strictly stronger* than the ordinal relation (E.7) below: (E.6) implies (E.7), and the converse fails. Single crossing survives an arbitrary strictly increasing transformation of either objective and a difference condition does not, so the two are ordered rather than incomparable: on $[0,1]$ the pair $f(a) = a^3$, $g(a) = a$ stands in the ordinal relation while $f - g$ rises over $[9/10, 1]$. And the sufficient route in which the switch gap weakly falls while the target gap strictly falls is strictly stronger again than (E.6), since it forbids exactly the adverse switch the net condition tolerates. The two-leg *ordinal* route below is a different sufficient condition again, constraining each leg separately against the intermediate objective rather than the aggregate; we do not claim an implication either way between it and (E.6).

The ordering condition is ordinal. Section 3 states the comparison under a strictly decreasing total gap. That is sufficient but stronger than necessary. Say that f is *strictly single-crossing lower* than g on A when, for all $a_L < a_H$ in A,

$$f(a_L) \leq f(a_H) \quad \Longrightarrow \quad g(a_L) < g(a_H). \tag{E.7}$$

Proposition E.4 (Ordinal form of the compliance ordering)**.** *If J_x^{FF} is strictly single-crossing lower than J_x^{MM} on $[0,1]$, then every maximizer of J_x^{FF} lies weakly below every maximizer of J_x^{MM}. A strictly decreasing Δ_x implies* (E.7)*, so the gap condition is one route to it. Relation* (E.7) *is transitive: if J_x^{FF} is strictly single-crossing lower than J_x^{TM}, and J_x^{TM} than J_x^{MM}, the conclusion follows for the pair (J_x^{FF}, J_x^{MM}).*

No continuity, concavity, compactness, or uniqueness enters, and the conclusion is about the argmax *correspondences* rather than about a chosen selection from either. The transitivity clause is what makes (E.5) usable rather than merely descriptive: the two components can be discharged one at a time against J_x^{TM}, with different tools and at different strengths, without a maximizer of J_x^{TM} ever being exhibited.

The relations here are, one for one, those of Amir and Rietzke (2025): their Definition 1 is the weak form below, their identity-difference order is the cardinal gap condition, their Lemma 2(ii) is the mixed composition, and the auxiliary problem they insert between two objectives is J_x^{TM}; we cite them for the method. Their Theorem 1(i) also supplies a converse on a one-dimensional choice set: the weak relation holds if and only if the argmax sets are strong-set ordered for every ordered pair of feasible sets, so the whole content sits on two-point sets (`sCLowerOn_iff_pairwise_meet_join`); the weak relation is not one sufficient condition among others but what the strong-set conclusion is.

Only one leg needs to be strict. Asking both legs for (E.7) demands the most from the leg that can supply the least, since the sender-response-switch component's sign is not fixed by the primitives at all. Write $f \preceq_{SC} g$ for the two-clause *weak* form: a weak rise in f forces a weak rise in g, and a strict rise forces a strict rise. Then

$$J_x^{FF} \preceq_{SC} J_x^{TM}, \qquad J_x^{TM} \prec_{SC} J_x^{MM}$$

already gives $J_x^{FF} \prec_{SC} J_x^{MM}$, so the conclusion of Proposition E.4 still orders *every* pair of maximizers. The switch component is asked only not to reverse the comparison; the target's own continuation motive carries the strictness. The mirror composition (strict switch, weak target) is available and is not the version we recommend.

Two cautions travel with the weak relation. On its own it orders the argmax *correspondences* in the strong set order, and hence their extremal elements, but not an arbitrary selection from each: two constant objectives satisfy it with every action maximizing both. That is precisely why one leg must remain strict. And the weak relation is not the weakest hypothesis delivering the conclusion: two argmax sets can be ordered while it fails away from the maximizers; its value is that it is selection-robust and does not depend on where the unknown maximizers sit.

The step for the direct effect of the sender's foresight is local. The comparison at a fixed target action, $\sigma^{\mathrm{M}}(x,a) \le \sigma^{\mathrm{dyn}}_{V_S}(x,a)$, does not need V_S monotone at every state. It needs only that the sender's continuation term, read at the current state as a function of the sanction, does not fall over feasible sanctions; global monotonicity of V_S is one way of supplying that. Weaker still, it suffices that the continuation term not prefer the dynamic best response to the myopic best response wherever the former lies below the latter: a condition on the two responses alone. Each of these is a weakening of its predecessor, so nothing that discharged the earlier form is lost.

Derivative and supplementary results

F Existence: constructions, the Schur/Perov machinery, and the long-run law

This appendix collects the technical constructions deferred from Section 3 and Appendix D.1: the alternative existence routes, the Schur-condition matrix computations, the purified (shock) constructions and the invariant-law machinery, and the discharge apparatus for the assumption residual clauses. None of it changes any statement in the main text.

The coupling matrix. With L_{TS}, L_{ST} and $\beta_{S,\mathrm{row}}$ as in Theorem F.1 below,

$$\mathbf{M} = \begin{pmatrix} \beta_T & L_{TS} \\ L_{ST} & \beta_{S,\mathrm{row}} \end{pmatrix},$$

and $\rho(\mathbf{M})$ is its spectral radius. When the four coefficients are non-negative and the two diagonal ones are below one, the boxed Schur inequality of Theorem F.1 is equivalent to $\rho(\mathbf{M}) < 1$ (`specRadius_lt_one_iff_schurDet`); the theorem itself imposes no sign on the coefficients.

The theorem, which Appendix D.2 states in words, is the following.

Theorem F.1 (Existence on invariant sets under the Schur condition)**.** *Fix closed, non-empty sets of value functions $\mathcal{C}_T$ and $\mathcal{C}_S$. Suppose that, for every sender value in $\mathcal{C}_S$, the sender's dynamic argmax is unique at every state and every feasible compliance level $a \in [0,1]$, and that the target argmax is unique at every value pair from the two sets. Together with the maintained continuity of the primitives, these conditions define continuous feasible selectors. Suppose also that the joint Bellman operator maps $\mathcal{C}_T \times \mathcal{C}_S$ into itself. Denote its two rows by $\mathcal{T}_T$ and $\mathcal{T}_S$. Suppose, for all value pairs $z = (\tilde{V}_T, \tilde{V}_S)$ and $z' = (\tilde{V}'_T, \tilde{V}'_S)$ from the two sets, that*

$$\|\mathcal{T}_T(z) - \mathcal{T}_T(z')\|_\infty \le \beta_T \|\tilde{V}_T - \tilde{V}'_T\|_\infty + L_{TS}\|\tilde{V}_S - \tilde{V}'_S\|_\infty, \tag{F.1}$$

$$\|\mathcal{T}_S(z) - \mathcal{T}_S(z')\|_\infty \le L_{ST} \|\tilde{V}_T - \tilde{V}'_T\|_\infty + \beta_{S,\mathrm{row}}\|\tilde{V}_S - \tilde{V}'_S\|_\infty, \tag{F.2}$$

where $\beta_T, \beta_{S,\mathrm{row}} < 1$ *and*

$$\boxed{L_{TS}L_{ST} < (1-\beta_T)(1-\beta_{S,\mathrm{row}})} \tag{F.3}$$

(no sign is imposed on the four coefficients; when they are non-negative, (F.3) *is equivalent to* $\rho(\mathbf{M}) < 1$*). Then the joint Bellman operator has a unique fixed point on the invariant product set in a weighted sup-norm. The associated optimal target policy and greatest sender response form an equilibrium* $E = (V_T, V_S, \alpha) \in \mathcal{E}$ *with* $(V_T, V_S) \in \mathcal{C}_T \times \mathcal{C}_S$ *and* $\alpha = \alpha_{V_T,V_S}$*, the target selector the two sets define. Every equilibrium* $E' = (V_T', V_S', \alpha') \in \mathcal{E}$ *with* $(V_T', V_S') \in \mathcal{C}_T \times \mathcal{C}_S$ *and* $\alpha' = \alpha_{V_T',V_S'}$ *coincides with* E*:* $V_T' = V_T$*,* $V_S' = V_S$ *and* $\alpha' = \alpha$*.*

A negative coefficient is allowed because it can only describe a degenerate invariant set. If $\beta_T < 0$ or $L_{ST} < 0$, the row bound at a fixed sender value forces the target set to be a single point, and symmetrically $L_{TS} < 0$ or $\beta_{S,\mathrm{row}} < 0$ makes the sender set a single point; the bounds then hold again with non-negative coefficients, and the Perov step applies unchanged. Both diagonal bounds are assumed in exchange: with non-negative coefficients $\beta_T < 1$ would follow from the other conditions, but without the signs it does not. The coincidence clause is uniqueness within the stated invariant sets and selector class; it is not global equilibrium uniqueness (`feedbackStackelbergTop_unique_in_carrier_schur`). The sender's argmax need be unique only at compliance levels in $[0, 1]$: the target chooses compliance there, and the target's Bellman operator reads the sender's response only there (`Params.DynamicArgmaxUniqueOnI`). Uniqueness at every real compliance level implies this (`DynamicArgmaxUnique.toOnI`), and the corner economies of Proposition I.5 obtain the weaker form with no bound on the curvature of their transitions (`SymbolicEconomyF.DAUOnI_on_cone`).

F.1 Alternative existence routes

Theorem F.1 uses a model-specific Schur-condition row-Lipschitz route, with the invariant set of state-Lipschitz value functions motivated by Curtat (1996). The literature offers several parallel sufficiency routes: Tarski monotone iteration on a lattice of value pairs (Balbus et al., 2014; Vives, 1990), Schauder on a sup-norm-compact Lipschitz ball (Raad and Woźny, 2019), measurable selection on a decomposable kernel (He and Sun, 2017), and the fixed-point route of the risk-sensitive overlapping-generations model (Jaśkiewicz and Nowak, 2014), which is a Glicksberg argument rather than a row-Lipschitz one. None is adopted as primary here: the lattice route needs an order-preservation structure the coupled operator does not have (Remark F.2); the Schauder and measurable-selection routes return existence without the greatest-selection convention or the closed-form constants the comparative statics use; and the risk-sensitive extension is not needed for time-additive preferences. The Schur-condition route is described in detail below. The general taxonomy of stationary Stackelberg-equilibrium concepts for discounted stochastic games is in Bucarey López et al. (2023), who formalize the stationary strong Stackelberg equilibrium (SSSE) and fixed-point equilibrium (FPE) notions, prove their existence and coincidence for classes of games including myopic follower strategy and team games, and provide counterexamples where existence fails in the general case. The restricted assumption set {Assumption 1,Assumption 2,Assumption 3} describes structural inputs; the Schur-condition construction of Theorem F.1 also requires the value-set, selector, invariance and row-bound hypotheses stated there, and the construction of Proposition B.1 discharges its own conditions from its fields. Neither makes the three assumptions alone a sufficient condition for existence.

Remark F.2 (Why the mixed-monotone order fails, and where). The lattice route is usually attempted twice: first under the pointwise product order on value pairs, and then, when that fails, under the *twisted* order that reverses one coordinate,

$$(V_T, V_S) \preceq_{\mathrm{tw}} (\tilde{V}_T, \tilde{V}_S) \quad \Longleftrightarrow \quad V_T \le \tilde{V}_T \ \text{ and } \ \tilde{V}_S \le V_S \ \text{ pointwise.}$$

The second attempt is the substantive one, because our pointwise refutations vary the target continuation with the sender continuation held fixed and so leave the twisted order untouched. Isotonicity of the joint Bellman map under $\preceq_{\mathrm{tw}}$ requires four coordinate signs. The target row's own-value sign holds, from $\beta_T \geq 0$ alone. Its cross sign does not.

The mechanism of the failure is worth stating, because it is not an artifact. The sender's continuation value enters the target's Bellman integrand only through the sender's best response. Raising that continuation makes the sender sanction more heavily, which costs the target current payoff but *raises* its continuation value, because a heavier sanction raises next period's entrenchment and the target's value is increasing in entrenchment. Writing L_T for the target's stage loss per unit of additional sanction, g_T for the lower slope of its value in the state, and d_S for the lower bound on the transition's sanction slope, the second effect dominates whenever

$$L_T \;<\; \beta_T \, g_T \, d_S,$$

which is the strict form of the entrenchment condition $c_T \leq \beta_T \, g_T \, d_{\mathrm{lo}}$ maintained throughout Section 3. That condition is the one under which the target tolerates sanctions rather than complying in the persistence certificate.

The lattice route therefore fails on the economies this paper is about. The refutation is a configuration: two ordered sender continuation values between which the sender's optimal response strictly shifts, a target maximizer, the difference-quotient inequalities and the strict entrenchment condition above (`not_antitone_T_T_joint_in_sender_value`). It defeats antitonicity of the target row on any domain containing that configuration, which is what the unrestricted twisted-order route needs. Three scope points. It does not show that every smaller value set fails: a value set on which the entrenchment condition is reversed, or on which the sender's response does not move between the two values (a constant shift of the continuation adds an action-independent term and moves nothing), is not addressed. It closes a proof *route*; existence is delivered by the contraction argument of Theorem F.1 and is untouched. And the refutation is empty when $\beta_S = 0$, where the sender is myopic, its response does not depend on its own continuation, and the target operator is constant in V_S, so the obstruction is a genuine consequence of two-sided forward-looking behavior rather than of the sequential timing.

F.2 The construction behind the theorem, in brief

The proof of Theorem F.1 is a joint Banach iteration on the value pair in an invariant set with an asymmetric weighted sup-norm, whose contraction modulus is governed by the Schur condition (F.3): the set is complete, two invariance conditions keep the operator inside it, the row estimates have the entries of $\mathbf{M}$ as coefficients, and the Perov 2×2 fixed point (Toda, 2021; the weighted-max-norm reduction of Bertsekas and Tsitsiklis, 1996, §2) yields the value pair from which the optimal target policy and the greatest sender response are read off. Three levels of the same route are kept apart in the formalization (the conditions at a supplied equilibrium, at the constructed fixed point, and over the whole invariant set), and the strongest is the checkable *ex ante* certificate (`Sanctions/Theory/Equilibrium/PerovExistence.lean`, `Sanctions/Theory/Certificates/TwoSidedRuleComparison.lean`; Appendix B). The fixed point is locally isolated inside its invariant set: a shape-conforming equilibrium within an enlarged norm ball at which the Schur condition still holds coincides with it. This is a local uniqueness statement inside a specified invariant set, with no genericity, manifold, nondegeneracy or finiteness claim about the global equilibrium set; the smallest invariant radii of the balls are $\|U_T\|_\infty/(1-\beta_T)$ and $\|U_S\|_\infty/(1-\beta_S)$. The anatomy of the iteration, the isolation statement and the coupled low-δ diagnostics are not reproduced in this appendix.

F.3 Contraction restoration under the shock

Machine-checked contraction restoration under the shock. The purified existence above is, in the cited references, a *topological* fixed point (a continuous self-map on a compact value class, closed by Fan–Glicksberg). A sharper statement of the mechanism on the sender side follows (the construction is mapped in Figure 8). Model the atomless shock as a smoothing operator on continuation values, $(\mathrm{smear}\, V)(y) = \mathbb{E}_\eta[V(y+\eta)]$ with η drawn from the shock density. Two properties of smear are exactly what an atomless smooth density delivers: it is sup-norm *nonexpansive* (averaging against a probability density, $\int \kappa = 1$), and it maps any bounded V to a state-*Lipschitz* function (mollifier smoothing); only the first is carried by the Lean structure `ShockedParams`, whose single proof field is nonexpansiveness. The first property alone shows, with no new regularity, that the shocked sender Bellman operator $V \mapsto \mathrm{bellmanS}(\alpha, \mathrm{smear}\, V)$ *remains a β_S-contraction*: the deterministic Hölder-$\frac{1}{2}$ obstruction is bypassed because the shock enters only through the nonexpansive continuation, so the contraction modulus is unchanged at $\beta_S < 1$, and the sender value is *constructed* by the Banach engine rather than postulated. This is a Banach, rather than Fan–Glicksberg, existence route under the shock. Nor is smear left abstract: the He–Sun shock is carried as a concrete *atomless probability kernel* κ on the transition (the He–Sun idiosyncratic component), with $(\mathrm{smear}\, V)(y) = \int V(\mathrm{clip}(y+\eta))\, \kappa(\mathrm{d}\eta)$, and its nonexpansiveness is *derived* from $\int \kappa = 1$ rather than assumed, so the shocked corner equilibrium holds from a genuine kernel primitive. The shocked equilibrium is an equilibrium of the *shocked* game (its Bellman equations carry smear), a sibling object to the deterministic Theorem D.1, and it too now holds: the shocked equilibrium structure, both shocked Bellman fixed-point equations (sender and target), and a non-vacuous shocked *corner* instance are established, with the two value functions *constructed* as Banach fixed points of the shocked sender and target operators. The *interior* (non-corner, state-varying leader) case is also now established: the slaved shocked operator $\Phi(V_S) = \mathrm{bellmanS_shock}(\mathrm{lead}(V_S), V_S)$ has modulus $\leq \beta_S + L_{U_S,a}\, L_\alpha$, where the continuation term contributes β_S (nonexpansive smear) and the policy term contributes $L_{U_S,a}\, L_\alpha$ (U_S Lipschitz in the target action, leader policy L_α-Lipschitz in V_S); under the condition $\beta_S + L_{U_S,a}\, L_\alpha < 1$ this is a Banach contraction, and its fixed point yields an interior shocked equilibrium with a state-varying leader policy. The contraction-specific quantity is L_α: deterministically the leader argmax is only Hölder-$\frac{1}{2}$ in V_S, but the atomless shock's mollifier-Lipschitz smoothing upgrades it to Lipschitz (the smear-regularity property), which the numerics confirm in the relevant range of shocks. In the canonical theorem it is derived from composite strong concavity and the Stackelberg-coupling bound; joint follower continuity supplies the objective continuity and the canonical leader selector. The full-width case $h = 1$ now discharges that follower-continuity input: the moving average is exactly affine on the physical state interval for every bounded continuous continuation, which yields a unique sender argmax and a jointly continuous greatest selection on $[0,1] \times \mathbb{R}$. At the P_3 economy, the same affine identity permits a stronger exact construction that bypasses the reusable leader-regularity theorem altogether. The two smeared continuation values are summarized by their endpoint spreads; eliminating the target spread gives a quartic equation for the sender spread. Lean isolates the economically relevant root in an exact rational interval, constructs the induced piecewise-affine leader policy and clipped greatest sender response, builds both bounded-continuous value functions, and verifies global target optimality and both Bellman fixed-point equations through every clipping region. The resulting full-width shocked equilibrium has no strong-concavity, cross-value/action coupling, invariant-set, or policy hypothesis. The *mechanism* of that upgrade is itself now established: a general lemma shows that for a c-strongly-concave objective whose two-maximizer difference is action-Lipschitz with constant L, the maximizers satisfy $c\,|a_f - a_g| \leq 2L$: a Lipschitz (not Hölder-$\frac{1}{2}$) argmax bound. The deterministic obstruction bounds the same difference by its *sup* ($\to \surd$, Hölder-$\frac{1}{2}$); the smear-

smoothed continuation makes it *action-Lipschitz* with constant $\propto \|V - V'\|$ (the smear_lipschitz property), turning the same inequality into the Lipschitz rate. That smear_lipschitz estimate is itself now discharged from a concrete atomless (uniform) kernel: the moving-average smoothing $(\operatorname{smear} U)(y) = \frac{1}{2h}\int_{y-h}^{y+h} U(\operatorname{clip} z)\,dz$ is state-Lipschitz with constant $\|U\|/h$, by the elementary window-shift bound $|\int_{y-h}^{y+h} g - \int_{y'-h}^{y'+h} g| \le 2\|g\|_\infty |y - y'|$. The argmax-core reduces lead_lip to a single action-direction estimate on the leader objective *difference*. The shock's σ- and V_T^{dyn}-Lipschitz bounds control the *magnitude* of that estimate, but its variation *in the leader's own action* is not supplied by the shock: because the sender response $\sigma(x, a)$ enters the leader objective and depends on a, this action-Lipschitz estimate (the Stackelberg coupling, isolated as the hypothesis h_obj_diff) is *consumed*, not derived. A reference sweep confirms this is not an artifact of our particular kernel: smoothing the continuation regularizes the *state* direction only (a uniform-Lipschitz-in-action integrand need not be jointly Lipschitz; Hinderer, 2005). Concretely, the moving-average smear is only C^1 (its second derivative would require a C^1 continuation, which value functions are not), so the action-direction cross-derivative an implicit-function / envelope derivation (the Lipschitz-policy route of Montrucchio, 1987) would need does not exist. The feedback-Stackelberg constructions we hold do not derive the analogous leader-side regularity from primitives either; where they need it, it is imposed. So lead_lip is the argmax-core applied to one consumed coupling estimate. The canonical interior theorem pins the leader objective to the genuine target payoff joint_T_top_integrand. Joint continuity of the shocked sender selector and strong concavity of this real composite objective at any positive modulus c_T then produce a unique canonical leader maximizer; Berge continuity, feasibility, and pointwise leader optimality are proved rather than separately assumed. For a general half-width, the remaining analytic surface is therefore exactly those two conditions plus the coupling estimate and the direct contraction condition $\beta_S + 2L_{\mathrm{lead}}/c_T < 1$, not an externally supplied leader map or a fixed numerical curvature. In the state-local full-width specialization, follower continuity is proved and the remaining surface contracts to composite leader strong concavity, the coupling estimate, and this same symbolic condition. The kernel side is now also inhabited by a *concrete* uniform-kernel parameter, and the smear's state-Lipschitz constant is the closed form $\|V\|/h$ rather than a named primitive. This does not discharge, or prove joint satisfiability of, the general three-condition analytic package. The conditional theorem is wired to the mechanism-active P_3 economy: once that package is supplied, it builds a purified equilibrium as a Banach fixed point with no Schur condition. Its abstract state-local $h = 1$ sibling discharges follower continuity but still consumes composite strong concavity, the coupling estimate, and $\beta_S + 2L_{\mathrm{lead}}/c_T < 1$. The older modulus-2 and $L_{\mathrm{lead}} < 1/2$ statement is only the arithmetic specialization at the printed example parameter vector. These conditional templates are not headline existence results.

The exact $h = 1$ construction is logically separate and stronger at that parameter vector. It solves the equilibrium rather than proving the abstract leader hypotheses over an operator domain. The selected sender endpoint spread is an exact root of a quartic in a rational bracket; the target spread is then recovered algebraically. These spreads determine a feasible continuous piecewise-affine leader policy, an exact clipped greatest sender response, and both bounded-continuous values. Lean verifies the sender fixed point, the target fixed point, and global target optimality through the lower-clipped, interior, and upper-clipped sender regions. Hence an unconditional, genuinely state-varying full-width P_3 equilibrium is exhibited.

The dynamic-over-myopic theorem still requires its separate basin inputs. At full width, monotonicity of the sanction rule is available twice. The reusable canonical-selector result follows from the change of variable $b = a - (27/10)x$, strict decreasing differences of the target objective, and interval dominance of the sender objective. At the explicit equilibrium there is also a direct proof: the leader policy is increasing, the sender vertex remains increasing after the leader response is

substituted, and clipping preserves monotonicity. Thus the equilibrium's own sanction rule, under the greatest selection, is monotone on all of $[0,1]$.

The separate dynamic floor is nevertheless impossible. The full-width sender selector is the clip of $\frac{9}{8}x - \frac{1}{12} - \frac{5}{12}\alpha(x) + (V_S(1) - V_S(0))/24$, and the top-state floor holds if and only if $V_S(1) - V_S(0) \geq 10\alpha(1) - 1$. The sender Bellman equation bounds the same spread above by $\frac{5}{8}(\alpha(1) - \alpha(0)) + \frac{35}{16}$, so a putative floor forces $\alpha(1) \leq 17/50$. The target Bellman equation gives $V_T(1) - V_T(0) \leq 60/11$. Combining these restrictions with the exact sender vertex makes a deviation at $x = 1$ to $a' = 1/2$ strictly profitable, contradicting target optimality. Hence no full-width shocked equilibrium has the top-state floor, and no non-empty basin ending at 1 satisfies the full-width floor component. In particular, the explicit equilibrium has a monotone sanction rule and a top-state sanction strictly below one. It therefore cannot be assembled with the floor into full pathwise ordering at P_3; the myopic basin inputs also remain open. The earlier test continuation $V_S(x) = -100x$ remains a useful disproof-first check, while the equilibrium argument closes the stronger fixed-point statement.

For a general half-width, the three-condition analytic package remains the reusable theorem's honest assumption surface. The stage curvature $\partial^2 U_T/\partial a^2 = -2$ does not by itself establish strong concavity of the composite leader objective, because the target action moves the clipped sender response and hence the continuation. An exact Lean counterexample at $x = 5/12$ shows that the static composite already violates 2-strong concavity on the rational actions 9/10, 37/40, and 19/20. Likewise, smoothing regularizes the state direction, not the leader-action direction controlled by the Stackelberg-coupling estimate. These facts explain why the abstract theorem retains those inputs; they are not hypotheses of the explicit full-width equilibrium. The deterministic model is the $\kappa \to \delta_0$ limit.

F.4 The derived dividing line, in general form

Proposition D.7 is stated for the equilibrium closed loop on the unit state space. The content is an order-topological fact about a continuous self-map, and it is proved at that level; the equilibrium statement substitutes $G = G_E$ and nothing else.

Proposition F.3 (Derived crossings of a continuous self-map)**.** *Let G be continuous on $[a,b]$ with $G(a) = a$ and $G(b) = b$, and suppose $a \leq G(x) \leq b$ for every $x \in [a,b]$. Let $a < x_L \leq x_H < b$ and suppose*

$$G(x) < x \quad \text{for all } x \in (a, x_L], \qquad x < G(x) \quad \text{for all } x \in [x_H, b).$$

No assumption is made about the sign of $G(x) - x$ on (x_L, x_H), and G is not assumed monotone anywhere. Put

$$t = \inf\{\, x \in [x_L, x_H] : x \leq G(x) \,\}, \qquad t' = \sup\{\, x \in [x_L, x_H] : G(x) \leq x \,\}.$$

Then $x_L < t \leq t' < x_H$, both are fixed points of G, and $G(x) < x$ for every $x \in [x_L, t)$ while $x < G(x)$ for every $x \in (t', x_H]$. Every orbit of G started in $[a,t)$ converges to a; every orbit started in $(t', b]$ converges to b; and $t = t'$ if and only if G has exactly one fixed point in $[x_L, x_H]$.

The construction is the whole proof. Continuity makes $\{x \in [x_L, x_H] : x \leq G(x)\}$ closed, so its infimum is attained and lies in it, giving $t \leq G(t)$; and every state below an infimum is outside the set, which *is* the statement $G(x) < x$ for $x < t$, whence $G(t) \leq t$ by continuity from the left. Because every fixed point of the window lies in the set, t is the least one. The mirror argument gives t', and $t \leq t'$ because t is itself a fixed point and therefore a member of the second set.

Two features are load-bearing rather than incidental, and both are settled by counterexample rather than left to the reader's confidence.

Continuity cannot be weakened. A map jumping from 0 to 1 at the middle of a window satisfies both endpoint conditions and has no fixed point in the window at all. Everything the proposition delivers is carried by continuity.

The interval $[t, t']$ can be non-degenerate. Take the window $[-1, 2]$ itself and the map $x \mapsto \min\{x, 2x\} + \max\{0, x - 1\}$, which is continuous there, falls strictly at -1, rises strictly at 2, and fixes every point of $[0, 1]$. Its two derived crossings over that window are 0 and 1. So the inequality $t \le t'$ admits no upgrade to equality, and the region between the crossings is a genuine gap in the conclusion rather than an artifact of how it is proved.

The orbit statements do not use monotonicity of G. Once G lies strictly below the diagonal throughout an interval, each iterate steps downward because of the sign, not because G preserves order; the orbit is then decreasing and bounded, its limit is a fixed point by continuity, and the strict sign forces that limit to the endpoint. This is what allows the proposition to be applied across a region where the sanction rule is known to be continuous but is not known to be monotone, which is the situation between two certified basins.

F.5 General closed-loop invariant laws and the affine specialization

Let $g : [0, 1] \to \mathbb{R}$ be a continuous closed-loop drift and let $T_{g,h}\mu$ be the law of

$$x' = \mathrm{clip}\big(g(x) + \eta\big), \qquad x \sim \mu, \quad \eta \sim \mathrm{Unif}[-h, h],$$

with $h > 0$ and independent noise. The formal result is stated at this general drift level. Continuity of g is enough for existence of an invariant probability law. If, in addition, there is an $L \in (0, 1)$ such that

$$|g(x) - g(y)| \le L|x - y| \quad \text{for all } x, y \in [0, 1],$$

then

$$d(T_{g,h}\mu, T_{g,h}\nu) \le L\, d(\mu, \nu), \qquad d(T^n_{g,h}\mu, \lambda) \le L^n d(\mu, \lambda),$$

and λ is the unique invariant probability law. This contraction condition is not an additional maintained assumption of the economic model; it identifies the conditional uniqueness-and-attraction region of the already general transition framework.

The proof does not use the Hopenhayn and Prescott (1992) stochastic-order argument, which needs an order on measures the current libraries lack. Instead, it constructs the dual Kantorovich–Rubinstein metric (Kantorovich and Rubinstein, 1958), in the duality form of Dudley (2002, §11.8), $d(\mu, \nu) = \sup_{f\ 1\text{-Lipschitz}}(\int f\, d\mu - \int f\, d\nu)$, and proves that the closed-loop adjoint maps 1-Lipschitz tests to L-Lipschitz tests. Uniqueness is genuine measure equality, not merely $d = 0$: Lipschitz ramps decreasing to interval indicators show that equal integrals against every Lipschitz test imply equality of the two probability measures.

The transfer operator and its adjoint identity are constructed directly. $T_{g,h}\mu$ is the pushforward of the product of μ and the box-noise law under the closed-loop-plus-noise map, and

$$\int f\, d(T_{g,h}\mu) = \int \big[\mathrm{movAvgReal}\ h\, f\big(g(x)\big)\big]\, d\mu(x)$$

follows from the change-of-variables formula and the Bochner Fubini theorem. Existence is obtained by the Krylov–Bogolyubov / Cesàro-average argument (Krylov and Bogolyubov, 1937; textbook form Stokey et al., 1989, Thm. 12.10). On the compact state space the Cesàro averages have a weak cluster point; the Feller adjoint identity makes that point invariant against every bounded continuous test, and the determining class upgrades this to measure-level fixed-point equality.

The affine drift is a specialization, not the primitive theorem. For $g_\delta(x) = (1-\delta)x + \delta$, the drift modulus is $L = 1-\delta$, the general operator is definitionally the affine box transfer, and the preceding bounds become

$$d(T_{\delta,h}\mu, T_{\delta,h}\nu) \le (1-\delta)d(\mu,\nu), \qquad d(T^n_{\delta,h}\mu, \lambda) \le (1-\delta)^n d(\mu,\lambda).$$

Equivalently, its dual pullback $(Sf)(x) = \text{movAvgReal}\ h\, f((1-\delta)x+\delta)$ is $(1-\delta)$-Lipschitz whenever f is 1-Lipschitz. Thus the chain with the familiar affine drift inherits the general theorem's unique, globally attracting invariant distribution.

The same construction covers the *endogenous* general-F chain by taking $g(x) = F(x, \sigma_{\text{real}}(x))$ whenever this equilibrium-induced drift is continuous. Existence then holds with no slope restriction. What does not follow automatically is uniqueness: that conclusion requires the displayed contraction bound. At the affine example economies with a positive stage slope the closed-loop factor is exactly $(1-\delta) + \delta \cdot \frac{9}{8} = 1 + \delta/8 > 1$: the slope $\frac{9}{8}$ of the sanction rule exceeds one precisely because the state–sanction cross-partial dominates the sender's concavity, the same $\frac{27}{20}$-vs-$\frac{6}{5}$ frontier that closes the distributional-equilibrium route. The contraction route is therefore provably unavailable there, while existence remains. Economically this is the appropriate boundary: an equilibrium chain with two invariant basins need not have a unique globally attracting law. What is established is only that this sufficient route to such a law is unavailable. Neither multiplicity of stationary laws nor metastability is claimed, and a blocked sufficient condition is not evidence for either.

F.6 Construction of the shocked (purified) existence (figure)

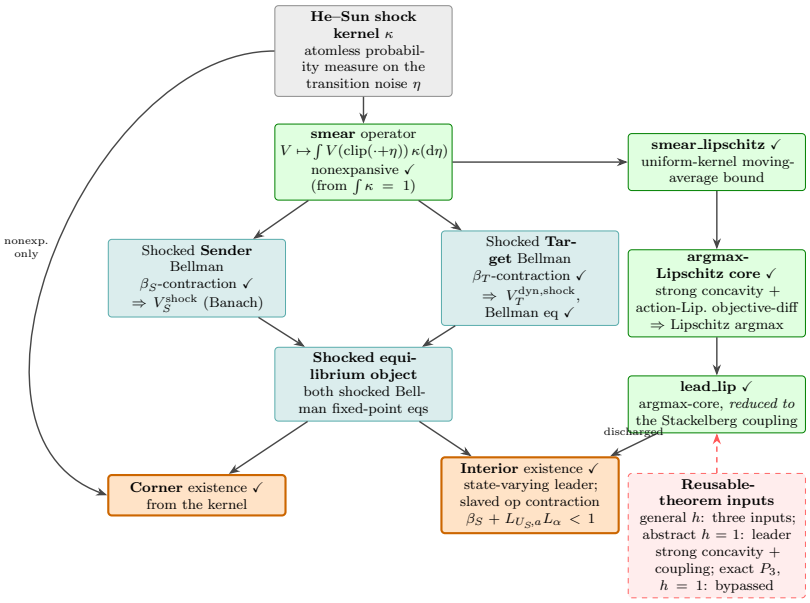


Figure 8: **Construction of the shocked (purified) existence.** Solid/colored boxes are established results (✓); the dashed red box records the reusable general interior theorem's remaining analytic hypotheses. From the He–Sun atomless kernel κ (grey) the smoothing operator smear is derived (green), with its nonexpansiveness proved from $\int \kappa = 1$. That nonexpansiveness alone makes the shocked sender and target Bellman operators β_S- and β_T-contractions (teal), whose Banach fixed points are the shocked value functions and satisfy the two shocked Bellman equations; these assemble the shocked-equilibrium object. Two existence theorems (orange) follow: the *corner* equilibrium directly from the kernel, and the *interior* (state-varying-leader) equilibrium from the slaved-operator contraction under the condition $\beta_S + L_{U_S,a} L_\alpha < 1$. The interior case's input lead_lip (leader argmax Lipschitz in V_S) is the argmax-Lipschitz core applied to the leader objective *difference*; the σ-Lipschitz and V_T^{dyn}-Lipschitz bounds (from smear_lipschitz, discharged from the uniform kernel by the moving-average window-shift estimate) bound the *magnitude* of that difference, but its variation in the leader's own action is the consumed *Stackelberg-coupling* estimate (part of the dashed-red package), so lead_lip holds given that estimate and composite strong concavity. Joint follower continuity and strong concavity also derive the canonical leader selector, including its feasibility, continuity, and pointwise optimality; at $h = 1$, exact affine smoothing supplies the former condition for every bounded-continuous continuation. Separately, the exact P_3, $h = 1$ construction solves the two-spread Bellman system and verifies the equilibrium directly, bypassing the dashed-red inputs without claiming them as general lemmas. The coupling is genuine: because the sender's response $\sigma(x, a)$ enters the leader objective and depends on the leader's own action a, the action-direction estimate is not supplied by the shock alone, and a reference sweep confirms it is not derivable from a smoother kernel (kernel smoothing regularizes the state direction only; the feedback-Stackelberg literature also assumes the analogous leader-side regularity). It is isolated as one estimate, exactly as the deterministic development isolates strong concavity. The deterministic model is the $\kappa \to \delta_0$ limit, where smear = id and the interior route is blocked.

G Assumption discharge routes

G.1 Stage complementarity and the derived tipping thresholds

No result on the class under the lifting condition of Section 5.1 uses it; Proposition G.4 below is what consumes it.

As the political state hardens, both players' marginal payoffs from sanctions can rise and eventually change sign. The following complementarity condition turns this change into derived tipping thresholds.

Assumption 4 (Stage complementarity in (s, x))**.** The stage payoffs are strictly supermodular in the sanction and the political state, uniformly in the compliance action:

$$\frac{\partial^2 U_S}{\partial s\,\partial x}(x,a,s) > 0 \qquad \text{and} \qquad \frac{\partial^2 U_T}{\partial s\,\partial x}(x,a,s) > 0 \qquad \text{for all } (x,a,s) \in [0,1]^3. \tag{G.1}$$

Economically, a harder state raises the sender's marginal value of sanctioning as sanction-supporting constituencies strengthen, and it raises the target incumbent's marginal payoff from being sanctioned through rally or habituation. The proofs use the corresponding increasing-differences condition in derivative-free form (Topkis, 1998); (G.1) is a smooth sufficient condition. It does not restrict the curvature of the target payoff in sanctions or the sign of the target's direct payoff from the political state. Nor does it, with the other sign conditions, deliver persistence on its own: sign conditions order slopes and bound no magnitude, and there is an economy satisfying every one of them in which every equilibrium sanctions at zero at every date (Appendix B); the magnitude condition that persistence needs is the sub-pivot slope condition (P5) of Theorem D.2.

Flip sets and derived pivots. Fix a sanction-floor parameter $\bar{s} \in (0,1]$ and a positive stage slope level m_S. On the general-F quantitative route, the requirement that the stage gain outweigh the continuation change is

$$\beta_S K_S \big|F(x,s_2) - F(x,s_1)\big| < m_S(s_2 - s_1)$$

for the ordered pairs used in the basin proof. Under (J.2), this reduces exactly to $\beta_S K_S \delta < m_S$. Denote the *target flip set* by $\mathcal{F}_T(\bar{s})$ and the *sender floor flip set at slope level* m by $\mathcal{F}_S(\bar{s}; m)$:

$$\mathcal{F}_T(\bar{s}) := \big\{x \in [0,1] : U_T(x,a,\cdot) \text{ is non-decreasing on } [\bar{s},1] \text{ for every } a \in [0,1]\big\}, \tag{G.2}$$

$$\mathcal{F}_S(\bar{s}; m) := \big\{x \in [0,1] : U_S(x,a,s_2) - U_S(x,a,s_1) \geq m(s_2 - s_1)\ \forall\, 0 \leq s_1 \leq s_2 \leq \bar{s},\ \forall a \in [0,1]\big\}. \tag{G.3}$$

Both sets quantify uniformly over compliance, so their definitions do not depend on the equilibrium policy. The boundary sign conditions are

$$1 \in \mathcal{F}_T(\bar{s}) \cap \mathcal{F}_S(\bar{s}; m_S), \qquad 0 \notin \mathcal{F}_T(\bar{s}) \cup \mathcal{F}_S(\bar{s}; m_S) : \tag{G.4}$$

the flip has happened at the top state and not at the bottom.

Proposition G.1 (Derived tipping thresholds)**.** *Under Assumption 4 and* (G.4)*, the pivots*

$$\bar{x}_T := \inf \mathcal{F}_T(\bar{s}), \qquad \bar{x}_S := \inf \mathcal{F}_S(\bar{s}; m_S) \tag{G.5}$$

exist, are interior ($0 < \bar{x}_T, \bar{x}_S \leq 1$), satisfy their own flip conditions, and the flip regions are exactly the upper intervals $\mathcal{F}_T(\bar{s}) = [\bar{x}_T, 1]$, $\mathcal{F}_S(\bar{s}; m_S) = [\bar{x}_S, 1]$.

The basin is therefore derived rather than imposed: its component thresholds are the infima of upper flip sets.

Proposition G.4 and Appendix G connect the derived basin to Theorems 4, D.2 and D.8. The static and dynamic sender thresholds, the per-action thresholds under the substitution between compliance and the sanction, an example economy with interior pivots, and the pivots' comparative statics are in Appendix H.7.

G.2 Interval dominance and the derived-basin composition

Remark G.2 (Interval dominance: the exact condition consumed). A1 is the exact ordinal field consumed by the persistence certificate: the interval dominance order of Quah and Strulovici (2009) for the policy-substituted family $\{\widetilde{J}_{V_T,V_S}(\cdot,x) : x \in \mathcal{B}\}$. It delivers a greatest sender argmax that is non-decreasing in the state. Single crossing in (s,x) implies interval dominance, so every single-crossing discharger remains sufficient. When the continuation is non-monotone, interval dominance can instead be obtained from single-peakedness in the sanction and an escalation-consistency condition. The latter is automatic wherever single crossing holds, but is a genuine primitive for general value sets when the continuation slope is unrestricted. An example economy discharges single-peakedness and escalation consistency even though single crossing fails, establishing that the weaker route is non-vacuous.

Proposition G.3 (General-F route through the transition-adjusted slope condition)**.** *Fix a Bellman-invariant set of value functions and pivots $\bar{x}_T, \bar{x}_S$ with $\bar{x} = \max(\bar{x}_T, \bar{x}_S) < 1$. Suppose every sender value in the set is K_S-Lipschitz and non-decreasing in the inherited state, the target combined slope condition holds, and the canonical myopic ceiling holds. For every value pair in the set, let α denote its induced target policy and assume that, for every $x \in [\bar{x}_S, 1]$ and $0 \le s_1 < s_2 \le 1$,*

$$U_S(x,\alpha(x),s_2) - U_S(x,\alpha(x),s_1) > \beta_S K_S \big|F(x,s_2) - F(x,s_1)\big|. \tag{GF-Chord}$$

Together with the row-contraction conditions for existence, these hypotheses construct the local general-F invariant-set certificate and hence deliver persistence, the dynamic-over-myopic comparison, and the general minimum-drift shock result.

The exact-displacement condition does two jobs. On the low sanction range its non-negative right-hand side implies strict stage-payoff monotonicity, which pins the level of the dynamic argmax. On the basin, its K_S-Lipschitz continuation bound makes the whole dynamic sender objective strictly increasing, hence supplies the ordinal interval-dominance field. No modulus of F in its sanction coordinate is introduced. The separate hypothesis that the sender's values in the invariant set are monotone is still load-bearing for the general-F headline certificate's dynamic-over-myopic comparison; (GF-Chord) does not silently remove it. If (F6-Q) and the composite stage-state Lipschitz bound hold, then K_S in (GF-Chord) can be set to the closed form (H.6).

The same exact-displacement argument survives the atomless moving-average shock. The box smear preserves the base value's K_S-Lipschitz constant, so the shocked sender objective is strictly increasing when the stage gain dominates $\beta_S K_S \operatorname{dist}(\operatorname{clip} F(x,s_1), \operatorname{clip} F(x,s_2))$. This yields shocked single crossing and monotonicity of the greatest shocked sender best response without an affine transition or a separate sanction-coordinate modulus for F. The exact-displacement condition is therefore the paper-facing general-F shocked route.

Proposition G.4 (Derived-basin route to Theorems 4, D.2 and D.8)**.** *Let the transition follow the general law* (1) *and rise under full sanctions below the derived basin, $x < F(x,1)$ for $x \in [0,\bar{x}]$ with $\bar{x} = \max(\bar{x}_T, \bar{x}_S)$ (so $\bar{x} < 1$). Suppose the sender half of Assumption 4 on the floor range*

$[0, \bar{s}]$ *(its derivative-free form suffices), the sender's top boundary sign* $1 \in \mathcal{F}_S(\bar{s}; m_S)$ *in* (G.4) *(the other signs are needed only for the pivot interiority of Proposition G.1, not here), Assumption 1 on the derived basin, row-sum contraction of the joint Bellman operator on a pair of invariant sets of value functions,* $\beta_T + L_{TS} < 1$ *and* $\beta_S + L_{ST} < 1$ *(stronger than the Schur condition of Section 3), and also:*

(i) (above-floor diagonal) *at every state* $x \geq \bar{x}_S$ *and every value pair in the invariant sets, the sender stage slope at the induced target policy clears the margin on the above-floor range:* $U_S(x, \alpha(x), s_2) - U_S(x, \alpha(x), s_1) \geq m_S(s_2 - s_1)$ *for* $\bar{s} \leq s_1 \leq s_2 \leq \max(x, \bar{x})$*;*

(ii) (monotone sender values) *every sender value* $V_S \in \mathcal{C}_S$ *is non-decreasing in the state;*

(iii) (myopic ceiling) *at every basin state the myopic equilibrium sanction is at most* $\max(x, \bar{x})$*.*

Then conclusions (i)–(iii) of Theorem D.2, the statewise ordering of the myopic sanction below the forward-looking one, and the shock entry of Theorem D.8 hold with the basin $[\max(\bar{x}_T, \bar{x}_S), 1]$ *given by the derived pivots of Proposition G.1. No target preference or sign condition is used (Assumption 2, the target half of Assumption 4, the target's top sign, monotone target values in the invariant set): the target enters only through its operator's row bound and its policy in the invariant sets, and its flip set only as the location* $\bar{x}_T$ *of the basin's lower end.*

The three goals read neither the target half of Assumption 4, nor the target's top sign, nor monotone target values in the invariant set, nor Assumption 3. With the target half, the target's top sign and monotone target values in the invariant set, the target combined slope condition of Assumption 2 is still a derived result: the stage slope is non-negative above the target pivot and the continuation slope is non-negative by monotonicity of those values; it is what conclusion (vi) of Theorem D.2 would need. The residual clauses, the general-F route of Proposition G.3, the discharger for a positive stage slope, and the route's scope limits are detailed below. The route is inhabited by an example economy with pivots $\bar{x}_T = 2/5$ and $\bar{x}_S = 1/2$, and the example is a family over the transition: its conclusions hold at every transition satisfying (F1)–(F4) whose full-sanction section is $F(y, 1) = y/2 + 1/2$ and whose lower bound on the sanction slope is at least $11/36$ (`sfFCert_three_goals_generalF`). The affine law at rate $\frac{1}{2}$ is one member (`sfCert_three_goals_generalF`); the quadratic-bump transition, with lower bound $47/100$ on the sanction slope and affine at no rate, is another (`sfqbCert_three_goals_generalF`, `sfqbParams_F_ne_affineF`). The corner section is an equality on the line $s = 1$, so at every member the realized full-sanction path is the affine one. These are existence certificates: they show that the route does not need the affine law, and they say nothing about transitions outside the family. The example's kinked sender payoff is essential because the diagonal-residual and myopic-ceiling clauses are incompatible near the top state for a smooth sender payoff with a Lipschitz derivative.

Dischargers for the residual clauses. Clause (iii) of Proposition G.4 has a stage-level discharger, and so has clause (i) when $\bar{x}_T \leq \bar{x}_S$. The discharger of (i) gives the diagonal on $[\bar{s}, x]$ at states $x \geq \bar{x}_S$ above $\bar{s}$, the form the earlier affine route read; this is clause (i) when $\bar{x}_T \leq \bar{x}_S$, since then $\max(x, \bar{x}) = x$, while for $\bar{x}_T > \bar{x}_S$ the slope on $[\bar{s}, \bar{x}]$ at states in $[\bar{x}_S, \bar{x})$ is not derived. That diagonal is derived from α-antitonicity on the basin (available under the substituted-objective slope condition discussed in Appendix I.5), the action-complementarity sign $\partial^2 U_S / \partial s \partial a \leq 0$, stage complementarity, and a single *pivot-action frontier* condition: margin-level slope conditions at the one action $\alpha(\bar{x}_S)$ along the diagonal states, so the per-basin-state quantification collapses to a one-action family. The myopic ceiling (iii) follows from strict stage decrease above the floor at the myopic action ("the myopic action is on the un-flipped side of the diagonal"); the bridge from this

stage-level condition to fully primitive form (the σ-difference between the stage and dynamic best responses) remains open. Complementing these, the next-state reparametrization $y = F(x, s)$, under which the continuation cancels from every slope comparison, gives monotonicity of the greatest next-state selection per fixed action from stage complementarity plus concavity of U_S in s alone (the Amir 1996 pattern), and under strict complementarity *every* selection from the stage argmax is monotone in the state.

Deriving the dynamic single crossing: the route-(b) convexity cone. The one sender-side dynamic input the primitives of the derived-basin route provably cannot deliver for free is the single crossing of the *forward-looking* objective $U_S(x, \alpha(x), s) + \beta_S V_S(F(x, s))$ in (sanction; state): single crossing is not preserved under sums, and at the affine transition the continuation term has the requisite increasing differences precisely when V_S is convex along the transition range (Topkis Thm 2.6.2). It is *derivable* from a range-local curvature cone: if every sender value in the value set is convex along the basin transition range $[\,\mathrm{lo}, 1]$ with $\mathrm{lo} \leq (1 - \delta)\bar{x}$, then together with stage complementarity, the compliance-placates sign, and α-antitonicity, the dynamic objective has increasing differences (hence single crossing, and a fortiori the certificate's interval-dominance condition of Remark G.2). The cone's economic content is *no diminishing returns to entrenchment on the post-tipping basin*: each further step of political hardening is worth weakly more than the last, the increasing-returns lock-in mechanism of the institutional path-dependence literature (Pierson, 2000), imposed only where that literature claims it. The contested and low-basin regions are unconstrained, so dual-basin path dependence is untouched. Two caveats: the cone excludes saturation of entrenchment returns strictly inside the basin (if saturation is the preferred semantics, the derivative-free form with its three existing dischargers remains canonical); and preservation of the cone under the Bellman map cannot remain range-local (invariance of convexity on $[\,\mathrm{lo}, 1]$ pulls in the strictly larger interval $[(1 - \delta)\mathrm{lo}, 1]$ and iteration expands the requirement to all of $[0, 1]$), so a *constructive* convex-cone invariant set would require the global commitment rejected above; the form stated here, which places the cone on a chosen value set, is the endpoint of the range-local route. Constant-V_S value sets (e.g. that of the auxiliary polynomial economy of Table 3, whose sender value is constant) lie in the cone with equality.

The Lipschitz route under a positive stage slope: single crossing *constructed* at the equilibrium. Because the convexity cone of the previous route is near-vacuous against the *concave* value functions standard dynamic programming delivers (Stokey et al., 1989, Thm 4.8), a second route discharges the single crossing for the genuinely *non-monotone* equilibrium value, using only Lipschitz regularity. Suppose the sender stage payoff has a strictly positive lower slope $m_S > 0$ in s on the *full* choice axis $[0, 1]$ at the equilibrium compliance (the *positive-stage-slope* setting, in which the marginal sanction benefit stays positive across $[0, 1]$), and that the transition-adjusted slope condition (GF-Chord) holds, with K_S the Lipschitz constant of the invariant set. Then the dynamic objective is strictly increasing in s on $[0, 1]$: the stage rise strictly exceeds the worst-case continuation fall $\beta_S K_S |F(x, s_2) - F(x, s_1)|$, an inequality that uses *only* that V_S is K_S-Lipschitz, never that it is monotone or convex. Under the affine law, the lower slope m_S and $\beta_S K_S \delta < m_S$ are a sufficient specialization. Strict monotonicity in s is the corner property (the sender plays the high-sanction boundary), and it implies the single-crossing clause of Assumption 1 a fortiori.

This is a property the *constructed* equilibrium *has*, not one imposed on it. The invariant set is the closed K_S-Lipschitz ball; the Bellman operator maps it into itself and the ball is closed in sup-norm, so the contraction's fixed point inherits K_S-Lipschitzness by construction (Stokey

et al., 1989, Cor. 1 to Thm 3.2). Single crossing is therefore *derived* from the positive-stage-slope condition and (GF-Chord), both stated over the model's existing primitives, and the certificate's interval-dominance field (Remark G.2) is filled by this derivation (the derived single crossing upcasts for free) rather than supplied as a separate hypothesis, so every high-basin result consuming the certificate inherits a derived condition. Unlike the convexity cone, this route admits the non-monotone equilibrium value the model actually produces.

The route's scope limit travels with it. It is confined to the positive-stage-slope setting; in the complementary setting (where the sender's stage payoff *falls* in s and only the forward-looking continuation gain carries the high-sanction corner), the two-sided $\pm K_S$ bound is too coarse to capture that gain at a non-monotone V_S, so in that setting the interval-dominance field of Remark G.2 remains assumption-backed rather than derived (Appendix G.1). The derivation above should not be read apart from this limit.

Remark G.5 (Lipschitz strict-slope margin as a cardinal sufficient condition). A cardinal sufficient condition for Assumption 1 is the Lipschitz strict-slope margin

$$U_S(x, a, s_2) - U_S(x, a, s_1) > \beta_S K_S |F(x, s_2) - F(x, s_1)|$$

for $s_1 < s_2$ in $[\max(x, \bar{x}), 1]$. Combined with $V_S \in \mathcal{C}_S$, this strict difference-quotient inequality implies the Milgrom-Shannon single-crossing clause of Assumption 1 (the stage gain strictly dominates the worst-case Lipschitz absorption of the continuation, so the increment of J_S in s is bounded below by $\text{gap} \cdot (s_2 - s_1) > 0$; hence single-crossing is satisfied with the constant-sign upper witness). The cardinal form is thus available to users with explicit numerical bounds, and it is more than an optional convenience: the *level* conclusion of Lemma C.1, that no sender maximizer lies below $\max(x, \bar{x})$, is proved from this same strict difference-quotient inequality imposed on the *complementary* range $[0, \max(x, \bar{x})]$ (equivalently, from the margin on the whole of $[0, 1]$), which bounds the stage gain against the worst-case Lipschitz absorption of a possibly non-monotone V_S. Under the affine law, a lower slope m_S and $\beta_S K_S \delta < m_S$ imply this exact inequality. The ordinal form is the canonical statement on which the proof otherwise depends.

Remark G.6 (Fixed-action interval dominance does not survive policy substitution automatically). Continuity and feasibility of the target policy are not enough to turn fixed-action interval dominance into Assumption 1. The machine-checked counterexample takes

$$G(s, x, a) = (x - a)s, \qquad \alpha(x) = 2x,$$

on sanctions $[0, 1]$ and states $[0, \frac{1}{2}]$. For every fixed action a, the family $G(\cdot, \cdot, a)$ satisfies interval dominance, and α is continuous and feasible. After substitution, however,

$$G(s, x, \alpha(x)) = -xs,$$

which violates interval dominance. A valid discharge must therefore derive an appropriate order or variation restriction on the realized policy, or verify interval dominance directly after substitution. The convex-value route remains valid precisely because it derives policy antitonicity before proving the realized objective's single crossing.

H Supporting robustness and boundary results

This section collects results that sharpen the main mechanisms without being needed to reconstruct the paper's central argument. Their formal counterparts and axiom dependencies are recorded in the formal-verification supplement.

H.1 Parametric robustness of the corner equilibrium

Proposition H.1 (Perpetual sanctions on a patience interval)**.** *There is a non-degenerate interval of sender discount factors $\beta_S \in [6/13,\ 4/7)$ such that, for every β_S in it, the corresponding economy has a deterministic affine transition and admits a stationary feedback Stackelberg equilibrium along which, from every initial state of the high basin and at every date, the realized sanction stays above a strictly positive floor, never falls, and the political state remains in the basin. In fact, the realized sanction is maximal at every date.*

This interval varies sender patience while holding the other primitives fixed. It establishes robustness in that direction, not an open neighborhood in the full parameter space.

H.1.1 Symbolic corner economies

Proposition H.2 (Symbolically parameterized certified corner economies)**.** *There is a class of economies whose sender and target stage payoffs carry nine free real coefficients, with the adjustment rate, both discount factors, and the basin geometry free as well, such that:*

(i) membership is defined by finitely many inequalities among these primitives, some strict and some weak, and the region they define is non-empty;

(ii) every economy in that region carries a complete compliance certificate in the legacy form and admits a stationary feedback Stackelberg equilibrium lying in invariant sets of value functions whose radius and slope are closed forms in the primitives, $K_S = (r+w)/\big(1-\beta_S(1-\delta)\big)$ and $g_T = (n+v)/\big(1-\beta_T(1-\delta)\big)$; an economy with a strictly interior basin bottom carries the Milgrom–Shannon/general-F certificate as well; and

(iii) along that equilibrium the independently optimized myopic equilibrium realizes no sanction at any feasible state, while the forward-looking equilibrium realizes the maximal sanction at every feasible state.

Proof. Three conditions pin the equilibrium. An implacability condition makes the forward-looking sender's dynamic objective strictly increasing in the sanction at every feasible pair, so its response is maximal. A stage condition keeps the slope of the sender's stage payoff in the sanction on the basin's low-sanction range above the per-unit continuation drag. A entrenchment condition covers the target's myopic sanction loss by its discounted continuation gain, making full non-compliance the target's derived Bellman argmax. Because both policies are at corners, the cross-Lipschitz coefficients vanish and the row conditions reduce to $\beta_T < 1$ and $\beta_S < 1$. The Bellman images are affine in the state, and the displayed constants of the invariant sets solve their fixed-point equations. Non-emptiness follows by choosing each coefficient strictly between bounds constructed from coefficients chosen earlier; the formal supplement records the corresponding Fourier–Motzkin elimination. □

The inequalities of (i) are of both kinds. The implacability and sender stage conditions are strict, and so are $0 < \delta < 1$, $\beta_S, \beta_T < 1$, $q, u_0, \ell > 0$ and $0 < \bar{s} < 1$ for the basin's sanction floor $\bar{s}$; the conditions $\beta_S, \beta_T \geq 0$, $q \leq r$, $u_0 + r \leq k$, $n \geq 0$, $\ell \leq c - n$, $0 \leq \bar{x} \leq \bar{s}$ and the entrenchment condition are weak (`SymbolicEconomy`).

These economies are corner-pinned. They do not contain the interior-sanction benchmark or the dual-basin construction of Appendix K.2.

The corner economies of Proposition I.5 contain these: each symbolic economy, with the state slope bounds $g_{\mathrm{lo}} = g_{\mathrm{hi}} = 1 - \delta$ and the lower bound on the sanction slope $d_{\mathrm{lo}} = \delta$, is one of

them (`SymbolicEconomy.toF`). Clauses (ii) and (iii) hold on all the corner economies as follows (`SymbolicEconomyF.symbolic_family_generalF`). At every corner economy an equilibrium lies in invariant sets with the closed-form constants $K_S = (r+w)/(1-\beta_S g_{\mathrm{hi}})$, $g_S = (r+w)/(1-\beta_S g_{\mathrm{lo}})$ and $g_T = (n+v)/(1-\beta_T g_{\mathrm{hi}})$, and it realizes the maximal sanction at every feasible state while the economy's own optimized myopic equilibrium realizes none, with no side condition (`regime_separation_symMyopic`); a corner economy with a strictly interior basin bottom carries the Milgrom–Shannon/general-F certificate. Against the canonical myopic equilibrium the separation is proved only under a strictly interior basin bottom and the strict substitution condition $u_0 + r < k$, which it reads at the top state. The legacy-form certificate of (ii) does not extend, because its consequences use the partial-adjustment law, and (i) is not claimed for the corner economies as a whole.

Proposition H.3 (Strict escalation)**.** *Fix any economy, any equilibrium, and any interval $[a, b)$ on which the equilibrium closed loop is invariant, and suppose that at every state of $[a, b)$ a feasible sanction above the state strictly raises it (local upward adjustment; it holds whenever the transition is strictly increasing in the sanction, or satisfies* (F5) *with $a \geq 0$). If the equilibrium sanction strictly exceeds the state throughout $[a, b)$, the political state strictly rises at every date. If, in addition, the sanction rule is strictly increasing on $[a, b)$, the realized sanction strictly rises at every date, from every start in $[a, b)$, forever.*

The two conditions are distinct: a rule can lie above the diagonal without being monotone, or be strictly monotone while lying below it. Because sanctions are bounded, strict escalation here means a strictly increasing convergent sequence, not unbounded growth.

H.2 Equilibrium feedback from sender patience

Proposition H.4 (Equilibrium value feedback from sender patience)**.** *Let E_1 and E_2 be two of the corner economies of Proposition I.5, which contain the symbolic economies of Proposition H.2, that share the transition F and the sender's stage payoff at the corner, $y \mapsto U_S(y, 0, 1)$. Write β_1, β_2 for their sender discount factors and $V_S^{\beta_i}$ for the sender value of any equilibrium of E_i whose sender value lies in the economy's certified invariant set; such an equilibrium exists at every corner economy (Proposition I.5). Along both equilibria the target policy and both realized actions remain at their corners. Then, for every state x, writing $F(x, 1)$ for the corner-pinned successor,*

$$V_S^{\beta_2}(x) - V_S^{\beta_1}(x) = (\beta_2 - \beta_1)\, V_S^{\beta_2}\big(F(x,1)\big) + \beta_1 \left[V_S^{\beta_2}\big(F(x,1)\big) - V_S^{\beta_1}\big(F(x,1)\big)\right]. \tag{H.1}$$

At the absorbing lock-in state,

$$(1-\beta_1)\left[V_S^{\beta_2}(1) - V_S^{\beta_1}(1)\right] = (\beta_2 - \beta_1) V_S^{\beta_2}(1), \tag{H.2}$$

and $V_S^{\beta}(1) = U_S(1, 0, 1)/(1-\beta)$; (H.2) and the sign comparison that follows need only a common corner payoff $U_S(1, 0, 1)$, not a common transition. Consequently, if $\beta_1 < \beta_2$, then $V_S^{\beta_1}(1) < V_S^{\beta_2}(1)$ if and only if $U_S(1, 0, 1) > 0$. The latter inequality holds at every corner economy, so the strict ranking is unconditional on these economies. Finally, the target equilibrium value is invariant to sender patience: if E_1 and E_2 share F, β_T and the target's stage payoff at the corner, $y \mapsto U_T(y, 0, 1) = -c + (n + v)y$, whatever their compliance costs ℓ and their sender payoffs, then $V_T^{(1)} = V_T^{(2)}$ for any two such equilibria.

Two remarks on the scope of the first two displays, because it is wider than the set of economies they are stated over.

No functional form for F is used. These economies carry a general transition, and (H.1) and (H.2) require only that both value functions solve the stationary Bellman equation along a common constant action pair, against a common stage payoff and, for the first, a common transition. The absorbing state that (H.2) needs is supplied by the diagonal condition $F(1,1) = 1$ rather than assumed, so it too is free of functional form. Both are verified at that generality, and the versions for the corner economies are special cases. On the symbolic economies of Proposition H.2, whose transition is affine, the identities are also proved for their closed-form values, with the shared adjustment rate in place of the shared transition.

The sign clause is an equivalence, and the direction it fails in is economically real. Greater sender patience raises the lock-in value *if and only if* the sender's corner flow is positive. Where a locked-in sanctions regime pays the sender a negative flow, patience makes the sender strictly worse off there, and no feature of the dynamics reverses that. What makes the ranking unconditional here is a property of the corner economies, their implacability condition, which forces the corner flow to be positive; it is not a property of the mechanism.

Proof. At an equilibrium whose sender value lies in the certified invariant set the sender sanctions fully at every state and the target maintains full non-compliance, since compliance costs it $\ell > 0$ per unit and does not move the sender off the corner; so both values solve corner Bellman equations along the action pair $(a, s) = (0, 1)$. Subtracting the two sender equations gives (H.1). At $x = 1$, the successor state is one, so the Bellman equation gives $V_S^\beta(1) = U_S(1, 0, 1) + \beta V_S^\beta(1)$ and hence (H.2). The sign comparison follows from this identity. Of the conditions on these economies only implacability is needed for $U_S(1, 0, 1) > 0$: with the slope bound $g_{\text{lo}} + d_{\text{lo}} \leq 1$ it places the worst stage disincentive $q + k - u_0$, which the sign condition $u_0 + r \leq k$ puts at or above $q + r$, below the corner state slope $r + w$, so $U_S(1, 0, 1) = u_0 - q/2 + r + w > u_0 + q/2 + r > 0$. The target value is invariant because its corner Bellman equation does not contain β_S and has exactly one bounded solution. Exact parameter intervals are reported in Table 5. □

H.3 The upper-boundary trade-off

Proposition H.5 (Ratchet dichotomy)**.** *Fix an economy in the model class, a sanction rule R taking values in $[0, 1]$, and any interval $[\bar{x}, 1]$ with $\bar{x} \leq 1$. Exactly one of the following holds.*

(i) $R(1) < 1$: the top sanction is interior. Then the ratchet floor $R(x) \geq \max\{x, \bar{s}\}$ fails on $[\bar{x}, 1]$, for every $\bar{s}$.

(ii) $R(1) = 1$. Then $F(1, R(1)) = 1$: the realized successor at the top state is the top state itself, whatever R does elsewhere.

Proof. In the first case, evaluating the floor at $x = 1$ would require $1 \leq R(1)$, a contradiction. In the second, the diagonal condition gives $F(1, R(1)) = F(1, 1) = 1$. □

Corollary H.6 (Endpoint obstruction to an upper-basin floor)**.** *Let R be any rule with $R(1) < 1$. Then no interval $[\bar{x}, 1]$ can satisfy*

$$\max\{x, \underline{s}\} \leq R(x) \qquad \text{for every } x \in [\bar{x}, 1].$$

Consequently, any equilibrium whose realized sanction is zero at state 1 fails the dynamic-floor ingredient of the upper-basin persistence certificate.

Proposition H.7 (Absorbing-boundary obstruction)**.** *Fix an economy in the model class and a candidate equilibrium. Suppose the sender's continuation value rises to the top state at rate at least m; shading the sanction below the corner by $1-s$ lowers the successor state by at least $\kappa(1-s)$; and the sender's current-period gain from that shading is at most $c(1-s)$. If*

$$c \leq \beta_S m \kappa,$$

then the sender's response is pinned at $s=1$ across a right-neighborhood of the target's action at $x=1$. Consequently, if the target's stage payoff strictly increases when its action rises from $\alpha(1)$ against the corner sanction, no such equilibrium exists.

At a pinned top sanction, the successor state is identical across the relevant target actions. The target's continuation term therefore cancels, leaving only its stage-payoff comparison. The proposition's inequality supplies sufficient conditions for that pinning.

H.4 A sufficient route to the weak switch leg

Appendix E.2 records that the sender-response-switch component is not signed by the primitives. That stands. What follows is a *sufficient* route, one way of supplying the weak leg from conditions a modeller can check, and not a derivation of the component's sign from the baseline.

Suppose that, across any two feasible sanctions, the *difference* in the sender's stage payoff is a function of the displacement from the sender's own myopic best response alone, so that compliance relocates the sender's stage problem without reshaping it, while an arbitrary compliance-dependent level is left free and cancels from every sanction comparison. Write $\Gamma_x(s) = \beta_S V_S\big(F(x,s)\big)$ for the discounted continuation term and $\Lambda(a) = \sigma^{\mathrm{dyn}}_{V_S}(x,a) - \sigma^{\mathrm{M}}(x,a)$ for the response gap. Say that Γ_x has *increasing translated chords* when the increment of Γ_x across a chord of any fixed length does not fall as the chord is translated toward larger sanctions. Then, provided the shift-translated responses remain feasible and the dynamic best response is taken as the greatest maximizer,

$$\Gamma_x \text{ has increasing translated chords} \quad \Longrightarrow \quad \Lambda \text{ does not widen in compliance,}$$

and the reverse chord condition makes Λ not narrow. An affine Γ_x satisfies both, leaving Λ exactly constant. The proof is an exchange argument (translate each maximizer by the shift and offer it in the other problem) and uses no derivative, no strong concavity, and no compactness. Convexity of Γ_x is one way to supply the chord condition, and on an interval with Γ_x continuous it is equivalent to it; the chord form is preferred because it names exactly the inequality the argument consumes.

The condition can be weakened further, and ordinally. Re-centring the sender's problem on its own myopic best response, so that the gap *is* the maximizer in the displacement variable, the comparison becomes a monotone-comparative-statics question, and what it requires is single-crossing lower dominance of the re-centered objective across the two compliance levels. Increasing chords imply that; the converse fails, since single crossing survives an arbitrary strictly increasing transformation and a difference condition does not. This is the same distinction Dekel et al. (2025) draw in a model of costly adjustment: monotonicity-type conditions govern the *direction* of adjustment, and curvature governs its *speed*. The comparison here is a direction result.

Only the chords the decision problem generates are needed. The chord condition above is stated over every chord of $[0,1]$, and for a continuous Γ_x on an interval that is convexity, so as written it is not a weakening of convexity at all. Tabor and Tabor (2009) explain why in general: every nontrivial linear inequality valid for all convex functions is valid *only* for convex functions, once it is imposed over a family of affine reparametrizations.

The exchange argument reads much less than that. It uses chords based at the two myopic best responses $\sigma^{\mathrm{M}}(x, a_L)$ and $\sigma^{\mathrm{M}}(x, a_H)$ only, and only shifts inside the interval those two compliance levels generate. Substituting $u = \sigma^{\mathrm{M}}(x, a_H) + z$ and writing $\Delta = \sigma^{\mathrm{M}}(x, a_L) - \sigma^{\mathrm{M}}(x, a_H)$ for the sanction gap, that restricted condition says exactly

$$u \longmapsto \Gamma_x(u + \Delta) - \Gamma_x(u) \quad \text{is non-decreasing on } [0, 1 - \Delta]. \tag{H.3}$$

A *single* lag, and that lag is the gap the two compliance levels themselves generate. The all-chord condition asks the same of every lag. So what buys the weakening is the restriction from all lags to one, and not a restriction of the reparametrization group.

The distinction is not decorative, because it changes what a separating example must look like. Tabor and Tabor's own escape ($x_1^2 - x_2^2$, which satisfies the spherical sub-mean-value inequality for every affine *similarity* yet is not convex) cannot serve here: it lives in $\mathbb{R}^2$ and works because the function is harmonic, whereas in one dimension the corresponding inequality is midpoint convexity, and a continuous midpoint-convex function on an interval is convex. Equation (H.3) gives the construction directly instead: a continuation term periodic with period Δ has vanishing Δ-difference, so it satisfies the condition however far from convex it is.

The weakening is therefore strict, and machine-checked to be. Take a sender whose stage loss is single-peaked at $3/4 - a/2$, so its stage argmax is a clamp falling in compliance and the gap at $a_L = 0$, $a_H = 1$ is exactly $\frac{1}{2}$; a transition that adjusts fully; and the $\frac{1}{2}$-periodic continuation value $\varphi(s) = \big|\,|s - \frac{1}{2}| - \frac{1}{4}\big|$. It satisfies Equation (H.3) and fails the all-chord condition, at base points $\frac{1}{4}$ and $\frac{1}{2}$ with chord length $\frac{1}{4}$, where the increment falls from $\frac{1}{4}$ to $-\frac{1}{4}$. So the implication from the all-chord form does not reverse.

Per the scope discipline this appendix keeps elsewhere, the example refutes a universal and licenses nothing further: it does not say which condition holds at any calibrated economy, nor that the weaker form is ever the binding one.

Three limits travel with all of this. Each implication is one-way, and a gap that fails to widen does not force any chord condition. The relevant shape is that of the *composite* Γ_x, not of V_S or of $F(x, \cdot)$ separately: curvature of one factor says nothing about the composite without a matching monotonicity of the other. And the response gap is *not* the switch component: it is one of five inputs to the switch-gap comparison, the remaining four being shape conditions on the target's complete sanction payoff over the reached sanctions.

That the ordering can genuinely fail is settled by an example economy whose transition is strictly concave in the sanction, non-affine at every δ, and carries genuine inheritance, and in which a fully forward-looking equilibrium and an independently optimized myopic one satisfy $\alpha^{MM}(x) = 0 < 1 = \alpha^{FF}(x)$ at every feasible state, with both Bellman equations holding exactly. Two things should be said about it, and no more. It licenses exactly one claim (that the baseline together with several natural monotonicity and optimality properties does not logically determine the two-sided ordering), and nothing about whether the intended ordering holds in typical or calibrated economies. And what it verifies is equilibrium-level: its stage-payoff state shifts are chosen so that the proposed values solve the Bellman equations exactly, and no claim is made that it satisfies the forms of Assumption 3 and its companions that quantify over sets of value functions.

H.5 Why the path comparison assumes a monotone sanction rule

The route taken here orders the state paths of the two equilibria by asking the forward-looking sanction rule $R^{FF}(x) = \sigma_{V_S}^{\mathrm{dyn}}\big(x, \alpha^{FF}(x)\big)$ to be non-decreasing in the state. That is a sufficient condition, not the exact one: what the induction actually consumes is a cross-order condition on the two closed-loop *state* maps $x \mapsto F(x, R(x))$, which a monotone rule discharges. The monotonicity

is assumed rather than derived, and the reason is a constraint the model imposes rather than a gap in the argument.

Write the composition informally as $R'(x) = \sigma_x + \sigma_a\,\alpha'(x)$, where σ_a is the sender's response to compliance. This model maintains $\sigma_a \le 0$ throughout (compliance placates), so the two effects pull against each other unless the target's own policy is non-increasing in the state. The baseline does not deliver that. A target with stage payoff $-(a-x)^2$, which does not mention the sanction at all, has $\alpha(x) = x$ as its genuine myopic stage argmax, strictly *increasing*. What that closes is the *sign-only* route: the closed-loop property cannot be read off the direction of α alone. A quantitative route survives, since the direct state effect may dominate the adverse compliance–sanction cross term even where α rises, but it turns on a comparison of magnitudes the baseline does not supply. The honest options are to assume the closed-loop property or to discharge it from a condition on the *composite*. The formalization takes the second: the single-crossing condition is imposed on the sender's objective with the target's policy already substituted in, which is precisely the formulation that never separates the two effects and so never walks into the composition trap.

The condition the induction actually consumes. Write $G_i(x) = F\big(x, R_i(x)\big)$ for the two closed-loop state maps. Stepping from a lower state x to an upper state $y \ge x$, the induction needs one inequality, $G_1(x) \le G_2(y)$, and nothing else. Say that G_1 is *cross-ordered* below G_2 on a region B when

$$x \in B, \quad y \in [0,1], \quad x \le y \quad \Longrightarrow \quad G_1(x) \;\le\; G_2(y). \tag{H.4}$$

Proposition H.8 (Stopped path order under cross-step domination)**.** *Fix two sanction rules mapping* $[0,1]$ *into itself, a region* $B \subseteq [0,1]$*, a feasible initial state, and a date* n*. If* (H.4) *holds and the lower path has in fact remained in* B *through step* n*, then the lower state path lies weakly below the upper one at* n*. If in addition the same cross-order condition holds for the two* rules *in place of the two closed loops, the sanction levied along the lower path at* n *is weakly below the one levied along the upper path.*

Neither rule is required to be monotone, no invariance of B is asserted, and B need not be closed or an interval. What replaces those is a hypothesis about the trajectory that was actually run, at the date in question.

Monotonicity of the upper rule, which Appendix H.5 uses, is one way to supply (H.4): it gives $G_1(x) \le G_2(x) \le G_2(y)$ through the transition's monotonicity in each argument. It is a strictly stronger way. Condition (H.4) is equivalent to G_1 being dominated on B by the greatest non-decreasing minorant of G_2, the pointwise infimum of G_2 over the states weakly above the current one, and that minorant coincides with G_2 exactly when G_2 is already monotone. Where it does not, the condition is correspondingly weaker.

That gap is inhabited. Under the lattice-meet transition $F(x,s) = \min\{x,s\}$, which satisfies (F1)–(F4) but not (F5) (it is insensitive to a sanction above the state: $F(0,1) = 0$), the rule $R(x) = \max\{x, 1-x\}$ is feasible and *not* monotone (it falls from 1 at $x = 0$ to $\frac{1}{2}$ at $x = \frac{1}{2}$), yet it never names a sanction below the state, so the closed loop is the identity and (H.4) holds trivially. The point is that rule monotonicity is a property of the rule, while what the comparison needs is a property of the rule composed with the transition; the two come apart wherever the transition is insensitive to the sanction.

The two path conclusions are stated separately on purpose. Ordering the states along the two paths does not by itself order the sanctions levied along them, because the two rules are evaluated at different states, which is why the sanction claim carries its own hypothesis rather than being read off the first.

The counterexample refutes derivability. It does *not* exhibit an economy in which the closed-loop rule actually fails to be monotone; that would also require the effect of compliance on the sender's return to sanctioning to overturn the direct state effect. The general shape of the obstruction is familiar from monotone comparative statics, where what a composition needs is the follower's response to the *leader's action* and not only to the state.

H.6 A sender payoff rising with the state, and a weaker persistence floor

For a fixed target policy, call the greatest maximizer of the sender's one-period problem at a state the *stage ceiling*.

Proposition H.9 (Dynamic floors require a sender payoff that rises with the state)**.** *Fix an economy in the model class, a continuous target policy, and the sender's continuation value determined by that policy. If the sender's stage payoff along the policy is non-increasing in the target's political state, then at every state the greatest maximizer of J_S is at most the stage ceiling. Equivalently, if at some state the equilibrium sanction strictly exceeds the stage ceiling, then there are feasible states $y_1 \le y_2$ and a feasible sanction s with*

$$U_S\big(y_1, \alpha(y_1), s\big) < U_S\big(y_2, \alpha(y_2), s\big).$$

The conclusion concerns the composed payoff $y \mapsto U_S(y, \alpha(y), s)$, not the partial effect of the state at fixed compliance. It also locates the required increase somewhere in the feasible state space, not necessarily at the state where the dynamic sanction exceeds the stage ceiling.

H.6.1 Persistence under a level floor

Proposition H.10 (Level-floor invariance and its cost)**.** *Suppose $R(x) \ge \bar{s}$ on $[\bar{x}, 1]$ with $\bar{x} \le \bar{s}$. Then $[\bar{x}, 1]$ is forward invariant under the closed-loop map of R, and the statewise ordering of dynamic over myopic sanctions and the finite-shock entry conclusion are unaffected. The level floor does not, however, entail the ratchet step $x \le F\big(x, R(x)\big)$ and therefore does not entail that the political state is non-decreasing: there are transitions and rules satisfying the level floor along which the state falls.*

Forward invariance follows from monotonicity and the diagonal condition:

$$\bar{x} = F(\bar{x}, \bar{x}) \le F(\bar{x}, \bar{s}) \le F(x, \bar{s}) \le F\big(x, R(x)\big).$$

For the partial-adjustment law with $\delta = 1/5$, $\bar{x} = \bar{s} = 1/2$, and a rule constant at $1/2$, the state falls from 1 to $9/10$. Thus the level floor preserves the basin but does not imply a monotone state path.

H.7 Further exposition

None of the passages below is a theorem statement or a hypothesis of a main-text result; each qualifies, illustrates or inventories one. Each is headed by what it is and where it stood.

The statement on the cube and the formalization on the line. *(From Section 2, after* (F1)–(F5)*.)*

One difference between this statement and the formalization should be recorded rather than left to inspection. States and sanctions are economically meaningful only on $[0, 1]$, and (F1) and

(F5) are accordingly stated there. The formalization, however, maintains (F2) and (F3) *globally* (F is monotone in each argument on all of $\mathbb{R}$, not merely on the cube) because a global monotonicity hypothesis is markedly easier to carry through the order arguments than its restriction. Read as an assumption on F it is strictly stronger than the displayed one. It costs no generality: a transition that is continuous, coordinatewise nondecreasing, and diagonal-fixing on the square extends to one with the same three properties on all of $\mathbb{R}^2$ and the same values on the square, by adding to $F(\operatorname{clip} x, \operatorname{clip} s)$ the two correction terms $\min\{0, \min\{x, s\}\}$ and $\max\{0, \max\{x, s\} - 1\}$, which vanish on the square and restore the diagonal outside it. That extension is machine-checked (`Foundations/TransitionExtension`); every difference-quotient condition stated on the square is inherited verbatim, since the extension *is* F there. The payoff functions and the stage selector extend by clipping their arguments. Both steps are assembled into a machine-checked constructor (`Postulates/ParamsOfCube`): from payoffs continuous on the cube, a selector feasible, continuous and argmax on the square, and a transition satisfying (F1)–(F4) on the square, it builds a formal economy that agrees with the data on the cube and square and inherits (F5) from it. So the formal statements quantify over a class no smaller than the displayed one, and every theorem stated over the formal economy applies to every cube-defined economy, evaluated where that economy is defined. The point is recorded because an assumption surface that is wider in the machine-checked statements than in the prose is the kind of gap a reader is entitled to be told about rather than to find.

Continuous payoffs whose myopic best response jumps. The formal economy asks of the myopic sender only a feasible stage maximizer $\sigma^{\mathrm{M}}(x, a)$, which exists whenever the payoffs are continuous; it does not ask that this choice vary continuously with the state and the compliance level. That continuity (`Params.StaticSelectorContinuous`) is read only by the results built on the myopic sender: the target's value against the myopic sender, the target-only case, the canonical and greatest myopic equilibria, and every comparison with the canonical myopic equilibrium, such as the three-goals conclusion of the route of Proposition G.4. Those results take it as a hypothesis, which every economy constructed in the formalization satisfies, and the class keeps it as a condition of its own, because its results on the four cases compare with the myopic sender. Every other result stated for the general game (in particular Theorem F.1, Theorem D.1, Theorem D.2, Corollary D.3, Theorem D.8, Theorem D.9, Theorem D.10, Theorem D.5, Proposition D.4, Proposition I.1, Proposition C.14 and Proposition B.4) holds for every economy with continuous payoffs that meets that result's own hypotheses; none of them reads this continuity. That includes an economy whose sender payoff is linear in the sanction with a slope whose sign depends on the state, so that the myopic best response jumps between no sanction and the full sanction. No such economy is constructed in the formalization, so this generality is stated, not yet illustrated by an example.

Why differentiability is stated on a neighborhood of the cube. *(From the footnote to the payoff domains in Section 2.)*

The Lean development states every quantitative bound it uses (Lipschitz constants, lower state slopes, curvature moduli) as a finite-difference inequality on the compact cube, and for the polynomial instances derives those bounds from the coefficients. Where a smooth route is used to *read off* such a constant, it needs the second partials to be continuous up to the boundary, which is why the hypothesis is stated on a neighborhood of the cube: continuity on $[0,1]^3$ together with C^2 on the open cube alone does not bound the second partials ($x \mapsto x^{3/2}$ is C^1 on $[0,1]$, C^∞ on $(0,1)$, and has unbounded second derivative). A reader who prefers a weaker maintained assumption may drop the differentiability altogether and take the finite-difference bounds as primitive; nothing else

changes.

Which result consumes which structural condition. *(From Appendix D.1.)*

The substitution between compliance and the sanction needs no concavity. *(From Appendix D.1, after A3.)*

A3 is not, however, what carries the substitution between compliance and the sanction. Decreasing differences of U_S in (s, a) orders the stage argmax *correspondence* in the strong set order, and while that does not order an arbitrary selection from it, it does order the *greatest* maximizer, which exists by Weierstrass alone. So the antitonicity of σ^{M} in a that the ordering results use follows from the cross-partial sign plus a greatest-selection convention, with no concavity and no uniqueness; uniqueness is merely one sufficient way to reach it, since a unique maximizer is trivially the greatest. This matters because uniqueness is a regularity condition rather than an economic primitive, and it fails exactly where a sender payoff is linear in the sanction or flat over a range.

Static and dynamic thresholds, substitution, and the constructed pivots. *(From the end of Appendix G.1.)*

Static and dynamic sender thresholds. The construction at slope level $m = 0$ gives the *static flip* $x_S^0 := \inf \mathcal{F}_S(\bar{s}; 0)$: the state at which sanctioning stops being a stage loss to the sender. Since a higher slope level is a more demanding condition, $x_S^0 \leq \bar{x}_S$: between the static flip and the dynamic pivot, the sender's stage incentive has already flipped but the transition-adjusted forward-looking drag (H.7) is not yet dominated. The band is indexed by the slope level m_S and is non-decreasing in it, since a higher level is a more demanding condition. Under the partial-adjustment law the drag per unit sanction is $\beta_S K_S \delta$, which vanishes in the myopic limit $\beta_S \to 0$, where the band closes; no comparative static in β_S is proved beyond that observation.

Compliance placates. Suppose compliance lowers the sender's marginal value of sanctioning:

$$\frac{\partial^2 U_S}{\partial s \, \partial a} \leq 0.$$

Define $x_S(a)$ as the infimum of the states at which the sender's difference-quotient inequality in (G.3) holds for that fixed action a, rather than uniformly over all actions. The resulting per-action threshold $a \mapsto x_S(a)$ is non-decreasing: against a more compliant target, a harder state is required before sanctioning clears the margin. The uniform flip set is therefore determined by the full-compliance slice, so $\bar{x}_S = x_S(1)$.

An example economy has $\bar{x}_T = 2/5$, $\bar{x}_S = 1/2$, and a target sanction slope that changes sign at the interior pivot. The construction uses a kinked sender payoff because its diagonal-residual and myopic-ceiling conditions cannot both hold smoothly near the top state.

Because the pivots are infima of upper flip sets, their comparative statics follow by set inclusion. A payoff change that raises the target's sanction slopes on the high range lowers $\bar{x}_T$, and the sender analogue shifts the interval between its static and dynamic thresholds. The sanction floor $\bar{s}$ remains a separate parameter in the construction; it may be set equal to the derived basin pivot only after the pivots have been computed.

Table 2: Which result consumes which structural condition.

Result	**Conditions consumed**
Theorem D.2 (persistence, general F)	the package (P1)–(P5): membership of the value sets (P1), the sender Lipschitz ceiling (P2), A1 in its value-set form (P3), the target combined slope condition A2 as (P4), the sub-pivot slope condition (P5); no A3. Dropping (P4) keeps feasibility and optimality, conclusions (iv)–(v), from the equilibrium object and loses only the preference conclusion (vi); that P4-free statement is a proved composition (Appendix B), not a second signature. A1 is read only for (iii): (i), (ii) and (iv) follow from (P1), (P2) and (P5).
Theorem D.8 (minimum-drift shock)	(P1)–(P3) and (P5), with no target condition, together with the transition's (F1)–(F4), full-sanction ascent below the basin (which gives $\bar{x} < 1$), and the temporary shock; the package alone is not the entry theorem. No A3.
Theorem D.9, Theorem D.10	the equilibrium, transition, floor and rule hypotheses stated in each: (F1)–(F4), with held-sanction ascent below the basin for entry and post-basin descent for exit in place of (F5); no A1–A3 hypothesis appears in their statements.
Proposition D.4 (long run)	the level condition at an equilibrium (the sanction rule at or above the state on the basin, with no floor) plus (F1)–(F5). (P1), (P2) and (P5) imply the level condition; neither A1 nor (P4) is read.
Theorem F.1 (Schur condition)	invariant sets of value functions; dynamic argmax uniqueness, the sender's at feasible compliance and the target's at every value pair; joint-operator invariance of those sets; the row bounds with the Schur condition (F.3) (Appendix F). A3 gives only stage-argmax uniqueness, neither the dynamic uniqueness assumed nor a substitute for the other premises.
Proposition G.4 (derived-basin route)	the sender half of Assumption 4 on the floor range and the sender's top boundary sign of (G.4) derive the sender pivot; the three-goals conclusion consumes in addition A1 on the derived basin, row-sum contraction ($\beta_T + L_{TS} < 1$, $\beta_S + L_{ST} < 1$), full-sanction ascent below the basin, and the diagonal-residual, monotone-sender-values and myopic-ceiling clauses (i)–(iii) (Appendix G); no target preference or sign condition and no A3. The target combined slope condition A2 can still be derived on the derived basin from the target half and the target's top sign, but the route does not read it.
Proposition B.1, Proposition B.2 (cases 00, 10)	the class's fields of Section 5.1, the concavity condition and the Riccati closure; as hypotheses of the results that read them, the myopic relief condition, interiority where $A/q < 1$ or a family closed form is read and elsewhere only myopic interiority $u_{\mathrm{hi}} < q$, the lifting condition for case 00 (the myopic equilibrium itself reads only the myopic lifting condition $u_{\mathrm{hi}} < k_{\mathrm{lo}}$) and the entrenchment condition in place of A2 for case 10.
Theorem 3 (case 11)	the same class fields, the myopic relief condition, myopic interiority $u_{\mathrm{hi}} < q$, the myopic lifting condition $u_{\mathrm{hi}} < k_{\mathrm{lo}}$, the entrenchment condition and a non-decreasing response at non-compliance, which the increasing-differences condition (B.4) implies.
Proposition B.3 (case 01)	the same class fields with interiority, the lifting condition, the sender-only compliance condition (T6), the forward-looking relief condition (RL) of Appendix C.4 and compliance neutrality.
Theorem 12	the union of the three rows above. None of the four uses A1–A3 as stated or Assumption 4: the class's signs $r_{\mathrm{lo}}, n_{\mathrm{lo}} \geq 0$ permit strict complementarity, and what it lacks is the boundary sign crossing.
Theorem 4	a sender value non-decreasing in the state; no equilibrium and no basin; none of A1–A3.
Proposition B.4	one comparison of the sender's continuation value at the successor state per lower sanction (no sanction at or below the greatest stage maximizer carries a larger continuation value at the successor state than it) together with one sanction at which the stage loss measured from the greatest stage maximizer is outweighed by the discounted continuation gain; the comparison alone can hold with equality and change no choice.
Interest-group polity (Appendix L)	derives the cross-partial signs of Assumption 4 and the curvature of A3 from region (R); A2 through the entrenchment-condition route at

The four cases: arithmetic form, sanction rules, and the one-period ranking. *(From the paragraph after Theorem 12.)*

Two arithmetic consequences follow from the constructed rules alone. For the limit of the state, writing L^{ij} for the four limits from a common start, $(L^{11}-L^{10})-(L^{01}-L^{00}) = L^{11}-L^{10} > 0$. At a common state x, writing $R^{ij}(x)$ for the four sanction rules, $R^{00} = R^{01} = 0$, $R^{10}(x) = \sigma^{\mathrm{M}}(x, 0)$ and $R^{11}(x) > \sigma^{\mathrm{M}}(x, 0)$ (the vertex $(u_0+rx)/q$ on the quadratic–linear family), so $(R^{11}-R^{10})(x)-(R^{01}-R^{00})(x) > 0$. Both are restatements of the rule identities, not interaction primitives. At every state the realized sanction is zero in the two cases in which the target is myopic and buys relief, equals the positive myopic sanction when only the target is forward-looking, and lies strictly between the myopic sanction and one when both are, above it by at least the foresight premium (B.3); with $a^{01}(x) > a^{00}(x)$, the sender's forward-looking incentive raises the sanction against a non-compliant forward-looking target and extracts more compliance from a myopic one. Eaton and Engers (1992) find that a more patient sender is more likely to succeed in their threat model; what is compared here is a pair of constructed equilibrium rules at a common state, not their toughness parameter or a threat stage, so the two statements are neighbors and not the same result. The one-period ranking $\ell_{\mathrm{hi}} < \underline{s}\, p_{\mathrm{lo}}$, under which full compliance beats full non-compliance in the one-period problem, is sufficient and not necessary for both forward-looking incentives to change behavior: on the enriched subclass, the increasing-differences condition (B.4) and the forward-looking relief condition (RL) alone give it, through a small compliance rather than full compliance, and (RL) does not imply the one-period ranking: the affine relief-only member satisfies (RL) as $6 < 12$ and fails the ranking as $6 < 4$.

Two boundaries of the sender-only case. *(From after Proposition B.3.)*

Two boundaries of the case are worth stating. First, Proposition B.3 gives $a^{00}(x) < a^{01}(x) \leq 1$ and no more: at an economy in the class with the lifting condition at equality ($A = k$) and no state–sanction cross terms, on the quadratic–linear family, the zero level is full compliance, $a^{01}(x) = 1$ at every state, while $a^{00} = 3/4$ (Appendix B). Relief from a forward-looking sender is bought at a strictly higher price, which can be the whole of compliance. Second, the entrenchment condition that makes the forward-looking target maintain non-compliance is sufficient, not necessary: with the same payoff shapes and the target's state valuation lowered until the condition fails, non-compliance is still the unique optimum at a constructed equilibrium (Appendix B). Where the condition fails the class's classification is silent, not reversed.

The Schur-condition existence route, summarized. *(From Appendix D.2.)*

Theorem F.1 (Appendix F) states the route: on closed, non-empty invariant sets of value functions, which the joint Bellman operator maps into themselves, with the sender argmax unique on the sender's invariant set at every compliance level in $[0, 1]$ and the target argmax unique at every value pair, if the two Bellman rows are Lipschitz in the value pair with coefficients $\beta_T, \beta_{S,\mathrm{row}} < 1$ on the diagonal and L_{TS}, L_{ST} off it, all four of any sign, and the Schur condition $L_{TS}L_{ST} < (1-\beta_T)(1-\beta_{S,\mathrm{row}})$ holds (for non-negative coefficients, equivalently $\rho(\mathbf{M}) < 1$), then the operator has a unique fixed point on these sets in a weighted sup-norm, the associated optimal target policy and the sender's dynamic best response form an equilibrium in $\mathcal{E}$ with values in these sets, and every equilibrium with values in these sets that plays the target selector those sets define coincides with it.

The Schur inequality permits one Bellman row sum to exceed one when the opposing cross-value coupling is sufficiently weak. Its uniqueness conclusion is confined to the stated invariant sets and selector restrictions; it is not global equilibrium uniqueness. The appendix states the theorem and

gives the Perov reduction and the exact constants (Appendix F).

Off-corner existence and the transition-smoothed game. *(From Appendix D.2, after Theorem D.1.)*

With an interior target action, each Bellman row depends on the other player's continuation value. The deterministic joint iteration need not preserve order and can cycle even when current argmaxes are unique. We call the game with an atomless transition component the *transition-smoothed game* ("purified" in earlier versions and in the formalization's names); the term refers to transition smoothing, not payoff purification. Under the regularity conditions in Appendix F, its joint operator is a contraction and supplies an off-corner equilibrium (He and Sun, 2017; Doraszelski and Satterthwaite, 2010). Deterministic off-corner existence is established for the example economies; the primitive model by itself supplies no global existence theorem of that form.

The benchmark economy. *(From Appendix I.6.)*

The first application shows that the conditions of Theorem D.2 are attained in an economy where continuation values affect observable choices: Theorem J.1 (Appendix J.2) exhibits, within the general-F model, an economy with positive discount factors for both players, a non-degenerate high basin and an equilibrium at which both value functions are nonconstant, the target maintains full non-compliance at the basin pivot while the sender's realized sanction there is strictly interior and strictly above its myopic level, every basin start keeps the realized sanction above the pivot and non-decreasing, and a finite temporary shock at the maximum sanction carries every initial state into the basin. The theorem establishes existence in the model class. The appendix strengthens its robustness in two directions: permanent sanctions survive on a non-degenerate interval of sender discount factors, and the symbolic corner economies free the payoff coefficients, adjustment rate, discount factors, and basin geometry (Propositions H.1 and H.2).

The off-corner boundary of primitive persistence. *(From the end of Appendix I.6.1.)*

The off-corner constructions locate the remaining boundary. A purified economy defined on the full state space has nonconstant values, a state-varying target policy, and a monotone sanction rule, but its top-state sanction is interior and therefore cannot satisfy a ratchet floor on a basin ending at one. It nevertheless ratchets strictly on an interior sub-basin. The supporting appendix states the general boundary results; Section 5.4 explains their economic content. Thus equilibrium existence and continuation values that change choices do not alone imply the stronger conditions of Theorem D.2 on a basin ending at one.

A local form of the conditions of the high-basin theorem, and no selection of a limit. *(From after Theorem D.2.)*

The same conclusions (i)–(iii) also follow from three conditions stated at the equilibrium pair alone (a Lipschitz modulus of the equilibrium sender value, one comparison per sub-pivot sanction against the pivot at the equilibrium policy, and interval dominance of the substituted objective) with no value sets and no value class (`standardCompliance_persistence_generalF_local_star`); P1–P3 and P5 supply those conditions at the pivot, and the local form constructs no equilibrium. The first two conditions alone give conclusions (i) and (ii), a non-decreasing state path and the target's optimality along it (`standardCompliance_persistence_generalF_level_star`): interval dominance is read only for (iii). The package does not select a limit: `persistence_package_not_unique_steady_state` fills every field of P1–P3 and P5 at an economy whose equilibrium closed

loop is the identity on the basin, so every basin state is steady and no state is a common limit of the basin's paths; attraction is the separate content of Proposition I.1 and Proposition I.3.

What the derived interval does not say. *(From after Proposition I.3.)*

The result neither places x_s in the interior nor describes starts above it. At the example economy $x_s = 1$, so the derived interval is the whole high basin and the proposition specializes to the absorbing-corner case. The three results therefore have distinct roles: Proposition D.4 proves convergence under the level condition of the high-basin theorem, Proposition I.1 selects an extremal limit, and Proposition I.3 obtains a common limit on a derived interval.

Corners at the dual-basin economy. *(From after Proposition D.7.)*

At this example economy the target chooses full non-compliance and the realized sanction lies at a corner on each basin, so both displayed long-run outcomes are corners. These are features of the construction, not of the general selection result, whose steady states may be interior. A sanction floor extending to the top state forces the sanction rule to equal one at the top state, $R(1) = 1$; that the rule sits at the ceiling throughout the high basin is a feature of this economy's payoffs, not a consequence of the floor (an equilibrium rule $(x + 1)/2$ satisfies such a floor with $R(x) < 1$ below the top). This absorbing corner is a specialization of the general result rather than the state-dependent ratchet motivating the mechanism.

The pathwise ordering at the Eaton–Engers corner economy. *(From after Corollary I.6.)*

At the Eaton–Engers corner economy, the myopic rule is

$$\sigma^{\mathrm{M}}\big(x, \alpha^{\mathrm{M}}(x)\big) = \max\Big\{0, x - \tfrac{1}{5}\Big\}, \tag{H.5}$$

so a high basin is not preserved. The pathwise ordering still holds there for the simpler reason that the dynamic sanction is one and every myopic sanction is feasible and hence no greater than one (`eeCert_pathwise_dominance`). The proof of the corollary, which needs basin preservation and so does not apply here, is in Appendix I.7.

The top of the state space, interior ratchets, and the open primitive problem. *(From Section 5.4.)*

The transition structure also creates an economic trade-off at the top of the state space. A ratchet floor on $[\bar{x}, 1]$ requires the realized sanction at state one to equal one. The diagonal condition then makes the top state absorbing. If the sender response is pinned at the corner across nearby target actions, those actions have the same successor state, so the target's continuation term cancels from its choice. The target policy at the boundary must then be justified by current payoffs. Propositions H.5 and H.7 state the kinematic and incentive parts of this boundary.

An interior sanction at the top state is therefore incompatible with that particular floor, but not with persistence in every form. A fixed level floor preserves the high basin without guaranteeing a monotone state path (Proposition H.10). The purified construction on the full state space goes further: its political state and realized sanction rise strictly on an interior sub-basin and converge to an interior steady state. The ratchet cannot be extended to the ceiling because its right-hand endpoint must be a fixed point. Off-corner ratcheting is thus possible, but it terminates before the absorbing top boundary.

A strict ratchet is needed neither for the political state to settle nor for the limit to be an extremal steady state; what it buys is a common long-run outcome for a whole interval, which on

an invariant interval is equivalent to uniqueness of the steady state there. Deriving that uniqueness from payoff and transition primitives on a pre-specified interval is the open primitive problem; Appendix H.8 records why the corner proof cannot simply be relaxed and what a sufficient condition must imply.

The North Korea mapping and what it leaves out. *(From Section 5, predictions and limitations.)*

The North Korea application is interpretive. One possible mapping is that tighter control raises the influence of actors who benefit from isolation (the model's political state), while resistance to opening that would weaken such control corresponds to the target's lower compliance. Haggard and Noland (2017) and United Nations Security Council (2024) document broader political control and resistance to opening, not the model's state or the sender's valuation of it directly. The same sources also document illicit foreign-exchange channels, sanctions evasion, and international enforcement and coordination problems, none of which is represented in the one-state baseline. The model isolates one domestic political feedback that can coexist with those mechanisms; it does not attribute North Korea's outcomes to the political-state mechanism alone.

The theorem on a finite temporary shock: the comparison in (iii), two boundaries, and an instantiation. *(From after Theorem D.9.)*

The comparison in (iii) is with the path under the sanction held at h: the path under the shock need not be monotone (an early sanction can push the state above a later one), but it dominates the held path at every date through T, and the held path is monotone and gains at least $\eta_h(\bar{x})$ per period below $\bar{x}$. Two boundaries delimit the statement, both machine-checked as refutations. A floor *equal* to the pivot need not enter in finite time: under the partial-adjustment law, a sanction held at $\bar{x} > 0$ from state 0 stays strictly below $\bar{x}$ at every date, so held-sanction ascent, which forces $\bar{x} < h$, cannot be weakened to $h = \bar{x}$. And there is no uniformly small duration: for every preassigned T and pivot in $(0, 1]$ there is a partial-adjustment law under which T periods of full sanction from 0 end below the pivot. The theorem also says nothing about the dates before T, at which the sanction is set exogenously and not chosen, and nothing about whether the shock was worth imposing.

Instantiation. At the example economy of the general-transition certificate route with a non-affine transition ($F(x, s) = x/2+s/2+(s-x)^2(1-s)/100$ on the square, pivot $1/10$), the constructed equilibrium's sanction rule is at or above $1/10$ on $[1/10, 1]$, and (ii), the maximal-shock case of (iii), and the general case of (iii) are all discharged (declarations in Appendix B).

Entry and exit durations compared. *(From after Theorem D.10.)*

The entry and exit durations are computed from different sections of the same transition, $F(\cdot, 1)$ below the threshold and $F(\cdot, 0)$ above it, so they need not coincide. Their difference is an asymmetry between two sufficient duration bounds; it is a difference of durations, not a band in the state space, and neither bound is a required duration. Under the partial-adjustment law they collapse to $\eta_\delta(\bar{x}) = \delta(1 - \bar{x})$ and $\eta^R_\delta(\bar{x}) = \delta\bar{x}$, so before rounding the exit bound exceeds the entry bound when $\bar{x} < \frac{1}{2}$ and falls below it when $\bar{x} > \frac{1}{2}$: the asymmetry is governed by which side of the midpoint the threshold sits on, not by δ, which cancels. (As in Corollary I.4, the comparison of the *integer* durations remains weak, since distinct bounds can share a ceiling; the specialization to the partial-adjustment law is machine-checked as `affine_entryBound_lt_exitBound_iff`, with `affine_fullReliefMinDrift_eq` for η^R_δ.) For a general transition under strict adjustment

and $0 < \bar{x} < 1$, the exit bound exceeds the entry bound if and only if $\bar{x}\,\eta_F^R(\bar{x}) < (1-\bar{x})\,\eta_F(\bar{x})$ (`entryBound_lt_exitBound_iff`). If the transition is symmetric under reflection of the square, $F(1-x, 1-s) = 1 - F(x, s)$, so that relief at x mirrors a shock at $1-x$, then $\eta_F^R(\bar{x}) = \eta_F(1-\bar{x})$, the exit duration at $\bar{x}$ is the entry duration at $1-\bar{x}$, and the exit bound exceeds the entry bound exactly when $\bar{x} < \frac{1}{2}$, the two coinciding at $\frac{1}{2}$ (`fullReliefMinDrift_eq_reflect`, `fullReliefExitHorizon_eq_reflect`, `entryBound_lt_exitBound_iff_lt_half`). The partial-adjustment law is reflection-symmetric, so its comparison above is a corollary. The class is larger: $F(x, s) = (x+s)/2 + (s-x)^3/10$, clipped off the square, is reflection-symmetric, adjusts strictly and is affine at no rate (`cubicReflectF_ne_affineF`, `reflectParams_entryBound_lt_exitBound_iff`). Both statements compare two sufficient durations of temporary shocks, not required durations or equilibrium behavior.

Why the class under the lifting condition and not corner economies. *(From the opening of Section 5.1.)*

(Several corner economies of Appendix I.6 below have the maximal sanction along equilibrium play. A corner on the equilibrium path does not make the sanction insensitive to compliance deviations; what the class under the lifting condition adds is an interior sanction at non-compliance and relief at full compliance, which is what lets the two forward-looking incentives be separated.)

The target-only case: the rate, and the sender's state–sanction cross term. *(From after Proposition B.2.)*

On the quadratic–linear family under interiority, which keeps the vertex $(u_0 + rx)/q$ below full sanction so that the myopic sanction rule is the vertex (without interiority the rule is the vertex clipped at full sanction), the rate in (iii) is a centered error bound toward the steady state $u_0/(q-r)$, obtained because the vertex rule is affine about it with slope r/q and the lower bound on the transition's sanction slope moves the state at least the fraction d_{lo} of the way; it is not a pairwise contraction of the closed loop (Appendix B). The vertex $(u_0 + rx)/q$, the rule under interiority, rises strictly at every positive state with the sender's state–sanction cross term r, and so does the steady state $u_0/(q-r)$ when the economy with the larger cross term satisfies interiority, which keeps $r < q$ there (Appendix B): in the interest-group interpretation of Kaempfer and Lowenberg (1988), a sender whose pro-sanction constituency responds more to the target's entrenchment imposes more, and entrenches the target further when the target maintains non-compliance. The comparison is across economies that share u_0 and q; the analogue when the sender is also forward-looking is not a theorem, because the constructed sender value would have to be pinned down across economies.

Continuity of the sanction rule. *(From after Corollary I.2.)*

Continuity of R_E is the mathematical requirement, and it is what Proposition I.1 assumes. One economic condition that supplies it is uniqueness of the maximizer of J_S at every physical state and compliance level: a single-valued argmax has a continuous selection, and the equilibrium already carries continuity of the target policy. We state the proposition with continuity directly so that the requirement is not confused with that particular sufficient condition.

Amplification is not claimed. *(From Appendix I.5.)*

Whether they *amplify* one another is a separate question. Theorem 12 settles its arithmetic form on the class under the lifting condition: the common-state cross-difference of the four sanction rules is strictly positive (`four_cell_rule_difference_at_state'`, Section 5.1), but that is an identity

of the constructed rules, not strategic complementarity, and no result below signs an interaction primitive.

Off the corner, in the discussion. *(From Section 5.4.)*

Off the corner, the Schur condition yields existence and uniqueness within specified invariant sets of value functions and an atomless transition component yields a purified equilibrium under stated regularity, while the deterministic joint Bellman operator need not contract or preserve order; deterministic off-corner existence is therefore established at example economies rather than for the full primitive class (Appendix H.8).

H.8 Scope notes: off-corner existence and the open primitive problem

The two passages below restrict conclusions that a reader of Section 5.4 might otherwise draw.

The distinction matters most off the corner. The Schur condition yields existence and uniqueness within specified invariant sets of value functions, and an atomless transition component yields a purified off-corner equilibrium under the stated regularity conditions. The deterministic joint Bellman operator, however, need not be a contraction or preserve order. This is consistent with the non-existence examples for stationary feedback Stackelberg equilibrium in finite-state games documented by Bucarey López et al. (2023). Accordingly, deterministic off-corner existence is established at example economies rather than for the full primitive class.

A strict ratchet is needed neither for the political state to settle nor for the limit to be shown to be an extremal steady state; order selection delivers both without it. What a strict condition buys is a *common* long-run outcome for a whole interval, and as Section 3 records, that is not a separate requirement but the same one written pointwise: for a continuous, non-decreasing closed loop mapping an interval into itself, a common limit from every start, uniqueness of the steady state there, and a strictly signed closed loop are equivalent. The monotonicity and continuity are not decorative: $G(x) = 1-x$ has a unique fixed point on $[0, 1]$ and the sign pattern, and almost every orbit alternates (`nonunique_fixedPoint_breaks_named_limit`, `monotone_orbit_tendsto_of_unique`).

Uniqueness inside the claimed interval is necessary. Local isolation of a candidate limit is insufficient because a second steady state elsewhere retains its own path. Proposition I.3 obtains uniqueness by ending the interval at the first steady state. The remaining question concerns an interval fixed before the equilibrium is solved.

For a pre-specified state interval (the whole high basin, for example), monotonicity of the sanction rule and the weak level condition do not exclude a second steady state. A primitive theorem covering such an interval would need to construct the joint equilibrium, derive a sanction level floor, and establish the direction of the sanction rule after substituting the target's endogenous policy. Monotone-argmax results can supply direction but not level (Milgrom and Shannon, 1994; Quah and Strulovici, 2009); a monotone decision rule also does not imply that the realized sanction at x is at least x, which is the condition needed for an upward state path. The corner proof cannot simply be relaxed, because the condition that removes cross-value feedback is also what pins the sender's response at the ceiling. What the equivalence adds is a target: any sufficient condition must imply uniqueness of the basin steady state. The open primitive problem is therefore to derive that uniqueness from payoff and transition primitives on a pre-specified interval.

Four scope examples. *Tied stage maximizers.* With stage payoff $(3/2)x - \frac{1}{2} - s^2(1-s)^2$, sender value $2y$, the affine transition at rate $\frac{1}{2}$ and $\beta_S = \frac{1}{2}$, the stage maximizers at state 0 are $\{0, 1\}$, the sender's continuation value at the successor state is $\Gamma(s) = s$, the stage loss measured from the stage

maximizer 0 is outweighed at $s' = \frac{1}{2}$ by the discounted continuation gain ($\frac{1}{16} < \frac{1}{4}$), and the forward-looking objective $-s^2(1-s)^2 + s/2$ has the unique maximizer 1: a comparison measured from a non-greatest stage maximizer orders only that maximizer (`tied_stage_maximizers_counterexample`).

An endpoint floor is not a whole-basin corner. The rule $R(x) = (x+1)/2$ satisfies $R(x) \geq x$ and $R(1) = 1$ with $R(x) < 1$ below the top, and is realized by an equilibrium with $U_S = -(s-(x+1)/2)^2$, $U_T = -a^2$, both values zero and $\alpha \equiv 0$; a floor reaching the top state forces $R(1) = 1$ and nothing more (`endpoint_floor_not_whole_basin_corner`: the equilibrium is constructed, both Bellman equations verified, and the sanction rule shown to be the unique maximizer of the sender's objective).

Full compliance in the sender-only case and *the failed entrenchment-condition converse*: the two economies in the paragraph after Proposition B.3 (`exists_sender_only_full_compliance`, `loss_offset_converse_counterexample`; coefficients and values in Appendix B). A third economy shows that the lifting condition at equality is also compatible with strictly *partial* compliance: with $(q, u_0, k, r, w, \ell, c, n, v) = (3, 1, 2, \frac{1}{2}, 1, 1, 4, 0, 30)$, the affine transition at rate $\frac{1}{2}$ and both discounts $\frac{1}{2}$, $A = 2 = k$ and the zero level is $a^*(x) = \frac{2}{3} + \frac{x}{4}$ at every state (`exists_equality_placabilit y_partial_compliance`). Equality in the lifting condition $A \leq k$ therefore characterizes nothing about the level.

H.9 Further exposition of the results

None of the passages below is a theorem statement or a hypothesis of a main-text result; each qualifies or interprets one. Each is headed by what it is and where it stood. The stage-complementarity route, which is a hypothesis block, moved on the author's word to Appendix G.1.

The sender's Lipschitz constant and the transition-adjusted slope condition, in full. *(From Appendix D.1; the display* (D.1) *stays in the main text and is not repeated with its label here.)*

Let $L^{\alpha}_{S,x}$ bound the inherited-state variation of the sender's *composite* stage payoff along a continuous target policy:

$$|U_S(x_2, \alpha(x_2), s) - U_S(x_1, \alpha(x_1), s)| \leq L^{\alpha}_{S,x}|x_2 - x_1|.$$

Under the optional transition regularity (F6-Q), denote the sender value's envelope Lipschitz bound by K_S:

$$K_S = \frac{L^{\alpha}_{S,x}}{1 - \beta_S \gamma_F}. \tag{H.6}$$

This is the standard value-function Lipschitz recursion (Stokey et al., 1989; Bertsekas and Tsitsiklis, 1996). The corresponding target bound uses the inherited-state modulus of the target's *closed-loop* successor map, which need not equal γ_F because the sender response also moves with the state. The target constant, selector moduli, curvature constants, and the distinction between primitives and automatically derived intermediates are collected in Appendix N. Under the partial-adjustment law, (H.6) specializes to $L^{\alpha}_{S,x}/[1 - \beta_S(1-\delta)]$.

The modulus γ_F does not bound the transition's response to the sanction. For sanction comparisons the exact general-F continuation allowance is instead

$$\beta_S K_S \big|F(x, s_2) - F(x, s_1)\big|. \tag{H.7}$$

Accordingly, the cardinal general-F discharger used below is the transition-adjusted slope condition

$$U_S(x, a, s_2) - U_S(x, a, s_1) > \beta_S K_S \big|F(x, s_2) - F(x, s_1)\big|, \qquad s_1 < s_2, \tag{H.8}$$

on the range used below. It requires no separate Lipschitz modulus of F in its sanction argument. Under the partial-adjustment law, a stage lower slope m_S makes (D.1) follow from $\beta_S K_S \delta < m_S$.

What (F6-Q) is and is not. *(From Section 2, after* (F6-Q)*.)*

Condition (F6-Q) is not part of the qualitative baseline (F1)–(F5). It is stated on the feasible square, and that is the form the formalization consumes (Appendix B); the global extension of the transition described above preserves a square-stated modulus on the square but not outside it, so (F6-Q) is not to be read as a global Lipschitz bound. It is the quantitative statement that, holding today's sanction fixed, inherited political differences are transmitted at most at rate γ_F; discounting must dominate that persistence. Under (J.2), $\gamma_F = 1-\delta$. Thus γ_F is the inherited-state persistence coefficient, not the sanction pass-through coefficient δ.

Why the derived interval has one steady state, and when it is degenerate. *(From Section 3, after Proposition I.3.)*

Uniqueness on $[\bar{x}, x_s]$ holds by construction because no state below the least steady state can itself be steady. The interval may nevertheless be degenerate: if $G_E(\bar{x}) = \bar{x}$, then $x_s = \bar{x}$. Positive length requires strict motion at the pivot. The example economy in Appendix I.6 satisfies this requirement because its sanction rule equals one against a pivot of 19/20.

What the three long-run results say about sanctions regimes, in full. *(From Section 3, the paragraph of that name.)*

Proposition I.1 says the inherited political state decides which such regime the economy reaches, and decides it at date zero: if the equilibrium sanction at the inherited state already exceeds that state, entrenchment builds and the economy settles at the *lowest* self-sustaining regime above where it began; if the sanction falls short, entrenchment erodes to the *highest* regime below. Where several regimes are reachable, history selects among them, and it selects within a single equilibrium: no change of expectations and no coordination on a different continuation is involved. The mechanism is path dependence, not indeterminacy. Whether several are reachable is a property of the economy (Corollary D.6 exhibits one where they are), and where only one is, the proposition says every start reaches it.

Proposition I.3 says more on a narrower range. From any state up to the first self-sustaining regime above the pivot, every path reaches that regime. History selects across regimes but not within an interval containing only one.

What none of the three delivers is a rate. They say where a sanctions regime ends up, not how long it takes to get there, and nothing in the argument bounds the transition. A regime that is self-sustaining in the limit may be reached only asymptotically, so the results speak to durability rather than to the period over which relief becomes implausible.

The derived dividing line: what it determines and the three kinds of threshold. *(From Appendix D.4, after Proposition D.7.)*

The two convergence regions therefore extend strictly on both sides: from $[0, \bar{x}_L]$ out to $[0, x_s^-)$, and from $[\bar{x}_H, 1]$ down to $(x_s^+, 1]$. Their endpoints are derived from the equilibrium. The interval $[x_s^-, x_s^+]$ need not collapse to a point: a continuous map can fix a whole subinterval, and multiple steady states rule out a common limit across it. Whether the two bounds coincide in the dual-basin economy remains undetermined. Continuity is essential for obtaining a middle steady state and is supplied there by uniqueness of the sender's dynamic argmax.

The notation distinguishes three objects. The stage-payoff tipping states $\bar{x}_T$ and $\bar{x}_S$ are infima of payoff flip sets. The basin boundaries $\bar{x}_L$ and $\bar{x}_H$ are inputs to the conditions for persistence and for convergence to zero. Only x_s^- and x_s^+ are steady states derived from the closed loop.

Strict comparison at a fixed target policy: what it needs and what it does not say. *(From Section 3.2, after Proposition B.4.)*

No concavity, differentiability, uniqueness or positive discount enters, and Γ need not be monotone: one comparison per lower sanction, $\Gamma(d) \leq \Gamma(m)$ for every feasible $d \leq m$, suffices (`all_stage_lt_dynamic_of_greatest_stage_improvement_of_wedge_le`), a non-decreasing Γ being one way to supply it; the strict increase of Γ that the first formalization assumed is in any case derived from primitives (sanctions build entrenchment and the sender values it). The stage loss must be measured from the greatest stage maximizer: measured from a lower tied one it orders only that one, and an economy with stage maximizers $\{0, 1\}$, the stage loss measured from 0, and the maximizer 1 of J_S shows the all-maximizer conclusion fails (Appendix B). The proposition orders the two sets of maximizers and says nothing about interiority: a strict gap is consistent with both sanctions at corners. It is discharged at the example economies as the property that no maximizer of J_S is stage-optimal (Appendix B).

The pathwise ordering: the scope of the corollary. *(From Section 3.2, after Corollary I.6.)*

The corollary is stated at the scope of the certificate theorem that proves it (Appendix B): the statewise inequality comes from the certificate's myopic ceiling against the sender's floor, not from the compliance ordering $\alpha \leq \alpha^{\mathrm{M}}$. It is the corollary's display at the certificate's own policies; it does not by itself compare sanctions at two different, endogenously evolving states, and the path conclusion uses that both paths stay in the basin, the myopic one by (I.1) and the forward-looking one by the certificate's floor $R_E(x) \geq \max\{x, \bar{s}\}$ (Step 1 of Appendix I.7). The extra hypothesis is substantive because the corollary compares different state paths. The Eaton–Engers corner economy, where the myopic rule does not preserve a high basin and the pathwise ordering holds for a simpler reason, is in Appendix H.7.

Adjustment speed and sender patience in the comparative statics of lock-in. *(From Appendix I.8.)*

Adjustment speed has a different role. For any equilibrium-derived sanction rule σ_{real}, strict adjustment gives

$$F\big(x, \sigma_{\mathrm{real}}(x)\big) = x \quad \Longleftrightarrow \quad \sigma_{\mathrm{real}}(x) = x.$$

Holding the rule fixed, two positive affine adjustment rates therefore have the same steady-state set. A higher rate changes the transition path and convergence time, but it changes a steady state only through its effect on equilibrium policies or equilibrium selection.

At a fixed continuation value and target policy, greater sender patience weakly raises the dynamic best response whenever the sender value is non-decreasing over reachable successor states. The corner economies also permit a total equilibrium comparison. There, policies remain fixed, the sender's absorbing-state payoff is positive, so its value is $U_S(1, 0, 1)/(1 - \beta_S)$. Greater sender patience therefore strictly raises its lock-in value, while the target value is unchanged because the sender action remains pinned. Proposition H.4 gives the exact direct-plus-continuation decomposition and Table 5 reports the parameter intervals over which the comparison applies.

The constants of the contracting closed loop. *(From Appendix D.8, after Proposition D.11.)*

The constants 14/23 and 35/47 are evaluated upper bounds, the values of the proved formula at those members, and not measured convergence rates or the sharp Lipschitz constants of G. The formula rests on the sharp form of the argmax stability lemma: a μ-strongly concave objective moves its maximizer at most L/μ under an L-Lipschitz perturbation; the factor-four form of the first formalization gave 43/46 at the affine member and 277/235 at the non-affine one, where it was inconclusive.

"Finite" means long enough for the economy, not few. *(From Appendix D.6, after Theorem D.8.)*

Nor is the duration small in any uniform sense: it is a sufficient bound that depends on the economy through $\eta_F(\bar{x})$, and for any preassigned number of periods there is a partial-adjustment law slow enough that a temporary shock of that length ends below the threshold. "Finite" means long enough for the economy, not few.

The relief duration: sufficiency, institutional designs, and the two durations. *(From Appendix D.8, after Theorem D.10.)*

The bound $\widehat{N}_F^R(\bar{x})$ is a sufficient duration for a temporary suspension of sanctions. It is not a minimal duration, and the theorem does not say that the equilibrium would choose such a shock. Sunset clauses, mandatory review dates, and explicit termination requirements are distinct institutional designs; the model speaks to them only insofar as a design commits to a zero-sanction path of the stated length, which none of them does automatically, and it does not establish their optimality or their effects on the governments' incentives. The entry and exit durations are computed from different sections of the same transition and need not coincide; their comparison, with its specialization to the partial-adjustment law, is in Appendix H.7.

Two limits of the post-relief statement. *(From Appendix D.8, after Proposition D.12.)*

A second limit is at the threshold itself: relief delivers $x \leq \bar{x}$ while the high region of Theorem D.8 is $[\bar{x}, 1]$, so the two regions share the point $\bar{x}$, and reaching it does not by itself select a distinct low outcome; a same-equilibrium contrast of two such outcomes would need disjoint basins or strict entry into a low attraction region, which this section does not claim. And the duration is conservative, being built from a worst-case drift over the whole basin rather than along the realized path.

What the entry theorem and its corollary do and do not say. *(From Appendix D.8, before Proposition K.1; it refers to Theorem D.8 and Corollary I.4.)*

The theorem combines exogenous entry with endogenous persistence. It does not say that equilibrium sanctions move every initial state into the high basin. It says that once a temporary sanction shock carries the state across the threshold, the equilibrium mechanism prevents the high-sanction regime from unwinding. The corollary compares transition technologies under the same exogenously set sanction; it is not a comparative static of the equilibrium sanction itself.

Asymmetric adjustment, inaction bands and path dependence. *(From Section 2, the primitives.)*

State-dependent and asymmetric adjustment rates remain allowed when they satisfy (F1)–(F5). Those conditions exclude an inaction band with $F(x, s) = x$ when $s \neq x$; under a fixed sanction s, the unique rest point of $F(\cdot, s)$ is $x = s$. Dual-basin path dependence instead requires the equilibrium sanction rule to cross the diagonal more than once. The constructions below exhibit invariant

directional basins; they do not establish multiple crossings as a general fact. Compliance affects the state only through its effect on the sender's equilibrium sanction. Asymmetric adjustment connects this mechanism to political path dependence (Pierson, 2000) without imposing an inaction region.

A1: direction, and why direction does not determine level. *(From Appendix D.1, after Assumption 1.)*

A1 is ordinal: after the endogenous target policy is substituted, it makes the sender's dynamic best response non-decreasing in the state. Single crossing is a stronger sufficient condition (Milgrom and Shannon, 1994); interval dominance also covers the non-monotone-continuation case in Remark G.2. The premise of Theorem 4 is different: it orders the sender value over successor states while holding the target policy fixed.

Direction does not determine level. To keep the sender response above $\max\{x, \bar{x}\}$, the stage gain from raising sanctions must dominate the largest continuation loss allowed by the value set. This is exactly the role of (D.1); under the partial-adjustment law it is implied by $\beta_S K_S \delta < m_S$. The proof of the high-basin theorem keeps the direction and level arguments separate (Appendices C and G).

A3: what stage concavity does and does not deliver. *(From Appendix D.1, after Assumption 3.)*

A3 imposes diminishing returns to sanctions, as in standard public-choice models (Kaempfer and Lowenberg, 1988; Eaton and Engers, 1992). It makes the myopic best response unique. It does not ensure uniqueness of the maximizer of J_S because the continuation term can offset stage curvature. The smooth existence route therefore imposes curvature on the complete dynamic objective where needed (Appendix N). A3 restricts the change in the stage slope, not its sign.

A3 is not what carries the substitution between compliance and the sanction: decreasing differences of U_S in (s, a) order the greatest stage maximizer with no concavity and no uniqueness (Appendix H.7).

The increasing-differences condition, and why both forward-looking incentives change behavior together. *(From Section 5.1, after Proposition B.1.)*

The increasing-differences condition says that the stage complementarity between the state and the sanction covers two forward-looking forces against a monotone response: saturation of the continuation, and cross-effectiveness of the transition. Under the partial-adjustment law, and for every additively separable transition with convex state sections, both vanish, and with a sender payoff convex in the state the condition reads $0 \leq r_{\mathrm{lo}}$. The quadratic–linear family is inhabited at an affine member and at a member whose transition is not affine at any rate; at both the condition is empty. Since τ_x, M_{cx} and M_{mix} are data of the two-sided extension of the class (Appendix C.4, panel A0), the condition, and every clause that assumes it, is stated for economies in that extension. The target's forward-looking incentive changes behavior here *because* the sender is responsive: the substitution rate covers the sender's whole dynamic marginal return to sanctioning, the sanction responds to compliance, and both forward-looking incentives change behavior together. The entrenchment condition does double duty, making the forward-looking target maintain noncompliance and making a sanction increase induced by the state–sanction complementarity a net gain for the target's value slope.

The sender-only case: the per-unit relief rate and the forward-looking relief condition. *(From Section 5.1, after Proposition B.3.)*

The forward-looking sender is more demanding than the myopic one (it still sanctions at $a^{00}(x)$, because of the foresight premium), so the myopic target pays more for relief; but it still buys it. Below a^{01}, at distance $\Delta = a^{01} - a$, the sanction is at least $k_{\mathrm{lo}}\Delta/M_{\mathrm{tot}}$, where $M_{\mathrm{tot}} = \bar{Q} + \beta_S(\sigma_C d_{\mathrm{hi}}^2 + K_S M_{\mathrm{cs}})$ bounds the curvature of the sender's dynamic objective in the sanction, the stage curvature ceiling $\bar{Q}$ plus the discounted continuation curvature through the transition's moduli (Appendix C.4). This bound is a per-unit relief rate, from the semiconvexity of the sender's objective read as a quadratic support at its maximizer, and the forward-looking relief condition (RL), $\ell_{\mathrm{hi}} M_{\mathrm{tot}} < k_{\mathrm{lo}} p_{\mathrm{lo}}$ ($\ell M_{\mathrm{tot}} < k(c-n)$ on the quadratic–linear family), says that the sanction pain avoided per unit of compliance, at the lowest price p_{lo}, exceeds the unit compliance cost: a strict inequality among the class's rates and nothing else.

The high-basin theorem: how the proof is organized and what each hypothesis does. *(From Section 3, after Theorem D.2.)*

The proof separates the direction and level of the sender's response. A1 makes the dynamic best response non-decreasing in the inherited state after the target policy has been substituted. The transition-adjusted slope condition in P5 prevents that response from falling below the basin floor. Together these conditions imply $s_t^E \geq x_t^E$; transition monotonicity then gives $x_{t+1}^E \geq x_t^E$ and preserves the basin. Under the partial-adjustment law, a familiar sufficient form of P5 is $\beta_S K_S \delta < m_S$, where m_S is a lower stage-payoff slope. A local form with no value sets, and an economy showing that the package selects no limit, are in Appendix H.7.

The target combined slope condition P4 is logically separate. It yields conclusion (vi), which verifies that the target's complete objective favors higher sanctions along the relevant range. Conclusions (i)–(iii) continue to hold under P1–P3 and P5 alone, and A1 is read only for (iii): (i), (ii) and (iv) hold under P1, P2 and P5. Conclusions (iv) and (v) record that both realized actions remain equilibrium-optimal. The proof is in Appendix C.

The same equilibrium supplies these conclusions for every initial state in the basin; its construction does not depend on x_0. The theorem is conditional on the stated equilibrium and the package P1–P5. Theorem D.1 and Proposition I.5 provide unconditional applications under narrower primitive conditions.

The long-run lock-in level: the argument in brief. *(From Section 3, after Proposition D.4.)*

The state sequence is increasing and bounded. Compactness and continuity then make every convergent sanction subsequence optimal at the state limit and consistent with its transition. Strict adjustment implies that any such sanction limit equals the state limit, so the entire sanction sequence converges there. Under the partial-adjustment law this identity is immediate from

$$s_t^E = \frac{x_{t+1}^E - (1-\delta)x_t^E}{\delta} \longrightarrow x^*.$$

Entry and exit as opposite-direction crossings. *(From the opening of Appendix D.8.)*

The entry result has a counterpart on the exit side. The two are opposite-direction threshold crossings under temporary shocks; whether one equilibrium sustains both basins is a further question that neither of them settles in general, and Proposition K.1 at the end of this section is the one economy at which it is settled. Theorem D.8 spends only the ascent half of (F5), in which sanctions above the inherited state strictly raise it, and only at the full sanction below the basin. The exit side spends only the descent half, in which sanctions below the inherited state strictly lower it, and only at the zero sanction inside the basin.

Three limits of the two-sided statement. *(From Appendix D.8, after Proposition K.1.)*

This is the same-equilibrium two-sided statement that Appendix D.8 declines to make in general: at this economy one equilibrium sustains both basins, and temporary shocks move between them. Three limits travel with it. The transition is affine. The two basins do not exhaust the state space, and nothing is said about the interval between them. And the statement is an existence certificate: it shows the two level conditions of the general composition can hold at one equilibrium, and nothing about other economies.

Predictions inside a high basin. *(From Section 5, predictions and limitations.)*

Inside a high basin satisfying the theorem's conditions, the model predicts non-decreasing realized sanctions, a weakly positive foresight premium of the sender, relative to a myopic sender at a common target policy, when the sender value is non-decreasing, and persistence after a sufficiently long temporary sanction shock. In the dual-basin construction, initial political conditions select between convergence to zero under low sanctions and lock-in under high sanctions. Faster state adjustment shortens the entry duration under the partial-adjustment law and changes the transition path; holding the sanction rule fixed, it does not change the steady-state set.

The conclusion's two paragraphs on policy paths and the long run. *(From Section 6.)*

The mechanism has distinct implications for policy paths. A forward-looking sender sanctions at least as heavily as a myopic sender at the same target policy when the political state raises its value. A temporary sanction shock can move the economy into the high basin, after which equilibrium behavior keeps the state there. When low and high basins coexist under one equilibrium, initial political conditions select between convergence to zero and lock-in.

The long-run results also determine where a monotone equilibrium path settles. Its initial direction selects the nearest steady state on that side, and every start between the basin pivot and the first steady state above it converges to that same state. This common limit is derived from the equilibrium rule; it need not be interior, and convergence need not have a uniform rate.

I Further results on persistence and the long run

This section collects results on persistence and the long run that the main text does not need: the order-selection block, the transition-strength corollary and a pointer to the two-sided statement; and, from the sections on the general results, post-entry non-termination and the selection boundary, the economic mechanism behind the basins, persistence from primitives, the pathwise ordering under basin preservation and the comparative statics of lock-in. Each statement is the one the formalization map cites.

I.1 The order-selection block (Appendix I.5)

Proposition I.1 (Order-selected steady state)**.** *Let $\mathcal{B}$ be preserved by the equilibrium closed loop $G_E(x) = F(x, R_E(x))$, and suppose that the closed loop is non-decreasing on $\mathcal{B}$ (a higher state today never leads to a lower state tomorrow, which holds whenever the equilibrium sanction rule R_E is non-decreasing), and that R_E is* continuous *on $\mathcal{B}$. Fix a start $x_0 \in \mathcal{B}$.*

(i) If $x_0 \le R_E(x_0)$, the political state path converges to the least *steady state of G_E at or above x_0.*

(ii) If $R_E(x_0) \le x_0$, it converges to the greatest *steady state of G_E at or below x_0.*

In either case the realized sanction converges to R_E evaluated at that limit. The two cases are exhaustive, and nothing is assumed about how many steady states $\mathcal{B}$ contains.

Corollary I.2 (Coincidence of the two limits)**.** *If in addition the transition satisfies strict directional adjustment* (F5)*, the realized sanction and the political state converge to the same value.*

Continuity of R_E is the requirement; uniqueness of the maximizer of J_S is one sufficient condition for it (Appendix H.7).

Order selection need not imply attraction of a whole basin. The selected limit may vary with the initial condition. For a monotone continuous closed loop on an invariant interval, every path converges to one common limit if and only if the interval contains a unique steady state; equivalently, G_E lies strictly off the diagonal on either side of that state. The next result obtains uniqueness by deriving the interval from the first steady state rather than imposing an additional condition.

Proposition I.3 (Attraction on a derived interval)**.** *Under the hypotheses of Proposition I.1 (an invariant basin, a non-decreasing closed loop and a continuous sanction rule, with no level condition), let x_s be the* least *steady state of G_E at or above the basin pivot $\bar{x}$. Then every equilibrium path started anywhere in $[\bar{x}, x_s]$ converges to x_s: a single limit, common to every start in that interval. Under strict directional adjustment* (F5) *the realized sanction converges to x_s as well.*

In plain terms: initial conditions determine the direction of movement and select an extremal steady state on that side. Different initial states in the same invariant interval need not converge to a common point: unless the interval runs only as far as the first steady state, in which case they do.

What the three results say about sanctions regimes. A steady state of G_E is a political state at which the sender's optimal sanction reproduces the entrenchment it inherits: escalation has no further return, and a lower sanction would concede entrenchment the sender wants kept. The regime sustains itself without either government changing its behavior.

Proposition I.1 says that the inherited political state selects, at date zero and within a single equilibrium, which self-sustaining regime is reached (path dependence, not indeterminacy), and Proposition I.3 that up to the first such regime above the pivot every start reaches it. None of the three results delivers a rate: they say where a sanctions regime ends up, not how long it takes (Appendix H.9).

I.2 The transition-strength corollary (Section 3.2)

Corollary I.4 (Transition strength and shock duration)**.** *Consider two transition laws F_1 and F_2 satisfying* (F1)–(F4) *and evaluated at the same threshold $0 \le \bar{x}$, below which the first rises under full sanctions, $x < F_1(x,1)$ for $x \in [0, \bar{x}]$ (so $\bar{x} < 1$). If*

$$F_1(x,1) \le F_2(x,1) \qquad \text{for every } x \in [0,\bar{x}],$$

then

$$\eta_{F_1}(\bar{x}) \le \eta_{F_2}(\bar{x}), \qquad \widehat{N}_{F_2}(\bar{x}) \le \widehat{N}_{F_1}(\bar{x}).$$

If the transition inequality is strict throughout $[0,\bar{x}]$, the drift inequality is strict. The duration inequality remains weak because distinct drift bounds can have the same integer ceiling.

Under the affine law,

$$\eta_\delta(\bar{x}) = \delta(1-\bar{x}), \qquad \widehat{N}_\delta(\bar{x}) = \left\lceil \frac{\bar{x}}{\delta(1-\bar{x})} \right\rceil.$$

Hence $\delta_1 \le \delta_2$ implies $\widehat{N}_{\delta_2}(\bar{x}) \le \widehat{N}_{\delta_1}(\bar{x})$.

I.3 The same-equilibrium two-sided statement (Appendix D.8)

The statement, Proposition K.1, is in Appendix K.2, with the definition of the dual-basin economy.

I.4 Post-entry non-termination and the selection boundary

Fix a date T with $x_T^E \in \mathcal{B}$ and restart the stationary equilibrium path from x_T^E. The proof of Theorem D.2 applies to this tail and gives

$$s_{T+n}^E \geq \bar{x}, \qquad s_{T+n}^E \leq s_{T+n+1}^E \quad \text{for every } n \geq 0.$$

For a cutoff $q < \bar{x}$, define

$$\tau_q^E(T) := \inf\{n \geq 0 : s_{T+n}^E \leq q\}, \qquad \inf \varnothing := +\infty.$$

The strict cutoff inequality and the inherited floor make the set empty, so $\tau_q^E(T) = +\infty$. A weak cutoff is insufficient: a path that stays exactly at its floor reaches that weak threshold. Only the floor is used, so the conclusion needs only the level condition, the sanction rule at or above the state on the basin (`equilibriumTop_tailF_no_relief_of_ratchet`); neither A1 nor the target condition enters.

Condition (E4) supplies the optimization content. At every tail date and against every feasible sanction $r \in [0, 1]$,

$$\begin{aligned} &U_S\big(x_{T+n}^E, \alpha(x_{T+n}^E), s_{T+n}^E\big) + \beta_S V_S\big(F(x_{T+n}^E, s_{T+n}^E)\big) \\ &\quad \geq U_S\big(x_{T+n}^E, \alpha(x_{T+n}^E), r\big) + \beta_S V_S\big(F(x_{T+n}^E, r)\big)\,. \end{aligned}$$

Thus non-termination is selected from an action set that includes relief and every reduction of the sanction. Without state-local dynamic-argmax uniqueness, this conclusion is specific to the greatest selection: the greatest maximizer stays above the floor, while another tied maximizer need not. With uniqueness, every maximizer equals the realized action and every other sanction is strictly worse. Lean verifies the tail restart, floor, empty first-passage set, optimality comparison, and uniqueness upgrade in `Sanctions/Theory/LockIn/PostEntryNonTermination.lean`.

I.5 Economic mechanism

Political lock-in combines two forward-looking incentives. A sender that values a more entrenched successor state weighs current sanctions against the entrenchment they leave. For feasible sanctions $s_L < s_H$, the dynamic sender prefers s_H to s_L exactly when

$$\beta_S[V_S(F(x, s_H)) - V_S(F(x, s_L))] > U_S(x, a, s_L) - U_S(x, a, s_H).$$

The discounted continuation gain must exceed the current payoff loss. This is the direct effect of the sender's foresight; it underlies the comparison of dynamic and myopic sanctions at a fixed target policy in Theorem 4.

The target faces the corresponding intertemporal trade-off. More compliance can reduce the current sanction but also erode a political constituency valued by the incumbent. A Bellman-optimal target may therefore comply less than a myopic target. The two incentives point the same way: lower compliance raises the sender's sanction (the indirect effect of the target's lower compliance), and the sender's continuation value raises the sanction relative to its myopic best response. Whether they *amplify* one another is a separate question, and no result below signs an interaction primitive (Appendix H.7).

Formation of the high basin. Under Assumption 4, the target and sender flip sets are upper intervals, and Proposition G.1 defines their infima as political pivots. Above their joint pivot, the sender's incentive supports a sanction floor and the target's current-plus-future objective favors the high-sanction direction. The target condition supplies preference evidence for the latter claim. Basin invariance itself follows from the sender's equilibrium floor and transition monotonicity.

I.6 Primitive applications and scope

The conditions of Theorem D.2 are attained in an economy where continuation values affect observable choices (Theorem J.1, Appendix J.2); its description and its two robustness extensions are in Appendix H.7.

I.6.1 Persistence from primitives

The high-basin theorem (Theorem D.2) starts from an equilibrium and value sets satisfying its conditions. A narrower class, whose economies we call the corner economies, permits both to be derived from payoff and transition primitives. Its transition law satisfies the model's maintained continuity, monotonicity, diagonal, and strict-adjustment conditions, together with a lower bound on its sanction slope and a two-sided bound on the corner section $x \mapsto F(x, 1)$. No upper bound on the sanction slope and no bound on the curvature of F are imposed, and it contains non-affine realized recursions, including members whose curvature in the sanction exceeds half the lower bound on their sanction slope.

Proposition I.5 (Primitive persistence over a general transition)**.** *For every corner economy (an economy of the general-F class above) there exists a stationary feedback Stackelberg equilibrium, lying in invariant sets constructed from the economy's own coefficients, such that from every initial state in the high basin, the political state remains in the basin at every date, the realized sanction never falls below the basin pivot, and the realized sanction sequence is non-decreasing.*

No equilibrium, value function, policy, single-crossing condition, sanction floor, or path property is assumed in this proposition. Those objects are constructed from inequalities among the primitives. The gain in primitive content comes with a precise limitation: the equilibrium sanction is maximal at every state. The result establishes persistence, but its non-decreasing sanction path is flat rather than a state-dependent ratchet.

No curvature bound is needed because the sender's choice is settled by monotonicity rather than by concavity. Every sender value in the constructed invariant set rises with the state at least at a fixed rate, every unit of sanction raises the next state by at least the lower bound on the transition's sanction slope, and this class's implacability condition makes the discounted value of that rise exceed the largest current cost of the unit. The sender's forward-looking objective is therefore strictly increasing in the sanction at every state and at every compliance level the target can choose, and full sanction is its unique maximizer, however curved the transition is. The constructed invariant set for the sender accordingly restricts only the level and the slope of the sender's value. If the corner section is in addition semiconcave with some modulus $M \geq 0$, the equilibrium sender value is semiconcave as well, with a modulus proportional to M; no curvature of F in the sanction is needed for this. This class contains a member whose transition is semiconcave with modulus $1/5$, and with no smaller modulus, both in the sanction and along the corner section; that modulus exceeds half the lower bound on the member's sanction slope, $3/20$, and the proposition holds there (Appendix B).

The off-corner boundary of this result is described in Appendix H.7 and Section 5.4.

I.7 Pathwise ordering under basin preservation

The corollary below compares the myopic equilibrium M with the equilibrium that a local invariant-set certificate of the high-basin route constructs. It is an existence statement about that constructed equilibrium, not a statement about every equilibrium satisfying the hypotheses of Theorem D.2.

Corollary I.6 (Pathwise ordering under basin preservation)**.** *Let the transition follow the general law* (1)*, let $\bar{x}_T, \bar{x}_S \in [0,1]$ with $\bar{x} < 1$, and take the following local invariant-set certificate: invariant sets $\mathcal{C}_T \times \mathcal{C}_S$ of value functions on which the sender's dynamic argmax and the target's argmax are unique, as in Theorem F.1, and on which the joint Bellman operator satisfies the row bounds of Theorem F.1 with non-negative coefficients and $\beta_T + L_{TS} < 1$, $\beta_{S,\mathrm{row}} + L_{ST} < 1$;* (P2) *and* (P3) *of Theorem D.2; and a sanction floor $\bar{s} \in [0,1]$, a slope m_S and a rate d such that, for every value pair and every $x \in [\bar{x}_S, 1]$, the sender's stage payoff at the induced target policy rises at rate at least m_S on $[0, \max\{x, \bar{s}\}]$, the transition's sanction slope is at most d there, and $\beta_S K_S d < m_S$. This rate form implies the pivot-star slope condition* (P5)*; the target condition* (P4) *is not needed. Suppose in addition that the sanction rule of the myopic equilibrium satisfies, at every $x \in \mathcal{B}$, the ceiling $\sigma^{\mathrm{M}}(x, \alpha^{\mathrm{M}}(x)) \le \max\{x, \bar{s}\}$, and that it preserves the basin:*

$$\sigma^{\mathrm{M}}(x, \alpha^{\mathrm{M}}(x)) \in \mathcal{B} \qquad \text{for every } x \in \mathcal{B}. \tag{I.1}$$

Then these hypotheses construct an equilibrium $E = (V_T, V_S, \alpha) \in \mathcal{E}$ with $V_T \in \mathcal{C}_T$ and $V_S \in \mathcal{C}_S$ such that, at every $x \in \mathcal{B}$,

$$\sigma^{\mathrm{M}}(x, \alpha^{\mathrm{M}}(x)) \le \sigma^{\mathrm{dyn}}_{V_S}(x, \alpha(x)),$$

and from every common initial state $x_0 \in \mathcal{B}$ the realized sanctions satisfy $s^M_t \le s^E_t$ for every $t \ge 0$.

The corollary is stated at the scope of the certificate theorem that proves it: the statewise inequality compares the myopic ceiling with the sender's floor and uses neither Theorem 4 nor the compliance ordering $\alpha \le \alpha^{\mathrm{M}}$, the certificate reads no condition on how the target ranks sanction levels, and the extra hypothesis (I.1) is substantive because two different state paths are compared (Appendix H.9; the proof is Appendix I.7). Both paths are evolved by the general transition.

Proof of Corollary I.6. The corollary is the certificate theorem `pathwise_dominance_general F`; its hypotheses are the fields of the core certificate `StandardComplianceLocalCoreCertificate`, listed in the statement, and basin preservation (I.1), a separate hypothesis of the theorem. Write $\hat{x} = \max\{x, \bar{s}\}$.

Step 1: the equilibrium, its floor and its direction. Row contraction gives $E = (V_T, V_S, \alpha) \in \mathcal{E}$ with $V_T \in \mathcal{C}_T$, $V_S \in \mathcal{C}_S$ and $\alpha = \alpha_{V_T, V_S}$, which is (P1) by construction (the formal statement exports membership of the invariant sets) (`exists_feedbackStackelbergTop_from_valueClass2_with_membership`). At $x \in \mathcal{B}$ the rate form gives the transition-adjusted slope condition (D.1) on $[0, \hat{x}]$ (`sender_transition_adjusted_chord_of_upperChordOn_lowerSlope`). Together with the K_S-Lipschitz bound (P2) it makes $J_S(x, \alpha(x), \cdot\,; V_S)$ strictly increasing on $[0, \hat{x}]$ (`senderDynObjective_strictMono_of_transition_adjusted_chord_and_V_lipschitz`), as in Lemma C.1(b) with $\bar{s}$ in place of $\bar{x}$. So the dynamic best response is at least $\hat{x}$: $R_E(x) \ge \hat{x} \ge x$ (`SenderPersistenceBasinTop.level_dyn_top`), and since $R_E \le 1$, R_E maps $\mathcal{B}$ into itself. By (P3), R_E is non-decreasing on $\mathcal{B}$ (`dynamicTopRealizedRule_mono_of_IDOFamilyOn`). Both facts are packaged in `exists_equilibrium_senderPersistenceBasinTop_generalF`.

Step 2: statewise. At $x \in \mathcal{B}$ the myopic ceiling and Step 1 give $\sigma^{\mathrm{M}}(x, \alpha^{\mathrm{M}}(x)) \le \hat{x} \le R_E(x) = \sigma^{\mathrm{dyn}}_{V_S}(x, \alpha(x))$ (`ruleDom_of_dynamic_floor_and_myopic_ceiling`).

Step 3: pathwise. Let $R_M(x) = \sigma^{\mathrm{M}}(x, \alpha^{\mathrm{M}}(x))$, and let x_t^M and x_t^E be the paths from $x_0 \in \mathcal{B}$ under R_M and R_E, both evolved by F. Each rule maps $\mathcal{B}$ into itself: R_M by (I.1) and R_E by Step 1. If $y \in \mathcal{B}$ and $R(y) \geq \bar{x}$, then $F(y, R(y)) \geq F(\bar{x}, \bar{x}) = \bar{x}$ by (F2)–(F4), so both paths stay in $\mathcal{B}$. Suppose $x_t^M \leq x_t^E$. Step 2 at x_t^M, monotonicity of R_E on $\mathcal{B}$, (F3) and (F2) give

$$x_{t+1}^M = F\big(x_t^M, R_M(x_t^M)\big) \leq F\big(x_t^M, R_E(x_t^E)\big) \leq F\big(x_t^E, R_E(x_t^E)\big) = x_{t+1}^E.$$

So $x_t^M \leq x_t^E$ at every date (`rulePathF_le_of_rule_le_and_upper_monoOn_basin`), and $s_t^M = R_M(x_t^M) \leq R_E(x_t^M) \leq R_E(x_t^E) = s_t^E$ (`ruleSanctionsF_le_of_rule_le_and_upper_monoOn_basin`).

The statewise inequality compares the myopic ceiling with the sender's floor; it does not go through Theorem 4 or the compliance ordering. Basin preservation is used only in Step 3, where the two paths are at different states. □

I.8 Comparative statics of lock-in

Consider two configurations with the same transition law and initial state. If the second sanction rule is non-decreasing and pointwise no lower than the first on the high basin, transition monotonicity implies

$$x_t^{(1)} \leq x_t^{(2)} \qquad \text{for every } t.$$

When both paths satisfy the conditions of Theorem D.2, their limiting lock-in states have the same order. Thus any primitive or equilibrium change that raises the sanction rule raises the state path, subject to the stated monotonicity and common-transition conditions.

Adjustment speed and patience enter differently. With the sanction rule held fixed, the steady-state set does not depend on a positive affine adjustment rate; and at a fixed continuation value and target policy greater sender patience weakly raises the dynamic best response, the corner economies permitting a total comparison (Proposition H.4, Table 5, Appendix H.9).

Part II
Examples: parametric families and numerical economies

The examples the main text uses

J Parametric families and the example economies in use

This section records the paper's parametric families, the benchmark economy at which the conditions of persistence and shock entry hold, and an economy in the class outside the quadratic–linear family. Further example economies, among them the roster and the mechanism-active P_3 economy, are in Appendix M.

J.1 The parametric examples

The main text names no functional form. Its parametric examples are collected here with the closed forms of its results.

The paper's examples are parametric: a family of payoffs with closed forms, a law of motion, and an interest-group polity.[1] The *quadratic–linear family* is

$$U_S(x,a,s) = -\tfrac{q}{2}s^2 + s\,(u_0 - ka + rx) + wx, \qquad U_T(x,a,s) = -\ell a + s\,(nx - c) + vx, \tag{J.1}$$

with $u_0, w, \ell, k, v > 0$, $r, n \geq 0$ and $c > n$, so that a sanction hurts the target at every state. A member is an economy in the class when its sanction cost is curved enough (Assumption 4), and then every bound of the class has a closed form; for instance $\underline{s} = u_0/q$ when $u_0 \leq q$. The cross terms r and n let entrenchment raise the sender's marginal return to sanctioning and soften the target's marginal pain. Compliance neutrality holds automatically, $U_S(x,a,0) = wx$.

The *partial-adjustment law* moves the state a fixed fraction of the way toward the current sanction. At adjustment speed δ,

$$F(x,s) = (1-\delta)x + \delta s, \qquad 0 < \delta < 1, \tag{J.2}$$

which satisfies (F1)–(F5) at every speed. The *interest-group polity* of Appendix L, with two groups in each country, derives both governments' payoffs from a within-period political contest; it is a structured example for the welfare results of Section 4.

Closed forms of the main text's results on the family. For every transition in the class, the myopic benchmark's compliance (Proposition 2) is $a^{00}(x) = (u_0 + rx)/k$; in the target-only case of Theorem 12 the sanction is $(u_0 + rx)/q$ and the state converges to $x^{10}_\infty = u_0/(q-r)$. Under the partial-adjustment law and with $u_0 \leq q$, the entrenchment condition (Assumption 5) reads

$$c \leq \frac{\beta_T\,\delta\,(v + nu_0/q)}{1 - \beta_T(1-\delta)}, \tag{J.3}$$

its strong form (Section 3.4) is (J.3) with $n = 0$, and the premium bound of Section 3.2 is

$$\frac{\beta_S\, w\, \delta}{q\big(1 - \beta_S(1-\delta)\big)}. \tag{J.4}$$

J.2 The benchmark economy with persistence and shock entry

The non-monotone (B2) economy of Table 3 is the first application of Appendix I.6; its statement, moved here from the main text, is the following.

Theorem J.1 (A benchmark with persistence and shock entry)**.** *Within the general-F model there exists an economy with $\beta_T, \beta_S > 0$, a non-degenerate high political basin $[\bar{x}, 1]$, and a stationary feedback Stackelberg equilibrium E such that:*

(i) both equilibrium value functions are nonconstant;

(ii) the target chooses full non-compliance at the basin pivot and the sender's realized sanction there is strictly interior;

[1] The verifications of the examples are in the rest of this section and in Appendix L; the numerical example economies, among them the dual-basin economy, are in Appendices K.1 and K.2.

(iii) the dynamic sanction at the pivot strictly exceeds the sanction in the myopic equilibrium;

(iv) from every $x_0 \in [\bar{x}, 1]$, the basin is invariant and the realized-sanction sequence stays above $\bar{x}$ and is non-decreasing; and

(v) maximum sanctions have a strictly positive minimum drift below the basin, so a temporary sanction shock of finite length moves every initial state into $[\bar{x}, 1]$, after which the conclusions in (iv) hold.

Proof. At the parameter vector of the non-monotone (B2) row of Table 3 (`b2Params`; the machine-checked statement is `b2NonMono_three_goals_dominance`), $(\beta_T, \beta_S) = (9/10, 1/2)$ and $\bar{x} = 1/2$. Both discounted value differences are nonzero. At the pivot, the target action is zero and the dynamic sanction is $7/8$, compared with $5/14$ in the myopic equilibrium. The same construction establishes basin invariance, the sanction floor, monotonicity, and positive minimum drift under a temporary sanction shock at the maximum sanction. □

J.3 An economy in the class outside the family

The class contains an economy that is quadratic–linear in no payoff coordinate and affine in no transition coordinate (`Examples/ChordEconomyQuarticWitness.lean`). Its payoffs are

$$U_S = -\tfrac{3}{2}s^2 - \tfrac{1}{8}s^4 + s\big(1 - 3a - \tfrac{1}{4}a^2 + \tfrac{1}{2}x + \tfrac{1}{8}x^2\big) + x - \tfrac{1}{10}x^2,$$

$$U_T = -a - \tfrac{1}{8}a^2 - 4s - \tfrac{1}{4}s^2 + \tfrac{1}{2}xs + \tfrac{1}{8}x^2 s + 30x - x^2,$$

the transition is `quadDiagF`, refuted against the affine law at every rate (`qw_F_ne_affineF`), the sender discounts at $\beta_S = 1/2$, and the myopic best response is the maximizer of the sender's strictly concave stage payoff, defined as an argmax rather than by a formula. Neither payoff is quadratic–linear at any coefficients (`qwU_S_ne_symU_S`, `qwU_T_ne_symU_T`), and ten of the lower and upper rate pairs of panel A0 are strictly split (`qw_pairs_split`). Each rate is a theorem on the cube: $w \in [4/5, 31/20]$, $\sigma_x = 1/20$, $\tau_x = 1/5$, $r \in [1/2, 3/4]$, $q = 3$, $\bar{Q} = 9/2$, $k \in [3, 7/2]$, $u \in [1, 13/8]$ for the sender; $\ell \in [1, 5/4]$, $p_{\mathrm{lo}} = 27/8$, $c = 9/2$, the envelopes $28 + \frac{3}{4}s$ and $30 + \frac{1}{2}s$ for the target; slopes $g, d \in [2/5, 3/5]$ with $M_x = M_{\mathrm{cs}} = 1/5$ and the other moduli zero; $\sigma_V = 1$. The target's envelope slopes are ordered $n_{\mathrm{lo}} = 3/4 > n_{\mathrm{hi}} = 1/2$, as two envelopes with different intercepts may be. At $\beta_T = 1/30$ the relief equilibrium exists and its sanction rule is zero at every state (`qwRelief_reliefEquilibrium`, `qwRelief_rule_zero`). At $\beta_T = 1/2$ the entrenchment condition and the increasing-differences condition hold, Theorem 12 holds in all four cases, and both forward-looking incentives change behavior at the constructed equilibrium (`qwDefiance_four_cells`, `qwDefiance_both_channels`). The economy is an existence certificate: it shows that the class is inhabited outside the family, and it says nothing about how common such economies are or about magnitudes.

K Numerical example economies

This section collects the two numerical economies the paper uses. The partial-adjustment example is a member of the quadratic–linear family at which the four-case theorem holds in all four cases, together with every coefficient vector near it. The dual-basin economy lies outside the class of the main text; at it one equilibrium has both a high and a low basin, and the initial state selects between them. The general results these economies illustrate are in Part I; the parametric families they belong to are in Appendix J.

K.1 The partial-adjustment example

The main text now states only parametric examples. The numerical example used in this appendix as *the partial-adjustment example* is the following member of the quadratic–linear family (J.1) with the partial-adjustment law (J.2). The *partial-adjustment example* has $q = k = 3$, $u_0 = w = \ell = 1$, $r = \frac{1}{2}$, $c = 4$, $n = 0$ and $v = 30$, transition $F(x, s) = (x + s)/2$ and both discount factors $\frac{1}{2}$. Every coefficient vector within $1/20$ of it, with $n \geq 0$ and the same transition and discount factors, is also an economy in the class, and there Theorem 12 holds in all four cases.

K.2 The dual-basin economy

This section defines the dual-basin economy and states its two-sided result. The *dual-basin economy* has payoffs

$$U_S(x, a, s) = a - \tfrac{1}{2}s^2 + s\big(2x - \tfrac{1}{2} - 2a\big), \qquad U_T(x, a, s) = \tfrac{19}{10}x - \tfrac{1}{5}a - \tfrac{9}{10}s,$$

transition $F(x, s) = (9x + s)/10$, the partial-adjustment law at speed $1/10$, and discount factors $\beta_T = 9/10$ and $\beta_S = 1/2$. The sender gains from compliance and pays a convex cost of sanctioning. The term $2xs$ is its government's political return to sanctioning an entrenched adversary, and compliance lowers its marginal value of sanctioning. The target incumbent values the political state at rate $19/10$, and compliance and sanctions both cost it. Because the term $2xs$ makes the sender's marginal return to sanctioning rise with the state, its best response in the one-period game against full non-compliance is no sanction at every state up to $1/4$ and the full sanction from $3/4$ up. Its marginal return to sanctioning a non-compliant target is negative at low states, so the economy lies outside the class of the main text.

Proposition K.1 (Both basins at one equilibrium)**.** *At the dual-basin economy there is an equilibrium in $\mathcal{E}$ such that, from every initial state $x_0 \in [0, 1]$: a temporary sanction shock at full sanctions, of length $\widehat{N}_F(19/20)$, is followed by permanent residence in $[19/20, 1]$ with the realized sanction equal to 1 at every later date; and a temporary suspension of sanctions of length $\widehat{N}_F^R(1/10)$ is followed by permanent residence in $[0, 1/10]$ with the realized sanction equal to 0 at every later date. Here $\widehat{N}_F^R(1/10)$ is $9/10$ divided by the smallest one-period fall $x - F(x, 0)$ over $x \in [1/10, 1]$, rounded up.*

The two lengths are 190 and 90 periods; they are upper bounds on what entry needs, valid from every start. The state under full sanctions from $x_0 = 0$ enters the high basin at $t = 29$, and the state under zero sanctions from $x_0 = 1$ enters the low basin at $t = 22$. In the equilibrium the target's value rises with the state at rate 10, so a sanction's pain $9/10$ exactly offsets the discounted value of the entrenchment it builds, $\beta_T \cdot \frac{1}{10} \cdot 10$: the target's objective does not depend on the sanction, and it maintains non-compliance in both basins. Appendix D.4 discusses the two basins, and the formalization map lists the declarations that prove the statement.

L A microfounded interest-group example in the Kaempfer–Lowenberg tradition

The analytic example economies of Appendix J are hand-chosen parameter vectors. This section instead supplies a public-choice interest-group polity in the Kaempfer and Lowenberg (1988) tradition, with two interest groups in each country, in which the stage payoffs are *computed* from interest-group primitives by the Grossman and Helpman (1994) weighted-sum aggregation rule,

and the stage complementarity (G.1) and the diminishing-returns curvature of Assumption 3 then follow, on the sign region defined below, from a *sign condition on interest-group effectiveness* rather than being posited directly at the stage level. The construction is machine-checked in `Theory.Microfoundations.FourGroupPolity` (the derivation and the general-transition bridge `toParamsF`, which takes the transition law and its five structural obligations as arguments) and instantiated in `Examples.FourGroupWitness`. The instance `toParams` under the partial-adjustment law, the same polity with its transition pinned to `affineF` δ, is the `affineF` δ instance of `toParamsF` and lives with the other affine instances in `Examples.AffineInstance.Microfoundations`.

L.1 Four groups, two per country

The sender and the target each contain two organized interest groups, four in all. On the sender side: a *pro-sanction coalition* J_S (strategic, security, or import-competing interests whose political standing is advanced by confrontation) and a *commercial lobby* K_S (exporters and investors harmed by the sanction). On the target side: a *regime-aligned bloc* J_T that rallies under external pressure, and an *opposition* K_T that the sanction's economic damage can empower. Each group g has a welfare W_g and a political weight θ_g (its political effectiveness); following Kaempfer and Lowenberg (1988), effectiveness is allowed to shift with the political state x and the sanction level s as the confrontation hardens. Each country's policymaker maximizes the Grossman and Helpman (1994, eq. (11)) weighted sum of its two groups' welfares, plus the resource terms that enter its objective at unit weight: on the sender side the sanctioning deadweight $k_S s^2$, the compliance–sanction cross term $w_{\mathrm{pl}} s a$ and the direct compliance benefit $\beta_S^{\mathrm{base}} a$; on the target side the compliance cost $k_T a^2$ and the sanction cost $d_s s$:

$$U_S = \sum_{g \in \{\mathsf{J}_S, \mathsf{K}_S\}} \theta_g W_g + \beta_S^{\mathrm{base}} a - k_S s^2 - w_{\mathrm{pl}}\, s\, a, \qquad U_T = \sum_{g \in \{\mathsf{J}_T, \mathsf{K}_T\}} \theta_g W_g - k_T a^2 - d_s\, s, \tag{L.1}$$

with group welfares $w_{\mathsf{J}_S} = g_J s$, $w_{\mathsf{K}_S} = -q_K s$, $w_{\mathsf{J}_T} = b_J x - c_J a$ and $w_{\mathsf{K}_T} = -b_K x + c_K a$. The unweighted terms are the ones the planner of Appendix L.6 keeps at unit weight; they are not political transfers.

L.2 Reduced stage payoffs

Taking the linear-quadratic specification of the group welfares and effectiveness schedules: the pro-sanction coalition's pull rising in the political state at rate ν (the source of the sender's *state–sanction complementarity*, $\theta_{\mathsf{J}_S} = \theta^0_{\mathsf{J}_S} + \nu x$), the regime bloc's and opposition's effectiveness rising in the sanction at rates ρ and μ respectively, a sanctioning-cost curvature k_S, a compliance cost k_T, and a cross-group *compliance–sanction* term w_{pl} by which target compliance dampens the marginal political value of sanctioning: the aggregation (L.1) collapses to

$$U_S(x, a, s) = \beta_S^{\mathrm{base}} a + \big(\theta^0_{\mathsf{J}_S} + \nu x\big) g_J\, s - \theta^0_{\mathsf{K}_S} q_K\, s - k_S s^2 - w_{\mathrm{pl}}\, s\, a, \tag{L.2}$$

$$U_T(x, a, s) = \big(\theta^0_{\mathsf{J}_T} + \rho s\big)\big(b_J x - c_J a\big) + \big(\theta^0_{\mathsf{K}_T} + \mu s\big)\big(-b_K x + c_K a\big) - k_T a^2 - d_s s. \tag{L.3}$$

Every coefficient in (L.2)–(L.3) is an interest-group primitive; nothing is chosen to make a monotonicity hold.

L.3 The stage assumptions follow from a sign condition on interest-group effectiveness

The cross-partials of (L.2)–(L.3) are read off directly:

$$\frac{\partial^2 U_S}{\partial s\,\partial x} = \nu\, g_J, \qquad \frac{\partial^2 U_S}{\partial s\,\partial a} = -w_{\text{pl}}, \qquad \frac{\partial^2 U_S}{\partial s^2} = -2k_S, \qquad \frac{\partial^2 U_T}{\partial s\,\partial x} = \rho\, b_J - \mu\, b_K. \tag{L.4}$$

Define the *sign region* $\mathcal{R}$ by the strict sign conditions

$$k_S > 0, \qquad \nu\, g_J > 0, \qquad \mu\, b_K < \rho\, b_J, \qquad w_{\text{pl}} \geq 0, \tag{L.5}$$

expressed entirely as inequalities among existing group primitives (no new named gap scalar). On $\mathcal{R}$:

- **Stage complementarity (Assumption 4).** On $\mathcal{R}$, under the sign condition, $\partial^2 U_S/\partial s\,\partial x = \nu g_J > 0$ is the sender's half of the stage complementarity, and $\partial^2 U_T/\partial s\,\partial x = \rho b_J - \mu b_K > 0$ says that the regime bloc's gain in effectiveness under sanctions strictly dominates opposition empowerment. Both halves of (G.1) hold strictly: this is exactly the case-split Kaempfer and Lowenberg leave with indeterminate sign, here resolved by the region $\mathcal{R}$. (In Lean: `U_S_senderStageIDStrict`, `U_T_targetStageIDStrict`, the derivative-free Topkis form that (G.1) implies.)
- **Assumption 3 (strict concavity of U_S in s) holds on $\mathcal{R}$ under the sign condition.** $\partial^2 U_S/\partial s^2 = -2k_S$, so the curvature floor is $c_\sigma = 2k_S > 0$ whenever $k_S > 0$. (In Lean: `U_S_senderChordConcaveInS`; the same $k_S > 0$ makes the clipped-vertex static sender selector a genuine stage argmax, `sigmaSel_isMaxOn`: the optimality coupling.)

Assumption 4 is thus a *theorem* on $\mathcal{R}$, not a posit, and the sign of the compliance–sanction cross term $\partial^2 U_S/\partial s\,\partial a = -w_{\text{pl}} \leq 0$ supplies the decreasing-differences input the sender response uses.

The welfare identities and sign conditions of Appendix L.6 need none of this: they hold for every vector of the eighteen coefficients, inside $\mathcal{R}$ or not (`FourGroupCoeffs`), and the planner-optimum theorem there is stated for interest-group polities but its proof reads, of $\mathcal{R}$, only $k_S > 0$ and $w_{\text{pl}} \geq 0$. The region is what the government game's stage assumptions need, so it still matters that $\mathcal{R}$ is inhabited without exhibiting a calibration, and it is. Of the eighteen group-effectiveness coefficients, twelve (the baseline weights $\theta^0_{J_S}, \theta^0_{K_S}, \theta^0_{J_T}, \theta^0_{K_T}$, the anti-sanction coefficient q_K, the direct compliance benefit base_S, the concession costs c_J, c_K, the opposition weight b_K, the opposition-empowerment slope μ, the target concavity k_T and the sanction-cost rate d_s) may be given *arbitrary* real values, and the remaining six are then obtained by choosing each strictly beyond a threshold assembled from those already fixed: k_S and w_{pl} above zero, the factors ν and g_J of the sender's state–sanction cross term above zero, the hawk-welfare weight b_J above zero, and finally the slope ρ of the regime bloc's effectiveness above $\mu b_K/b_J$, which is exactly the third sign condition in (L.5). No magnitude is selected anywhere in the construction (in Lean: `exists_fourGroup_of_free`, with `NonemptyFourGroup` as a one-line corollary). $\mathcal{R}$ is thus a sign region on interpretable primitives rather than a condition whose satisfiability rests on one parameter vector, which is the appropriate standard for a hypothesis class over which universally quantified results range.

We state exactly what stage complementarity does *not* buy on its own. Proposition G.1 derives the tipping pivots $\bar{x}_T, \bar{x}_S$ from Assumption 4 *together with* the boundary sign conditions (G.4), and Assumption 2 follows from those pivots, in the derivation after Proposition G.4, only if the top sign $1 \in \mathcal{F}_T(\bar{s})$ holds: the target's stage payoff must already be non-decreasing in the sanction at the

hardest political state. The conditions (L.5) that define $\mathcal{R}$ restrict cross-partials and the sanction curvature; they leave the *level* of the target's sanction slope, in particular the sanction-cost rate d_s, entirely free, and so do not deliver (G.4). On $\mathcal{R}$ the pivots are therefore derived comparative statics wherever the boundary signs independently hold, and are unavailable where they do not.

They do not hold at the calibrated certificate economy `fgcParams` of Appendix L.4, and this is machine-checked rather than conjectured. There the induced target stage payoff is $U_T(x, a, s) = 3x - \frac{1}{5}a + s\left(x - \frac{14}{5}\right)$: the target's state–sanction cross-partial $\rho b_J - \mu b_K = 1 > 0$ is genuine (Assumption 4 does hold), but the sanction-cost level $d_s = \frac{14}{5}$ swamps it, so the sanction slope $x - \frac{14}{5} \leq -\frac{9}{5}$ is strictly negative at every state and action. Every chord in (G.2) therefore falls, and in particular the top sign fails at every non-degenerate floor $\bar{s} < 1$ ($1 \notin \mathcal{F}_T(\bar{s})$: `fgcCert_one_not_mem_targetFlipSet`), while the degenerate floor $\bar{s} = 1$ leaves only the trivial chord and so puts 0 in the flip set, destroying the bottom sign $0 \notin \mathcal{F}_T$ instead; either way the boundary bundle (G.4) is contradictory (`fgcCert_stageFlip_MS_boundary_infeasible`). The derived-basin route therefore does *not* fire at this economy: neither the pivots of Proposition G.1 nor the derivation of Assumption 2 along that route is available here. The route is not thereby vacuous (a separate economy designed for it inhabits it, in the non-vacuity paragraph of Appendix G.1), but the paper refutes it at *this* economy rather than quietly assuming it.

Assumption 2 at `fgcParams` is instead supplied by the mutually exclusive *entrenchment-condition* route, and supplied from the same derived group primitives: the target's strictly negative stage slope is bounded by the loss rate $c_T = \frac{14}{5} > 0$ (numerically the derived sanction cost d_s) and is offset by the continuation gain $g_T = \frac{80}{11}$ carried on the target's invariant set of value functions, the entrenchment condition $c_T = \frac{14}{5} \leq \frac{36}{11} = \beta_T g_T \delta$ being discharged outright (Lean: the fields `cT`, `gT`, `hT_loss_margin` of `fourGroupCertCertificate`; Appendix L.4). Stage complementarity and the sign of the compliance–sanction cross term continue to do their work (on the sender side, in the comparative statics, and in the interest-group interpretation of the mechanism), but what they do not do at this calibration is generate the basin, which the entrenchment condition generates instead. Finally, what is *not* discharged unconditionally from (L.5) alone is the *dynamic* content of Assumption 1 and Assumption 2, single-crossing of the forward-looking objectives once the continuation value is included, together with joint MPE existence; these are supplied on an invariant set of value functions by a model-specific row-Lipschitz / Schur-condition route (Section 3 and Appendix F).

L.4 A concrete instance and its long-run behavior

The Lean parameter vector `fourGroupWitnessParams` fixes

$$\nu g_J = 3, \quad \rho b_J - \mu b_K = \tfrac{9}{10}, \quad w_{\mathrm{pl}} = \tfrac{3}{2}, \quad c_\sigma = 2k_S = 1, \quad \delta = \tfrac{1}{5},\ \beta_T = \tfrac{9}{10},\ \beta_S = \tfrac{1}{2}, \tag{L.6}$$

all four region conditions (L.5) holding strictly. The machine-checked headline `fourGroupWitnessParams_three_goals_conditional` delivers the dynamic-over-myopic comparison on the basin, conditional on the value class, with the inputs of the derived-basin route feeding it discharged from (L.4).

The conditional headline is now discharged unconditionally. The value-class residual that `fourGroupWitnessParams_three_goals_conditional` leaves open is closed by a certificate-complete sibling, another interest-group polity. At the tuned instance `fgcParams` (the Grossman and Helpman (1994) interest-group polity with $\delta = \frac{1}{2}$, $\beta_T = \frac{9}{10}$, $\beta_S = \frac{1}{2}$ and the region conditions holding strictly), every field of the core certificate `StandardComplianceLocalCoreCertificate` is discharged from the interest-group primitives: the sender high-sanction condition with $\beta_S K_S \delta =$

$\frac{1}{3} < \frac{3}{2} = m_S$, and the target entrenchment condition with $c_T = \frac{14}{5} \leq \frac{36}{11} = \beta_T g_T \delta$, on the slope-pinned invariant sets

$$\mathcal{C}_S = \{V : \|V\|_\infty \leq 5,\ V(y) - V(x) = \tfrac{4}{3}(y - x) \text{ for } 0 \leq x \leq y \leq 1\},$$
$$\mathcal{C}_T = \{V : \|V\|_\infty \leq 28,\ V(y) - V(x) = \tfrac{80}{11}(y - x) \text{ for } 0 \leq x \leq y \leq 1\}$$

(each is a Lipschitz ball whose lower slope floor on the whole physical interval equals its Lipschitz ceiling, so the set is pinned to one slope; the radii are the discounted stage bounds $\frac{5/2}{1-1/2} = 5$ and $\frac{14/5}{1-9/10} = 28$; `fgcCarrierS_eq`, `fgcCarrierT_eq`). The resulting headline `fourGroupCert_three_goals_dominance` (axiom-clean) is *unconditional*: it *constructs* the stationary feedback Stackelberg equilibrium E on those invariant sets and delivers the dynamic-over-myopic comparison (each side at its own equilibrium's target action) $\sigma^{\mathrm{M}}(x, \alpha^{\mathrm{M}}(x)) \leq \sigma^{\mathrm{dyn}}_{V_S}(x, \alpha(x))$ for every $x \in [\frac{1}{10}, 1]$, with no value-class certificate taken as a hypothesis. This is the first *microfounded* instance of the core certificate: existence and the three goals hold at a public-choice-derived stage economy, not only at the analytic example economies of Appendix I.6.

What the certificate-complete economy exhibits, and what it does not. At `fgcParams` the two Bellman fixed points are exact and affine, $V_S(x) = \frac{11}{3} + \frac{4}{3}x$ and $V_T(x) = \frac{52}{11} + \frac{80}{11}x$ (`fgc_VS_star_fixed`, `fgc_VT_star_fixed`), and the derived actions of the constructed equilibrium are full non-compliance and the maximal sanction at every physical state (`fgc_`α`_on_eq_zero`, `fgc_`σ`_dyn_zero_eq_one`: the joint objective, with the sender's dynamic best response substituted, is maximized at full non-compliance, and at full non-compliance the sender's dynamic best response is 1). Its dynamic-over-myopic comparison is an equilibrium comparison of the constructed equilibrium with the myopic equilibrium, in which both governments re-optimize; it is not the sender's foresight premium at a common compliance level, since the myopic sanction $\min(1, 2 + x)$ and the dynamic best response at full non-compliance are both 1. This economy supplies microfounded non-vacuity of the core certificate and the welfare comparison of Appendix L.6. Whether either government's forward-looking incentive changes its behavior at a given state is a separate question that this example economy is not used to answer; the uniform statement that both forward-looking incentives change behavior belongs to the class under the lifting condition (Proposition B.1 and its form without the one-period ranking condition after Theorem 12). The earlier calibration `fourGroupWitnessParams` of Appendix L.4, whose bistability was explored numerically only, supports no machine-checked statement about its continuation values or the size of any foresight premium, and none is made here. Displaying the indirect effect of the target's lower compliance at an interior target policy remains the role of the mechanism-active P_3 economy (Appendix I.6).

What this example adds. The interest-group example *derives* the two stage-level primitives, Assumption 4 (stage complementarity) and Assumption 3 (strict sender concavity in the sanction), from public-choice primitives, so the "flip" in the cross-partials is a consequence of the regime bloc's gain in effectiveness under sanctions exceeding opposition empowerment rather than an analyst's posit. What it does *not* do is obtain Assumption 2 from Assumption 4: the derived-basin route of Proposition G.4 needs the boundary sign conditions (G.4) *in addition* to complementarity, and the region conditions (L.5) constrain cross-partials and curvature, not the *level* of the target's sanction slope (Appendix L.3). At the calibration `fourGroupWitnessParams` the target half of Assumption 2 is carried inside the value-class certificate that the conditional headline assumes; at the certificate-complete `fgcParams` it is discharged outright on the mutually exclusive entrenchment-condition route ($c_T = \frac{14}{5} > 0$), the boundary bundle of the derived-basin route

being machine-checked infeasible there. That instance also discharges dynamic single-crossing and joint MPE existence on its invariant set of value functions and *constructs* the equilibrium (`fourGroupCert_three_goals_dominance`). This is the decoupled $L_{ST} = 0$ route; a coupled three-goals/Schur certificate would need analytic discharges of the conditions on the target's invariant set that are not available on the sets examined, and none is claimed.

The grounding is a *trade*, not a one-directional subsumption, relative to the Eaton–Engers example economy of Appendix I.6: that economy derives the *transition* sign from a two-group Kaempfer–Lowenberg market-clearing polity (the assignment of states to the lock-in basin and to the basin of convergence to zero is an equilibrium output, not the bare partial-adjustment law), whereas the interest-group example derives the *stage payoffs* but takes the partial-adjustment transition as given. The two constructions ground complementary halves of the model; neither subsumes the other.

L.5 The pressure schedules from a Becker contest

The market-clearing polity above takes the groups' pressure schedules as monotone primitives. They too can be derived. Let the two target-side groups choose pressure levels at quadratic cost, with political influence additive in the pressure difference: the canonical pressure-competition structure of Becker (1983). Best responses are then *dominant strategies* with closed-form equilibrium pressure $p_i^* = \max\{0,\, S_i(s)/(2c_{p_i})\}$, where the stake $S_i(s)$ is the group's effectiveness-weighted marginal benefit of entrenchment net of income-incidence erosion: both read off the interest-group payoff fields already introduced. The three Kaempfer–Lowenberg effects (rally, incidence, empowerment) become *definitions*, equilibrium contest objects rather than assumed schedules, and the market-clearing map they induce is exactly the partial-adjustment transition used throughout, so the lock-in condition becomes a *cost-adjusted* form of the third sign condition of region (R), $c_{p_J}(\mu b_K) < c_{p_K}(\rho b_J)$, degenerating literally to that sign condition at equal pressure costs. The only new primitives are the two pressure-cost coefficients and two incidence-exposure slopes; the lock-in conclusion is obtained by *applying* the existing market-clearing theorem, and at the example's numbers the contest-induced transition coincides with the partial-adjustment law: the transition used throughout the paper is, at the example economy, itself a contest-equilibrium object.

L.6 Aggregate welfare: construction of W, the two comparison paths, and the period-by-period accounting

Section 4 states the welfare results on the class; this subsection carries the polity's instance of them and the derivation behind it. Aggregate welfare, its closed form at the example economy, the two state paths, the per-period gap and every discounted constant below are machine-checked (`Theory.Microfoundations.FourGroupPlanner`, `Examples.FourGroupPlannerWitness` and, for the primitive identity, its sign conditions and the two counterexamples, `Examples.FourGroupWelfareSign`, and, for the planner's optimum at every polity and transition, `Theory.Microfoundations.FourGroupPlannerOptimum`; Appendix B). The accounting identities, the sign conditions and the stationary comparison are stated over the coefficients alone, with no region condition (`FourGroupCoeffs`), and a polity reaches them through `FourGroup.toCoeffs`; the one-line stationarity reduction $x = F(x,s) \iff x = s$ is immediate from (J.2).

Construction of aggregate welfare. The Grossman and Helpman (1994) rule (L.1) weights each group's welfare by that group's political effectiveness θ_g. In the reduced payoffs (L.2)–(L.3) those weights appear not as bare constants but as the *schedules* $\theta_{\mathsf{J}_S}(x) = \theta^0_{\mathsf{J}_S} + \nu x$, $\theta_{\mathsf{K}_S} = \theta^0_{\mathsf{K}_S}$,

$\theta_{\mathsf{J}_T}(s) = \theta^0_{\mathsf{J}_T} + \rho s$ and $\theta_{\mathsf{K}_T}(s) = \theta^0_{\mathsf{K}_T} + \mu s$. Aggregate welfare is the *same* aggregation with the political weights removed: each schedule, not merely its intercept, set identically to one, so the effectiveness slopes ν, ρ, μ vanish along with them. Nothing else changes: the group-borne resource terms already enter (L.2)–(L.3) at unit weight, multiplied by no θ_g, so they carry through untouched. Substituting $\theta_{\mathsf{J}_S} \equiv \theta_{\mathsf{K}_S} \equiv \theta_{\mathsf{J}_T} \equiv \theta_{\mathsf{K}_T} \equiv 1$ into (L.2)–(L.3) and summing the two countries' group welfares gives

$$W(x,a,s) = \big(g_J\, s - q_K\, s + \beta_S^{\text{base}}\, a - k_S s^2 - w_{\text{pl}}\, s\, a\big) + \big((b_J - b_K)\, x + (c_K - c_J)\, a - k_T a^2 - d_s s\big), \quad \text{(L.7)}$$

the first bracket the two sender groups' summed welfare, the second the two target groups'. Every coefficient in (L.7) is an existing group primitive of (L.2)–(L.3); no scalar is introduced. Subtracting (L.7) from the sum of (L.2) and (L.3) and collecting group by group returns (B.7) coefficient by coefficient in all eighteen fields: pure algebra, no side condition. Evaluating (L.7) at the two corners the dynamics single out gives $W(0,0,0) = 0$ and $W(1,0,1) = (g_J - q_K) + (b_J - b_K) - (k_S + d_s)$, and comparing the two is the flip boundary reported in Section 4. For any planner discount factor $\beta \in [0,1)$, capitalizing a stationary flow gives

$$\mathcal{W}^{\text{stat}}_\beta(x,a,s) = \frac{W(x,a,s)}{1-\beta}.$$

Therefore

$$\mathcal{W}^{\text{stat}}_\beta(0,0,0) - \mathcal{W}^{\text{stat}}_\beta(1,0,1) = \frac{(k_S + d_s) - \big[(g_J - q_K) + (b_J - b_K)\big]}{1-\beta},$$

so the discounted stationary ranking has exactly the same parametric boundary as the flow ranking. Lean proves the capitalization identity, the discounted if-and-only-if boundary, and the exact primitive gap as `stationaryPlannerValue_eq`, `stationaryPlannerValue_lockin_dominated_iff`, and `stationaryPlannerValue_relief_gain`.

At the certificate-complete instance `fgcParams` of Appendix L.4 the sender groups' sanction benefits net out ($g_J = q_K = 1$), the target groups' net entrenchment value is $b_J - b_K = 1$, the direct compliance flows cancel ($\beta_S^{\text{base}} = 0$, $c_J = c_K$) and compliance curvature is absent ($k_T = 0$), so (L.7) collapses to

$$W(x,a,s) = x - \tfrac{14}{5} s - \tfrac{1}{2} s^2 - \tfrac{11}{5}\, s\, a : \quad \text{(L.8)}$$

entrenchment is worth x, and sanctioning burns the target's sanction cost $d_s = \frac{14}{5}$, the sender's deadweight $k_S s^2 = \frac{1}{2} s^2$ and the compliance–sanction transfer $w_{\text{pl}}\, s\, a = \frac{11}{5} sa$. On the physical cube $[0,1]^3$, (L.8) obeys $|W| \le \frac{13}{2}$: the uniform bound the discounting step below consumes.

Stationary points of the transition. Under the partial-adjustment law (J.2) with $\delta \in (0,1)$, $x = F(x,s)$ holds iff $x = s$: the stationary points are exactly the diagonal triples (s,a,s), $(s,a) \in [0,1]^2$. Evaluating (L.8) along that diagonal, the entrenchment credit and the sanction cost collapse into a single linear term,

$$W(s,a,s) = -\tfrac{9}{5} s - \tfrac{1}{2} s^2 - \tfrac{11}{5}\, s\, a \;\le\; 0 = W(0,0,0),$$

each of the three terms being nonpositive on the cube, with *strict* inequality whenever $s > 0$ (the first is then strictly negative). So no stationary point does better than the origin, and every stationary point carrying *positive sanctions* is strictly worse. That weak-with- strict-above-zero statement is exactly what is claimed and no more: the zero-sanction stationary points $(0,a,0)$ all attain 0 as well, so what is pinned is that *sanctioning* at a stationary point is what destroys planner value, not that the origin is the unique maximizer. The lock-in corner $(1,0,1)$, the absorbing point of the certified play, sits $\frac{23}{10}$ below the origin in flow terms.

The two comparison paths. At `fgcParams`, $\delta = \frac{1}{2}$, so (J.2) reads $F(x,s) = (x+s)/2$. The *equilibrium* path is the orbit of the certified play, not of an assumed policy: the derived sender corner $\sigma_{V_S}^{\mathrm{dyn}}(x,0) = 1$ against the derived leader policy $\alpha \equiv 0$ (full non-compliance), both pinned for every value function in the sender's certified invariant set and every state in $[0,1]$ by the certificate behind `fourGroupCert_three_goals_dominance`, so $x_{t+1} = F(x_t,1) = (x_t+1)/2$. The *comparison* path shuts sanctions off, $s \equiv 0$, $a \equiv 0$ (a *feasible* policy, not an equilibrium one, which is all an inefficiency claim requires), so $x_{t+1} = F(x_t,0) = x_t/2$. Both are affine contractions and solve in closed form from any start x_0:

$$x_t^E = 1 - (1-x_0)\,2^{-t} \uparrow 1, \qquad x_t^0 = x_0\,2^{-t} \downarrow 0. \tag{L.9}$$

Both orbits stay in $[0,1]$. The cube-wide policy identities proved for the exact equilibrium (`fgcEquilibriumTargetPath_eq_zero`, `fgcEquilibriumSenderPath_eq_one`) identify the equilibrium path with equilibrium play from *every* start $x_0 \in [0,1]$; for a basin start $x_0 \in [\frac{1}{10},1]$ the orbit is increasing and also remains in the high basin $[\frac{1}{10},1]$ used by the separate three-goals certificate, an additional invariance fact and not the domain of the welfare theorem.

Period-by-period accounting and the discounted gap. Substituting (L.9) into (L.8), the equilibrium flow is $W(x_t^E,0,1) = x_t^E - \frac{33}{10} = -\frac{23}{10} - (1-x_0)2^{-t}$ (entrenchment worth x_t^E against the per-period resource burn $k_S + d_s = \frac{33}{10}$) and the comparison flow is $W(x_t^0,0,0) = x_0 2^{-t}$, so the per-period gap is

$$W(x_t^0,0,0) - W(x_t^E,0,1) = \tfrac{23}{10} + 2^{-t}, \tag{L.10}$$

in which the start x_0 cancels *identically*: a permanent flow gain of $\frac{23}{10}$ from abandoning the sanctions war, plus a vanishing transition bonus. Now discount at any $\beta \in [0,1)$ and sum. Along any cube-valued path the series $\sum_t \beta^t W$ converges absolutely by geometric comparison against $\frac{13}{2}\sum_t \beta^t$, so both path values are well defined and may be summed term by term. Each is then a pair of geometric series, with ratios β (the constant flow) and $\beta/2$ (the transition term inherited from the contraction factor $1-\delta = \frac{1}{2}$ in (L.9)):

$$V^E(x_0;\beta) = -\frac{23}{10(1-\beta)} - \frac{1-x_0}{1-\beta/2}, \qquad V^0(x_0;\beta) = \frac{x_0}{1-\beta/2}. \tag{L.11}$$

Differencing (L.11), the two $(1-\beta/2)^{-1}$ terms recombine, $x_0 + (1-x_0) = 1$, and the start drops out, leaving

$$V^0(x_0;\beta) - V^E(x_0;\beta) = \frac{23}{10(1-\beta)} + \frac{1}{1-\beta/2},$$

the exact discounted gap of Section 4, an identity for every start $x_0 \in [0,1]$ and every $\beta \in [0,1)$; both terms are strictly positive on that range, which is the strict-dominance form.

The stationary comparison. Let g_J and q_K be the sender groups' per-unit sanction benefit and loss, b_J and b_K the target groups' state valuations, k_S the sender's sanctioning-cost curvature, and d_s the target's sanction-cost rate. The benefit of maximal entrenchment and sanctions is $(g_J - q_K) + (b_J - b_K)$; its resource cost is $k_S + d_s$. For a planner discount factor $\beta \in [0,1)$, define discounted welfare at a constant state-action-sanction triple by $\mathcal{W}_\beta^{\mathrm{stat}}(x,a,s) = \sum_{t=0}^{\infty} \beta^t W(x,a,s)$.

Proposition L.1 (Discounted stationary welfare boundary)**.** *For every vector of interest-group coefficients, inside region (R) or not, and every $\beta \in [0,1)$,*

$$\mathcal{W}_\beta^{\mathrm{stat}}(0,0,0) - \mathcal{W}_\beta^{\mathrm{stat}}(1,0,1) = \frac{(k_S + d_s) - \big[(g_J - q_K) + (b_J - b_K)\big]}{1-\beta}.$$

Consequently, the stationary zero path strictly planner-dominates the stationary lock-in path if and only if

$$(g_J - q_K) + (b_J - b_K) < k_S + d_s.$$

Discounting capitalizes the flow gap without changing its sign, and the example interest-group economy satisfies the inequality. No region condition is read (`FourGroupCoeffs.stationaryPlannerValue_relief_gain`, `FourGroupCoeffs.stationaryPlannerValue_lockin_dominated_iff`). The stationary comparison is not the equilibrium comparison, which runs along moving paths from a common start; that comparison is the next proposition.

The same accounting in the primitives, and the sign. The two closed forms are the case $\delta = \frac{1}{2}$, $b_J - b_K = 1$, $(k_S + d_s) - (g_J - q_K) = \frac{33}{10}$ of an identity that holds for every vector of the eighteen coefficients and every transition of the paper's class. No sign condition of region (R) enters its proof, so it also covers coefficient vectors outside the region, such as those in which opposition empowerment dominates the regime bloc's gain in effectiveness under sanctions (`FourGroupCoeffs`). Write $H = (k_S + d_s) - (g_J - q_K) - (b_J - b_K)$ for the stationary gap and $B = b_J - b_K$ for the target groups' net valuation of the state.

Proposition L.2 (The welfare gap in primitives)**.** *For every vector of interest-group coefficients, every transition $F\colon [0,1]^2 \to [0,1]$ satisfying* (F2)–(F3)*, every planner discount $\beta \in [0,1)$ and every start $x_0 \in [0,1]$, let $e_{t+1} = F(e_t, 1)$ and $y_{t+1} = F(y_t, 0)$, with $e_0 = y_0 = x_0$, be the corner path of full sanctions against full non-compliance and the zero-sanction path, and let $z_t = 1 - e_t + y_t$, so that $z_t \in [0,1]$ and $z_0 = 1$. Then*

$$V^0(x_0;\beta) - V^E(x_0;\beta) = \frac{H}{1-\beta} + B\sum_{t=0}^{\infty} \beta^t z_t.$$

If the transition also satisfies (F1)*,* (F4) *and strict adjustment* (F5)*, the difference is positive for every $\beta \in [0,1)$ if and only if $k_S + d_s > g_J - q_K$ and $k_S + d_s \geq (g_J - q_K) + (b_J - b_K)$. Under the partial-adjustment law* (J.2) *with $\delta \in (0,1)$ the two paths are $x_t^0 = x_0(1-\delta)^t$ and $x_t^E = 1 - (1-x_0)(1-\delta)^t$, the residual is $z_t = (1-\delta)^t$, and*

$$V^0(x_0;\beta) - V^E(x_0;\beta) = \frac{(k_S + d_s) - (g_J - q_K) - (b_J - b_K)}{1-\beta} + \frac{b_J - b_K}{1 - \beta(1-\delta)};$$

this difference is positive if and only if $[(k_S + d_s) - (g_J - q_K) - (b_J - b_K)](1 - \beta(1-\delta)) + (b_J - b_K)(1-\beta) > 0$, and it is positive for every $\beta \in [0,1)$ if and only if $k_S + d_s > g_J - q_K$ and $k_S + d_s \geq (g_J - q_K) + (b_J - b_K)$.

Proof. At full non-compliance $W(x, 0, s) = (g_J - q_K - d_s)s - k_S s^2 + (b_J - b_K)x$, so the per-period gap between the two paths is $H + Bz_t$. Both paths stay in $[0,1]$, where the flows are bounded, and summing termwise gives the identity (`FourGroupCoeffs.pathValue_cornerF_gap`). Under strict adjustment the corner path rises to 1 and the zero-sanction path falls to 0, so their separation $1 - z_t$ starts at 0, stays in $[0,1]$ and tends to one; with $C = (k_S + d_s) - (g_J - q_K)$, a discounted series $\sum_t \beta^t\big(C - B(1 - z_t)\big)$ of that kind is positive at every $\beta \in [0,1)$ exactly when $C > 0$ and $B \leq C$ (`DiscountedSign.tsum_pos_all_discounts_iff`, `pathValue_cornerF_gap_pos_all_discounts_iff_of_params`). Under the partial-adjustment law the residual is $(1-\delta)^t$ and the start cancels; two geometric series give the closed form (`pathValue_affineCorner_gap`). Clearing the positive denominators gives the fixed-β condition (`affineCornerGap_pos_iff`). At $\beta = 0$ the numerator is $(k_S + d_s) - (g_J - q_K)$, and as $\beta \to 1$ it tends to $\delta[(k_S + d_s) - (g_J - q_K) - (b_J - b_K)]$, which gives

the all-β condition (`affineCornerGap_pos_all_discounts_iff`). The welfare theorems cited are stated over the coefficients and read no region condition; the polity forms are their instances through `FourGroup.toCoeffs`. □

The conjunction "stationary gap positive and $b_J - b_K \geq 0$" is sufficient and not necessary: a zero stationary gap with a positive transition bonus, or a negative $b_J - b_K$, can still give a positive difference. Since $1 \leq \sum_t \beta^t z_t \leq 1/(1-\beta)$, the difference is at least $H/(1-\beta) + B$ when $B \geq 0$ and at least $[(k_S + d_s) - (g_J - q_K)]/(1-\beta)$ when $B < 0$ (`pathValue_cornerF_gap_ge_of_nonneg`, `pathValue_cornerF_gap_ge_of_neg`). A lower bound on the transition's sanction slope bounds the residual sum from above and so gives no lower bound in the first case; a lower bound $g \geq 0$ on the transition's state slope along the two endpoint sections sharpens the first bound's transition term to $B/(1-\beta g)$ (`pathValue_cornerF_gap_ge_of_nonneg_of_cornerChords`). The all-discount condition needs no affine rate: it holds under the qualitative direction of (F5) with the continuity of (1), which is the paper's maintained class, and under its quantitative form, a positive lower bound on the sanction slope, such as the family's $d_{\text{lo}} > 0$, which adds a geometric rate (`pathValue_cornerF_gap_pos_all_discounts_iff`). Without strict adjustment the necessity direction fails: the identity transition keeps both paths at the start and gives $[(k_S + d_s) - (g_J - q_K)]/(1-\beta)$ whatever $b_J - b_K$ is (`not_necessary_without_strict_adjustment`).

Region (R) does not determine the sign. Neither inequality of Proposition L.2 is a region-(R) condition, and the government game does not settle them either. Rescale the pro-sanction coalition: multiply its per-unit benefit g_J by some $\phi \neq 0$ and divide its weight schedule, $\theta^0_{\mathsf{J}_S}$ and ν, by the same ϕ (`hawkRescale`). The product $(\theta^0_{\mathsf{J}_S} + \nu x) g_J$ does not change, so both government payoffs, the static sender selector, every region-(R) condition and the induced game under every transition are those of the original polity, and every equilibrium of the one is an equilibrium of the other (`hawkRescale_U_S`, `hawkRescale_U_T`, `hawkRescale_toParamsF`). The planner, who counts the hawks' benefit at unit weight, sees $W + (\phi - 1) g_J s$, so under every transition into $[0,1]$ the gap between the zero-sanction path and the corner path of full sanctions against full non-compliance falls by exactly $(\phi - 1) g_J/(1-\beta)$ (`hawkRescale_cornerF_gap`), while under (F2)–(F3) the gap is at most $(|H| + |B|)/(1-\beta)$ in absolute value (`abs_cornerF_gap_le`). Hence, for every polity whose hawks gain from sanctions, $g_J > 0$, some $\phi > 0$ makes the gap strictly negative at every planner discount, every start and every transition satisfying (F2)–(F3) (`exists_hawkRescale_cornerF_gap_neg`); for every polity whose hawks lose, $g_J < 0$, which region (R) allows when $\nu < 0$, some $\phi > 0$ makes it strictly positive (`exists_hawkRescale_cornerF_gap_pos`). Each statement goes one way: when the hawks gain, the game never shows that the zero-sanction path is at least as good for the planner as persistent full sanctions, and when they lose, it never shows the reverse.

Next to the example economy, the rescaling with $\phi = 5$ raises g_J from 1 to 5 and scales the weight schedule down to $\theta^0_{\mathsf{J}_S} = \frac{3}{5}$, $\nu = \frac{1}{5}$, so that $(\frac{3}{5} + \frac{x}{5}) \cdot 5 = 3 + x$ is exactly what it was (`fourGroupCertHawkScaled_eq_hawkRescale`). The induced game is `fgcParams` itself and the constructed equilibrium is an equilibrium of the new polity verbatim (`fourGroupCertHawkScaled_toParams`), while the planner sees $W + 4s$: $(k_S + d_s) - (g_J - q_K) = -\frac{7}{10}$, the gap falls by $4/(1-\beta)$ (`fourGroupCertHawkScaled_gap_shift`), and the value of the specified feasible zero-sanction path less that of equilibrium play is

$$-\frac{17}{10(1-\beta)} + \frac{1}{1-\beta/2} = -\frac{14 + 3\beta}{20(1-\beta)(1-\beta/2)} < 0$$

at every $\beta \in [0,1)$ (`fourGroupCertHawkScaled_gap_neg`): same government game, same equilibrium, opposite welfare ranking. The reversal concerns the same specified zero-sanction alternative,

not the hawk-scaled planner's own optimum. Bundled, the two polities induce `fgcParams` and their planners, who maximize aggregate welfare, rank the equilibrium path and the same feasible zero-sanction alternative oppositely at every discount and every physical start (`fourGroupCert_welfare_not_identified_by_game`): government payoff functions and equilibrium behavior do not by themselves determine the welfare ranking of these two paths. A second polity changes only the direct sanction cost to $d_s = \frac{1}{10}$, so region (R) again holds and the cost stays positive: the difference is $\frac{3}{5}$ at $\beta = 0$, zero at $\beta = \frac{3}{4}$ and $-\frac{24}{11}$ at $\beta = \frac{9}{10}$ (`fourGroupCertPositiveCost_gap_at_nine_tenths`); the difference equals $(3 - 4\beta)/(5(1-\beta)(1-\beta/2))$, positive exactly when $\beta < \frac{3}{4}$ (`fourGroupCertPositiveCost_gap_sign_iff`), so even with positive resource costs the relative welfare of the two specified paths reverses with the planner's discount factor. Both are existence certificates: they refute the universal "region (R) implies a positive gap" and say nothing about magnitudes. Politically weighted payoff coefficients therefore do not identify the comparison by aggregate welfare; the region conditions constrain cross-partials and curvature, not the resource burn against the groups' benefits.

From a feasible comparison to the planner value. For a polity, a transition F and an initial state x_0, let $\mathcal{P}(x_0)$ be the set of cube-valued path triples $(x_t, a_t, s_t)_{t\geq 0}$ starting at x_0 and satisfying the transition law $x_{t+1} = F(x_t, s_t)$, and define

$$\mathcal{W}^\star_\beta(x_0) := \sup_{(x,a,s)\in\mathcal{P}(x_0)} \sum_{t=0}^{\infty} \beta^t W(x_t, a_t, s_t).$$

On the cube $|W|$ is at most a bound in the coefficients (`plannerWBound`; 13/2 at the certificate-complete polity, whose welfare function is (L.8)), so this supremum is finite. (The bound is a property of the polity's coefficients; it has no counterpart for the interest-group microfoundations of the family below, whose welfare coefficients are unbounded across microfoundations.) Write $\varphi(a) = (\beta_S^{\text{base}} + c_K - c_J)a - k_T a^2$ for the compliance part of W, a^* for a maximizer of φ on $[0,1]$, which exists by compactness (`exists_complianceFlow_max`), and $Y(x_0) = \sum_t \beta^t y_t$ for the discounted state along the zero-sanction path from x_0.

Proposition L.3 (The planner's optimum and the exact loss)**.** *(i) For every interest-group polity, every transition $F\colon [0,1]^2 \to [0,1]$ satisfying* (F2)–(F3)*, every $\beta \in [0,1)$ and every $x_0 \in [0,1]$, write $B = b_J - b_K$ and suppose either (a) $B \geq 0$, the transition has an upper bound L on its sanction slope, $F(x, s_2) - F(x, s_1) \leq L(s_2 - s_1)$ for $s_1 \leq s_2$, and an upper bound λ on its state slope along the zero-sanction section, $F(y_2, 0) - F(y_1, 0) \leq \lambda(y_2 - y_1)$ for $y_1 \leq y_2$, with $\beta\lambda < 1$, and*

$$g_J - q_K + \frac{\beta B L}{1 - \beta\lambda} \leq d_s;$$

or (b) $B \leq 0$ and $g_J - q_K \leq d_s$. Then the zero-sanction path with the constant compliance a^* attains the planner's optimum,

$$\mathcal{W}^\star_\beta(x_0) = B\,Y(x_0) + \frac{\varphi(a^*)}{1-\beta},$$

and the optimum exceeds the value of the corner path of full sanctions against full non-compliance by $H/(1-\beta) + B\sum_t \beta^t z_t + \varphi(a^*)/(1-\beta)$, with H and z_t as in Proposition L.2. Under the partial-adjustment law (J.2)*, for either sign of B, the condition $g_J - q_K + \beta\delta B/(1-\beta(1-\delta)) \leq d_s$ gives $\mathcal{W}^\star_\beta(x_0) = Bx_0/(1-\beta(1-\delta)) + \varphi(a^*)/(1-\beta)$.*

(ii) At the certificate-complete interest-group economy, for every $x_0 \in [0,1]$ and $\beta \in [0,1)$, the zero-sanction path attains the planner's optimum, $\mathcal{W}^\star_\beta(x_0) = x_0/(1-\beta/2)$, and

$$\mathcal{W}^\star_\beta(x_0) - V^E(x_0;\beta) = \frac{23}{10(1-\beta)} + \frac{1}{1-\beta/2} > 0,$$

where V^E is the discounted welfare of the play of the constructed equilibrium.

The planner condition of (i) says that the first unit of sanction does not pay socially once the political shift it causes is counted. A unit of sanction gives the sender's groups $g_J - q_K$ net and raises next period's state by at most L; along the zero-sanction path that follows, the planner values the higher state at most $\beta B/(1-\beta\lambda)$ per unit when $B \geq 0$ and at most zero when $B \leq 0$, since a planner with $B < 0$ counts a higher state as a loss. If these together do not cover the target's direct sanction cost d_s, the zero-sanction path is optimal: the planner gains nothing by sanctioning. The quadratic cost $k_S s^2$ and the compliance–sanction cross term only strengthen the conclusion.

Proof. (i) Let $v(x) = BY(x) + \varphi(a^*)/(1-\beta)$. At every feasible triple, $W(x,a,s) + \beta v(F(x,s)) - v(x)$ is the sum of the political term $(g_J - q_K - d_s)s + \beta B\big(Y(F(x,s)) - Y(F(x,0))\big)$ and $\varphi(a) - \varphi(a^*) - k_S s^2 - w_{\mathrm{pl}} sa$, which is nonpositive by the choice of a^* and the polity's signs $k_S > 0$ and $w_{\mathrm{pl}} \geq 0$ (`bellman_le_of_politicalMargin`). Where the political term is nonpositive at every feasible state and sanction, the supersolution bound of `Foundations/DiscountedChordSum.lean` bounds every feasible value by $v(x_0)$, and the zero-sanction path with compliance a^* attains it (`plannerOptimalValue_eq_of_politicalMargin`). For $B \geq 0$, Y is non-decreasing, two zero-sanction paths stay within λ^t times their initial distance, and so $Y(F(x,s)) - Y(F(x,0)) \leq Ls/(1-\beta\lambda)$, which with the condition of (a) makes the political term nonpositive (`zeroStateSum_sub_le`, `plannerOptimalValue_eq_of_upperChords`; $\lambda \geq 0$ is derived, not assumed). For $B \leq 0$ a sanction raises the state, which a planner with $B \leq 0$ values at no more than zero, so (b) suffices and no slope bound is used (`plannerOptimalValue_eq_of_nonpos`). Under the partial-adjustment law $Y(x) = x/(1-\beta(1-\delta))$ exactly (`plannerOptimalValue_eq_affineF`). The loss is the accounting identity of Proposition L.2 plus the capitalized compliance flow (`plannerOptimalValue_loss_of_upperChords`, `plannerOptimalValue_loss_of_nonpos`, `plannerOptimalValue_loss_affineF`).

(ii) At the certificate-complete economy $B = 1$, $g_J = q_K$ and $\varphi \equiv 0$, and the planner condition of (i) under the partial-adjustment law at $\delta = \frac{1}{2}$ reads $\beta/(2-\beta) \leq \frac{14}{5}$, which holds at every $\beta \in [0,1)$; so (i) gives the optimum (`plannerOptimalValue_eq_affineF`, `fgcPlannerOptimalValue_eq_general`). The verification at the example economy is direct. Let $v_0(x) = x/(1-\beta/2)$. At every feasible triple, $W(x,a,s) + \beta v_0(F(x,s)) - v_0(x) = s\big(\frac{\beta}{2-\beta} - \frac{14}{5}\big) - \frac{s^2}{2} - \frac{11}{5}sa \leq 0$, with equality at $s = 0$ (`fgc_planner_bellman_upper`). Multiplying by β^t and summing over $t < N$ along any feasible path telescopes to $\sum_{t<N} \beta^t W_t + \beta^N v_0(x_N) \leq v_0(x_0)$; the terminal term vanishes because the state stays in $[0,1]$, and the series converges by the cube bound, so every feasible value is at most $v_0(x_0)$ (`fgc_plannerPath_value_le`, through the generic verification lemma of `Foundations/InfiniteHorizonVerification`). The zero-sanction path attains $v_0(x_0)$ by (L.11), so the supremum equals it (`fgcPlannerOptimalValue_eq`), and the exact difference in (L.11) gives the loss of equilibrium play relative to the specified feasible zero-sanction path (`fgc_plannerOptimalValue_gap_eq_equilibrium`). □

At the certificate-complete economy the optimum is attained by the zero-sanction path, but not uniquely: at $s = 0$ every compliance choice ties, and at $\beta = 0$ future controls are not ranked.

The planner condition is sufficient, and it fails at the hawk-scaled and positive-cost polities above. At the first, $g_J - q_K = 4 > \frac{14}{5} = d_s$, so it fails at every discount; at the second, where

$d_s = \frac{1}{10}$, the condition under the partial-adjustment law reads $\beta/(2-\beta) \le \frac{1}{10}$ and fails for $\beta > \frac{2}{11}$. There sanctions can pay socially: at the first polity at every discount, and at the second at $\beta = \frac{9}{10}$, the corner path of full sanctions against full non-compliance is worth strictly more to the planner than the zero-sanction path at full non-compliance (`fourGroupCertHawkScaled_gap_neg`, `fourGroupCertPositiveCost_gap_at_nine_tenths`). At both polities $\beta_S^{\text{base}} = 0$, $c_J = c_K$ and $k_T = 0$, so the compliance part φ of W is identically zero and every zero-sanction path is worth the same. Since the corner path is feasible, an immediate consequence is that no zero-sanction path attains the planner's optimum there; this step is an inference from the two comparisons, not a separate machine-checked result.

Evaluating the exact gap at $\beta_S = \frac{1}{2}$ and at $\beta_T = \frac{9}{10}$ returns the two capitalized numbers reported there. The planner condition in general form is Proposition 10, and the sensitivity of the verdict to how W is defined is Proposition C.15; the verdict itself, a lock-in that is politically rational and socially wasteful, is the sanctions-persistence analogue of the inefficient-persistence verdicts in the dynamic political-economy tradition (Acemoglu and Robinson, 2008; Battaglini and Coate, 2008; Hassler et al., 2003).

Interest-group microfoundations of the quadratic–linear family under the lifting condition. Everything above, the cube bound 13/2 included, concerns affine interest-group polities that sit next to one example economy. The remainder of this section leaves them. The same non-identification holds on the whole family of Section 5.1, through a separate accounting object that leaves the interest-group polity of this appendix untouched (that polity has no direct state benefit wx and requires strict signs on the state–sanction cross terms, while the family allows $r = 0$ and $n = 0$). An *interest-group microfoundation* of a family member's government payoffs consists of target groups with state valuations $b_J, b_K > 0$ and a compliance transfer $\kappa > 0$ between them, baseline political weights $t_J, t_K > 0$ with a common sanction slope $\nu_T \ge 0$, and sender groups gaining $g_H > 0$ and losing $Q > 0$ per unit of sanction, with weights $(u_0 + Q + rx)/g_H$ and one, subject to

$$t_J b_J - t_K b_K = v, \qquad \nu_T(b_J - b_K) = n, \qquad \kappa(t_J - t_K) = \ell.$$

The direct state benefit wx, the sanctioning cost $\frac{q}{2}s^2$, the compliance–sanction cross term kas and the target's sanction cost cs are kept at unit weight. Every weight is strictly positive on the feasible cube.

Lemma L.4 (Interest-group microfoundations: representation and existence)**.** *For every interest-group microfoundation the politically weighted group sums equal $U_S(x,a,s)$ and $U_T(x,a,s)$ at every state and action, so the government game and every constructed equilibrium are the member's own, and aggregate welfare is*

$$W(x,a,s) = \big(w + b_J - b_K\big)x + \big(g_H - Q - c\big)s - \tfrac{q}{2}s^2 - kas.$$

Every member has a microfoundation with any prescribed net sanction coefficient $\eta = g_H - Q - c$ and any $b_J - b_K > 0$. Microfoundations with $b_J = b_K$ exist exactly at the members with $n = 0$, and there at every η.

Proof. The representation is algebra in the three displayed equations (`govS_eq`, `govT_eq`, `welfare_eq`); existence takes $b_K = 1$, $b_J = 1 + B$, $\kappa = 2\ell b_J/v$, $t_K = v/(2B)$, $t_J = t_K + \ell/\kappa$, $\nu_T = n/B$, $Q = |c + \eta| + 1$ and $g_H = Q + c + \eta$ (`exists_welfareCompletion`); these are choices inside an existence proof, not restrictions on the class, and the class does not require $b_J > b_K$. Since $\nu_T(b_J - b_K) = n$, equal valuations force $n = 0$; and at $n = 0$ the record $b_J = b_K = 1$,

$t_K = 1$, $t_J = 1 + v$, $\kappa = \ell/v$, $\nu_T = 0$, with the same Q and g_H, is a microfoundation at every η, whatever the transition and the state–sanction cross term r (`exists_completion_bJ_eq_bK_iff`, `exists_equalValuation_completion`). The microfoundation of Corollary L.6 is of this kind. □

The family has its own version of Proposition 9, with these microfoundations in place of the accounting representations used there.

Proposition L.5 (Welfare non-identification on the family)**.** *Take a member of the quadratic–linear family in the class that satisfies the lifting condition, the entrenchment condition and the forward-looking relief condition* (RL)*, and at which the sender's response at full non-compliance at its constructed value is non-decreasing in the state, as under the increasing-differences condition* (B.4) *(Appendix C.4). There are two interest-group microfoundations such that, at every initial state and every common planner discount factor $\beta \in [0, 1)$, discounted aggregate welfare along the four constructed equilibria is ranked*

$$11 \succ 10 \succ 00 \sim 01 \quad \textit{by one}, \qquad 00 \sim 01 \succ 10 \succ 11 \quad \textit{by the other}.$$

Proof of Proposition L.5. For the two ranking microfoundations take $b_J - b_K = 1$, so that the state coefficient is $K = w + 1 > 0$. The constructed equilibria of cases 00 and 01 have the zero sanction and one state path, and $W(x, a, 0)$ does not depend on a, so they tie. Along the other two the target maintains non-compliance, the target-only sanction is the stage vertex, at least u_0/q, and the forward-looking sanction exceeds it by at least the foresight premium (B.3) at every date, while the state paths are ordered $00 \leq 10 \leq 11$. With $\eta = q + 1$ every flow comparison is strictly positive in the direction of more sanctions; with η below $-(w + 1)/\min\{u_0/q, \text{premium}\} - 1$ it is strictly negative, since a state difference is at most one. The flows are bounded, so the strict termwise comparison survives discounting, including at $\beta = 0$ (`exists_completions_reversing_regime_ranking_of_response_monotone`; the form under the increasing-differences condition, `exists_completions_reversing_regime_ranking`, is its corollary). □

Corollary L.6 (A planner-efficient boundary)**.** *Let a member have $r = n = 0$. Then there is a microfoundation with $W = U_S$ such that, at the planner discount β_S, the play of every equilibrium in $\mathcal{E}$ at which the target maintains full non-compliance (full non-compliance and its sanction rule, at every state) attains the planner's optimum against every time-dependent plan over both actions, from every start.*

Proof. Take $b_J = b_K = 1$, $t_J = 1 + v$, $t_K = 1$, $\kappa = \ell/v$, $Q = 1$ and $g_H = 1 + c + u_0$; then $W = U_S$, the sender's constructed value satisfies $W(x, a, s) + \beta_S V_S(F(x, s)) \leq U_S(x, 0, s) + \beta_S V_S(F(x, s)) \leq V_S(x)$ with equality at full non-compliance and the dynamic best response to it, and the bounded verification theorem of `Foundations/InfiniteHorizonVerification.lean` concludes (`exists_planner_efficient_completion`). That inequality holds on all of $r = n = 0$. Every full-non-compliance equilibrium has the constructed values and plays the constructed response, so the compared plan is that equilibrium's policy and sanction rule, with neither the entrenchment condition nor the increasing-differences condition (`exists_planner_efficient_completion_of_policy_zero`); the form at the constructed equilibrium under those two conditions is `exists_planner_efficient_completion_at_equilibrium`, the flatness the increasing-differences condition forces at $r = 0$ is `flat_of_margin_of_r_eq_zero`, and the inhabitant is `placableBoundary_planner_efficient_at_equilibrium`. The statement concerns one microfoundation and one planner discount: it is an existence boundary for any universal strict-inefficiency sentence, not generic efficiency. □

Such an equilibrium exists exactly when full non-compliance is a target best reply at the constructed values; under the entrenchment condition and a non-decreasing response the constructed equilibrium of Proposition B.1 is one. The corollary needs neither the entrenchment condition nor the increasing-differences condition. Through the increasing-differences condition it reaches only flat transitions: at $r = 0$ that condition forces a flat transition certificate, $M_{\text{mix}} = 0$ and $M_{\text{cx}}\, g_{\text{hi}} = 0$, as under the partial-adjustment law and every additively separable transition with convex state sections. The boundary is inhabited by the member with $r = n = 0$ whose transition is the partial-adjustment law (Appendix B); no member with $r = n = 0$, a non-flat transition and a full-non-compliance equilibrium is exhibited.

The quantifier order is: for every member there exist two microfoundations that work for every start and every planner discount. The microfoundations depend on the member; no claim is made about every microfoundation, about all equilibria, or about magnitudes. Three accounting choices carry the statement and are choices, not consequences of the government game: wx counted at unit social weight; compliance as a pure transfer between the two target groups, which is what makes the constructed equilibria of cases 00 and 01 tie; and group benefits unrestricted in size, with the inverse political weight compensating. The microfoundations retain the compliance–sanction cross term kas at unit weight, while the compliance transfer κa cancels between the two target groups; counting a term at unit weight is a convention, not a demonstration that it is a resource loss. A bounded class is treated in Proposition L.7 below. A ranking of the constructed equilibria is not a planner-efficiency verdict: under the first microfoundation a planner would want the maximal sanction, not the interior equilibrium one, so the forward-looking equilibrium is still planner-inefficient there; Corollary L.6 is what shows that inefficiency itself is not universal.

What a restriction on the microfoundations identifies. With cardinal group benefits free and political weights compensating, non-identification is what one should expect. The next proposition says what a restriction buys. Along the constructed equilibria of the two non-compliance cases compliance is zero, so the welfare gap is affine in the net sanction coefficient; the restriction is on the sender side only, and the target side of the microfoundation is held fixed.

Proposition L.7 (Partial identification with fixed exposure and bounded hawk weights)**.** *Let a member satisfy the entrenchment condition, with the sender's response at full non-compliance at its constructed value non-decreasing in the state (as under the increasing-differences condition* (B.4)*; Appendix C.4), fix an initial state and a planner discount $\beta \in [0, 1)$, and let ΔX, ΔS and ΔS_2 be the discounted differences between the constructed equilibria of the forward-looking and the target-only cases in the state, the sanction and the squared sanction.*

(i) For every interest-group microfoundation, $V^{11} - V^{10} = K\,\Delta X + \eta\,\Delta S - \frac{q}{2}\Delta S_2$ with $K = w + b_J - b_K$ and $\eta = g_H - Q - c$; the three differences do not depend on the microfoundation, and $\Delta S \geq m/(1-\beta) > 0$ with m the foresight premium (B.3)*.*

(ii) Fix the commercial lobby's exposure $Q = \bar{Q} > 0$ and require the hawks' political weight to satisfy $\underline{\theta} \leq (u_0 + \bar{Q} + rx)/g_H \leq \bar{\theta}$ at every feasible state, with $0 < \underline{\theta}$, $0 < \bar{\theta}$. This class is non-empty if and only if $\underline{\theta}\,(u_0 + \bar{Q} + r) \leq \bar{\theta}\,(u_0 + \bar{Q})$, and on it η ranges over exactly $\big[(u_0 + \bar{Q} + r)/\bar{\theta} - \bar{Q} - c,\ (u_0 + \bar{Q})/\underline{\theta} - \bar{Q} - c\big]$, every value being attained with the target side, hence K, unchanged.

(iii) With the target side fixed, the set of welfare gaps over the class is the closed interval with endpoints $K\Delta X - \frac{q}{2}\Delta S_2 + \eta\,\Delta S$ at the two ends of that range. On a non-empty class, among its microfoundations with the given state coefficient K, a set that contains those with the

given target side, the equilibrium of the forward-looking case ranks strictly first at every microfoundation exactly when the lower endpoint is positive, and strictly last exactly when the upper endpoint is negative; both strict rankings occur exactly when the endpoints have opposite strict signs, and then at microfoundations that keep the whole target side. An endpoint at zero leaves a weak ordering, and the interval $\{0\}$ *a tie.*

(iv) A weight band alone need not identify the sign. For any band straddling one, $\underline{\theta} < 1 < \bar{\theta}$*, unrestricted exposure permits both rankings: with the target side kept, one microfoundation with weights inside the band ranks the equilibrium of the forward-looking case strictly above that of the target-only case and another ranks it strictly below. Nothing is claimed for a band that does not straddle one.*

Proof. (i) is Lemma L.4 along paths on which $as = 0$, with bounded flows summed termwise, and the foresight premium at every date (`regimeValue11_sub_10_eq`, `dS_ge`). (ii): the weight rises with the state because $r \geq 0$, so the two bounds are $(u_0 + \bar{Q} + r)/\bar{\theta} \leq g_H \leq (u_0 + \bar{Q})/\underline{\theta}$; the representation equations involve neither g_H nor Q, so any g_H in that range can be installed without touching the target side (`hawkBounded_iff`, `hawkBounded_nonempty_iff`, `hawkBounded_eta_mem`, `exists_withSender_hawkBounded_eta`, whose attaining microfoundation is the given one with g_H and Q replaced and every other field kept). (iii) follows from (i), (ii) and $\Delta S > 0$ (`gap_mem_interval`, `gap_attained_withSender`, `forall_gap_pos_iff`, `forall_gap_neg_iff`, `exists_both_signs_iff`, and `exists_withSender_both_signs_of_straddle`, which names the two microfoundations). (iv): with $g_H = (u_0 + Q)/\underline{\theta}$ the weights stay in the band once Q is large and $\eta = (1/\underline{\theta} - 1)Q + u_0/\underline{\theta} - c \to +\infty$; with $g_H = (u_0 + Q + r)/\bar{\theta}$ they stay in the band and $\eta \to -\infty$ (`exists_withSender_weightBounded_eta_gt`, `exists_withSender_weightBounded_eta_lt`, `exists_withSender_weightBounded_reversal`; each names its microfoundation as the given one with g_H and Q replaced). Each of the theorems on the constructed equilibria cited is stated under the increasing-differences condition and is the corollary of a sibling that assumes only the non-decreasing response (`regimeValue11_sub_10_eq_of_response_monotone`, `dS_ge_of_response_monotone`, `gap_mem_interval_of_response_monotone`, `gap_attained_withSender_of_response_monotone`, `forall_gap_pos_iff_of_response_monotone`, `forall_gap_neg_iff_of_response_monotone`, `exists_both_signs_iff_of_response_monotone`, `exists_withSender_both_signs_of_straddle_of_response_monotone`, `exists_withSender_weightBounded_reversal_of_response_monotone`); the theorems on the microfoundations alone read neither the entrenchment condition nor the increasing-differences condition. □

The restricted class is a definition over the microfoundation's own data, not a hypothesis on the governments' game: payoffs and constructed equilibria are untouched. It is one restriction among many; holding the target side fixed is substantive, and the proposition identifies the ranking of two constructed equilibria, not a planner optimum. The interval is conditional on accounting information the governments' behavior does not supply; its sharpness does not supply it either. What it shows is where the information has to come from: group exposures, not political weights alone.

Further examples and constants

M Further example economies

This section holds example economies that the main text does not need: the roster of example economies, the payoff bookkeeping of the mechanism-active P_3 economy, the non-compliance core

outside the class, interior compliance at a positive sanction, and national welfare at the dual-basin economy. The auxiliary polynomial economy of the roster is a deliberately simple economy used to test the machinery of Bellman-invariant sets of value functions and of the joint operator (non-emptiness and invariance of the sets, dynamic argmax uniqueness, a decoupled Schur contraction). It is *not* an example economy at which Assumptions 1 to 3 hold jointly: its sender payoff is linear in the sanction and fails Assumption 3, and no headline theorem is applied to it. Logical non-vacuity of the full headline package is supplied by the complete certificate economies of the roster, including the Eaton–Engers and interest-group constructions. Its parameter vector and verified invariant-set checks are not reproduced here.

M.1 Roster of example economies

Table 3 separates logical non-vacuity, unconditional applications of the headline results and mechanism displays. Two conditional existence diagnostics (the low-δ purified window and the coupled Schur condition, both at the P_3 economy with $\delta = \frac{1}{100}$) deliver no equilibrium and carry no result of this paper; their declarations are indexed in Appendix B.

Table 3: Roster of example economies. The reported fractions evaluate certified conditions at illustrative economies and are not calibrated empirical magnitudes.

Economy	**What it certifies**	**Coupling & policies**	**Caveat**	**Where**
Auxiliary polynomial economy	Invariant set of value functions, unique dynamic argmaxes, and a Schur-contractive joint operator.	Decoupled; $\alpha \equiv 1$ and $\sigma_{\text{dyn}} \equiv 1$ on the invariant set.	Linear sender payoff violates A3; no headline-theorem application.	Appendix J
Eaton–Engers certificate	Two-sided mechanism-active full-package application: both values are strictly increasing, and continuation reverses each player's named stage ranking over sanctions.	Full local invariant-set certificate; unconditional persistence, ordering and shock conclusions.	Decoupled cross-value rows; affine transition; corner policies and corner-pinned realized sanctions.	Appendix I.6
Quadratic bump	Same certificate at a non-affine transition.	Decoupled; corner policies.	Realized sanctions corner-pinned.	Section 3
Integrated nonlinear interior	One exact nonlinear equilibrium with positive discount factors, interior target and sender actions, strict target sanction preference, and the full persistence floor.	target policy $\alpha \equiv 1/2$; sender sanction rule $(x+1)/2$; quadratic-bump transition.	Both equilibrium values are zero, so neither forward-looking incentive changes behavior; logical integration, not a substantive calibration.	Appendix I.6

Continued on next page

Economy	What it certifies	Coupling & policies	Caveat	Where
Integrated nonlinear target-active	The same nonlinear interior geometry with $V_T(x) = x$; a stage-profitable compliance deviation is strictly rejected by the full target Bellman objective, while the complete persistence floor still holds.	target policy $\alpha \equiv 1/2$; sender sanction rule $(x+1)/2$; quadratic-bump transition.	$V_S = 0$ and cross-value coupling remains absent; only the target's forward-looking incentive changes behavior in this construction.	Appendix I.6
Low-basin (dovish)	Mirror low-basin theorem non-vacuous; both invariant directional basins exhibited.	Decoupled; non-affine transition.	No interior policy display.	Appendix D.4
Non-monotone (B2)	Mechanism-active full-package application: positive discounting, nonconstant values, exact nonzero foresight premiums, an interior realized sender sanction, and the full persistence/minimum-drift package; single crossing fails but interval dominance holds.	At the pivot, $(a^E, s^E) = (0, 7/8)$ while the myopic equilibrium has $(a^M, s^M) = (1, 5/14)$.	Cross-value rows are decoupled and the transition is affine; no interior target policy or nonlinear transition in this application.	Appendix I.6 and Remark G.2
Interest-group microfounded	Stage assumptions derived from interest-group primitives.	Decoupled; corner policies.	Region (R); strictly interior basin pivot.	Appendix L
Derived-basin example	Inhabits the derived-basin route via its constructor.	Decoupled; target stage payoff flips at interior pivot 2/5.	Kinked sender payoff essential (smooth impossible).	Appendix G.1
Mechanism-active P_3	At $h = 1$, an explicit unconditional equilibrium has nonconstant values, a monotone sanction rule, and a strict foresight premium at the top state.	State-varying leader and sender policies; affine transition; purified box kernel.	Every $h = 1$ equilibrium violates the top-state dynamic floor, so the specialization cannot furnish the three-goals/pathwise-ordering package.	Appendix I.6
Dual-basin selection	Both invariant directional basins certified at one economy and the same equilibrium; no limit point is named.	target row exactly decoupled at the entrenchment-condition boundary; leader policy derived at the corner.	sender argmax interior only between basins.	Appendix D.4

Continued on next page

Economy	What it certifies	Coupling & policies	Caveat	Where
Quartic economy in the class	The class inhabited outside the quadratic–linear family: the four cases at $\beta_T = 1/2$, the relief equilibrium at $\beta_T = 1/30$.	sender payoff quartic in the sanction, target payoff quadratic in each coordinate; non-affine transition; myopic best response defined as an argmax.	An existence certificate; no genericity or magnitude claim.	Appendix J.3

M.2 The P_3 example: payoff and cross-partial bookkeeping

$$U_S = a - \tfrac{3}{5}s^2 - s\big(\tfrac{1}{10} + \tfrac{1}{2}a - \tfrac{27}{20}x\big) + \tfrac{2}{5}x, \qquad U_T = -a^2 + 2xa - a - s\big(\tfrac{1}{2} - \tfrac{1}{2}a - x\big),$$

keeping $\delta = \frac{1}{5}$, $\beta_S = \frac{1}{2}$, $\beta_T = \frac{9}{10}$ and the affine transition. The stronger curvature $-\frac{3}{5}s^2$ makes the sender's myopic best response $\sigma^{\mathrm{M}} = \mathrm{clip}\big(\frac{9}{8}x - \frac{1}{12} - \frac{5}{12}a\big)$ *interior* on the lower basin; the target's state–compliance term $+2xa$ makes compliance interior and reluctant; the sender's stake $+\frac{2}{5}x$ keeps V_S state-dependent independently of the sanction. None of the four changes a structural cross-partial: the state–sanction cross-partial signs $\partial^2 U_S/\partial s\partial x = \frac{27}{20} > 0$, $\partial^2 U_T/\partial s\partial x = 1 > 0$, the sign of the compliance–sanction cross term $\partial^2 U_S/\partial s\partial a = -\frac{1}{2} \le 0$, and the s-concavity all carry over, and the conditional three-goals conclusion carries over verbatim.

What the non-affine examples certify, and the transition-free ordering goal. The certified corner recursion is $x_{t+1} = F(x_t, 1)$, so only the corner section $y \mapsto F(y, 1)$ ever executes along the certified path. The quadratic-bump transition's bump carries a factor $(1 - s)$ that vanishes there (`quadBumpF_corner_isAffine`): that example exercises the incentive layer under a non-affine transition while its realized path is the affine one. The low-basin mirror and the wobble economy bend the corner section itself (`quadBumpLowF_corner_not_affine`, `wobbleEconF10_corner_not_affine`). All three transitions are refuted against the affine law at *every* weight, not only at the economy's own rate (`quadBumpF_ne_affineF`, `quadBumpLowF_ne_affineF`, `ratchetF_ne_affineF`); the non-affine member of the family under the lifting condition is refuted the same way (`quadDiagF_ne_affineF`). Separately, the certificate route's ordering goal, the statewise ordering of the forward-looking sanction above the myopic one, names no transition and never needed the affine law: `goal2_dominance_generalF` reaches it with the transition entering only as a lower bound d_{lo} on its sanction slope on the target leg and a range-restricted upper bound d on that slope on the sender leg, at the cost of one sign hypothesis $0 \le g_T$ that the affine route does not need because there the transition step is an equality. The two rates cannot be collapsed into one: a two-sided bound on the sanction slope at a single rate, with the diagonal condition, pins $F(x, s) = (1 - d)x + ds$. The affine law is the instance $d_{\mathrm{lo}} = d = \delta$ of `goal2_dominance_generalF`, applied directly at the affine examples (no separate affine wrapper is needed).

Open-set robustness at the affine member of the family under the lifting condition. The quadratic–linear family under the lifting condition is inhabited on a box of coefficients (a statement in the family's coefficients, not in the class's rates). Write the coefficient vector as $(q, u_0, k, r, w, \ell, c, n, v)$; the affine member's is $(3, 1, 3, 1/2, 1, 1, 4, 0, 30)$. Let B be the set of coefficient vectors with each coordinate within $1/20$ of the affine member's, except that the target's state–sanction cross term keeps its sign domain, $n \in [0, 1/20]$: B is a neighborhood of the member relative to $n \ge 0$, not a Euclidean ball around it, and it contains the full-dimensional open sub-box with $n > 0$. The transition is the affine law at rate $1/2$, the discounts are $1/2$ and $\sigma_V = 1$ throughout,

so this is a statement about payoff coefficients, not an open-set statement in function space. Every point of B is an economy in the class, and interiority, the lifting condition, the one-period ranking and the strong entrenchment condition hold there as well: at the worst corners, interiority reads $u_0 + r + (r + w)/3 \le 2.27 < 2.95 \le q$, the lifting condition the same against k, the one-period ranking $\ell q \le 3.21 < 3.70 \le u_0(c - n)$, the strong entrenchment condition $c \le 4.05 < 9.98 \le v/3$, the concavity condition $1/8 < q$, and the Riccati closure $1/8 + (r + 1/8)^2/(q - 1/8) \le 1$. On all of B the increasing-differences condition is vacuous (the transition is flat), both forward-looking incentives change behavior at the composed equilibrium, in the target-only case the state converges to $u_0/(q - r)$, and $L_{\text{loop}} = 1/2 + 2(r + 1/8)/(q - 1/8) < 1$, so the closed loop is a contraction with one globally stable steady state. The sanction-section modulus is zero, so every point of B is also an economy in the enriched subclass, on which $M_{\text{tot}} = q$ and the forward-looking relief condition (RL) follows from $2\ell \max\{q, k\} \le 1281/200 < 2301/200 \le k(c - n)$ at the worst corners; hence in the sender-only case the state converges to zero and all four cases of Theorem 12 are ordered at every point of B.

M.3 The non-compliance core outside the class

Proposition M.1 (The non-compliance core outside the class)**.** *There is an economy whose target payoff is that of no economy in the class, with an equilibrium in $\mathcal{E}$ that has full non-compliance at every state and a nondecreasing sanction rule, at which every equilibrium in $\mathcal{E}$ with a nondecreasing sanction rule has full non-compliance at every state.*

The economy has the transition $F(x, s) = (x + s)/2$, both discount factors $\frac{1}{2}$, the quadratic–linear sender payoff $U_S = x - \frac{3}{2}s^2 + (1 - 3a)s$ without a state–sanction cross term, and the target payoff $U_T = 3x - s - C(a)$ with compliance cost $C(a) = a + 10a(1 - a)$. The cost is at least a but not monotone, $C(3/4) = 21/8 > C(1) = 1$, so the class's requirement that every compliance increment cost at least $\ell_{\text{lo}} > 0$ fails and no economy in the class has this target payoff (`oc_not_chord`). The equilibrium has $V_S = 4x/3 + 16/27$, $V_T = 4x$, full non-compliance and the response $\max\{0, 4/9 - a\}$; the target's objective is $4x - C(a)$ whatever the sanction, so full non-compliance is its unique optimum (`ocEquilibrium`, `oc_argmax_unique`). The conditions of the non-compliance core (Appendix C.6) hold with $g_{\text{lo}} = d_{\text{lo}} = \frac{1}{2}$, $v = 3$ and $c = \ell = 1$, and the strong entrenchment condition binds: $\beta_T d_{\text{lo}} v/(1 - \beta_T g_{\text{lo}}) = 1 = c$ (`oc_core_policy_zero`, `defiance_core_outside_chord_class`). The economy is an existence certificate: the core's extra generality over the class is not empty, and nothing is said about how common such economies are. The transition is affine in this example; the core it instantiates is general in F, and the generality exhibited is in the target's payoff.

M.4 Interior compliance at a positive sanction

Proposition M.2 (A strict interior optimum in the class)**.** *There is an economy in the class satisfying interiority, the myopic relief condition, the lifting condition and compliance neutrality, with an equilibrium in $\mathcal{E}$ in which the target's compliance* $1/14$ *is its unique optimum at every state, its objective falls below that optimum by at least* $(a - 1/14)^2/2$ *at every compliance a, and the sanction is* $47/126 > 0$ *at every state.*

The economy is the affine member of the family under the lifting condition of Appendix M.2 with r, c, v and β_T set to 0, 5/2, 10 and 1/4, and the target's payoff replaced by $U_T = 10x - \frac{5}{2}s - a - \frac{1}{2}a^2$, whose compliance cost is quadratic. Its compliance rates lie between $\ell_{\text{lo}} = 1$ and $\ell_{\text{hi}} = 2$, so the economy is in the class but outside the family, whose cost is linear (`qcChord`,

qc_interior, qc_stageRelief, qc_placable, qc_neutral). The equilibrium has $\alpha \equiv 1/14$, $V_S = 4x/3 + 2209/5292$ and $V_T = 80x/7 - 557/882$ (qcEquilibrium, qc_rule). The growth bound is qc_growth, so no first-order condition or grid is used (qc_argmax_unique, strict_interior_equilibrium_in_chord_class). On the family the cost is linear and interior compliance is an equilibrium only through indifference; the curvature of the cost is what makes the interior optimum strict. The economy is an existence certificate, with an affine transition, and says nothing about how common such economies are.

M.5 National welfare at the dual-basin economy

At the dual-basin economy each government's payoff is its nation's welfare plus a political part that grows with the political state. Define the sender's and the target's national welfare by

$$W_S(a,s) = a - \tfrac{1}{2}s^2 - \tfrac{1}{2}s - 2as, \qquad W_T(a,s) = -\tfrac{1}{5}a - \tfrac{9}{10}s,$$

so that $U_S(x,a,s) = W_S(a,s) + 2xs$ and $U_T(x,a,s) = W_T(a,s) + \frac{19}{10}x$. Both political parts vanish at the state $x = 0$ and rise with the state.

Proposition M.3 (Entrenchment against both nations)**.** *At the dual-basin economy:*

(i) At the state $x = 0$, each government's payoff equals its nation's welfare at every action. At every state $x \le 1/4$ the sender's myopic best response to every compliance $a \ge 0$ is no sanction.

(ii) There is an equilibrium in $\mathcal{E}$ such that, from every initial state, after a temporary shock that holds the sanction at 1 for at least $\widehat{N}_F(19/20)$ periods, at every later date the state lies in $[19/20, 1]$, the target maintains full non-compliance and the sanction is 1. In every such period the sender's nation has welfare -1 and the target's $-9/10$, against 0 for both at full trade, so at every discount factor $\beta \in [0,1)$ their discounted welfare is $-1/(1-\beta)$ and $-(9/10)/(1-\beta)$. The sender government's payoff in every such period is $2x_t - 1 \ge 9/10$, above its payoff 0 at full trade. At its own discount factor, the target government's payoff from the state x at which the shock ends is $10x$, the same as under permanent relief from that state.

(iii) Replace the sender's payoff by its nation's welfare W_S, with everything else unchanged. The resulting economy has an equilibrium in $\mathcal{E}$ at which the sanction is zero at every state. Every equilibrium in $\mathcal{E}$ of that economy whose target value rises with the state at exactly rate 10 and is bounded by 19 in absolute value, the set of target values used to construct Proposition K.1, has zero sanctions at every state.

Proof of Proposition M.3. (i) The identities $U_S = W_S + 2xs$ and $U_T = W_T + \frac{19}{10}x$ hold at every state and action, and both political parts are zero at $x = 0$ and non-decreasing in x for $s \ge 0$ (dbU_S_split, dbU_T_split, db_political_parts_zero_at_full_trade, db_political_parts_monotone). The sender's myopic best response is the clipped vertex $\max\{0, \min\{1, 2x - \frac{1}{2} - 2a\}\}$, which is 0 when $x \le 1/4$ and $a \ge 0$ (dbσ_eq_zero_of_le).

(ii) Take an equilibrium of the dual-basin economy whose values lie in the value sets of the high-basin certificate, as for Proposition K.1. After the shock, the resumed path stays in $[19/20, 1]$, where the target's policy is full non-compliance and the realized sanction is 1. Hence the national welfare in each period is $W_S(0,1) = -1$ and $W_T(0,1) = -9/10$, while both are 0 at full trade, and the geometric sums give $-1/(1-\beta)$ and $-(9/10)/(1-\beta)$. The sender government's payoff is

$U_S(x,0,1) = 2x - 1 \geq 9/10$ on the basin, against $U_S(0,0,0) = 0$. Under permanent full sanctions from a state z the state is $1-(1-z)(9/10)^n$, so at $\beta_T = 9/10$ the target government's payoff is

$$\sum_{n\geq 0} \left(\tfrac{9}{10}\right)^n \left[\tfrac{19}{10}\left(1-(1-z)\left(\tfrac{9}{10}\right)^n\right) - \tfrac{9}{10}\right] = \tfrac{19}{10}\left(10-(1-z)\tfrac{100}{19}\right) - 9 = 10z.$$

Under permanent relief from z the state is $z(9/10)^n$, and the payoff is $\sum_{n\geq 0}(9/10)^n \tfrac{19}{10} z(9/10)^n = \tfrac{19}{10} z \cdot \tfrac{100}{19} = 10z$ (`db_entrenchment_loss_after_episode`).

(iii) With U_S replaced by W_S, the sender's stage payoff against full non-compliance, $W_S(0,s) = -\tfrac{1}{2}s^2 - \tfrac{1}{2}s$, does not depend on the state, is 0 at $s=0$ and negative at every $s \in (0,1]$. Against a target policy of full non-compliance at every state, such a sender's value is identically zero and its forward-looking best response is 0 at every state (`dynamicTopRealizedRule_eq_zero_of_stateFree`, a general statement in `Theory/Welfare/RepresentativeSender.lean`). The values $V_S \equiv 0$ and $V_T(x) = 10x$ with full non-compliance and no sanction form an equilibrium in $\mathcal{E}$: $10x = \tfrac{19}{10}x + \tfrac{9}{10}\cdot 10 \cdot \tfrac{9x}{10}$, and the target's objective $10x - a/5$ is maximal at $a=0$ (`dualRepEquilibriumTop`, `dualRep_rule_eq_zero`). For an equilibrium whose target value is $V_T(x) = c + 10x$, the target's objective is $\tfrac{19}{10}x - \tfrac{1}{5}a - \tfrac{9}{10}s + \tfrac{9}{10}\big(c + (9x+s)\big) = 10x - \tfrac{1}{5}a + \tfrac{9}{10}c$: a sanction costs the target exactly what the entrenchment it builds is worth to it, so the objective does not depend on the sender's choice, the target maintains full non-compliance, and the sender side is as above (`dualRep_rule_eq_zero_of_carrier`). □

N Closed-form parametric constants and the assumption-surface floor

The partial-adjustment law. The simplest transition in the class is the partial-adjustment law (J.2). Its state and sanction slopes equal $1-\delta$ and δ at both ends of their bounds, and every curvature modulus is zero. Where the lower and upper bounds on either slope coincide, the transition is of the form (J.2) on the square (Proposition C.9). Under a fixed sanction s, the unique rest point of every transition in the class is $x = s$, because (F1)–(F5) exclude a range of inaction.

Status of the constants in derivative-free form. The derivative-free parametric-argmax results of Klatte and Kummer (2002) and the set-valued analysis of Aubin and Frankowska (1990) provide the appropriate route from *complete-objective* strong concavity and objective-difference bounds to Lipschitz selectors. Stage curvature by itself is not such a route. The complete target-objective modulus c_T, where invoked, is the scalar form of the diagonal strict concavity condition of Rosen (1965).

This appendix lists the closed-form constants entering Assumptions 1 to 3 and the Schur condition (F.3). The paper-facing primitives are the discount and transition objects $\beta_T, \beta_S \in [0,1)$ and $F\colon [0,1]^2 \to [0,1]$ satisfying (F1)–(F5); the stage payoffs $U_T, U_S\colon [0,1]^3 \to \mathbb{R}$ (continuous and twice continuously differentiable on $(0,1)^3$); the basin pivots $\bar{x}_T, \bar{x}_S \in [0,1]$, derived from U_T, U_S via the threshold characterization in Appendix I.5; and the first-derivative bounds $L_{U_T}, L_{U_S}, L_{U_a}$ used in the sanction- and action-sensitivity estimates, defined as suprema of $|\partial U_T/\partial s|$, $|\partial U_S/\partial s|$, and $|\partial U/\partial a|$ on $[0,1]^3$.

The value-function state bound has its own numerator. Let $L^{\alpha}_{S,x}$ bound the inherited-state variation of $U_S(x,\alpha(x),s)$, and let γ_F be the fixed-sanction transition modulus in (F6-Q). In primitive applications both are discharged from derivative and transition bounds; neither is a sanction slope, and under the affine law $\gamma_F = 1-\delta$. The target-side analogue (a closed-loop state

modulus γ_T and the Lipschitz constant $K_T = L^{\mathrm{cl}}_{T,x}/(1-\beta_T\gamma_T)$ of the target's invariant set in the optional target-Lipschitz route) does not enter the existence construction for the class under the lifting condition, whose invariant set for the target is a norm ball with a lower-slope floor and no Lipschitz clause. It is instantiated on that class after the fact: with γ_T the sharp loop constant $g_{\mathrm{hi}} + d_{\mathrm{hi}}L_{\mathrm{resp}}/\mu$ and $L_U = \max\{v_{\mathrm{hi}}, v_{\mathrm{hi}} + n_{\mathrm{hi}}\} + cL_{\mathrm{resp}}/\mu$ ($v + n + cL_{\mathrm{resp}}/\mu$ on the quadratic–linear family), the equilibrium target value is $K_T = L_U/(1-\beta_T\gamma_T)$-Lipschitz whenever $\beta_T\gamma_T < 1$ (`VTstar_lipschitz`, `Examples/PlacableSenderTargetLipschitz.lean`).

N.1 Derived and separately verified constants

The following list distinguishes algebraically derived bounds from positive curvature or selector moduli that must be verified where used:

- $K_S = L^{\alpha}_{S,x}/(1-\beta_S\gamma_F)$, closed-form Lipschitz constant of the sender's invariant set (derived from $V_S \in \mathrm{Lip}_{K_S}$ by Banach iteration under $\beta_S\gamma_F < 1$; the recursion is Hinderer (2005, Thm. 4.2(c)) and its inheritance by the invariant set is Corollary 1 to Theorem 3.2 of Stokey et al. (1989)).

- $c_\sigma = \inf_{[0,1]^3}\left(-\partial^2 U_S/\partial s^2\right) > 0$, the stage strong-concavity modulus imposed in Assumption 3. It controls the static sender problem only.

- No general dynamic-selector modulus L_σ is derived from c_σ. Whenever an existence theorem uses a Lipschitz dynamic sender selector, its named analytic package must instead verify strong concavity of the complete objective J_S and the relevant parametric objective-difference bound (Klatte and Kummer, 2002; Doraszelski and Pakes, 2007).

- $K^{\mathrm{IFT}}_T, K^{\mathrm{IFT}}_S \geq 0$ objective-difference moduli of the joint target objective in the value-pair coordinates: a sup-norm perturbation of size ε in V_T (respectively V_S) changes the objective's variation between two actions by at most $K^{\mathrm{IFT}}\varepsilon$ times the action distance. This is the objective-difference input of `argmax_lipschitz_of_strongConcave_lipschitzDiff`, verified by each analytic package that uses it. It is not a finite-dimensional Hessian bound, and smoothness of the stage payoff on a neighborhood of the cube does not supply it: a uniformly strongly concave objective can have a maximizer that is not Lipschitz in sup-norm perturbations of the objective.

- $c_T > 0$ joint-target complete-objective strong-concavity modulus in a, verified as part of the leader-regularity package wherever it is used; continuity of second derivatives alone does not imply its positivity.

- $L_{\alpha,V_T} = K^{\mathrm{IFT}}_T/c_T$, $L_{\alpha,V_S} = K^{\mathrm{IFT}}_S/c_T$, Implicit-Function Lipschitz constants of the target equilibrium policy in the value-pair coordinates. The canonical derivation from C^2 target stage payoffs plus an Arrow-Hurwicz-Uzawa constraint qualification (without an interiority hypothesis on the optimal compliance choice) is Rincón-Zapatero and Santos (2009), which proves continuous differentiability of the value function and uniqueness of the KKT multipliers; the IFT gradient strong-monotonicity in derivative-free form taken as a primitive here is the gradient-monotonicity content of this uniqueness statement. The encyclopedic reference for the parametric-argmax Lipschitz stability results from which this derivative-free form descends is Klatte and Kummer (2002, Ch. 2 + Ch. 7).

Every parametric constant above is either (a) algebraically derived from verified moduli via a closed-form expression (e.g. K_S, L_{α,V_*}), or (b) a bound verified from the complete objective by

the analytic package that uses it (the curvature moduli c_σ, c_T and the objective-difference moduli K_T^{IFT}, K_S^{IFT}); where such a bound is read off a smooth stage payoff, the derivative bound must be verified on the closed cube, since differentiability on the open cube does not bound curvature at the boundary. The positive curvature moduli c_σ and c_T must be verified from the corresponding stage or complete objective; compactness only guarantees that a continuous curvature bound attains its infimum, not that the infimum is positive. The optional moduli $L^{\alpha}_{S,x}, \gamma_F$ must themselves be discharged from the specified payoff, transition, and selector bounds; they are not silently identified with sanction derivatives. The structural assumptions Assumptions 1 to 3 are qualitative; they enter the paper-facing surface as conditions on the payoff functions and the basin, not as numerical primitives.

Table 2. Derived and separately verified intermediates.

Symbol	Closed form / interpretation
K_S	$L^{\alpha}_{S,x}/(1-\beta_S\gamma_F)$, Lipschitz constant of the Sender's invariant set under (F6-Q) and $\beta_S\gamma_F < 1$; under the affine law, $\gamma_F = 1-\delta$.
L_σ	Optional dynamic Sender-selector modulus; available only from complete-objective strong concavity plus a parametric objective-difference bound, not from A3 alone.
$K_T^{\mathrm{IFT}}, K_S^{\mathrm{IFT}}$	Cross-Hessian bounds on the joint Target integrand.
$L_{\alpha,V_T}, L_{\alpha,V_S}$	K_T^{IFT}/c_T, K_S^{IFT}/c_T, IFT Lipschitz of α in (V_T, V_S).
L_{TS}, L_{ST}	Row-Lipschitz off-diagonals of the joint Bellman operator (Section 3).
R_T, R_S	Sup-norm bounds on the balls of the invariant sets; the smallest invariant radii are $\|U_T\|_\infty/(1-\beta_T)$ and $\|U_S\|_\infty/(1-\beta_S)$ (Appendix F.2).
$\bar{x}_T, \bar{x}_S$	High-basin tipping thresholds: infima of the flip sets (G.2)–(G.3) (Proposition G.1); $\bar{x} = \max(\bar{x}_T, \bar{x}_S)$.
x_S^0	Static Sender flip ($m = 0$ level); contested band $[x_S^0, \bar{x}_S)$ (§G.1).
$\bar{x}_T^*, \bar{x}_S^*$	Low-basin ceilings: suprema of the dual flip sets (§D.4); $\bar{x}_* = \min(\bar{x}_T^*, \bar{x}_S^*)$.
$d_F(x), \eta_F(\bar{x})$	Full-sanction drift $d_F(x) = F(x,1) - x$ and its positive minimum on $[0, \bar{x}]$.
$\widehat{N}_F(\bar{x})$	Explicit general-transition bound $\lceil \bar{x}/\eta_F(\bar{x}) \rceil$ on the worst-case shock-entry duration under (F1)–(F4) and full-sanction ascent on $[0, \bar{x}]$.
$\eta_\delta(\bar{x}), \widehat{N}_\delta(\bar{x})$	$\delta(1-\bar{x})$ and $\lceil \bar{x}/[\delta(1-\bar{x})] \rceil$, the affine specialization of the general minimum-drift bound. The sharper affine choice is $N_*(\bar{x}, \delta) = \lceil \log(1-\bar{x})/\log(1-\delta) \rceil$ (Theorem D.8 and Corollary I.4).

The notation choice R_T, R_S for the sup-norm bounds of the invariant sets (in place of the older M_T, M_S) avoids collision with $M = (\alpha^M, \sigma^M)$, the myopic equilibrium (Definition 1).

N.2 Certified comparative-static windows

The full-adjustment comparative statics at the corner economies hold on exact parameter intervals rather than only at isolated calibration points. Table 5 records the intervals and the constraint that binds at each edge.

At the certified equilibria the value functions also have exact state derivatives, the shadow prices of political entrenchment. They are 82/11 for the Target and 16/3 for the Sender at the Eaton–Engers economy, and 80/11 and 4/3 at the interest-group economy. Each obeys the boundary-envelope recursion

$$\text{slope} = U_x + \beta(1-\delta)\,\text{slope}.$$

The positive signs certify that the equilibrium incumbents value a harder political state, while the exact magnitudes are example-specific rather than general predictions.

Parameter	Certified window	What binds at the endpoints
Eaton–Engers economy		
Sender patience β_S	$[6/13,\ 4/7)$	The continuation drag must cover the worst stage drop at the lower edge and remain below the strict foresight premium at the upper edge.
Target patience β_T	$[598/709,\ 1)$	The Target entrenchment condition binds at the lower edge; the upper edge is the discount bound.
Adjustment speed δ	$[3/7,\ 2/3)$	The Sender corner pin binds below and the strict foresight premium binds above.
Interest-group economy		
Sender patience β_S	$[0,\ 1)$	The whole range is certified because the defiance-corner stage slope is already positive.
Target patience β_T	$[14/17,\ 1)$	The Target condition and the entrenchment condition bind at the lower edge.
Adjustment speed δ	$[7/27,\ 1)$	The Target condition and the invariant-set membership condition bind together at the lower edge.

Table 5: Certified parameter windows for the full-adjustment comparative statics. One deep parameter varies in each row while the other primitives remain at their example values. Closed endpoints correspond to weak conditions; open endpoints to strict conditions or the discount bound.

Part III
Literature

O Related literature in detail

This appendix gives the full positioning summarized in Section 1: the public-choice analysis of sanctions, endogenous political power and conflict, hysteresis, the closest antecedent, and the analytical methods.

This paper connects the public-choice analysis of sanctions with work on endogenous political power, dynamic commitment, and hysteresis. Its distinctive object is a slow target-side political state that sanctions change and both governments value.

O.1 Public choice, sanctions, and duration

The public-choice starting point is Kaempfer and Lowenberg (1988), who explain sanctions through the political pressure and effectiveness of organized interest groups. Halcoussis et al. (2021) emphasize that sanctions may retain domestic political functions after their direct economic leverage has diminished, while Galtung (1967) supplies the classic rally mechanism and Eaton and Engers (1992) study sanctions effectiveness without an endogenous political state. We make this political feedback dynamic and strategic: the target chooses compliance, the sender chooses sanctions, and both choices are derived from continuation values generated by the same political state.

The four-case classification is what separates this paper from the two lines it builds on. Eaton and Engers (1992) already have both parties discounting, with possibly different discount factors, under complete information, and derive the outcome from relative toughness: a party's patience together with its own cost of bearing sanctions; in their once-and-for-all case a more patient sender

can even lengthen the delay before compliance. The payoff-relevant state there is the rival's standing choice and there is no political state, so patience changes what is extracted and how long a confrontation lasts. In the public-choice line, Kaempfer and Lowenberg (1988) let the sanction level clear a static political market among interest groups, and Kaempfer et al. (2004) study a dictator's choice of power, loyalty and repression under sanctions; here both governments' choices are derived from continuation values over a political state that the sanction itself moves. The four cases pose a question of their own: holding non-discount primitives fixed, which government's forward-looking incentive produces non-compliance, and which changes its price. Our answer is that the target's forward-looking incentive decides between accommodation and non-compliance, while the sender's decides between extracting compliance and sustaining entrenchment.

The model characterizes how long sanctions last once imposed. Its qualitative prediction is that, after a temporary sanction shock carries the state across the political pivot, sanctions do not fall within the equilibrium basin. This focus complements empirical work on sanction duration and termination (Bolks and Al-Sowayel, 2000; Attia et al., 2020), including the sender–target termination dataset of Attia and Grauvogel (2023), in which gradual lifting is common.

Two empirical findings delimit the political setting the model describes. Ahn and Ludema (2020) find that a targeted regime shields the firms it values, absorbing the sanction's cost through procurement terms so that the burden falls elsewhere; we keep the government's political objective, the constituencies it protects, and aggregate welfare as separate objects for that reason, and the interest-group polity of Appendix L does so explicitly. Escribà-Folch and Wright (2010) find that sanctions destabilize personalist rulers, who lose the external revenue that funds patronage, while single-party and military regimes absorb them through taxation, reallocated spending and repression. The target's positive state valuation $v_{\text{lo}} > 0$ of Section 3.1 describes the second kind of setting, in which an incumbent can absorb the sanction and the political state x measures the power of its anti-concession constituencies. The sender's stake $w_{\text{lo}} > 0$ is a separate restriction, on the sender's own constituencies, motivated by the interest-group literature above.

O.2 Political persistence and strategic investment

The political state is closest in interpretation to de facto power in Acemoglu and Robinson (2008): influence that formal institutions do not allocate, which there is a per-period investment by the group that holds it (Acemoglu and Robinson, 2008, p. 272) and here a slow stock moved by the sender's sanctions. It is also close to models in which current choices shape future political power: extending the franchise changes the future median voter and asset redistribution can consolidate democracy (Acemoglu and Robinson, 2001), current decisions determine the future distribution of political power (Acemoglu et al., 2012), institutional states chosen by current power holders are shifted by shocks (Acemoglu et al., 2015), and policy shapes future power (Bai and Lagunoff, 2011). The difference is that two governments, with separate discount factors, act on the same slow target-side state. Two of these papers name, as directions for future work, "capital-like state variables so that some subcomponents of the state have autonomous dynamics" (Acemoglu et al., 2012, Section VI) and relations between countries (Acemoglu et al., 2015, concluding section). The present model takes up a case of both: a political state with its own law of motion that another country's instrument moves. The model shares with the commitment-problem literature (Powell, 2006) the feature that current actions shift future political power in the absence of commitment. There the parties bargain over transfers and the settlement's enforcement fails; here each government sets its own instrument, compliance or the sanction, against a political state that the sanction moves. The model's dual-basin mechanism is related to Hassler et al. (2003). The political state here is itself shaped by forward-looking strategic choices.

The closest conflict-theoretic benchmark is the information-centered framework of Acemoglu and Wolitzky (2024), in which mistrust, misperception and misunderstanding drive conflict in a security dilemma, its repeated version, and a deterrence model. Two comparisons place the present paper. The first is what carries incentives across dates. In their security dilemma and its repeated version it is a belief about the opponent's type; the repeated version assumes infinitely impatient players to rule out dynamic incentives from strategic experimentation, and names forward-looking leadership as a direction for future research. In the dynamic deterrence result they report from Wolitzky (2023, Theorem 2) it is the threat of switching to a worse equilibrium after a publicly observed signal, which requires sufficiently patient players. Here it is an observed, payoff-relevant political state under complete information and perfect monitoring, and the results compare constructed equilibria across the four cases to isolate what each government's forward-looking incentive does. The second is the stage structure, and here the benchmark shares our logic: in their deterrence model with a perfectly observed action the first mover's hawkish action is the better reply to any fixed response, and yet the anticipated response induces restraint; likewise, in the class under the lifting condition, compliance is costly at a fixed sanction and the sender's anticipated response makes accommodation the target's unique optimum in the one-period problem. The difference is that the sender ranks the two diagonal outcomes the other way, so the two governments do not share a preference for the restrained outcome (Appendix O.4), and that both internalize an observed political state which sanctions move. The constructed forward-looking equilibrium reverses the accommodation of the one-period problem for that reason. In Acemoglu and Wolitzky (2014) conflict spirals end from inside: Bayesian players with limited memory come to treat a long spiral as uninformative about the opponent's type and experiment with cooperation. Here information is complete, and what resumed equilibrium play does after a temporary shock depends on the state the shock leaves behind.

A separate literature shows that valuing the future can sustain conflict under complete information. In Skaperdas and Syropoulos (1996) and Garfinkel and Skaperdas (2000), parties who would cooperate or settle in a one-shot interaction arm or fight once today's outcome determines tomorrow's resources. Powell (2004) gives a condition on continuation security values under which no equilibrium of a stochastic game is efficient; in Powell (2012) sufficiently patient factions fight in every period to forestall an adverse shift in power, and in Krainin and Wiseman (2016) enough patience makes war occur in every subgame perfect equilibrium. Powell (2013) makes the distribution of power a continuous state chosen by the government. We share the premise that the one-shot interaction ends in accommodation and the dynamic one need not. Three things differ. Those models give the players a common discount factor, or none, and where the players do differ in patience (the long-lived litigant facing one-period challengers of McBride and Skaperdas (2009, Appendix B)), only one party's patience is varied. The closest antecedents are Eaton and Engers (1992), who classify outcomes by the target's cost of sanctions and its discount factor across Markov-perfect steady states, and Eaton and Engers (1999), who report that a more patient sender is more likely to succeed while the target's patience works against it; in both the sender is committed between reviews, which is why their sanctions are never suffered in equilibrium. Feedback Stackelberg games also have existence results organized by patience: Bressan and Jiang (2020) under a far-sighted leader or a narrow-sighted follower, and Bucarey López et al. (2023) re-solved at three corners of the discount factors. What is new here is the object: our sender re-optimizes every period against the political state its own sanction has built, which is why sanctions are imposed and persist. That greater patience can deepen a political distortion is published: investment in de facto power rises with the common discount factor (Acemoglu and Robinson, 2008, Proposition 2), populist bias rises with the politicians' common discount factor (Acemoglu et al., 2013, Proposition 12), patience makes a Pareto-inefficient state stable (Acemoglu et al., 2012, Example 1), and sufficiently patient factions

fight (Powell, 2012). A common discount factor can also act non-monotonically, as on whether a society experiments with institutions (Acemoglu et al., 2015, Proposition 7). What is new here is that the two governments' discount factors are separate and do different things: more target patience moves the constructed equilibrium from relief to non-compliance, while a forward-looking sender, against a myopic one, raises the price of relief on one side and the intensity of conflict on the other. Sanctions theory has a standing objection. Under complete information and divisible demands the target concedes at the threat stage, so imposed sanctions are instances where bargaining failed or information was incomplete (Drezner, 2003); the same conclusion is an implication of the binary model of Eaton and Engers (1992). The answer is what is proved. At identical primitives, under stated conditions and with complete information throughout, the realized sanction in the constructed stationary equilibria is zero in the two cases with a myopic target and positive in the two with a forward-looking one. The reason is that the sender has a direct benefit from sanctioning and the sanction builds entrenchment that the forward-looking target values more than the relief it forgoes. The premise that fails is that sanctions are a deadweight loss to both. The state here is political power within the target polity, moved by the coercive instrument itself and valued by both incumbents. Under the conditions of Proposition C.8, neither incumbent gains from an unconditional temporary suspension of sanctions followed by the same full-non-compliance equilibrium. The condition has the form of the participation conditions of the conflict literature (Garfinkel and Skaperdas, 2000; McBride and Skaperdas, 2009), and it presupposes transferable utility when read as a side payment; it concerns uncompensated relief. Compliance-contingent relief lies inside the class through the lifting condition (compliance ends the sanction) and resembles negotiated agreements of the kind the 2015 Iran arrangement provided. The relief of the proposition is unconditional and set exogenously. This places it closer to Acemoglu et al. (2010), where a civilian elite keeps the army too small to end a civil war, because an army strong enough to end it could stage a coup once the war is over and the elite cannot commit not to downsize it.

Relative to classic hysteresis and lock-in models (Dixit, 1989; Krugman, 1991; Arthur, 1989), the slow state is moved by optimizing agents in all four settings. In those models persistence comes from a sunk cost, an externality or an increasing return. Here two strategic governments move the state, one of them choosing the state-moving instrument while internalizing the other's response, so that persistence is an equilibrium property of a game. The dual-basin picture itself is known: Krugman's real-root case, in which initial conditions alone select the long-run outcome; the equilibrium of Hassler et al. (2003) with two basins inside one equilibrium, built by splicing two rules; and the stable-state mapping of Acemoglu et al. (2012), which sends each initial state to the state it settles in. What the paper adds is the state space and the instrument: the basins here are basins of a political state moved by a coercive instrument that one government chooses against the other's response.

History operates through that inherited state while the equilibrium rule is held fixed. The dual-basin result therefore differs from expectations-driven equilibrium selection: it varies the initial state and holds fixed the equilibrium on which the governments coordinate. The distinction also separates the paper's path dependence from models in which several Markov equilibria coexist.

O.3 Closest antecedent and analytical method

The political-entrenchment motivation is also related to the earlier working paper by Gong and Zhong (2025). That paper studies a target-only dynamic program with a reduced-form sanctions regime, reports comparative statics in the target's patience among other parameters, and supplements the model with sanction-duration evidence. With one decision maker it cannot ask what the sender's patience does. The present paper makes the sender a forward-looking player with its own

discount factor and solves the two-government feedback Stackelberg game. It obtains the separate roles of the two discount factors and the threshold discount factor under the nine tightness conditions, which make the transition affine and yield a threshold independent of the sender's patience. It adds the high-basin and shock results, the welfare analysis, and the formal verification. The papers share a public-choice motivation and the partial-adjustment law as a leading example; they have distinct decision problems, empirical content, and results.

Methodologically, the paper combines a public-choice microfoundation, a feedback Stackelberg Bellman system, and order methods for equilibrium policies and basin dynamics (Başar and Olsder, 1999; Krawczyk and Petkov, 2018; Stokey et al., 1989; Curtat, 1996; Bertsekas, 2012; Topkis, 1998; Milgrom and Shannon, 1994; Quah and Strulovici, 2009; Krishnamurthy, 2023). Interval dominance delivers a monotone greatest response in cases not covered by classical single crossing. The analysis combines that direction result with a separate sanction-level condition, because monotonicity of a decision rule alone does not imply an upward political path.

Monotone iteration and extremal steady states. The order-theoretic step behind Proposition I.1 is classical. Iteration of an increasing map from a lower solution converges to the minimal fixed point above it, and from an upper solution to the maximal fixed point below it (Amann, 1976, Theorem 6.1 and Corollary 6.2); the statement we use is the one-dimensional case, where continuity on a compact interval supplies the compactness his theorem assumes. The paper connects this classical step to a forward-looking political equilibrium, derives the relevant invariance and monotonicity conditions from equilibrium restrictions, and separates convergence of the political state from convergence of the equilibrium sanction. Terminology for the economic sense of stability follows Stokey et al. (1989).

The closest primitive-level comparison is Huang et al. (2025), who construct a locator function whose sign determines the direction of motion, read steady states and their stability off it, and obtain convergence of every path to a common limit when the locator is single-crossing from above. Their setting is single-agent dynamic optimization, in which an envelope step produces the locator. Carrying that step to a Stackelberg game with endogenous compliance and a general transition is open, and it is the natural route to a primitive-level strengthening.

O.4 Conflict under complete information

In this model, conflict is sustained by a political state that the conflict moves and both governments value, under complete information and perfect monitoring.[2] In the information-based account, misperception and misunderstanding, with mistrust, play central roles in the onset, continuation and cessation of conflict (Acemoglu and Wolitzky, 2024). In Acemoglu and Wolitzky (2014), spirals that misperceptions start end from inside, as Bayesian players with limited memory come to treat a long spiral as uninformative. Table 6 compares the two accounts interpretively.

The political state that sustains sanctions can also end them. A temporary suspension of sanctions lowers the state, and in any economy a resumed sanction rule that stays at or below the state after the shock delivers durable de-escalation (Section 3.3). No belief is revised: the conflict ends because the state has moved.

Along deterministic equilibrium paths, the political-state mechanism also produces no cycles under a sanction rule that does not fall with the state. State and sanction paths move in one direction and converge, and a path that returns to a state stays there. The result covers the

[2]The absence of cycles, persistence after a suspension of sanctions and the comparison with history-dependent strategies are proved in Appendices H and I and Appendix C.8.

Table 6: Two accounts of conflict dynamics. The left column paraphrases the information-based account; the right column summarizes results of this paper at their stated scope.

	Information-based account	Political-state mechanism (this paper)
What links dates	A belief about the opponent's type, or a threat to switch equilibria	An observed political state that the conflict moves
What sustains conflict	Mistrust, misperception, misunderstanding	A sender that gains from sanctioning a non-compliant target, and a forward-looking target that values the entrenchment sanctions build at least as much as the relief it forgoes (Theorem 3); under the strong entrenchment condition threats cannot coerce it (Section 3.4)
What ends it	Belief revision: a long spiral comes to look uninformative, and players experiment with cooperation (Acemoglu and Wolitzky, 2014)	A temporary suspension of sanctions after which the resumed sanction rule stays at or below the state (Section 3.3); in the constructed equilibria of the four cases, no sanction arises when the target does not value the future state (Theorem 12)
Cycles	Recurring spirals, started by misperceptions, under limited memory (Acemoglu and Wolitzky, 2014); traps or eventual peace under other information structures (Acemoglu and Wolitzky, 2024, Section 4)	None along deterministic equilibrium paths under a sanction rule that does not fall with the state
What would have to differ	Mistrust, misperception, misunderstanding, for instance through mediation (Acemoglu and Wolitzky, 2024, Section 4.4)	The state, when the resumed sanction rule keeps sanctions at or below it after a suspension of sanctions; in the special case of Proposition 6, the target's patience and valuation of the state; the sender's stake changes the sanction level under non-compliance (Section 3.2)

constructed equilibria of the four cases and every equilibrium in $\mathcal{E}$ whose sanction rule is non-decreasing, on any economy. Within the model, a cycle along an equilibrium path needs a sanction rule that falls with the state somewhere. The cycles of Acemoglu and Wolitzky (2014), spirals that misperceived actions start, end and restart, come from noise and limited memory.

Read as comparative statics, the results say what would have to differ for sanctions to end under the political-state mechanism. With information already complete, the levers are the state and the target's incentives. Under the strong entrenchment condition, threats cannot coerce the target, since relief is the harshest treatment the sender can impose (Section 3.4). A temporary suspension of sanctions has a lasting effect if it leaves the state in a basin where resumed equilibrium play keeps sanctions at zero. It has no effect on the long-run limit where the equilibrium law of motion has an explicit Lipschitz constant below one. In the special case of Proposition 6, the target's patience, with its valuation of the state, its cost and its sanction pain, decides whether a full-non-compliance equilibrium exists. The sender's stake raises the sanction under non-compliance above the myopic sanction (Section 3.2), and non-compliance persists even with a myopic sender (case 10 of Theorem 12).

Timing and information in the sanctions game

The game is a sequential game of complete information. Each period the sender observes the target's compliance before responding, and both observe the political state. Two questions relate it to models of conflict, the first being current action incentives. On the class, at any fixed sanction more compliance strictly lowers the target's payoff, so in payoffs non-compliance is its better reply, as the hawkish action is for the first mover of a deterrence game (Acemoglu and Wolitzky, 2024). What pays for compliance is the sender's response. In the myopic equilibrium the target's unique optimum is the least compliance that ends the myopic sanction (Proposition 2). On the quadratic–linear family (Appendix J.1), the target strictly prefers this accommodation to non-compliance met by the myopic sanction, while the sender ranks those two outcomes the other way. On the family, the two governments therefore have opposed interests over accommodation.

The second question is what carries incentives across dates: the political state that the instrument itself moves. In the constructed forward-looking equilibrium the target's non-compliance and the sender's sanction both maximize objectives that include the state's continuation value, so a credible current incentive to accommodate coexists with dynamic non-compliance. The governments' only actions are compliance and the sanction. Against each stationary equilibrium studied here, no pure history-dependent deviation is profitable, nor is any finite mixture of them.

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